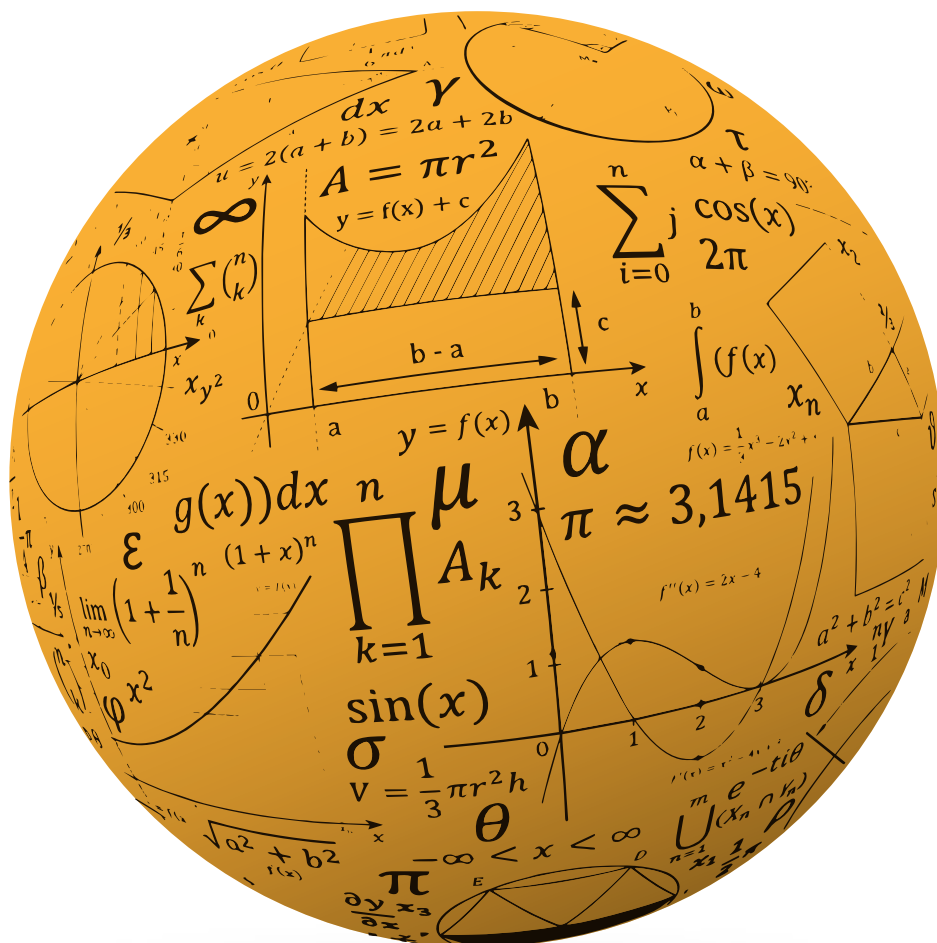


Vedantu's Tatva Practice Book

Mathematics

Volume 2

Class 12 **JEE**



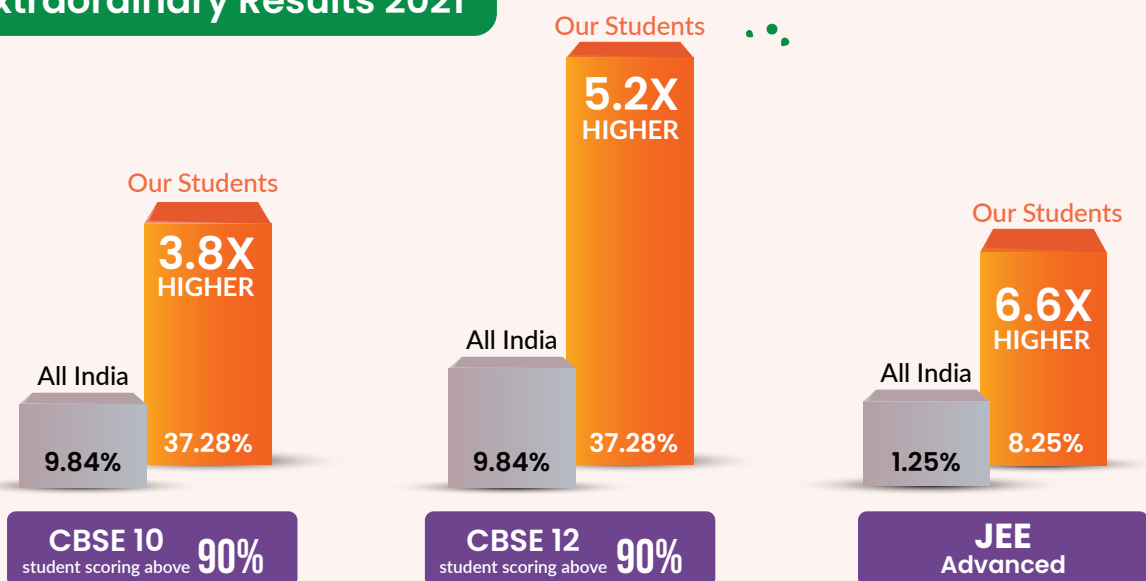
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Class 12 **JEE**



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How to use your Tatva Practice Book

4. Fundamental Principle of Counting :

If an event can occur in 'm' different ways following which another event can occur in 'n' different ways following which another event can occur in 'p' different ways then



SCAN CODE
Permutations

1

Scan the QR Code in each chapter's theory section to view micro concept videos related to the chapter, on the Vedantu app.

2

Solve all types of exercise questions based on the latest JEE pattern.

Exercise - 2:

1. If every element of a third order determinant of value Δ is multiplied by 5, then the value of new determinant is:

(JEE 2019)

- (a) Δ (b) 5Δ
(c) 25Δ (d) 125Δ

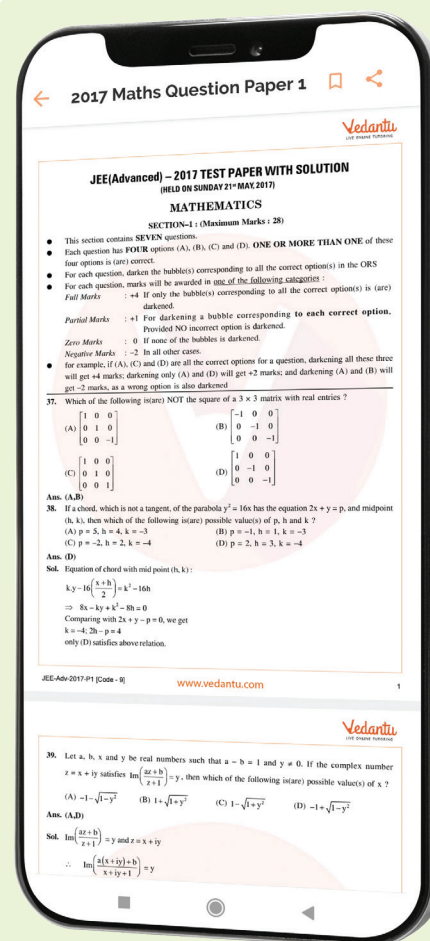
Answer Key

CHAPTER-1: DETERMINANTS
Exercise-1: Basic Objective



3

Scan the QR code in the Answers Section to view detailed solutions for all exercise questions.



For extra exam preparation content, visit the Vedantu app. You can find previous years' JEE papers with solutions and detailed analysis.

Founder's Message

Dear Student,

I am delighted to present to you a Ready Reckoner and an amazing book to guide you for your exams-'TATVA'. Tatva—which means the 'Core' is fully aligned with the culture, the mission, and the vision of Vedantu and therefore it gives me immense pleasure and joy to share this book with you. We at Vedantu have always believed in revolutionizing the teaching and learning process and always speedily progressed in the direction of bringing superior quality education and content to your table. Tatva is a step forward in this direction. This book is your guide, your practice guru, and your companion in moving towards your dreams. The book is a result of the consistent effort, diligence, and research by our experienced team of subject experts and teachers.

This book has been customized with curated content to suit the needs of JEE aspirants like you and guide you on the right path to cracking JEE and optimizing your efficiency. Tatva is a comprehensive amalgamation of important concepts, theories, derivations, definitions, solved examples, concept videos, practice questions, and important questions. We have ensured that high-quality content and the right ingredients are in place in this booklet to help you climb up the success ladder.

A few guiding points to optimally use Tatva with a planned approach:

Tatva equips you with Theory, Concept Videos, and Solved examples to help you revise concepts, mark your notes, walk you through the entire summary, and eventually makes you capable of clearing all your conceptual doubts all by yourself.

We suggest revision of theory followed by practice of solved examples.

Practice relevant questions daily after finishing Vedantu lectures and session assignments. We believe that a daily dose of Tatva will keep all your exam blues at bay.

Use the Tatva booklet to mark notes so that it always comes in handy for last-minute revision sessions before your exams. Notes should include key points of theory, solved examples, and some questions which you couldn't solve in the first attempt.

Exercise 1 and Exercise 2 of JEE Tatva deal with basic questions and those which can be asked or already asked in JEE Main. Similarly, Exercise 3 and Exercise 4 deal with JEE Advanced level questions. We recommend you solve basic JEE Main questions before moving to JEE Advanced level questions.

Before wrapping up, the practice mantra: "Don't practice until you get it right. Practice until you can't get it wrong."

We strongly believe in you and your capabilities. So believe in yourself because success is only one step away. Wishing that your talent shines bright. All the very best!

Anand Prakash

Founder and Academic Head, Vedantu



Anand Prakash Sir has been a pioneer in producing Top Ranks in JEE/NEET and Olympiads. He has personally taught and mentored AIR 1, 6, 7 (JEE Advanced), AIR-1, 7, 9(AIIMS), and thousands of more students who have successfully cleared these competitive exams in the last few years.

Credits

*“Happiness lies in the joy of achievement
and the thrill of creative effort.”*

—Franklin D. Roosevelt

Tatva is the brainchild of a group of creative Vedans who have strived tirelessly to weave success stories for you. We extend our heartfelt gratitude to the superb team of Vedans who give wings to the vision of Vedantu, starting with our leaders who have been guiding and encouraging us at every step of the way:

Vamsi Krishna Sir, Anand Prakash Sir and Pulkit Jain Sir

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Mathematics Team

We truly appreciate all the Master Teachers of Vedantu whose relentless efforts helped us translate this vision into reality. Our heartfelt gratitude to our creative content developers and the typesetting team, who have put in their hard work, insight, and eagerness to nurture and execute Tatva into ‘your ready handbook’ and bring a positive learning experience to you.

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The journey of bringing Tatva to life, from an idea to the book you are holding, would not have been possible without the extensive support of our diligent Operations Team, our amazing Academic Team, our dedicated team of Teachers, and our talented Tech Team.

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INDEFINITE INTEGRATION

INDEFINITE INTEGRATION



Integration is the inverse process of differentiation. That is, the process of finding a function, whose differential coefficient is known, is called integration.

If the differential coefficient of $F(x)$ is $f(x)$,

i.e. $\frac{d}{dx}[F(x)] = f(x)$, then we say that the **antiderivative** or

integral of $f(x)$ is $F(x)$, written as $\int f(x)dx = F(x)$,

Here $\int dx$ is the notation of integration, $f(x)$ is the integrand, x is the variable of integration and dx denotes the integration with respect to x .

1. INDEFINITE INTEGRAL

1.1 Definition

We know that if $\frac{d}{dx}[F(x)] = f(x)$, then $\int f(x)dx = F(x)$.

Also, for any arbitrary constant C ,

$$\frac{d}{dx}[F(x) + C] = \frac{d}{dx}[F(x)] + 0 = f(x).$$

$$\therefore \int f(x)dx = F(x) + C,$$

This shows that $F(x)$ and $F(x) + C$ are both integrals of the same function $f(x)$. Thus, for different values of C , we obtain different integrals of $f(x)$. This implies that the integral of $f(x)$ is not definite. By virtue of this property $F(x)$ is called the indefinite integral of $f(x)$.

1.2 Properties of Indefinite Integration

$$1. \quad \frac{d}{dx} \left[\int f(x)dx \right] = f(x)$$

$$2. \quad \int f'(x)dx = \int \frac{d}{dx}[f(x)]dx = f(x) + c$$

$$3. \quad \int k f(x)dx = k \int f(x)dx, \text{ where } k \text{ is any constant}$$

4. If $f_1(x), f_2(x), f_3(x), \dots$ (finite in number) are functions of x , then

$$\int [f_1(x) \pm f_2(x) \pm f_3(x) \dots]dx$$

$$= \int f_1(x)dx \pm \int f_2(x)dx \pm \int f_3(x)dx \pm \dots$$

5. If $\int f(x)dx = F(x) + c$

$$\text{then } \int f(ax \pm b)dx = \frac{1}{a}F(ax \pm b) + c$$

6. Suppose I and J are intervals, $g: J \rightarrow I$ is differentiable and $f: I \rightarrow \mathbb{R}$ has integral with primitive F . Then $(f \circ g).g' : J \rightarrow \mathbb{R}$ has an integral and

$$\int ((f \circ g).g')(x)dx = \int f(g(x))g'(x)dx = F(g(x)) + c$$

1.3 Standard Formulae of Integration

The following results are a direct consequence of the definition of an integral.

$$1. \quad \int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1.$$

$$2. \quad \int \frac{1}{x} dx = \log |x| + C$$

$$3. \quad \int e^x dx = e^x + C$$

$$4. \quad \int a^x dx = \frac{a^x}{\log_e a} + C.$$

$$5. \quad \int \sin x dx = -\cos x + C$$

$$6. \quad \int \cos x dx = \sin x + C$$

$$7. \quad \int \sec^2 x dx = \tan x + C$$

$$8. \quad \int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$9. \quad \int \sec x \tan x dx = \sec x + C$$



$$10. \int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + C.$$

$$11. \int \tan x \, dx = -\log |\cos x| + C = \log |\sec x| + C.$$

$$12. \int \cot x \, dx = \log |\sin x| + C$$

$$13. \int \sec x \, dx = \log |\sec x + \tan x| + C$$

$$14. \int \operatorname{cosec} x \, dx = \log |\operatorname{cosec} x - \cot x| + C$$

$$15. \int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C ; |x| < 1$$

$$16. \int \frac{dx}{1+x^2} = \tan^{-1} x + C$$

$$17. \int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} |x| + C ; |x| > 1$$

$$= \frac{1}{n} f(t) = \frac{1}{n} f(x^n) + c$$

(iii) When the integrand is of the form $[f(x)]^n \cdot f'(x)$, we put $f(x) = t$ and $f'(x) dx = dt$.

$$\text{Thus, } \int [f(x)]^n f'(x) dx = \int t^n dt = \frac{t^{n+1}}{n+1} = \frac{[f(x)]^{n+1}}{n+1} + c$$

(iv) When the integrand is of the form $\frac{f'(x)}{f(x)}$, we put $f(x) = t$ and $f'(x) dx = dt$.

$$\text{Thus, } \int \frac{f'(x)}{f(x)} dx = \int \frac{dt}{t} = \log |t| = \log |f(x)| + c$$

2.2 Some Special Integrals

$$1. \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$2. \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$$

$$3. \int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C$$

$$4. \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + C$$

$$5. \int \frac{dx}{\sqrt{x^2 + a^2}} = \log \left| x + \sqrt{x^2 + a^2} \right| + C$$

$$6. \int \frac{dx}{\sqrt{x^2 - a^2}} = \log \left| x + \sqrt{x^2 - a^2} \right| + C$$

$$7. \int \sqrt{a^2 - x^2} \, dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$$

$$8. \int \sqrt{x^2 + a^2} \, dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log \left| x + \sqrt{x^2 + a^2} \right| + C$$

$$9. \int \sqrt{x^2 - a^2} \, dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log \left| x + \sqrt{x^2 - a^2} \right| + C$$

2. INTEGRATION BY SUBSTITUTION

2.1 Method of Substitution

By suitable substitution, the variable x in $\int f(x) dx$ is changed into another variable t so that the integrand $f(x)$ is changed into $F(t)$ which is some standard integral or algebraic sum of standard integrals.

There is no general rule for finding a proper substitution and the best guide in this matter is experience.

However, the following suggestions will prove useful.

(i) If the integrand is of the form $f'(ax+b)$, then we put

$$ax + b = t \text{ and } dx = \frac{1}{a} dt.$$

$$\text{Thus, } \int f'(ax+b) dx = \int f'(t) \frac{dt}{a}$$

$$= \frac{1}{a} \int f(t) dt = \frac{f(t)}{a} = \frac{f(ax+b)}{a} + c$$

(ii) When the integrand is of the form $x^{n-1} f'(x^n)$, we put $x^n = t$ and $nx^{n-1} dx = dt$.

$$\text{Thus, } \int x^{n-1} f'(x^n) dx = \int f'(t) \frac{dt}{n} = \frac{1}{n} \int f'(t) dt$$



2.3 Integrals of the Form

$$(a) \int f(a^2 - x^2) dx,$$

$$(b) \int f(a^2 + x^2) dx,$$

$$(c) \int f(x^2 - a^2) dx,$$

$$(d) \int f\left(\frac{a-x}{a+x}\right) dx,$$

Working Rule

Integral

$$\int f(a^2 - x^2) dx,$$

Substitution

$$x = a \sin \theta \text{ or } x = a \cos \theta$$

$$\int f(a^2 + x^2) dx,$$

$$x = a \tan \theta \text{ or } x = a \cot \theta$$

$$\int f(x^2 - a^2) dx,$$

$$x = a \sec \theta \text{ or } x = a \operatorname{cosec} \theta$$

$$\int f\left(\frac{a-x}{a+x}\right) dx \text{ or } \int f\left(\frac{a+x}{a-x}\right) dx \quad x = a \cos 2\theta$$

2.4 Integrals of the Form

$$(a) \int \frac{dx}{ax^2 + bx + c},$$

$$(b) \int \frac{dx}{\sqrt{ax^2 + bx + c}},$$

$$(c) \int \sqrt{ax^2 + bx + c} dx$$

Working Rule

(i) Make the coefficient of x^2 unity by taking the coefficient of x^2 outside the quadratic.

(ii) Complete the square in the terms involving x , i.e. write $ax^2 + bx + c$ in the form

$$a \left[\left(x + \frac{b}{2a} \right)^2 \right] - \frac{(b^2 - 4ac)}{4a}.$$

(iii) The integrand is converted to one of the nine special integrals.

(iv) Integrate the function.

2.5 Integrals of the Form

$$(a) \int \frac{px + q}{ax^2 + bx + c} dx,$$

$$(b) \int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx,$$

$$(c) \int (px + q) \sqrt{ax^2 + bx + c} dx$$

Working Rule

$$(a) \int \frac{px + q}{ax^2 + bx + c} dx$$

Put $px + q = \lambda(2ax + b) + \mu$ or

$px + q = \lambda(\text{derivative of quadratic}) + \mu$.

Comparing the coefficient of x and constant term on both sides, we get

$$p = 2a\lambda \text{ and } q = b\lambda + \mu \Rightarrow \lambda = \frac{p}{2a} \text{ and } \mu = \left(q - \frac{bp}{2a} \right). \text{ Then}$$

integral becomes

$$\begin{aligned} & \int \frac{px + q}{ax^2 + bx + c} dx \\ &= \frac{p}{2a} \int \frac{2ax + b}{ax^2 + bx + c} dx + \left(q - \frac{bp}{2a} \right) \int \frac{dx}{ax^2 + bx + c} \\ &= \frac{p}{2a} \log | ax^2 + bx + c | + \left(q - \frac{bp}{2a} \right) \int \frac{dx}{ax^2 + bx + c} \end{aligned}$$

$$(b) \int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx$$

In this case the integral becomes

$$\begin{aligned} & \Rightarrow \int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx \\ &= \frac{p}{2a} \int \frac{2ax + b}{\sqrt{ax^2 + bx + c}} dx + \left(q - \frac{bp}{2a} \right) \int \frac{dx}{\sqrt{ax^2 + bx + c}} \\ &= \frac{p}{a} \sqrt{ax^2 + bx + c} + \left(q - \frac{bp}{2a} \right) \int \frac{dx}{\sqrt{ax^2 + bx + c}} \end{aligned}$$

$$(c) \int (px + q) \sqrt{ax^2 + bx + c} dx$$

The integral in this case is converted to





$$\int (px + q)\sqrt{ax^2 + bx + c} \, dx = \frac{p}{2a} \int (2ax + b)\sqrt{ax^2 + bx + c} \, dx$$

$$+ \left(q - \frac{bp}{2a}\right) \int \sqrt{ax^2 + bx + c} \, dx$$

$$= \frac{p}{3a} (ax^2 + bx + c)^{3/2} + \left(q - \frac{bp}{2a}\right) \int \sqrt{ax^2 + bx + c} \, dx$$

2.6 Integrals of the Form

$$\int \frac{P(x)}{\sqrt{ax^2 + bx + c}} \, dx, \text{ where } P(x) \text{ is a polynomial in } x \text{ of degree } n \geq 2.$$

Working Rule:

$$\text{Write } \int \frac{P(x)}{\sqrt{ax^2 + bx + c}} \, dx =$$

$$= (a_0 + a_1x + a_2x^2 + \dots + a_{n-1}x^{n-1})$$

$$\sqrt{ax^2 + bx + c} + k \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

where $k, a_0, a_1, \dots, a_{n-1}$ are constants to be determined by differentiating the above relation and equating the coefficients of various powers of x on both sides.

2.7 Integrals of the Form

$$\int \frac{x^2 + 1}{x^4 + kx^2 + 1} \, dx \quad \text{or} \quad \int \frac{x^2 - 1}{x^4 + kx^2 + 1} \, dx,$$

where k is a constant positive, negative or zero.

Working Rule

- Divide the numerator and denominator by x^2 .
- Put $x - \frac{1}{x} = z$ or $x + \frac{1}{x} = z$, whichever substitution, on differentiation gives, the numerator of the resulting integrand.
- Evaluate the resulting integral in z
- Express the result in terms of x .

2.8 Integrals of the Form

$$\int \frac{dx}{P\sqrt{Q}}, \text{ where } P, Q \text{ are linear or quadratic functions of } x.$$

Integral **Substitution**

$$\int \frac{1}{(ax + b)\sqrt{cx + d}} \, dx \quad cx + d = z^2$$

$$\int \frac{dx}{(ax^2 + bx + c)\sqrt{px + q}} \quad px + q = z^2$$

$$\int \frac{dx}{(px + q)\sqrt{ax^2 + bx + c}} \quad px + q = \frac{1}{z}$$

$$\int \frac{dx}{(ax^2 + b)\sqrt{cx^2 + d}} \quad x = \frac{1}{z}$$

2.9 Integrals of the Form

$$\int R \left(x, x^{\frac{p_1}{q_1}}, x^{\frac{p_2}{q_2}}, \dots, x^{\frac{p_k}{q_k}} \right) dx$$

To evaluate $\int R \left(x, x^{\frac{p_1}{q_1}}, x^{\frac{p_2}{q_2}}, \dots, x^{\frac{p_k}{q_k}} \right) dx$ where R is a rational

function of its variables $x, x^{\frac{p_1}{q_1}}, \dots, x^{\frac{p_k}{q_k}}$, put $x = t^n$ where n is the L.C.M of the denominator of the fractions $p_1/q_1, p_2/q_2, \dots, p_k/q_k$.

3. INTEGRATION BY PARTIAL FRACTIONS

Integrals of the type $\int \frac{p(x)}{g(x)}$ can be integrated by resolving

the integrand into partial fractions. We proceed as follows:
Check degree of $p(x)$ and $g(x)$.

If degree of $p(x) \geq$ degree of $g(x)$, then divide $p(x)$ by $g(x)$ till its degree is less, i.e. put in the

form $\frac{p(x)}{g(x)} = r(x) + \frac{f(x)}{g(x)}$ where degree of $f(x) <$ degree of $g(x)$.

CASE 1: When the denominator contains non-repeated linear factors. That is



$$g(x) = (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n).$$

In such a case write $f(x)$ and $g(x)$ as:

$$\frac{f(x)}{g(x)} = \frac{A_1}{(x - \alpha_1)} + \frac{A_2}{(x - \alpha_2)} + \dots + \frac{A_n}{(x - \alpha_n)}$$

where $A_1, A_2, \dots, A_n$ are constants to be determined by comparing the coefficients of various powers of x on both sides after taking L.C.M.

CASE 2: When the denominator contains repeated as well as non-repeated linear factor. That is

$$g(x) = (x - \alpha_1)^2(x - \alpha_3) \dots (x - \alpha_n).$$

In such a case write $f(x)$ and $g(x)$ as:

$$\frac{f(x)}{g(x)} = \frac{A_1}{x - \alpha_1} + \frac{A_2}{(x - \alpha_1)^2} + \frac{A_3}{x - \alpha_3} + \dots + \frac{A_n}{(x - \alpha_n)}$$

where $A_1, A_2, \dots, A_n$ are constants to be determined by comparing the coefficients of various powers of x on both sides after taking L.C.M.

Note: Corresponding to repeated linear factor $(x - a)^r$ in the denominator, a sum of r partial fractions of the

type $\frac{A_1}{x - a} + \frac{A_2}{(x - a)^2} + \dots + \frac{A_r}{(x - a)^r}$ is taken.

CASE 3: When the denominator contains a non repeated quadratic factor which cannot be factorised further:

$$g(x) = (ax^2 + bx + c)(x - \alpha_3)(x - \alpha_4) \dots (x - \alpha_n).$$

In such a case express $f(x)$ and $g(x)$ as:

$$\frac{f(x)}{g(x)} = \frac{A_1x + A_2}{ax^2 + bx + c} + \frac{A_3}{x - \alpha_3} + \dots + \frac{A_n}{x - \alpha_n}$$

where $A_1, A_2, \dots, A_n$ are constants to be determined by comparing the coefficients of various powers of x on both sides after taking L.C.M.

CASE 4: When the denominator contains a repeated quadratic factor which cannot be factorised further: That is

$$g(x) = (ax^2 + bx + c)^2(x - \alpha_3)(x - \alpha_6) \dots (x - \alpha_n)$$

In such a case write $f(x)$ and $g(x)$ as

$$\frac{f(x)}{g(x)} = \frac{A_1x + A_2}{ax^2 + bx + c} + \frac{A_3x + A_4}{(ax^2 + bx + c)^2} + \frac{A_5}{x - \alpha_3} + \dots + \frac{A_n}{(x - \alpha_n)}$$

where $A_1, A_2, \dots, A_n$ are constants to be determined by comparing the coefficients of various powers of x on both sides after taking L.C.M.

CASE 5: If the integrand contains only even powers of x

- (i) Put $x^2 = z$ in the integrand.
- (ii) Resolve the resulting rational expression in z into partial fractions
- (iii) Put $z = x^2$ again in the partial fractions and then integrate both sides.

4. INTEGRATION BY PARTS

The process of integration of the product of two functions is known as integration by parts.

For example, if u and v are two functions of x ,

$$\text{then } \int (uv) dx = u \int v dx - \int \left(\frac{du}{dx} \cdot \int v dx \right) dx.$$

In words, integral of the product of two functions = first function $\times$ integral of the second – integral of (differential of first $\times$ integral of the second function).

Working Hints

- (i) Choose the first and second function in such a way that derivative of the first function and the integral of the second function can be easily found.
- (ii) In case of integrals of the form $\int f(x) \cdot x^n dx$, take x^n as the first function and $f(x)$ as the second function.
- (iii) In case of integrals of the form $\int (\log x)^n \cdot 1 dx$, take 1 as the second function and $(\log x)^n$ as the first function.
- (iv) Rule of integration by parts may be used repeatedly, if required.
- (v) If the two functions are of different type, we can choose the first function as the one whose initial comes first in the word "ILATE", where
I — Inverse Trigonometric function
L — Logarithmic function
A — Algebraic function
T — Trigonometric function
E — Exponential function.
- (vi) In case, both the functions are trigonometric, take that function as second function whose integral is simple. If both the functions are algebraic, take that function as first function whose derivative is simpler.



- (vii) If the integral consists of an inverse trigonometric function of an algebraic expression in x , first simplify the integrand by a suitable trigonometric substitution and then integrate the new integrand.

4.1 Integrals of the Form:

Where the initial integrand reappears after integrating by parts.

Working Rule

- Apply the method of integration by parts twice.
- On integrating by parts second time, we will obtain the given integrand again, put it equal to I .
- Transpose and collect terms involving I on one side and evaluate I .

4.2 Integrals of the Form

$$\int e^x [f(x) + f'(x)] dx$$

Working Rule

- Split the integral into two integrals.
- Integrate only the first integral by parts, i.e.

$$\begin{aligned} & \int e^x [f(x) + f'(x)] dx \\ &= \int e^x f(x) dx + \int e^x f'(x) dx \\ &= \left[f(x) \cdot e^x - \int f'(x) \cdot e^x dx \right] + \int e^x f'(x) dx \\ &= e^x f(x) + C. \end{aligned}$$

4.3 Integrals of the Form

$$\begin{aligned} & \int (f(x) + xf'(x)) dx \\ &= \int f(x) dx + \int x f'(x) dx \\ &= \int f(x) dx + \left[x f(x) - \int 1 \cdot f(x) dx \right] = xf(x) + C \end{aligned}$$

5. INTEGRATION OF VARIOUS TRIGONOMETRIC FUNCTIONS

5.1 Integral of the Form

$$(a) \int \frac{dx}{a + b \cos x} \quad (b) \int \frac{dx}{a + b \sin x}$$

$$(c) \int \frac{dx}{a + b \cos x + c \sin x}$$

Working Rule

- (i) Put $\cos x = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$ and $\sin x = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$ so that the given

integrand becomes a function of $\tan \frac{x}{2}$.

- Put $\tan \frac{x}{2} = z \Rightarrow \frac{1}{2} \sec^2 \frac{x}{2} dx = dz$
- Integrate the resulting rational algebraic function of z
- In the answer, put $z = \tan \frac{x}{2}$.

5.2 Integrals of the Form

$$(a) \int \frac{dx}{a + b \cos^2 x} \quad (b) \int \frac{dx}{a + b \sin^2 x}$$

$$(c) \int \frac{dx}{a \cos^2 x + b \sin x \cos x + c \sin^2 x}$$

Working Rule

- Divide the numerator and denominator by $\cos^2 x$.
- In the denominator, replace $\sec^2 x$, if any, by $1 + \tan^2 x$.
- Put $\tan x = z \Rightarrow \sec^2 x dx = dz$.
- Integrate the resulting rational algebraic function of z .
- In the answer, put $z = \tan x$.



5.3 Integrals of the Form

$$\int \frac{a \cos x + b \sin x}{c \cos x + d \sin x} dx$$

Working Rule

- (i) Put Numerator = λ (denominator) + μ (derivative of denominator)
 $a \cos x + b \sin x = \lambda (c \cos x + d \sin x) + \mu (-c \sin x + d \cos x)$.
- (ii) Equate coefficients of $\sin x$ and $\cos x$ on both sides and find the values of λ and μ .
- (iii) Split the given integral into two integrals and evaluate each integral separately, i.e.

$$\int \frac{a \cos x + b \sin x}{c \cos x + d \sin x} dx =$$

$$\lambda \int 1 dx + \mu \int \frac{-c \sin x + d \cos x}{c \cos x + d \sin x} dx = \lambda x + \mu \log |c \cos x + d \sin x|.$$

- (iv) Substitute the values of λ and μ found in step 2.

5.4 Integrals of the Form

$$\int \frac{a + b \cos x + c \sin x}{e + f \cos x + g \sin x} dx$$

Working Rule

- (i) Put Numerator = l (denominator) + m (derivative of denominator) + n
 $a + b \cos x + c \sin x = l (e + f \cos x + g \sin x) + m (-f \sin x + g \cos x) + n$
- (ii) Equate coefficients of $\sin x$, $\cos x$ and constant term on both sides and find the values of l , m , n .
- (iii) Split the given integral into three integrals and evaluate each integral separately, i.e.

$$\int \frac{a + b \cos x + c \sin x}{e + f \cos x + g \sin x} dx$$

$$= l \int 1 dx + m \int \frac{-f \sin x + g \cos x}{e + f \cos x + g \sin x} dx + n \int \frac{dx}{e + f \cos x + g \sin x}$$

$$= lx + m \log |e + f \cos x + g \sin x| + n \int \frac{dx}{e + f \cos x + g \sin x}$$

- (iv) Substitute the values of l , m , n found in Step (ii).

5.5 Integrals of the Form

$$\int \sin^m x \cos^n x dx$$

Working Rule

- (i) If the power of $\sin x$ is an odd positive integer, put $\cos x = t$.
- (ii) If the power of $\cos x$ is an odd positive integer, put $\sin x = t$.
- (iii) If the power of $\sin x$ and $\cos x$ are both odd positive integers, put $\sin x = t$ or $\cos x = t$.
- (iv) If the power of $\sin x$ and $\cos x$ are both even positive integers, then express it as sines or cosines of multiple angles. Further integrate term by term.
- (v) If the sum of powers of $\sin x$ and $\cos x$ is an even negative integer, put $\tan x = z$.

5.6 Integrating $\int \tan^m x \sec^n x dx$

1. When m is odd and any n , rewrite the integrand in terms of $\sin x$ and $\cos x$:

$$\tan^m x \sec^n x dx = \left(\frac{\sin x}{\cos x} \right)^m \left(\frac{1}{\cos x} \right)^n dx$$

$$= \frac{\sin^{m-1} x}{\cos^{n+m} x} \sin x dx$$

and then substitute $u = \cos x$, $du = -\sin x dx$

$$\sin^2 x = 1 - \cos^2 x = 1 - u^2.$$

2. Alternatively, if m is odd and $n \geq 1$ move one factor of $\sec x$ to the side so that you can see $\sec x \tan x dx$ in the integral, and substitute $u = \sec x$. $du = \sec x \tan x dx$ and $\tan^2 x = \sec^2 x - 1 = u^2 - 1$.
3. If n is even with $n \geq 2$, move one factor of $\sec^2 x$ to the side so that you can see $\sec^2 x dx$ in the integral, and substitute $u = \tan x$, $du = \sec^2 x dx$ and $\sec^2 x = 1 + \tan^2 x = 1 + u^2$.
4. When m is even and $n = 0$ – that is the integrand is just an even power of tangent – we can still use the $u = \tan x$ substitution, after using $\tan^2 x = \sec^2 x - 1$ (possibly more than once) to create a $\sec^2 x$.





6. REDUCTION FORMULA

Reduction formulae makes it possible to reduce an integral depending on the index $n > 0$, called the order of the integral, to an integral of the same type with smaller index. (i.e. To reduce the integral into similar integral of order less than or greater than given integral). Application of reduction formula is given with the help of some examples.

6.1 Reduction Formula for $\int \sin^n x \, dx$

$$\begin{aligned} \text{Let } I_n &= \int \sin^n x \, dx = \int \sin^{n-1} x \sin x \, dx \\ &\quad \text{I} \quad \quad \text{II} \\ &= -\sin^{n-1} x \cos x + \int (n-1) \sin^{n-2} x \cos^2 x \, dx \\ &= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) \, dx \\ &= -\sin^{n-1} x \cos x + (n-1) \int (\sin^{n-2} x - \sin^n x) \, dx \\ &= -\sin^{n-1} x \cos x + (n-1) I_{n-2} - (n-1) I_n \\ \therefore n I_n &= -\sin^{n-1} x \cos x + (n-1) I_{n-2} \\ \Rightarrow I_n &= -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2} \\ \text{Thus, } \int \sin^n x \, dx &= \frac{-\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx \end{aligned}$$

6.2 Reduction Formula for $\int \cos^n x \, dx$

$$\begin{aligned} \text{Let } I_n &= \int \cos^n x \, dx = \int \cos^{n-1} x \cos x \, dx \\ n I_n &= \cos^{n-1} x \sin x + (n-1) I_{n-2} \\ \text{or } \int \cos^n x \, dx &= \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx \end{aligned}$$

6.3 Reduction Formula for $\int \tan^n x \, dx$

$$\begin{aligned} I_n &= \frac{\tan^{n-1} x}{n-1} - I_{n-2} \\ \int \tan^n x \, dx &= \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x \, dx \end{aligned}$$



SOLVED EXAMPLES

Example – 1

Evaluate : $\int \left(x^3 + 5x^2 - 4 + \frac{7}{x} + \frac{2}{\sqrt{x}} \right) dx$

Sol. $\int \left(x^3 + 5x^2 - 4 + \frac{7}{x} + \frac{2}{\sqrt{x}} \right) dx$

$$= \int x^3 dx + \int 5x^2 dx - \int 4 dx + \int \frac{7}{x} dx + \int \frac{2}{\sqrt{x}} dx$$

$$= \int x^3 dx + 5 \cdot \int x^2 dx - 4 \cdot \int 1 \cdot dx + 7 \cdot \int \frac{1}{x} dx + 2 \cdot \int x^{-1/2} dx$$

$$= \frac{x^4}{4} + 5 \cdot \frac{x^3}{3} - 4x + 7 \log |x| + 2 \left(\frac{x^{1/2}}{1/2} \right) + C$$

$$= \frac{x^4}{4} + \frac{5}{3} x^3 - 4x + 7 \log |x| + 4\sqrt{x} + C$$

Example – 2

Evaluate : $\int e^{x \log a} + e^{a \log x} + e^{a \log a} dx$

Sol. We have,

$$\int e^{x \log a} + e^{a \log x} + e^{a \log a} dx$$

$$= \int e^{\log a^x} + e^{\log x^a} + e^{\log a^a} dx$$

$$= \int (a^x + x^a + a^a) dx$$

$$= \int a^x dx + \int x^a dx + \int a^a dx$$

$$= \frac{a^x}{\log a} + \frac{x^{a+1}}{a+1} + a^a \cdot x + C.$$

Example – 3

Evaluate : $\int \frac{x^4}{x^2+1} dx$

Sol. $\int \frac{x^4}{x^2+1} dx$

$$= \int \frac{x^4 - 1 + 1}{x^2 + 1} dx = \int \frac{x^4 - 1}{x^2 + 1} + \frac{1}{x^2 + 1} dx$$

$$= \int (x^2 - 1) dx + \int \frac{1}{x^2 + 1} dx = \frac{x^3}{3} - x + \tan^{-1} x + C$$

Example – 4

Evaluate : $\int \frac{2^x + 3^x}{5^x} dx$

Sol. $\int \frac{2^x + 3^x}{5^x} dx$

$$= \int \left(\frac{2^x}{5^x} + \frac{3^x}{5^x} \right) dx$$

$$= \int \left[\left(\frac{2}{5} \right)^x + \left(\frac{3}{5} \right)^x \right] dx = \frac{(2/5)^x}{\log_e 2/5} + \frac{(3/5)^x}{\log_e 3/5} + C$$

Example – 5

Evaluate : $\int x^3 \sin x^4 dx$

Sol. We have

$$I = \int x^3 \sin x^4 dx$$

Let $x^4 = t \Rightarrow d(x^4) = dt$

$$\Rightarrow 4x^3 dx = dt \Rightarrow dx = \frac{1}{4x^3} dt$$

$$I = \frac{1}{4} \int \sin t dt = -\frac{\cos t}{4} + C = -\frac{\cos(x^4)}{4} + C$$



Example – 6

Evaluate : $\int \frac{x}{x^4 + x^2 + 1} dx$

Sol. We have,

$$I = \int \frac{x}{x^4 + x^2 + 1} dx = \int \frac{x}{(x^2)^2 + x^2 + 1} dx$$

Let $x^2 = t$, then, $d(x^2) = dt$

$$\Rightarrow 2x dx = dt$$

$$\Rightarrow dx = \frac{dt}{2x}$$

$$I = \int \frac{x}{t^2 + t + 1} \cdot \frac{dt}{2x}$$

$$= \frac{1}{2} \int \frac{1}{t^2 + t + 1} dt$$

$$= \frac{1}{2} \int \frac{1}{\left(t + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dt$$

$$= \frac{1}{2} \cdot \frac{1}{\frac{\sqrt{3}}{2}} \tan^{-1} \left(\frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) + C$$

$$= \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2t+1}{\sqrt{3}} \right) + C = \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x^2+1}{\sqrt{3}} \right) + C.$$

Example – 7

Evaluate : $\int \sqrt{x^2 + 2x + 5} dx$

Sol. We have,

$$\int \sqrt{x^2 + 2x + 5}$$

$$= \int \sqrt{x^2 + 2x + 1 + 4} dx = \int \sqrt{(x+1)^2 + 2^2} dx$$

$$= \frac{1}{2} (x+1) \sqrt{(x+1)^2 + 2^2} + \frac{1}{2} \cdot (2)^2 \log |(x+1)$$

$$+ \sqrt{(x+1)^2 + 2^2}| + C$$

$$= \frac{1}{2} (x+1) \sqrt{x^2 + 2x + 5} + 2 \log |(x+1) + \sqrt{x^2 + 2x + 5}| + C$$

Example – 8

Evaluate : $\int \frac{1}{x^2 - x + 1} dx$

Sol. $\int \frac{1}{x^2 - x + 1} dx$

$$= \int \frac{1}{x^2 - x + \frac{1}{4} - \frac{1}{4} + 1} dx$$

$$= \int \frac{1}{(x - 1/2)^2 + 3/4} dx$$

$$= \int \frac{1}{(x - 1/2)^2 + \left(\sqrt{3}/2\right)^2} dx$$

$$= \frac{1}{\sqrt{3}/2} \tan^{-1} \left(\frac{x - 1/2}{\sqrt{3}/2} \right) + C$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x - 1}{\sqrt{3}} \right) + C.$$

Example – 9

Evaluate : $\int \frac{1}{\sqrt{9 + 8x - x^2}} dx$

Sol. $\int \frac{1}{\sqrt{9 + 8x - x^2}} dx$

$$= \int \frac{1}{\sqrt{-\{x^2 - 8x - 9\}}} dx$$

$$= \int \frac{1}{\sqrt{-\{x^2 - 8x + 16 - 25\}}} dx$$

$$= \int \frac{1}{\sqrt{5^2 - (x - 4)^2 - 5^2}} dx$$

$$= \int \frac{1}{\sqrt{5^2 - (x - 4)^2}} dx = \sin^{-1} \left(\frac{x - 4}{5} \right) + C$$



Example – 10

Evaluate : $\int \frac{2x+3}{\sqrt{x^2+4x+1}} dx$

Sol.
$$\int \frac{2x+3}{\sqrt{x^2+4x+1}} dx$$

$$= \int \frac{(2x+4)-1}{\sqrt{x^2+4x+1}} dx$$

$$= \int \frac{2x+4}{\sqrt{x^2+4x+1}} dx - \int \frac{1}{\sqrt{x^2+4x+1}} dx$$

$$= \int \frac{dt}{\sqrt{t}} - \int \frac{1}{\sqrt{(x+2)^2 - (\sqrt{3})^2}} dx, \text{ where } t = x^2 + 4x + 1$$

$$= 2\sqrt{t} - \log |(x+2) + \sqrt{x^2+4x+1}| + C$$

$$= 2\sqrt{x^2+4x+1} - \log |x+2 + \sqrt{x^2+4x+1}| + C$$

Example – 11

Evaluate : $\int (x-5)\sqrt{x^2+x} dx$

Sol Let $(x-5) = \lambda \cdot \frac{d}{dx} (x^2+x) + \mu$. Then,
 $x-5 = \lambda(2x+1) + \mu$.
 Comparing coefficients of like powers of x , we get

$$1 = 2\lambda \text{ and } \lambda + \mu = -5 \Rightarrow \lambda = \frac{1}{2} \text{ and } \mu = -\frac{11}{2}$$

$$\int (x-5)\sqrt{x^2+x} dx$$

$$= \int \left(\frac{1}{2}(2x+1) - \frac{11}{2} \right) \sqrt{x^2+x} dx$$

$$= \int \frac{1}{2}(2x+1) \sqrt{x^2+x} dx - \frac{11}{2} \int \sqrt{x^2+x} dx$$

$$= \frac{1}{2} \int (2x+1)\sqrt{x^2+x} dx - \frac{11}{2} \int \sqrt{x^2+x} dx$$

$$= \frac{1}{2} \int \sqrt{t} dt - \frac{11}{2} \int \sqrt{\left(x+\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2} dx \text{ where}$$

$$t = x^2 + x$$

$$= \frac{1}{2} \cdot \frac{t^{3/2}}{3/2} - \frac{11}{2} \left[\frac{1}{2} \left(x + \frac{1}{2}\right) \sqrt{\left(x + \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2} \right]$$

$$- \frac{1}{2} \cdot \left(\frac{1}{2}\right)^2 \log \left[\left(x + \frac{1}{2}\right) + \sqrt{\left(x + \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2} \right] + C$$

$$= \frac{1}{3} t^{3/2} - \frac{11}{2} \left[\frac{2x+1}{4} \sqrt{x^2+x} - \frac{1}{8} \ln \left| \left(x + \frac{1}{2}\right) + \sqrt{x^2+x} \right| \right] + C$$

$$= \frac{1}{3} (x^2+x)^{3/2}$$

$$- \frac{11}{2} \left[\frac{2x+1}{4} \sqrt{x^2+x} - \frac{1}{8} \ln \left| \left(x + \frac{1}{2}\right) + \sqrt{x^2+x} \right| \right] + C$$

Example – 12

$$\int \frac{1-x^2}{1+x^2+x^4} dx$$

Sol.
$$\int \frac{-\left(1-\frac{1}{x^2}\right)dx}{x^2+\frac{1}{x^2}+1} \quad (\text{Dividing numerator and denominator by } x^2)$$

Put $x + \frac{1}{x} = t$

$$\Rightarrow -\int \frac{dt}{t^2-1}$$

$$= -\frac{1}{2} \ln \left| \frac{t-1}{t+1} \right| + C$$

$$= -\frac{1}{2} \ln \left| \frac{x+\frac{1}{x}-1}{x+\frac{1}{x}+1} \right| + C$$



Example – 13

Evaluate : $\int \frac{1}{x^4 + 1} dx$

Sol. We have,

$$I = \int \frac{1}{x^4 + 1} dx$$

$$\Rightarrow I = \int \frac{\frac{1}{x^2}}{x^2 + \frac{1}{x^2}} dx$$

$$\Rightarrow I = \frac{1}{2} \int \frac{\frac{2}{x^2}}{x^2 + \frac{1}{x^2}} dx$$

$$\Rightarrow I = \frac{1}{2} \int \frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} - \frac{1 - \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} dx$$

$$\Rightarrow I = \frac{1}{2} \int \frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} dx - \frac{1}{2} \int \frac{1 - \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} dx$$

$$\Rightarrow I = \frac{1}{2} \int \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 2} dx$$

$$\Rightarrow I = -\frac{1}{2} \int \frac{1 - \frac{1}{x^2}}{\left(x + \frac{1}{x}\right)^2 - 2} dx$$

Putting $x - \frac{1}{x} = u$ in 1st integral and $x + \frac{1}{x} = v$ in 2nd integral, we get

$$I = \frac{1}{2} \int \frac{du}{u^2 + (\sqrt{2})^2} - \frac{1}{2} \int \frac{dv}{v^2 - (\sqrt{2})^2}$$

$$= \frac{1}{2\sqrt{2}} \tan^{-1}\left(\frac{u}{\sqrt{2}}\right) - \frac{1}{2} \frac{1}{2\sqrt{2}} \log \left| \frac{v - \sqrt{2}}{v + \sqrt{2}} \right| + C$$

$$= \frac{1}{2\sqrt{2}} \tan^{-1}\left(\frac{x - 1/x}{\sqrt{2}}\right) - \frac{1}{4\sqrt{2}} \log \left| \frac{x + 1/x - \sqrt{2}}{x + 1/x + \sqrt{2}} \right| + C$$

$$= \frac{1}{2\sqrt{2}} \tan^{-1}\left(\frac{x^2 - 1}{\sqrt{2}x}\right) - \frac{1}{4\sqrt{2}} \log \left| \frac{x^2 - \sqrt{2}x + 1}{x^2 + x\sqrt{2} + 1} \right| + C$$

Example – 14

Evaluate : $\int x \log(1+x) dx$

Sol. $\int x \log(1+x) dx$

$$= \log(x+1) \cdot \frac{x^2}{2} - \int \frac{1}{x+1} \cdot \frac{x^2}{2} dx$$

$$= \frac{x^2}{2} \log(x+1) - \frac{1}{2} \int \frac{x^2}{x+1} dx$$

$$= \frac{x^2}{2} \log(x+1) - \frac{1}{2} \int \frac{x^2 - 1 + 1}{x+1} dx$$

$$= \frac{x^2}{2} \log(x+1) - \frac{1}{2} \int \frac{x^2 - 1}{x+1} + \frac{1}{x+1} dx$$

$$= \frac{x^2}{2} \log(x+1) - \frac{1}{2} \left[\int \left((x-1) + \frac{1}{x+1} \right) dx \right]$$

$$= \frac{x^2}{2} \log(x+1) - \frac{1}{2} \left[\frac{x^2}{2} - x + \log|x+1| \right] + C$$

Example – 15

Evaluate $\int \frac{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}}{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}} dx$

Sol. $\int \frac{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}}{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}} dx$

$$= \int \frac{\sin^{-1} \sqrt{x} - (\pi/2 - \sin^{-1} \sqrt{x})}{\pi/2} dx$$

$$\{ \because \sin^{-1} \theta + \cos^{-1} \theta = \pi/2 \}.$$



$$\Rightarrow I = \frac{2}{\pi} \int (2 \sin^{-1} \sqrt{x} - \pi/2) dx$$

$$I = \frac{4}{\pi} \int \sin^{-1} \sqrt{x} dx - \int 1 \cdot dx$$

$$I = \frac{4}{\pi} \int \sin^{-1} \sqrt{x} dx - x + c \quad \dots (i)$$

Let $x = \sin^2 \theta$, then $dx = 2 \sin \theta \cos \theta d\theta = \sin 2\theta d\theta$

$$\therefore \int \sin^{-1} \sqrt{x} dx = \int \theta \cdot \sin 2\theta d\theta$$

applying integration by parts

$$\int \sin^{-1} \sqrt{x} dx = -\theta \cdot \frac{\cos 2\theta}{2} + \int \frac{1}{2} \cos 2\theta d\theta$$

$$= \frac{-\theta}{2} \cdot \cos 2\theta + \frac{1}{4} \sin 2\theta$$

$$= \frac{-1 \cdot \theta}{2} \cdot (1 - 2 \sin^2 \theta) + \frac{1}{2} \cdot \sin \theta \cdot \sqrt{1 - \sin^2 \theta}$$

$$= \frac{-1}{2} \sin^{-1} \sqrt{x} (1 - 2x) + \frac{1}{2} \cdot \sqrt{x} \sqrt{1 - x} \quad \dots (ii)$$

from (i) and (ii)

$$I = \frac{4}{\pi} \left\{ \frac{-1}{2} (\sin^{-1} \sqrt{x}) (1 - 2x) + \frac{1}{2} \sqrt{x} \sqrt{1 - x} \right\} - x + c$$

$$= \frac{2}{\pi} \{ \sqrt{x - x^2} - (1 - 2x) \sin^{-1} \sqrt{x} \} - x + c$$

Example – 16

If $\int f(x) dx = \psi(x)$, then $\int x^5 f(x^3) dx$ is equal to

$$(a) \frac{1}{3} \left[x^3 \psi(x^3) - \int x^2 \psi(x^3) dx \right] + C$$

$$(b) \frac{1}{3} x^3 \psi(x^3) - 3 \int x^3 \psi(x^3) dx + C$$

$$(c) \frac{1}{3} x^3 \psi(x^3) - \int x^2 \psi(x^3) dx + C$$

$$(d) \frac{1}{3} \left[x^3 \psi(x^3) - \int x^3 \psi(x^3) dx \right] + C$$

Ans. (c)

Sol. Given, $\int f(x) dx = \psi(x)$

$$\text{Let } I = \int x^5 f(x^3) dx$$

$$\text{put } x^3 = t$$

$$\Rightarrow x^2 dx = \frac{dt}{3} \quad \dots (i)$$

$$\therefore I = \frac{1}{3} \int t f(t) dt$$

$$= \frac{1}{3} \left[t \int f(t) dt - \int \left\{ \frac{d}{dt}(t) \int f(t) dt \right\} dt \right]$$

[Integration by parts]

$$= \frac{1}{3} \left[t \psi(t) - \int \psi(t) dt \right]$$

$$= \frac{1}{3} \left[x^3 \psi(x^3) - 3 \int x^2 \psi(x^3) dx \right] + c \text{ from } \dots (i)$$

$$= \frac{1}{3} x^3 \psi(x^3) - \int x^2 \psi(x^3) dx + c$$

Example – 17

Evaluate $\int (e^{\log x} + \sin x) \cos x dx$.

Sol. $\int (e^{\log x} + \sin x) \cos x dx$

$$I = \int (x + \sin x) \cos x dx$$

$$I = \int x \cos x dx + \frac{1}{2} \int \sin 2x dx$$

$$= x \sin x - \int \sin x + x - \frac{1}{4} \cos 2x + c$$

$$= x \sin x + \cos x - \frac{1}{4} \cos 2x + c$$



Example – 18

Evaluate

$$(i) \int e^x \left(\frac{1 + \sin x \cos x}{\cos^2 x} \right) dx \quad (ii) \int e^{2x} \left(\frac{1 + \sin 2x}{1 + \cos 2x} \right) dx$$

Sol. (i) $I = \int e^x \left(\frac{1 + \sin x \cos x}{\cos^2 x} \right) dx$

$$I = \int e^x \left\{ \frac{1}{\cos^2 x} + \frac{\sin x \cos x}{\cos^2 x} \right\} dx$$

$$I = \int e^x \{ \tan x + \sec^2 x \} dx$$

M-I: $I = \int \frac{e^x}{\cos^2 x} \cdot \tan x dx + \int e^x (\sec^2 x) dx$

$$I = \tan x \cdot e^x - \int \sec^2 x \cdot e^x dx + \int e^x \cdot \sec^2 x dx + c$$

$$I = e^x \tan x + c.$$

M-II: $\int e^x (f(x) + f'(x)) dx = e^x f(x) + c$

$$= e^x \tan x + c$$

(ii) $I = \int e^{2x} \left\{ \frac{1 + \sin 2x}{1 + \cos 2x} \right\} dx$

$$= \int e^{2x} \left\{ \frac{1 + 2 \sin x \cos x}{2 \cos^2 x} \right\} dx$$

$$= \int e^{2x} \left\{ \frac{1}{2 \cos^2 x} + \frac{2 \sin x \cos x}{2 \cos^2 x} \right\} dx$$

$$= \int e^{2x} \left\{ \frac{1}{2} \sec^2 x + \tan x \right\} dx$$

M-I: $I = \int \frac{e^{2x}}{\cos^2 x} \cdot \tan x dx + \frac{1}{2} \int e^{2x} \cdot \sec^2 x dx$

$$= \tan x \cdot \frac{e^{2x}}{2} - \int \sec^2 x \cdot \frac{e^{2x}}{2} dx + \frac{1}{2} \int e^{2x} \cdot \sec^2 x dx$$

$$I = \frac{1}{2} e^{2x} \cdot \tan x + c.$$

M-II: $I = \int \frac{1}{2} e^{2x} (2 \tan x + \sec^2 x) dx$

$$\therefore \int e^{g(x)} (f(x) \cdot g'(x) + f'(x)) dx = e^{g(x)} f(x) + c$$

$$\therefore I = \frac{1}{2} e^{2x} \tan x + c$$

Example – 19

Evaluate $\int \frac{(x-1)e^x}{(x+1)^3} dx.$

Ans. $\frac{e^x}{(x+1)^2} + c$

Sol. $I = \int \frac{(x-1)e^x}{(x+1)^3} dx = \int \frac{(x+1-2)e^x}{(x+1)^3} dx$

$$= \int e^x \left[\frac{1}{(x+1)^2} + \frac{-2}{(x+1)^3} \right] dx$$

$$\therefore \int e^x [f(x) + f'(x)] dx = e^x f(x) + c$$

$$= \frac{e^x}{(x+1)^2} + c$$

Example – 20

Evaluate $\int \frac{3x-5}{(3x-2)(x+1)^2} dx$

Sol. Let $\frac{3x-5}{(3x-2)(x+1)^2} = \frac{A}{3x-2} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$

$$\therefore 3x-5 = A(x+1)^2 + B(3x-2)(x+1) + C(3x-2)$$

Putting $x = \frac{2}{3}$, we get $A = -\frac{27}{25}$

Putting $x = -1$, we get $C = \frac{8}{5}$;

Putting $x = 0$, we get $B = \frac{9}{25}$

$$\therefore I = \int \left[-\frac{27}{25} \cdot \frac{1}{3x-2} + \frac{9}{25} \cdot \frac{1}{(x+1)} + \frac{8}{5} \cdot \frac{1}{(x+1)^2} \right] dx$$

$$= -\frac{9}{25} \log |3x-2| + \frac{9}{25} \log |x+1| - \frac{8}{5} \cdot \frac{1}{(x+1)} + c.$$



Example – 21

Evaluate $\int \frac{dx}{(x-1)(x^2+1)}$

Sol. Let $\frac{1}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$

$$= \frac{A(x^2+1) + (Bx+C)(x-1)}{(x-1)(x^2+1)}$$

or $1 = A(x^2+1) + (Bx+C)(x-1)$

Putting $x = 1$, we get $A = \frac{1}{2}$;

Putting $x = 0$, we get $A - C = 1 \therefore C = A - 1 = -\frac{1}{2}$

Putting $x = -1$, we get $2A - 2(-B+C) = 1$

or $1 + 2B - 2\left(-\frac{1}{2}\right) = 1 \therefore B = -\frac{1}{2}$

Now $\frac{1}{(x-1)(x^2+1)} = \frac{1}{2(x-1)} - \frac{1}{2} \cdot \frac{x+1}{x^2+1}$

$$\therefore \int \frac{dx}{(x-1)(x^2+1)} = \int \frac{1}{2(x-1)} dx - \frac{1}{2} \int \frac{x}{x^2+1} dx - \frac{1}{2} \int \frac{dx}{x^2+1}$$

$$= \frac{1}{2} \log |x-1| - \frac{1}{4} \log (x^2+1) - \frac{1}{2} \tan^{-1} x + c$$

Example – 22

Find $\int \frac{x^4}{(x-1)(x^2+1)} dx$

Sol. $x^3 - x^2 + x - 1 \overline{) x^4}$

$$\begin{array}{r} x^4 - x^3 + x^2 - x, \\ + \quad - \quad + \\ \hline x^3 - x^2 + x - 1 \\ + \quad - \quad + \\ \hline 1 \end{array}$$

$$\therefore \frac{x^4}{(x-1)(x^2+1)} = x + 1 + \frac{1}{(x-1)(x^2+1)}$$

$$\therefore \int \frac{x^4}{(x-1)(x^2+1)} dx = \int (x+1) dx + \int \frac{dx}{(x-1)(x^2+1)}$$

$$= \frac{x^2}{2} + x + \left(\int \frac{1}{2(x-1)} - \frac{x}{2(x^2+1)} - \frac{1}{2} \frac{1}{x^2+1} \right) dx$$

(By Partial Fraction)

$$= \frac{x^2}{2} + x + \frac{1}{2} \log |x-1| - \frac{1}{4} \log |x^2+1| - \frac{1}{2} \tan^{-1} x + c$$

Example – 23

Evaluate $\int \frac{x^3+3x+2}{(x^2+1)^2(x+1)} dx$

Sol. $I = \int \frac{x(x^2+1)+2(x+1)}{(x^2+1)^2(x+1)} dx$

$$= \int \frac{x}{(x^2+1)(x+1)} dx + 2 \int \frac{dx}{(1+x^2)^2} \quad \dots (1)$$

Let $\frac{x}{(1+x)(1+x^2)} = \frac{A}{1+x} + \frac{Bx+C}{1+x^2}$

$\therefore x = A(1+x^2) + (Bx+C)(1+x)$

Putting $x = -1$, we get $A = -\frac{1}{2}$

Putting $x = 0$, we get, $0 = A + C \Rightarrow C = -A = \frac{1}{2}$

Putting $x = 1$, we get $1 = 2A + 2(B+C)$

$$= 2A + 2B + 2C = -1 + 2B + 1 \therefore B = \frac{1}{2}$$

$$\therefore \int \frac{x}{(1+x)(1+x^2)} dx = \int \left(-\frac{1}{2(1+x)} + \frac{\frac{1}{2}x + \frac{1}{2}}{1+x^2} \right) dx$$

$$= -\frac{1}{2} \log |1+x| + \frac{1}{2} \int \frac{x}{1+x^2} dx + \frac{1}{2} \int \frac{dx}{1+x^2}$$



$$= -\frac{1}{2} \log |1+x| + \frac{1}{4} \log (1+x^2) + \frac{1}{2} \tan^{-1} x \quad \dots (2)$$

To evaluate : $\int \frac{dx}{(1+x^2)^2}$, put $x = \tan \theta$

$$\begin{aligned} \text{Then, } \int \frac{dx}{(1+x^2)^2} &= \int \frac{\sec^2 \theta}{(1+\tan^2 \theta)^2} d\theta \\ &= \int \cos^2 \theta d\theta = \int \frac{1+\cos 2\theta}{2} d\theta \\ &= \frac{1}{2} \left[\theta + \frac{\sin 2\theta}{2} \right] = \frac{1}{2} [\theta + \sin \theta \cos \theta] \\ &= \frac{1}{2} \left[\tan^{-1} x + \frac{x}{\sqrt{1+x^2}} \cdot \frac{1}{\sqrt{1+x^2}} \right] \\ &= \frac{1}{2} \tan^{-1} x + \frac{1}{2} \frac{x}{1+x^2} \end{aligned}$$

Now from (1),

$$I = \frac{3}{2} \tan^{-1} x - \frac{1}{2} \log |1+x| + \frac{1}{4} \log (1+x^2) + \frac{x}{1+x^2} + c$$

Example – 24

Evaluate $\int \frac{dx}{\sin x + \sin 2x}$

Sol. Let the given integral be I , then

$$\begin{aligned} I &= \int \frac{dx}{\sin x + 2 \sin x \cos x} = \int \frac{dx}{\sin x (1 + 2 \cos x)} \\ &= \int \frac{\sin x dx}{\sin^2 x (1 + 2 \cos x)} = \int \frac{\sin x dx}{(1 - \cos^2 x) (1 + 2 \cos x)} \end{aligned}$$

Put $\cos x = y$ so that $-\sin x dx = dy$.

$$\begin{aligned} \therefore I &= \int \frac{-dy}{(1-y^2)(1+2y)} \\ &= - \int \frac{dy}{(1-y)(1+y)(1+2y)} \end{aligned}$$

Now we break $\frac{1}{(1-y)(1+y)(1+2y)}$ into partial fractions

$$\begin{aligned} \text{Let } \frac{1}{(1-y)(1+y)(1+2y)} &= \frac{A}{1-y} + \frac{B}{1+y} + \frac{C}{1+2y} \\ &= \frac{A(1+y)(1+2y) + B(1-y)(1+2y) + C(1-y)(1+y)}{(1-y)(1+y)(1+2y)} \end{aligned}$$

$$\therefore 1 = A(1+y)(1+2y) + B(1-y)(1+2y) + C(1-y)(1+y)$$

$$\text{Putting } y = 1, \text{ we get } A = \frac{1}{6}; \text{ putting } y = -\frac{1}{2}, \text{ we get } C = \frac{4}{3}$$

$$\text{Putting } y = -1, \text{ we get } B = -\frac{1}{2}$$

$$\begin{aligned} \text{Now } I &= - \int \left[\frac{1}{6(1-y)} - \frac{1}{2(1+y)} + \frac{4}{3(1+2y)} \right] dy \\ &= -\frac{1}{6} \log |1-y| + \frac{1}{2} \log |1+y| - \frac{2}{3} \log |1+2y| + c \\ &= -\frac{1}{6} \log (1-\cos x) + \frac{1}{2} \log (1+\cos x) \\ &\quad - \frac{2}{3} \log |1+2 \cos x| + c \end{aligned}$$

Example – 25

Evaluate $\int \sin^{-11/3} x \cos^{-1/3} x dx$

Sol. Here, $\int \sin^{-11/3} x, \cos^{-1/3} x dx$

$$\text{i.e., } \left(-\frac{11}{3} - \frac{1}{3} \right) = -4$$

$$\therefore I = \int \frac{\cos^{-1/3}}{\sin^{-11/3} x \cdot \sin^4 x} dx = \int (\cot^{-1/3} x) (\operatorname{cosec}^2 x)^2 \cdot dx$$

$$I = \int (\cot^{-1/3} x) (1 + \cot^2 x) \operatorname{cosec}^2 x dx.$$

$$\{\text{let } \cot x = t, -\operatorname{cosec}^2 x dx = dt\}$$

$$= - \int t^{-1/3} (1+t^2) dt = - \int (t^{-1/3} + t^{5/3}) dt$$

$$= - \left\{ \frac{3}{2} t^{2/3} + \frac{3}{8} t^{8/3} \right\} + c$$

$$= - \left\{ \frac{3}{2} (\cot^{2/3} x) + \frac{3}{8} (\cot^{8/3} x) \right\} + c.$$



Example – 26

Evaluate : $\int \frac{1}{1 + \sin x + \cos x} dx$

Sol. $I = \int \frac{1}{1 + \sin x + \cos x} dx$

$$= \int \frac{1}{1 + \frac{2 \tan x/2}{1 + \tan^2 x/2} + \frac{1 - \tan^2 x/2}{1 + \tan^2 x/2}} dx$$

$$= \int \frac{1 + \tan^2 x/2}{1 + \tan^2 x/2 + 2 \tan x/2 + 1 - \tan^2 x/2} dx$$

$$= \int \frac{\sec^2 x/2}{2 + 2 \tan x/2} dx$$

Putting $\tan \frac{x}{2} = t$ and $\frac{1}{2} \sec^2 \frac{x}{2} dx = dt$, we get

$$I = \int \frac{1}{t+1} dt = \log |t+1| + C = \log \left| \tan \frac{x}{2} + 1 \right| + C$$

Example – 27

Evaluate : $\int \frac{3 \sin x + 2 \cos x}{3 \cos x + 2 \sin x} dx$

Sol. $I = \int \frac{3 \sin x + 2 \cos x}{3 \cos x + 2 \sin x} dx$

Let $3 \sin x + 2 \cos x = \lambda \cdot \frac{d}{dx} (3 \cos x + 2 \sin x) +$

$$\mu (3 \cos x + 2 \sin x)$$

$$\Rightarrow 3 \sin x + 2 \cos x = \lambda (-3 \sin x + 2 \cos x) +$$

$$\mu (3 \cos x + 2 \sin x)$$

Comparing the coefficients of $\sin x$ and $\cos x$ on both sides, we get

$$-3\lambda + 2\mu = 3 \text{ and } 2\lambda + 3\mu = 2$$

$$\Rightarrow \mu = \frac{12}{13} \text{ and } \lambda = -\frac{5}{13}$$

$$\therefore I = \int \frac{\lambda(-3 \sin x + 2 \cos x) + \mu(3 \cos x + 2 \sin x)}{3 \cos x + 2 \sin x} dx$$

$$= \mu \int 1 \cdot dx + \lambda \int \frac{-3 \sin x + 2 \cos x}{3 \cos x + 2 \sin x} dx$$

$$= \mu x + \lambda \int \frac{dt}{t}, \text{ where } t = 3 \cos x + 2 \sin x$$

$$= \mu x + \lambda \ln |t| + C$$

$$= \frac{12}{13} x + \frac{-5}{13} \ln |3 \cos x + 2 \sin x| + C$$

Example – 28

Evaluate : $\int \frac{3 \cos x + 2}{\sin x + 2 \cos x + 3} dx$

Sol. We have,

$$I = \int \frac{3 \cos x + 2}{\sin x + 2 \cos x + 3} dx$$

Let $3 \cos x + 2 = \lambda (\sin x + 2 \cos x + 3) +$

$$\mu (\cos x - 2 \sin x) + v$$

Comparing the coefficients of $\sin x$, $\cos x$ and constant term on both sides, we get

$$\lambda - 2\mu = 0, 2\lambda + \mu = 3, 3\lambda + v = 2$$

$$\Rightarrow \lambda = \frac{6}{5}, \mu = \frac{3}{5} \text{ and } v = -\frac{8}{5}$$

$$\therefore I = \int \frac{\lambda(\sin x + 2 \cos x + 3) + \mu(\cos x - 2 \sin x) + v}{\sin x + 2 \cos x + 3} dx$$

$$\Rightarrow I = \lambda \int dx + \mu \int \frac{\cos x - 2 \sin x}{\sin x + 2 \cos x + 3} dx +$$

$$v \int \frac{1}{\sin x + 2 \cos x + 3} dx$$

$$\Rightarrow I = \lambda x + \mu \log |\sin x + 2 \cos x + 3| + v I_1, \text{ where}$$

$$I_1 = \int \frac{1}{\sin x + 2 \cos x + 3} dx$$

Putting, $\sin x = \frac{2 \tan x/2}{1 + \tan^2 x/2}$, $\cos x = \frac{1 - \tan^2 x/2}{1 + \tan^2 x/2}$ we get

$$I_1 = \int \frac{1}{\frac{2 \tan x/2}{1 + \tan^2 x/2} + \frac{2(1 - \tan^2 x/2)}{1 + \tan^2 x/2} + 3} dx$$



$$= \int \frac{1 + \tan^2 x / 2}{2 \tan x / 2 + 2 - 2 \tan^2 x / 2 + 3(1 + \tan^2 x / 2)} dx$$

$$= \int \frac{\sec^2 x / 2}{\tan^2 x / 2 + 2 \tan x / 2 + 5} dx$$

Put $\tan x / 2 = t$

$$\sec^2 x / 2 \cdot \frac{1}{2} dx = dt$$

$$\sec^2 x / 2 dx = 2dt = 2 \int \frac{dt}{t^2 + 2t + 5} = 2 \int \frac{dt}{(t+1)^2 + (2)^2}$$

$$= \tan^{-1} \left(\frac{t+1}{2} \right) + C$$

$$= 2 \tan^{-1} \left(\frac{\tan \left(\frac{x}{2} \right) + 1}{2} \right) + C$$

Example – 29

Integrate $\frac{1}{1 - \cot x}$ or $\frac{\sin x}{\sin x - \cos x}$.

Sol. Let $I = \int \frac{\sin x}{\sin x - \cos x} dx$

Again, let $\sin x = A(\cos x + \sin x) + B(\sin x - \cos x)$ then
 $A + B = 1$ and $A - B = 0$

$$\Rightarrow A = \frac{1}{2}, B = \frac{1}{2}$$

$$\therefore I = \int \frac{\frac{1}{2}(\cos x + \sin x) + \frac{1}{2}(\sin x - \cos x)}{(\sin x - \cos x)} dx$$

$$= \frac{1}{2} \int \frac{\cos x + \sin x}{\sin x - \cos x} dx + \frac{1}{2} \int 1 dx + c$$

$$= \frac{1}{2} \log(\sin x - \cos x) + \frac{1}{2} x + c$$

Example – 30

Evaluate $\int \sin^3 x \cdot \cos^5 x dx$

Sol. $I = \int \sin^3 x \cdot \cos^5 x dx$

Let $\cos x = t \Rightarrow -\sin x dx = dt$

$$I = - \int (1 - t^2) \cdot t^5 dt$$

$$I = \int t^7 dt - \int t^5 dt = \frac{t^8}{8} - \frac{t^6}{6} + c$$

$$I = \frac{\cos^8 x}{8} - \frac{\cos^6 x}{6} + c$$

Method II: $I = \int R^3 (1 - R^2)^2 dR$, if $\sin x = R$, $\cos x dx = dR$.

$$I = \int R^3 dR - \int 2R^5 dR + \int R^7 dR$$

$$I = \frac{\sin^4 x}{4} - \frac{2 \sin^6 x}{6} + \frac{\sin^8 x}{8} + c$$

Example – 31

Evaluate

(i) $\int \frac{1}{\sin(x-a) \cos(x-b)} dx$

(ii) $\int \frac{1}{\cos(x-a) \cos(x-b)} dx$

Sol. (i) $I = \int \frac{1}{\sin(x-a) \cos(x-b)} dx$

$$I = \frac{\cos(a-b)}{\cos(a-b)} \cdot \int \frac{dx}{\sin(x-a) \cos(x-b)}$$

$$= \frac{1}{\cos(a-b)} \cdot \int \frac{\cos\{(x-b) - (x-a)\}}{\sin(x-a) \cos(x-b)} dx$$

$$= \frac{1}{\cos(a-b)}$$

$$\cdot \int \left\{ \frac{\cos(x-b) \cdot \cos(x-a)}{\sin(x-a) \cos(x-b)} + \frac{\sin(x-b) \cdot \sin(x-a)}{\sin(x-a) \cos(x-b)} \right\} dx$$



$$I = \frac{1}{\cos(a-b)} \int \{\cot(x-a) + \tan(x-b)\} dx$$

$$I = \frac{1}{\cos(a-b)} \{\log|\sin(x-a)| - \log|\cos(x-b)|\} + c$$

$$I = \frac{1}{\cos(a-b)} \log_e \left| \frac{\sin(x-a)}{\cos(x-b)} \right| + c$$

$$\begin{aligned} \text{(ii)} \quad I &= \int \frac{1}{\cos(x-a) \cos(x-b)} dx \\ &= \frac{1}{\sin(a-b)} \int \frac{\sin(a-b) dx}{\cos(x-a) \cos(x-b)} \\ &= \frac{1}{\sin(a-b)} \int \frac{\sin\{(x-b)-(x-a)\}}{\cos(x-a) \cos(x-b)} dx \\ &= \frac{1}{\sin(a-b)} \int \left\{ \frac{\sin(x-b) \cos(x-a)}{\cos(x-a) \cos(x-b)} - \frac{\cos(x-b) \sin(x-a)}{\cos(x-a) \cos(x-b)} \right\} dx \\ &= \frac{1}{\sin(a-b)} \int \{\tan(x-b) - \tan(x-a)\} dx \\ &= \frac{1}{\sin(a-b)} [-\log|\cos(x-b)| + \log|\cos(x-a)|] + c \\ &= \frac{1}{\sin(a-b)} \left[\log \left| \frac{\cos(x-a)}{\cos(x-b)} \right| \right] + c \end{aligned}$$

Example – 32

If $\int \frac{\sin x}{\sin(x-\alpha)} dx = Ax + B \log \sin(x-\alpha) + C$, then the

value of (A, B) is

- (a) $(-\sin \alpha, \cos \alpha)$ (b) $(\cos \alpha, \sin \alpha)$
(c) $(\sin \alpha, \cos \alpha)$ (d) $(-\cos \alpha, \sin \alpha)$

Ans. (b)

So $\int \frac{\sin x}{\sin(x-\alpha)} dx$

Let $x - \alpha = t \Rightarrow x = t + \alpha \Rightarrow dx = dt$

$$I = \int \frac{\sin(t+\alpha)}{\sin t} dt$$

$$\begin{aligned} I &= \int \frac{\sin t \cos \alpha + \cos t \sin \alpha}{\sin t} dt \\ &= \int \cos \alpha dt + \int \sin \alpha \cot t dt \\ &= t \cos \alpha + \sin \alpha \ln \sin t + c \\ &= (x - \alpha) \cos \alpha + \sin \alpha \ln \sin(x - \alpha) + c \\ &\Rightarrow A = \cos \alpha, B = \sin \alpha \end{aligned}$$

Example – 33

$\int \frac{dx}{\cos x - \sin x}$ is equal to

(a) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} - \frac{3\pi}{8} \right) \right| + C$

(b) $\frac{1}{\sqrt{2}} \log \left| \cot \left(\frac{x}{2} \right) \right| + C$

(c) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} - \frac{\pi}{8} \right) \right| + C$

(d) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{3\pi}{8} \right) \right| + C$

Ans. (d)

Sol. $\int \frac{dx}{\cos x - \sin x} = \frac{1}{\sqrt{2}} \int \frac{dx}{\cos \left(x + \frac{\pi}{4} \right)}$

$$\int \frac{dx}{\cos x - \sin x} = \frac{1}{\sqrt{2}}$$

$$\int \sec \left(x + \frac{\pi}{4} \right) dx = \frac{1}{\sqrt{2}} \log \tan \left(\frac{x}{2} + \frac{3\pi}{8} \right) + c$$

Example – 34

$\int \frac{\sin^8 x - \cos^8 x}{(1 - 2 \sin^2 x \cos^2 x)} dx$ is equal to:

(a) $\frac{1}{2} \sin 2x + c$

(b) $-\frac{1}{2} \sin 2x + c$

(c) $-\frac{1}{2} \sin x + c$

(d) $-\sin^2 x + c$

Ans. (b)

Sol. $= \int \frac{\sin^8 x - \cos^8 x}{1 - 2 \sin^2 x \cos^2 x} dx$



$$= \int \frac{(\sin^4 x - \cos^4 x)(\sin^4 x + \cos^4 x)}{1 - 2\sin^2 x \cos^2 x} dx$$

$$= \int \frac{(\sin^2 x - \cos^2 x)(1 - 2\sin^2 x \cos^2 x)}{1 - 2\sin^2 x \cos^2 x} dx$$

$$= \int -\cos 2x dx$$

$$= \frac{-1}{2} \sin 2x + c$$

$$I = \int \frac{(t^2 + 1) 2t dt}{\{(t^2 - 1)^2 + 3(t^2 - 1) + 3\}\sqrt{t^2}}$$

$$\Rightarrow I = 2 \int \frac{(t^2 + 1)}{t^4 + t^2 + 1} dt = 2 \int \frac{1 + \frac{1}{t^2}}{t^2 + \frac{1}{t^2} + 1} dt$$

$$\Rightarrow I = 2 \int \frac{du}{u^2 + (\sqrt{3})^2} \text{ where } t - \frac{1}{t} = u.$$

Example – 35

Evaluate : $\int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}} dx$

Sol. Let $I = \int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}} dx$

Putting $x+1 = t^2$, and $dx = 2t dt$, we get

$$\Rightarrow I = \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{u}{\sqrt{3}} \right) + C = \frac{2}{\sqrt{3}} \tan^{-1} \left\{ \frac{t - \frac{1}{t}}{\sqrt{3}} \right\} + C$$

$$\Rightarrow I = \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t^2 - 1}{t\sqrt{3}} \right) + C = \frac{2}{\sqrt{3}} \tan^{-1} \left\{ \frac{x}{\sqrt{3}(x+1)} \right\} + C$$



EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Integral as an anti-derivative

1. $\int \left(x - \frac{1}{x}\right) dx, (x > 0)$ equals

(a) $\frac{x^3}{3} - \frac{3}{2}x^2 + 3\log x + \frac{1}{2x^2} + C$

(b) $\frac{x^4}{4} - \frac{3}{2}x^2 + 3\log x + \frac{1}{2x^2} + C$

(c) $\frac{x^4}{4} + 3\log x + \frac{1}{2x^2} + C$

(d) none of these

2. The value of $\int \left(\frac{6}{1+x^2} + 10^x\right) dx$ is

(a) $6 \tan^{-1} x + 10^x \log_e 10 + C$

(b) $6 \tan^{-1} x + \frac{10^x}{\log_e 10} + C$

(c) $3 \tan^{-1} x + \frac{10^x}{\log_e 10} + C$

(d) none of these

3. $\int (\tan x + \cot x)^2 dx$ is equal to

(a) $\tan x - \cot x + c$ (b) $\tan x + \cot x + c$

(c) $\cot x - \tan x + c$ (d) none of these

4. $\int \frac{x^2 + \cos^2 x}{(1+x^2)\sin^2 x} dx$ is equals to

(a) $\tan^{-1} x + \cot x + c$ (b) $\tan^{-1} x - \cot x + c$

(c) $\cot^{-1} x - \tan x + c$ (d) $-\tan^{-1} x - \cot x + c$

5. $\int \left(\frac{2^x - 5^x}{10^x}\right) dx$ is equal to

(a) $\frac{2^{-x}}{\log_e 2} - \frac{5^{-x}}{\log_e 5} + c$ (b) $\frac{2^x}{\log_e 2} - \frac{5^x}{\log_e 5} + c$

(c) $\frac{2^x}{\log_e 2} + \frac{5^x}{\log_e 5} + c$ (d) $\frac{5^{-x}}{\log_e 5} - \frac{2^{-x}}{\log_e 2} + c$

6. $\int \sec^2(ax+b) dx$ equals

(a) $\tan(ax+b) + C$ (b) $\frac{1}{2} \tan x + C$

(c) $\frac{1}{a} \tan(ax+b) + C$ (d) none of these

7. $\int \frac{dx}{\sin^2 x \cos^2 x}$ is equal to

(a) $\tan x + \cot x + C$ (b) $(\tan x + \cot x)^2 + C$

(c) $\tan x - \cot x + C$ (d) $(\tan x - \cot x)^2 + C$

8. $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx$ is equal to

(a) $\tan x + \cot x + 3x + c$ (b) $\tan x + \cot x - 3x + c$

(c) $\tan x - \cot x - 3x + c$ (d) $\tan x - \cot x + 3x + c$

Integrations by substitution

9. $\int \sqrt{\frac{1+x}{1-x}} dx$ equals

(a) $\sin^{-1} x + \sqrt{1-x^2} + c$ (b) $\sin^{-1} x + \sqrt{x^2-1} + c$

(c) $\sin^{-1} x - \sqrt{1-x^2} + c$ (d) $\sin^{-1} x - \sqrt{x^2-1} + c$



10. $\int \frac{(\sin^{-1} x)^3}{\sqrt{1-x^2}} dx$ is equal to
- (a) $\frac{(\sin^{-1} x)^2}{2} + c$ (b) $\frac{(\sin^{-1} x)^3}{3} + c$
- (c) $\frac{\sin^{-1} x}{x} + c$ (d) $\frac{(\sin^{-1} x)^4}{4} + c$
11. $\int \sec^n x \tan x dx$ is equal to
- (a) $\frac{\sec^n x}{n} + c$ (b) $\frac{\sec^2 x}{2} + c$
- (c) $\frac{\tan x}{n} + c$ (d) $\frac{(\sec^n x) \tan x}{n} + c$
12. $\int \frac{\cos^3 x}{\sin^2 x + \sin x} dx$ is equal to
- (a) $\log |\cos x| - \sin x + c$ (b) $\log |\sin x| - \sin x + c$
- (c) $\log |\sin x| + \cos x + c$ (d) $\log |\cos x| - \cos x + c$
13. $\int \frac{\log_e x}{x\sqrt{1+\log_e x}} dx =$
- (a) $(1 + \log_e x)^{3/2} + c$
- (b) $\frac{2}{3} (1 + \log_e x) (\log_e x - 2) + c$
- (c) $\frac{2}{3} (1 + \log_e x)^{1/2} (\log_e x - 5) + c$
- (d) $\frac{2}{3} (1 + \log_e x)^{1/2} (\log_e x - 2) + c$
14. $\int \frac{1}{x \log x} dx$ is equal to
- (a) $\log |x \log x| + C$ (b) $\log |\log x + x| + C$
- (c) $\log x + C$ (d) $\log |\log x| + C$
15. $\int \sec^2 x \cos(\tan x) dx$ equals
- (a) $\sin(\cos x) + C$ (b) $\sin(\tan x) + C$
- (c) $\operatorname{cosec}(\tan x) + C$ (d) none of these
16. $\int \tan^n x \sec^2 x dx$ equals
- (a) $\frac{\tan^{n-1} x}{n-1} + C$ (b) $\frac{\tan^{n+1} x}{n+1} + C$
- (c) $\tan^{n+1} x + C$ (d) none of these
17. $\int \frac{\sin 2x}{1 + \cos^4 x} dx$ is equal to
- (a) $\cos^{-1}(\cos^2 x) + c$ (b) $\sin^{-1}(\cos^2 x) + c$
- (c) $\cot^{-1}(\cos^2 x) + c$ (d) none of these
18. $\int \frac{dx}{x + \sqrt{x}}$ equals
- (a) $2 \log(\sqrt{x} - 1) + c$ (b) $2 \log(\sqrt{x} + 1) + c$
- (c) $\tan^{-1} x + c$ (d) none of these
19. $\int \frac{x^5}{\sqrt{1+x^3}} dx$ equals
- (a) $\frac{2}{9} (x^3 - 2) \sqrt{1+x^3} + c$ (b) $\frac{2}{9} (x^3 + 2) \sqrt{1+x^3} + c$
- (c) $(x^3 + 2) \sqrt{1+x^3} + c$ (d) none of these
20. $\int \frac{dx}{3x^2 + 2x + 1}$ equals
- (a) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{3x+1}{\sqrt{2}} \right) + C$
- (b) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{3x-1}{\sqrt{2}} \right) + C$
- (c) $-\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{3x-1}{\sqrt{2}} \right) + C$
- (d) none of these



21. $\int \frac{dx}{\sqrt{3-5x-x^2}}$ equals

- (a) $\sin^{-1} \left(\frac{2x+5}{\sqrt{37}} \right) + c$ (b) $\cos^{-1} \left(\frac{2x+5}{\sqrt{37}} \right) + c$
 (c) $\sin^{-1} (2x+5) + c$ (d) none of these

22. $\int \frac{e^x}{e^{2x} + 5e^x + 6} dx$ equals

- (a) $\log \left(\frac{e^x + 3}{e^x + 2} \right) + c$ (b) $\log \left(\frac{e^x + 2}{e^x + 3} \right) + c$
 (c) $\frac{1}{2} \log \left(\frac{e^x + 2}{e^x + 3} \right) + c$ (d) none of these

23. $\int \frac{x^2}{x^2 - 1} dx$ equals

- (a) $x + \log \sqrt{\frac{x-1}{x+1}} + c$ (b) $x + \log \sqrt{\frac{x+1}{x-1}} + c$
 (c) $x + \log \left| \frac{x-1}{x+1} \right| + c$ (d) $x + \log \left| \frac{x+1}{x-1} \right| + c$

24. If m is a non-zero number and

$\int \frac{x^{5m-1} + 2x^{4m-1}}{(x^{2m} + x^m + 1)^3} dx = f(x) + c$, then $f(x)$ is

- (a) $\frac{x^{5m}}{2m(x^{2m} + x^m + 1)^2}$ (b) $\frac{x^{4m}}{2m(x^{2m} + x^m + 1)^2}$
 (c) $\frac{2m(x^{5m} + x^{4m})}{(x^{2m} + x^m + 1)^2}$ (d) $\frac{(x^{5m} - x^{4m})}{2m(x^{2m} + x^m + 1)^2}$

25. $\int \frac{dx}{1+e^x} =$

- (a) $\log_e \left(\frac{e^x + 1}{e^x} \right) + c$ (b) $\log_e \left(\frac{e^x}{e^x + 1} \right) + c$
 (c) $x + \log_e (e^x + 1) + c$ (d) $e^x + x + c$

26. $\int \sin^9 x \cdot \cos^3 x dx$ is equal to

- (a) $\frac{\sin^{10} x}{10} - \frac{\sin^{12} x}{12} + c$ (b) $\frac{\cos^6 x}{6} - \frac{\cos^8 x}{8} + c$
 (c) $\frac{\cos^6 x}{6} - \frac{\sin^8 x}{8} + c$ (d) none of these

27. If $\int \frac{\sin^4 x}{\cos x} dx = \frac{1}{2} \log_e \left(\frac{1+\sin x}{1-\sin x} \right) - g(x) + c$ where $g(x)$ equals.

- (a) $\frac{1}{3} \sin^3 x + \sin x$ (b) $\frac{1}{3} \cos^3 x + \cos x$
 (c) $\frac{1}{3} \sin^3 x - \sin x$ (d) $\frac{1}{3} \cos^3 x - \cos x$

28. $\int \frac{dx}{4\sin^2 x + 4\sin x \cos x + 5\cos^2 x}$ is equal to

- (a) $\tan^{-1} \left(\tan x + \frac{1}{2} \right) + c$ (b) $\frac{1}{4} \tan^{-1} \left(\tan x + \frac{1}{2} \right) + c$
 (c) $4 \tan^{-1} \left(\tan x + \frac{1}{2} \right) + c$ (d) none of these

29. The integral $\int \frac{\sin^2 x \cos^2 x}{(\sin^3 x + \cos^3 x)^2} dx$ equal to:

- (a) $\frac{1}{(1 + \cot^3 x)^2} + c$ (b) $-\frac{1}{3(1 + \tan^3 x)^2} + c$
 (c) $\frac{\sin^3 x}{(1 + \cos^3 x)^2} + c$ (d) $-\frac{\cos^3 x}{3(1 + \sin^3 x)^2} + c$

30. If $\int \tan^4 x dx = \lambda \tan^3 x + \mu \tan x + x + C$, then

- (a) $\lambda = \frac{1}{3}$ (b) $\mu = 1$
 (c) $\lambda = -\frac{1}{3}$ (d) none of these



Standard algebraic formats

31. $\int \frac{2x-3}{x^2+3x-18} dx$ is equal to

(a) $\log|x^2+3x-18| - \frac{3}{2} \log\left|\frac{x-3}{x+6}\right| + C$

(b) $\log|x^2+3x-18| - \frac{2}{3} \log\left|\frac{x-3}{x+6}\right| + C$

(c) $\log|x^2+3x-18| + \frac{2}{3} \log\left|\frac{x-3}{x+6}\right| + C$

(d) None of these

32. Evaluate $\int \frac{x^2+x+5}{x^2-x-1} dx$

(a) $x + \log|x^2-x-1| + \frac{7}{\sqrt{5}} \log\left|\frac{2x-1-\sqrt{5}}{2x-1+\sqrt{5}}\right| + C$

(b) $x + \log|x^2-x-1| + \frac{7}{2\sqrt{5}} \log\left|\frac{2x-1-\sqrt{5}}{2x-1+\sqrt{5}}\right| + C$

(c) $x + \log|x^2-x-1| + \frac{14}{\sqrt{5}} \log\left|\frac{2x-1-\sqrt{5}}{2x-1+\sqrt{5}}\right| + C$

(d) None of these

33. Evaluate $\int \frac{x^3+x+1}{x^2-1} dx$

(a) $\frac{x^2}{2} + \log|x^2-1| + \log\left|\frac{x-1}{x+1}\right| + C$

(b) $\frac{x^2}{2} + \log|x^2-1| + \frac{1}{2} \log\left|\frac{x-1}{x+1}\right| + C$

(c) $x^2 + \log|x^2-1| + \frac{1}{2} \log\left|\frac{x-1}{x+1}\right| + C$

(d) None of these

34. Evaluate $\int \frac{x^2+2}{x^4+4} dx$

(a) $\tan^{-1}\left(\frac{x-\frac{2}{x}}{2}\right) + C$ (b) $\frac{1}{4} \tan^{-1}\left(\frac{x-\frac{2}{x}}{2}\right) + C$

(c) $\frac{1}{2} \tan^{-1}\left(\frac{x-\frac{2}{x}}{2}\right) + C$ (d) None of these

35. Evaluate $\int \frac{dx}{(x^2+2x+6)}$

(a) $\tan^{-1}\left(\frac{x+1}{\sqrt{5}}\right) + C$

(b) $\frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{x+1}{\sqrt{5}}\right) + C$

(c) $\frac{1}{\sqrt{6}} \tan^{-1}\left(\frac{x+1}{\sqrt{5}}\right) + C$

(d) None of these

36. Evaluate the following $\int \frac{dx}{\sqrt{2ax-x^2}}$

(a) $\sin^{-1}\left(\frac{a-x}{a}\right) + c$

(b) $\frac{1}{2} \sin^{-1}\left(\frac{x-a}{a}\right) + c$

(c) $\sin^{-1}\left(\frac{x-a}{a}\right) + c$

(d) $2 \sin^{-1}\left(\frac{x-a}{a}\right) + c$



37. Evaluate $\int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}} dx$

(a) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{3(x+1)}} \right) + c$

(b) $\tan^{-1} \left(\frac{x}{\sqrt{3(x+1)}} \right) + c$

(c) $\frac{\sqrt{3}}{2} \tan^{-1} \left(\frac{x}{\sqrt{3(x+1)}} \right) + c$

(d) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x}{3(x+1)} \right) + c$

38. Evaluate $\int \frac{dx}{(x-2)\sqrt{x^2-4}}$

(a) $\frac{1}{2} \sqrt{\frac{x+2}{x-2}} + c$ (b) $-\frac{1}{2} \sqrt{\frac{x+2}{x-2}} + c$

(c) $\frac{1}{2} \sqrt{\frac{x-2}{x+2}} + c$ (d) $-\frac{1}{2} \sqrt{\frac{x-2}{x+2}} + c$

39. Evaluate $\int \frac{dx}{(1+x^2)\sqrt{1-x^2}}$

(a) $-\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{2}x}{\sqrt{1-x^2}} \right) + c$

(b) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{2}x}{\sqrt{1-x^2}} \right) + c$

(c) $-\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{1-x^2}}{\sqrt{2}x} \right) + c$

(d) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{1-x^2}}{\sqrt{2}x} \right) + c$

40. $\int \sqrt{1+x-2x^2} dx$ equals

(a) $\frac{1}{8}(4x-1)\sqrt{1+x-2x^2} + \frac{9\sqrt{2}}{32} \sin^{-1} \left(\frac{4x-1}{3} \right) + C$

(b) $\frac{1}{8}(4x-1)\sqrt{1+x-2x^2} - \frac{9\sqrt{2}}{32} \sin^{-1} \left(\frac{4x-1}{3} \right) + C$

(c) $\frac{1}{8}(4x-1)\sqrt{1+x-2x^2} + \frac{\sqrt{2}}{32} \sin^{-1} \left(\frac{4x+1}{3} \right) + C$

(d) None of these

Integration by partial fractions

41. $\int \frac{x^3+3}{(x+1)(x^2+1)} dx$ equals.

(a) $x + \log_e |x+1| - \log_e (x^2+1) + \cot^{-1} x + c$

(b) $x - \log_e |x+1| + \log_e (x^2+1) + \tan^{-1} x + c$

(c) $x + \log_e |x+1| - \log_e (x^2+1) + \tan^{-1} x + c$

(d) $x - \log_e |x+1| - \log_e (x^2+1) - \tan^{-1} x + c$

42. $\int \frac{x^2+1}{(x-1)(x-2)} dx$ equals

(a) $\log \left| \frac{(x-2)^5}{(x-1)^2} \right| + C$ (b) $x + \log \left| \frac{(x-2)^5}{(x-1)^2} \right| + C$

(c) $x + \log \left| \frac{(x-1)^5}{(x-2)^5} \right| + C$ (d) none of these

43. The value of $\int \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)}$ is

(a) $\frac{1}{b^2-a^2} \left[b \tan^{-1} \frac{x}{b} - a \tan^{-1} \frac{x}{a} \right] + C$

(b) $\frac{1}{b^2-a^2} \left[a \tan^{-1} \frac{x}{b} - b \tan^{-1} \frac{x}{a} \right] + C$

(c) $\frac{1}{b^2-a^2} \left[b \tan^{-1} \frac{x}{b} + a \tan^{-1} \frac{x}{a} \right] + C$

(d) none of these



Integrations by parts

44. $\int \log x \, dx$ is equal to

- (a) $x(\log x) - 1 + c$ (b) $x(\log x) - x + c$
(c) $x(\log x) - 1/x + c$ (d) $x(\log x) + c$

45. $\int \frac{\sin^{-1} \sqrt{x}}{\sqrt{1-x}} \, dx$ equals

- (a) $2[\sqrt{x} - \sqrt{1-x} \sin^{-1} \sqrt{x}] + c$
(b) $2[\sqrt{x} + \sqrt{1-x} \sin^{-1} \sqrt{x}] + c$
(c) $[\sqrt{x} - \sqrt{1-x} \sin^{-1} \sqrt{x}] + c$
(d) none of these

46. $\int x^2 e^x \, dx$ is equal to

- (a) $x^2 e^x - 2[e^{2x} - x e^x] + c$
(b) $x^2 e^x - 2[e^x - x e^x] + c$
(c) $x^2 e^x - 2[x e^{2x} - e^x] + c$
(d) $x^2 e^x - 2[x e^x - e^x] + c$

47. $\int (\log x)^2 \, dx$ is equal to

- (a) $x(\log x)^2 - 2[x \log x - x] + c$
(b) $x(\log x)^2 - 2[\log x - x] + c$
(c) $x(\log x)^2 - 2[\log x^2 - x] + c$
(d) $x(\log x)^2 - 2[\log x - 2x] + c$

48. $\int \frac{\sin^{-1} x}{(1-x^2)^{3/2}} \, dx$ is equal to

- (a) $\frac{x}{\sqrt{1-x^2}} \sin^{-1} x + \frac{1}{2} \log |\sqrt{1-x^2}| + c$
(b) $\frac{x}{\sqrt{1-x^2}} \sin^{-1} x + \frac{1}{2} \log |1-x^2| + c$
(c) $\frac{x}{\sqrt{1-x^2}} \sin^{-1} x + \frac{1}{4} \log |\sqrt{1-x^2}| + c$
(d) none of these

49. The value of $\int x \sec x \tan x \, dx$ is

- (a) $x \sec x + \log |\sec x + \tan x| + c$
(b) $x \sec x - \log |\sec x + \tan x| + c$
(c) $x \sec x + \log |\sec x - \tan x| + c$
(d) none of these

50. $\int \frac{x - \sin x}{1 - \cos x} \, dx$ is equal to

- (a) $-x \cot \frac{x}{2} + c$ (b) $\cot \frac{x}{2} + c$
(c) $-\cot \frac{x}{2} + c$ (d) none of these

51. $\int e^x \frac{(1-x)^2}{(1+x^2)^2} \, dx$ is equal to

- (a) $\frac{1}{x^2+1} + c$ (b) $\frac{e^x}{x^2+1} + c$
(c) $\frac{e^x - 1}{x^2+1} + c$ (d) $\frac{1 - e^x}{x^2+1} + c$

52. $\int \frac{x e^x}{(x+1)^2} \, dx$ is equal to

- (a) $\frac{e^x}{(x+1)^2} + c$ (b) $\frac{e^x}{x+1} + c$
(c) $\frac{e^x}{(x+1)^3} + c$ (d) none of these

53. $\int e^x \frac{x-1}{(x+1)^3} \, dx$ equals

- (a) $-\frac{e^x}{x+1} + C$ (b) $\frac{e^x}{x+1} + C$
(c) $\frac{e^x}{(x+1)^2} + C$ (d) $-\frac{e^x}{(x+1)^2} + C$



54. $\int \left\{ \frac{(\log x - 1)}{1 + (\log x)^2} \right\}^2 dx$ is equal to

(a) $\frac{x}{x^2 + 1} + C$ (b) $\frac{\log x}{(\log x)^2 + 1} + C$

(c) $\frac{x}{(\log x)^2 + 1} + C$ (d) $\frac{x e^x}{1 + x^2} + C$

55. The integral $\int \left(1 + x - \frac{1}{x} \right) e^{x + \frac{1}{x}} dx$ is equal to :

(a) $-x e^{x + \frac{1}{x}} + c$ (b) $(x - 1) e^{x + \frac{1}{x}} + c$

(c) $x e^{x + \frac{1}{x}} + c$ (d) $(x + 1) e^{x + \frac{1}{x}} + c$

Standard trigonometric Formats

56. $\int \sin^2(x/2) dx$ equals

(a) $\frac{1}{2}(x + \sin x) + c$ (b) $\frac{1}{2}(x + \cos x) + c$

(c) $\frac{1}{2}(x - \sin x) + c$ (d) none of these

57. $\int \frac{\cos 2x - \cos 2\theta}{\cos x - \cos \theta} dx$ is equal to

(a) $2(\sin x + x \cos \theta) + C$

(b) $2(\sin x - x \cos \theta) + C$

(c) $2(\sin x + 2x \cos \theta) + C$

(d) $2(\sin x - 2x \cos \theta) + C$

58. $\int \frac{1 - \cos 2x}{1 + \cos 2x} dx =$

(a) $\tan x - x + C$

(b) $x + \tan x + C$

(c) $x - \tan x + C$

(d) $-x - \cot x + C$

59. If the integral $\int \frac{5 \tan x}{\tan x - 2} dx = x + a \ln |\sin x - 2 \cos x| + k$, then a is equal to

(a) -1 (b) -2

(c) 1 (d) 2

60. $\int \sqrt[3]{\frac{\sin^n x}{\cos^{n+6} x}} dx$, $n \in \mathbb{N}$ is equal to

(a) $\frac{3}{n}(\tan x)^{\frac{n}{3}+1} + c$ (b) $\frac{3}{3+n}(\tan x)^{\frac{n}{3}+1} + c$

(c) $\frac{3}{n}(\cos x)^{n+1} + c$ (d) none of these

61. $\int \frac{\sin^4 x}{\cos^8 x} dx$ is equal to

(a) $\frac{(1 + \tan^5 x)}{5} - \frac{\tan^7 x}{7} + c$

(b) $\frac{\tan^5 x}{5} + \frac{\tan^7 x}{7} + c$

(c) $\frac{\tan^5 x}{7} + \frac{\tan^7 x}{5} + c$

(d) none of these

62. $\int \frac{\sin^{10} x}{\cos^{12} x} dx =$

(a) $10 \tan^9 x + C$

(b) $\frac{\tan^{11} x}{11} + C$

(c) $\frac{\tan^{11} x}{11} + C$

(d) none of these

63. $\int \sqrt{\frac{1 + \cos x}{1 - \cos x}} dx$ equals

(a) $\log \cos \left(\frac{x}{2} \right) + C$

(b) $2 \log \sin \left(\frac{x}{2} \right) + C$

(c) $2 \log \sec \left(\frac{x}{2} \right) + C$

(d) none of these



64. $\int \frac{1}{\sin(x-a)\sin(x-b)} dx$ is equal to

(a) $\sin(b-a) \log \left| \frac{\sin(x-b)}{\sin(x-a)} \right| + C$

(b) $\operatorname{cosec}(b-a) \log \left| \frac{\sin(x-a)}{\sin(x-b)} \right| + C$

(c) $\operatorname{cosec}(b-a) \log \left| \frac{\sin(x-b)}{\sin(x-a)} \right| + C$

(d) $\sin(b-a) \log \left| \frac{\sin(x-a)}{\sin(x-b)} \right| + C$

65. The value of $\sqrt{2} \int \frac{\sin x \, dx}{\sin\left(x - \frac{\pi}{4}\right)}$ is

(a) $x - \log \left| \cos x \left(x - \frac{\pi}{4} \right) \right| + C$

(b) $x + \log \left| \cos \left(x - \frac{\pi}{4} \right) \right| + C$

(c) $x - \log \left| \sin \left(x - \frac{\pi}{4} \right) \right| + C$

(d) $x + \log \left| \sin \left(x - \frac{\pi}{4} \right) \right| + C$

66. $\int \frac{dx}{\cos x + \sqrt{3} \sin x}$ equals

(a) $\log \tan \left(\frac{x}{2} + \frac{\pi}{12} \right) + C$ (b) $\log \tan \left(\frac{x}{2} - \frac{\pi}{12} \right) + C$

(c) $\frac{1}{2} \log \tan \left(\frac{x}{2} + \frac{\pi}{12} \right) + C$ (d) $\frac{1}{2} \log \tan \left(\frac{x}{2} - \frac{\pi}{12} \right) + C$

Numerical Value Type Questions

67. $\int \frac{e^{5 \log_e x} - e^{4 \log_e x}}{e^{3 \log_e x} - e^{2 \log_e x}} dx = \frac{x^k}{k} + c$. Then k is equal to

68. $\int \left(\frac{1-x}{1+x} \right)^2 dx = x - a \log |x+1| - \frac{b}{x+1} + c$. Then a + b is equal to

69. If $\int \frac{dx}{1+\tan x} = px + q \log_e |\cos x + \sin x| + c$ then p + q equals.

70. $\int \frac{x+1}{x^2+x+3} dx = \frac{1}{a} \ln |x^2+x+3| + \frac{1}{\sqrt{b}} \tan^{-1} \left(\frac{2x+1}{\sqrt{11}} \right) + c$

then (b - a) equals

71. If $\int \frac{2x+3}{(x-1)(x^2+1)} dx = \log \left[|x-1|^{5/2} (x^2+1)^{-5/a} \right] - \frac{1}{2} \tan^{-1} x + C$

where C is any arbitrary constant, then a is equal to

72. If $\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx = a \sin \sqrt{x} + c$, then a equals

73. $\int \frac{x^2}{1+x^6} dx = \frac{1}{k} \tan^{-1} x^3 + c$. Then k is equal to

74. $\int \frac{x+1}{\sqrt{x+2}} dx =$

$\frac{a}{b} (x+2)^{3/2} - a\sqrt{x+2} + c$ then ab equals

75. $\int \sqrt{1+x-2x^2} dx = \frac{1}{2a} (4x-1) \sqrt{1+x-x^2} + \frac{9\sqrt{2}}{b} \sin^{-1} \left(\frac{4x-1}{3} \right) + c$.

Then b/a is equal to



76. $\int \frac{(x-x^3)^{1/3}}{x^4} dx = -\frac{3}{k} \left(\frac{1}{x^2} - 1 \right)^{4/3} + c$. Then k is equal to

77. $\int \frac{\log x}{x^2} dx = -\frac{1}{x} (a + \log bx) + c$. (a, b $\in$ Integers). Then a + b equal to

78. $\int x^3 (\log x)^2 dx = \frac{1}{p} x^4 [q (\log x)^2 + r \log x + 1] + c$.
Then p + q + r is equal to

79. $\int \frac{x + \sin x}{1 + \cos x} dx = \frac{x}{a} \tan \left(\frac{x}{b} \right) + C$. Then b - a is equal to

80. If $\int \sin 2x \sin 3x dx$ equals $\frac{1}{k} (5 \sin x - \sin 5x) + c$. Then k is equal to



EXERCISE - 2 : PREVIOUS YEAR JEE MAIN QUESTIONS

1. The integral $\int \frac{dx}{x^2(x^4+1)^{3/4}}$ equal (2015)
- (a) $-(x^4+1)^{1/4} + c$ (b) $-\left(\frac{x^4+1}{x^4}\right)^{1/4} + c$
- (c) $\left(\frac{x^4+1}{x^4}\right)^{1/4} + c$ (d) $(x^4+1)^{1/4} + c$
2. The integral $\int \frac{dx}{(x+1)^{3/4}(x-2)^{5/4}}$ is equal to (2015/Online Set-1)
- (a) $4\left(\frac{x+1}{x-2}\right)^{1/4} + C$ (b) $-\frac{4}{3}\left(\frac{x+1}{x-2}\right)^{1/4} + C$
- (c) $-\frac{4}{3}\left(\frac{x-2}{x+1}\right)^{1/4} + C$ (d) $4\left(\frac{x-2}{x+1}\right)^{1/4} + C$
3. If $\int \frac{\log(t+\sqrt{1+t^2})}{\sqrt{1+t^2}} dt = \frac{1}{2} (g(t))^2 + C$, where C is constant, then $g(2)$ is equal to : (2015/Online Set-2)
- (a) $\frac{1}{\sqrt{5}} \log(2+\sqrt{5})$ (b) $\frac{1}{2} \log(2+\sqrt{5})$
- (c) $2 \log(2+\sqrt{5})$ (d) $\log(2+\sqrt{5})$
4. The integral $\int \frac{2x^{12}+5x^9}{(x^5+x^3+1)^3} dx$ is equal to : (2016)
- (a) $\frac{x^{10}}{2(x^5+x^3+1)^2} + C$ (b) $\frac{x^5}{2(x^5+x^3+1)^2} + C$
- (c) $\frac{-x^{10}}{2(x^5+x^3+1)^2} + C$ (d) $\frac{-x^5}{(x^5+x^3+1)^2} + C$
- Where C is an arbitrary constant.
5. If $\int \frac{dx}{\cos^3 x \sqrt{2 \sin 2x}} = (\tan x)^A + C(\tan x)^B + k$, where k is a constant of integration, then A+B+C equals : (2016/Online Set-1)
- (a) $\frac{21}{5}$ (b) $\frac{16}{5}$
- (c) $\frac{10}{7}$ (d) $\frac{27}{10}$
6. The integral $\int \frac{dx}{(1+\sqrt{x})\sqrt{x-x^2}}$ is equal to : (2016/Online Set-2)
- (where C is a constant of integration.)
- (a) $-2\sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}} + C$ (b) $-2\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} + C$
- (c) $-\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} + C$ (d) $2\sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}} + C$
7. Let $I_n = \int \tan^n x dx$, ($n > 1$). If $I_4 + I_6 = a \tan^5 x + bx^5 + C$, where C is a constant of integration, then the ordered pair (a, b) is equal to: (2017)
- (a) $\left(-\frac{1}{5}, 1\right)$ (b) $\left(\frac{1}{5}, 0\right)$
- (c) $\left(\frac{1}{5}, -1\right)$ (d) $\left(-\frac{1}{5}, 0\right)$
8. The integral $\int \sqrt{1+2 \cot x (\operatorname{cosec} x + \cot x)} dx \left(0 < x < \frac{\pi}{2}\right)$ is equal to (where C is a constant of integration) (2017/Online Set-1)
- (a) $4 \log\left(\sin \frac{x}{2}\right) + C$ (b) $2 \log\left(\sin \frac{x}{2}\right) + C$
- (c) $2 \log\left(\cos \frac{x}{2}\right) + C$ (d) $4 \log\left(\cos \frac{x}{2}\right) + C$



9. If $f\left(\frac{3x-4}{3x+4}\right) = x+2$, $x \neq -\frac{4}{3}$, and

$\int f(x) dx = A \log |1-x| + Bx + C$, then the ordered pair

(A, B) is equal to :

(where C is a constant of integration)

(2017/Online Set-2)

(a) $\left(\frac{8}{3}, \frac{2}{3}\right)$ (b) $\left(-\frac{8}{3}, \frac{2}{3}\right)$

(c) $\left(-\frac{8}{3}, -\frac{2}{3}\right)$ (d) $\left(\frac{8}{3}, -\frac{2}{3}\right)$

10. The integral

$\int \frac{\sin^2 x \cos^2 x}{(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x)^2} dx$ is

(2018)

(a) $\frac{-1}{1+\cot^3 x} + C$ (b) $\frac{1}{3(1+\tan^3 x)} + C$

(c) $\frac{-1}{3(1+\tan^3 x)} + C$ (d) $\frac{1}{1+\cot^3 x} + C$

(Where C is a constant of integration)

11. If $f\left(\frac{x-4}{x+2}\right) = 2x+1$, $(x \in \mathbb{R} - \{-1, -2\})$, then $\int f(x) dx$ is

equal to

(where C is a constant of integration)

(2018/Online Set-1)

(a) $12 \log_e |1-x| + 3x + C$

(b) $-12 \log_e |1-x| - 3x + C$

(c) $12 \log_e |1-x| - 3x + C$

(d) $-12 \log_e |1-x| + 3x + C$

12. If $\int \frac{2x+5}{\sqrt{7-6x-x^2}} dx = A\sqrt{7-6x-x^2} + B \sin^{-1}\left(\frac{x+3}{4}\right) + c$

(where C is a constant of integration), then the ordered pair (A, B) is equal to :

(2018/Online Set-2)

(a) (2, 1) (b) (-2, -1)

(c) (-2, 1) (d) (2, -1)

13. If $\int \frac{\tan x}{1+\tan x + \tan^2 x} dx = x - \frac{k}{\sqrt{A}} \tan^{-1}\left(\frac{k \tan x + 1}{\sqrt{A}}\right) + c$,

(C is a constant of integration), then the ordered pair (K, A) is equal to :

(2018/Online Set-3)

(a) (2, 1) (b) (-2, 3)

(c) (2, 3) (d) (-2, 1)

14. $\int \frac{\sin \frac{5x}{2}}{\sin \frac{x}{2}} dx$ is equal to :

(where c is a constant of integration.)

(8-04-2019/Shift-1)

(a) $2x + \sin x + 2 \sin 2x + c$

(b) $x + 2 \sin x + 2 \sin 2x + c$

(c) $x + 2 \sin x + \sin 2x + c$

(d) $2x + \sin x + \sin 2x + c$

15. If $\int \frac{dx}{x^3(1+x^6)^{2/3}} = x f(x) (1+x^6)^{1/3} + C$, where C is a

constant of integration, then the function $f(x)$ is equal to:

(08-04-2019/Shift-2)

(a) $\frac{3}{x^2}$ (b) $-\frac{1}{6x^3}$

(c) $-\frac{1}{2x^2}$ (d) $-\frac{1}{2x^3}$

16. The integral $\int \sec^{\frac{2}{3}} x \operatorname{cosec}^{\frac{4}{3}} x dx$ is equal to:

(9-04-2019/Shift-1)

(a) $-3 \tan^{-\frac{1}{3}} x + C$ (b) $-\frac{3}{4} \tan^{-\frac{4}{3}} x + C$

(c) $-3 \cot^{-\frac{1}{3}} x + C$ (d) $3 \tan^{-\frac{1}{3}} x + C$

(Here C is a constant of integration)



17. If $\int e^{\sec x} (\sec x \tan x f(x) + (\sec x \tan x + \sec^2 x)) dx = e^{\sec x} f(x) + C$, then a possible choice of $f(x)$ is:

(9-04-2019/Shift-2)

- (a) $\sec x + \tan x + C$ (b) $\sec x - \tan x - C$
(c) $\sec x + 2 \tan x - C$ (d) $x \sec x + \tan x + C$

18. If $\int x^5 e^{-x^2} dx = g(x) e^{-x^2} + c$, where c is a constant of integration, then $g(-1)$ is equal to:

(10-4-2019/Shift-2)

- (a) -1 (b) 1
(c) $-\frac{5}{2}$ (d) $-\frac{1}{2}$

19. The integral $\int \frac{2x^3 - 1}{x^4 + x} dx$ is equal to : (Here C is a constant of integration)

(12-04-2019/Shift-1)

- (a) $\frac{1}{2} \log_e \left| \frac{x^3 + 1}{x^2} \right| + C$ (b) $\frac{1}{2} \log_e \left| \frac{(x^3 + 1)^2}{x^3} \right| + C$
(c) $\log_e \left| \frac{x^3 + 1}{x} \right| + C$ (d) $\log_e \left| \frac{x^3 + 1}{x^2} \right| + C$

20. Let $\alpha \in \left(0, \frac{\pi}{2}\right)$ be fixed. If the integral

$$\int \frac{\tan x + \tan \alpha}{\tan x - \tan \alpha} dx = A(x) \cos 2\alpha + B(x) \sin 2\alpha + C,$$

where C is a constant of integration, then the functions $A(x)$ and $B(x)$ are respectively:

(12-04-2019/Shift-2)

- (a) $x + \alpha$ and $\log_e |\sin(x + \alpha)|$
(b) $x - \alpha$ and $\log_e |\sin(x - \alpha)|$
(c) $x - \alpha$ and $\log_e |\cos(x - \alpha)|$
(d) $x + \alpha$ and $\log_e |\sin(x - \alpha)|$

21. For $x^2 \neq n\pi, n \in \mathbb{N}$ (the set of natural numbers), the

$$\text{integral } \int x \sqrt{\frac{2 \sin(x^2 - 1) - \sin 2(x^2 - 1)}{2 \sin(x^2 - 1) + \sin 2(x^2 - 1)}} dx \text{ is equal to :}$$

(where c is a constant of integration)

(9-01-2019/Shift-1)

- (a) $\log_e \left| \frac{1}{2} \sec^2(x^2 - 1) \right| + c$
(b) $\frac{1}{2} \log_e |\sec(x^2 - 1)| + c$
(c) $\frac{1}{2} \log_e \left| \sec^2 \left(\frac{x^2 - 1}{2} \right) \right| + c$
(d) $2 \log_e \left| \sec \left(\frac{x^2 - 1}{2} \right) \right| - c$

22. If $f(x) = \int \frac{5x^8 + 7x^6}{(x^2 + 1 + 2x^7)^2} dx, (x \geq 0)$, and $f(0) = 0$, then

the value of $f(1)$ is

(9-01-2019/Shift-2)

- (a) $-\frac{1}{2}$ (b) $-\frac{1}{4}$
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

23. Let $n \geq 2$ be a natural number and $0 < \theta < \frac{\pi}{2}$ then

$$\int \frac{(\sin^n \theta - \sin \theta)^{\frac{1}{n}} \cos \theta}{\sin^{n+1} \theta} d\theta \text{ is equal to:}$$

(10-1-2019/Shift-1)

- (a) $\frac{n}{n^2 - 1} \left(1 - \frac{1}{\sin^{n-1} \theta} \right)^{\frac{n+1}{n}} + C$
(b) $\frac{n}{n^2 + 1} \left(1 - \frac{1}{\sin^{n-1} \theta} \right)^{\frac{n+1}{n}} + C$
(c) $\frac{n}{n^2 - 1} \left(1 + \frac{1}{\sin^{n-1} \theta} \right)^{\frac{n+1}{n}} + C$
(d) $\frac{n}{n^2 - 1} \left(1 - \frac{1}{\sin^{n+1} \theta} \right)^{\frac{n+1}{n}} + C$

(where C is a constant of integration)



24. If $\int x^3 e^{-4x^3} dx = \frac{1}{48} e^{-4x^3} f(x) + C$, where C is a constant of integration, then $f(x)$ is equal to:
- (10-01-2019/Shift-2)
- (a) $-2x^3 - 1$ (b) $-4x^3 - 1$
(c) $-2x^3 + 1$ (d) $4x^3 + 1$
25. If $\int \frac{\sqrt{1-x^2}}{x^4} dx = A(x) (\sqrt{1-x^2})^m + C$, for a suitable chosen integer m and a function A(x), where C is a constant of integration, then $(A(x))^m$ equals :
- (11-01-2019/Shift-1)
- (a) $\frac{-1}{27x^9}$ (b) $\frac{-1}{3x^3}$
(c) $\frac{1}{27x^6}$ (d) $\frac{1}{9x^4}$
26. If $\int \frac{x+1}{\sqrt{2x-1}} dx = f(x) \sqrt{2x-1} + C$, where C is a constant of integration, then $f(x)$ is equal to :
- (11-01-2019/Shift-2)
- (a) $\frac{1}{3}(x+1)$ (b) $\frac{2}{3}(x+2)$
(c) $\frac{2}{3}(x-4)$ (d) $\frac{1}{3}(x+4)$
27. The integral $\int \cos(\log_e x) dx$ is equal to (Where C is a constant of integration)
- (12-01-2019/Shift-1)
- (a) $\frac{x}{2} [\sin(\log_e x) - \cos(\log_e x)] + C$
(b) $x [\cos(\log_e x) + \sin(\log_e x)] + C$
(c) $\frac{x}{2} [\cos(\log_e x) + \sin(\log_e x)] + C$
(d) $x [\cos(\log_e x) - \sin(\log_e x)] + C$
28. The integral $\int \frac{3x^{13} + 2x^{11}}{(2x^4 + 3x^2 + 1)^4} dx$ is equal to (where C is a constant to integration)
- (12-01-2019/Shift-2)
- (a) $\frac{x^4}{6(2x^4 + 3x^2 + 1)^3} + C$ (b) $\frac{x^{12}}{6(2x^4 + 3x^2 + 1)^3} + C$
(c) $\frac{x^4}{(2x^4 + 3x^2 + 1)^3} + C$ (d) $\frac{x^{12}}{(2x^4 + 3x^2 + 1)^3} + C$
29. If $\int \sin^{-1} \left(\sqrt{\frac{x}{1+x}} \right) dx = A(x) \tan^{-1}(\sqrt{x}) + B(x) + C$, where C is a constant of integration, then the ordered pair $(A(x), B(x))$ can be :
- (3-09-2020/Shift-2)
- (a) $(x+1, -\sqrt{x})$ (b) $(x-1, -\sqrt{x})$
(c) $(x+1, \sqrt{x})$ (d) $(x-1, \sqrt{x})$
30. The integral $\int \left(\frac{x}{x \sin x + \cos x} \right)^2 dx$ is equal to (where C is a constant of integration) :
- (4-09-2020/Shift-1)
- (a) $\tan x - \frac{x \sec x}{x \sin x + \cos x} + C$
(b) $\sec x - \frac{x \tan x}{x \sin x + \cos x} + C$
(c) $\sec x + \frac{x \tan x}{x \sin x + \cos x} + C$
(d) $\tan x + \frac{x \sec x}{x \sin x + \cos x} + C$
31. If $\int (e^{2x} + 2e^x - e^{-x} - 1) e^{(e^x + e^{-x})} dx$
 $= g(x) e^{(e^x + e^{-x})} + c$ where c is a constant of integration, then $g(0)$ is equal to:
- (5-09-2020/Shift-1)
- (a) 2 (b) e
(c) 1 (d) e^2



32. If $\int \frac{\cos \theta}{5 + 7 \sin \theta - 2 \cos^2 \theta} d\theta = A \log_e |B(\theta)| + C$ where c is

a constant of integration, then $\frac{B(\theta)}{A}$ can be:

(5-09-2020/Shift-2)

(a) $\frac{5(2 \sin \theta + 1)}{\sin \theta + 3}$

(b) $\frac{5(\sin \theta + 3)}{2 \sin \theta + 1}$

(c) $\frac{2 \sin \theta + 1}{\sin \theta + 3}$

(d) $\frac{2 \sin \theta + 1}{5(\sin \theta + 3)}$

33. If $\int \frac{\cos x}{\sin^3 x (1 + \sin^6 x)^{\frac{1}{2}}} dx = f(x) (1 + \sin^6 x)^{\frac{1}{2}} + c$,

where c is a constant of integration, then $\lambda f\left(\frac{\pi}{3}\right)$ is equal

to:

(8-01-2020/Shift-1)

(a) $-\frac{9}{8}$

(b) $\frac{9}{8}$

(c) 2

(d) -2

34. If $f'(x) = \tan^{-1}(\sec x + \tan x)$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$

and $f(0) = 0$, then $f(1)$ is equal to:

(9-01-2020/Shift-1)

(a) $\frac{\pi + 1}{4}$

(b) $\frac{\pi + 2}{4}$

(c) $\frac{1}{4}$

(d) $\frac{\pi - 1}{4}$

35. The integral $\int \frac{dx}{(x+4)^{\frac{8}{7}}(x-3)^{\frac{6}{7}}}$ is equal to: (where C is a

constant of integration)

(9-01-2020/Shift-1)

(a) $-\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + C$

(b) $\frac{1}{2}\left(\frac{x-3}{x+4}\right)^{\frac{3}{7}} + C$

(c) $\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + C$

(d) $-\frac{1}{13}\left(\frac{x-3}{x+4}\right)^{\frac{13}{7}} + C$

36. If $\int \frac{d\theta}{\cos^2 \theta (\tan 2\theta + \sec 2\theta)} = \lambda \tan \theta + 2 \log_e |f(\theta)| + C$

where C is constant of integration, then the ordered pair

$(\lambda, f(\theta))$ is equal to:

(9-1-2020/Shift-2)

(a) $(-1, 1 - \tan \theta)$

(b) $(-1, 1 + \tan \theta)$

(c) $(1, 1 + \tan \theta)$

(d) $(1, 1 - \tan \theta)$

37. If $\int \frac{dx}{(x^2 + x + 1)^2} = a \tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + b\left(\frac{2x+1}{x^2 + x + 1}\right) + C$,

$x > 0$ where C is the constant of integration, then the

value of $9(\sqrt{3}a + b)$ is equal to _____.

(27-08-2021/Shift-1)

38. If $\int \frac{2e^x + 3e^{-x}}{4e^x + 7e^{-x}} dx = \frac{1}{14}(ux + v \log_e(4e^x + 7e^{-x})) + C$,

where C is a constant of integration, then $u + v$ is equal

to _____.

(27-08-2021/Shift-2)

39. The integral $\int \frac{1}{\sqrt[4]{(x-1)^3(x+2)^5}} dx$ is equal to

(where C is a constant of integration)

(31-08-2021/Shift-1)

(a) $\frac{3}{4}\left(\frac{x+2}{x-1}\right)^{\frac{5}{4}} + c$

(b) $\frac{3}{4}\left(\frac{x+2}{x-1}\right)^{\frac{1}{4}} + c$

(c) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{\frac{1}{4}} + c$

(d) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{\frac{5}{4}} + c$

40. $\int \frac{\sin x}{\sin^3 x + \cos^3 x} dx = \alpha \log_e |1 + \tan x| + \beta \log_e$

$|1 - \tan x + \tan^2 x| + \gamma \tan^{-1}\left(\frac{2 \tan x - 1}{\sqrt{3}}\right) + C$,

When C is constant of integration, then the value of

$18(\alpha + \beta + \gamma^2)$ is

(31-08-2021/Shift-2)



41. If $\int \frac{\cos x - \sin x}{\sqrt{8 - \sin 2x}} dx = a \sin^{-1} \left(\frac{\sin x + \cos x}{b} \right) + c$ where c is a constant of integration, then the ordered pair (a, b) is equal to (24-02-2021/Shift-1)

- (a) $(3, 1)$ (b) $(1, -3)$
(c) $(1, 3)$ (d) $(-1, 3)$

42. The value of the integral

$$\int \frac{\sin \theta \cdot \sin 2\theta (\sin^6 \theta + \sin^4 \theta + \sin^2 \theta) \sqrt{2 \sin^4 \theta + 3 \sin^2 \theta + 6}}{1 - \cos 2\theta} d\theta$$

is:

(where c is a constant of integration)

(25-02-2021/Shift-1)

- (a) $\frac{1}{18} [9 - 2 \cos^6 \theta - 3 \cos^4 \theta - 6 \cos^2 \theta]^{\frac{3}{2}} + c$
(b) $\frac{1}{18} [11 - 18 \cos^2 \theta + 9 \cos^4 \theta - 2 \cos^6 \theta]^{\frac{3}{2}} + c$
(c) $\frac{1}{18} [9 - 2 \sin^6 \theta - 3 \sin^4 \theta - 6 \sin^2 \theta]^{\frac{3}{2}} + c$
(d) $\frac{1}{18} [11 - 18 \sin^2 \theta + 9 \sin^4 \theta - 2 \sin^6 \theta]^{\frac{3}{2}} + c$

43. The integral $\int \frac{e^{3 \log_e 2x} + 5e^{2 \log_e 2x}}{e^{4 \log_e x} + 5e^{3 \log_e x} - 7e^{2 \log_e x}} dx, x > 0$ is equal

to:

(where c is a constant of integration)

(25-02-2021/Shift-2)

- (a) $4 \log_e |x^2 + 5x - 7| + c$
(b) $\log_e \sqrt{x^2 + 5x - 7} + c$
(c) $\frac{1}{4} \log_e |x^2 + 5x - 7| + c$
(d) $\log_e |x^2 + 5x - 7| + c$

44. For the real numbers α, β, γ and δ if

$$\int \frac{(x^2 - 1) + \tan^{-1} \left(\frac{x^2 + 1}{x} \right)}{(x^4 + 3x^2 + 1) \tan^{-1} \left(\frac{x^2 + 1}{x} \right)} dx$$

$$= \alpha \log_e \left(\tan^{-1} \left(\frac{x^2 + 1}{x} \right) \right) + \beta \tan^{-1} \left(\frac{\gamma(x^2 - 1)}{x} \right)$$

$$+ \delta \tan^{-1} \left(\frac{x^2 + 1}{x} \right) + C$$

where C is an arbitrary constant, then the value of $10(\alpha + \beta\gamma + \delta)$ is equal to (16-03-2021/Shift-2)

45. The integral $\int \frac{(2x - 1) \cos \sqrt{(2x - 1)^2 + 5}}{\sqrt{4x^2 - 4x + 6}} dx$ is equal to :

(where c is a constant of integration)

(18-03-2021/Shift-1)

- (a) $\frac{1}{2} \cos \sqrt{(2x - 1)^2 + 5} + c$
(b) $\frac{1}{2} \sin \sqrt{(2x + 1)^2 + 5} + c$
(c) $\frac{1}{2} \cos \sqrt{(2x + 1)^2 + 5} + c$
(d) $\frac{1}{2} \sin \sqrt{(2x - 1)^2 + 5} + c$



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

Objective Questions I [Only one correct option]

1. Evaluate : $\int \frac{\sin x \, dx}{\sqrt{3 \sin^2 x + 4 \cos^2 x}}$

(a) $-\ln(\cos x + \sqrt{3 + \cos^2 x}) + c$

(b) $\ln(\cos x + \sqrt{3 + \cos^2 x}) + c$

(c) $-\ln(\sin x + \sqrt{3 + \sin^2 x}) + c$

(d) None of these

2. If $\int \log(x^2 + x) \, dx = x \log|x| + (x+1) \log|x+1| + k$, then k equals

(a) $2x + \log|x+1| + c$ (b) $2x - \log|x+1| + c$

(c) constant (d) None

3. If $I = \int \cos \theta \log\left(\tan \frac{\theta}{2}\right) d\theta$, then I equals

(a) $\sin \theta \log(\tan \theta/2) + \theta + c$

(b) $\cos \theta \log(\tan \theta/2) + \theta + c$

(c) $\sin \theta \log(\tan \theta/2) - \theta + c$

(d) None of these

4. If $\int g(x) \, dx = g(x)$ then $\int g(x)(f(x) + f'(x)) \, dx$ is equal to

(a) $g(x)f(x) - g(x)f'(x) + c$ (b) $g(x)f'(x) + c$

(c) $g(x)f(x) + c$ (d) $g(x)f^2(x) + c$

5. If $\int (x^3 - 2x^2 + 5) e^{3x} \, dx = e^{3x}(Ax^3 + Bx^2 + Cx + 13/9)$ then which of the following statement is incorrect :

(a) $3C = 2$ (b) $A + B + \frac{2}{3} = 0$

(c) $C + 2B = 0$ (d) $A + B + C = 0$

6. $\int \frac{e^{2 \tan^{-1} x} (1+x)^2}{(1+x^2)} \, dx$ is equal to

(a) $xe^{\tan^{-1} x} + c$

(b) $xe^{2 \tan^{-1} x} + c$

(c) $2xe^{2 \tan^{-1} x} + c$

(d) none of these

7. If $I = \int \sin^7 x \, dx$, then I equals

(a) $-\cos x + \cos^3 x - \cos^5 x + \frac{1}{7} \cos^7 x + c$

(b) $-\cos x + \cos^3 x - \frac{3}{5} \cos^5 x + \frac{1}{7} \cos^7 x + c$

(c) $\tan x - \tan^3 x + \sin x - \frac{1}{3} \cos^3 x + c$

(d) $-\cos x + \cos^5 x - \frac{1}{3} \cos^3 x + \frac{1}{7} \cos^7 x + c$

8. If $\int \frac{dx}{(x^2 - 4)\sqrt{x+1}} = \frac{1}{k} \log \left| \frac{\sqrt{x+1} - \sqrt{3}}{\sqrt{x+1} + \sqrt{3}} \right|$

$-\frac{1}{2} \tan^{-1} \sqrt{x+1} + c$ then k equals

(a) $2\sqrt{3}$

(b) $4\sqrt{3}$

(c) $\frac{1}{4\sqrt{3}}$

(d) none of these

9. Evaluate $\int \frac{(x+2) \, dx}{(x^2 + 3x + 3)\sqrt{x+1}}$

(a) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3} \sqrt{x+1}}{-x} \right) + c$

(b) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3} \sqrt{x+1}}{x} \right) + c$

(c) $\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3} \sqrt{x+1}}{-x} \right) + c$

(d) $\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3} \sqrt{x+1}}{x} \right) + c$



10. Evaluate : $\int \frac{\sqrt{x-1}}{x\sqrt{x+1}} dx$

(a) $\ln(x + \sqrt{x^2 + 1}) + \sec^{-1} x + c$

(b) $\ln(x - \sqrt{x^2 + 1}) - \sec^{-1} x + c$

(c) $\ln(x + \sqrt{x^2 - 1}) - \sec^{-1} x + c$

(d) None of these

11. Let $g(x)$ be an antiderivative of $f(x)$. Then $\ln(1+(g(x))^2)$ is an antiderivative for :

(a) $\frac{2f(x) \cdot g(x)}{1+(f(x))^2}$

(b) $\frac{2f(x) \cdot g(x)}{1+(g(x))^2}$

(c) $\frac{2f(x)}{1+(f(x))^2}$

(d) None

12. If $I_n = \int (\ln x)^n dx$ then $I_n + n I_{n-1} =$

(a) $(\ln x)^n$

(b) $x (\ln x)^n$

(c) $x^n \ln x$

(d) $x (\ln x)^{n-1}$

13. If $I_n = \int \cos^n x dx$ then $I_n - \left(\frac{n-1}{n}\right) I_{n-2} =$

(a) $\cos^{n-1} x \cdot \sin x + c$

(b) $\frac{1}{n} (\cos^{n-1} x \sin x) + c$

(c) $\frac{1}{n} (\cos^n x \sin x) + c$

(d) None of these

14. If $I = \int \frac{dx}{(a^2 - b^2 x^2)^{3/2}}$

(a) $\frac{x}{\sqrt{a^2 - b^2 x^2}} + c$

(b) $\frac{x}{a^2 \sqrt{a^2 - b^2 x^2}} + c$

(c) $\frac{ax}{\sqrt{a^2 - b^2 x^2}} + c$

(d) none of these

15. If $I = \int \frac{x^3 + x}{x^4 - 9} dx$ then I equals :

(a) $\frac{1}{4} \log |x^4 - 9| + \frac{1}{12} \log \left| \frac{x^2 + 3}{x^2 - 3} \right| + c$

(b) $\frac{1}{4} \log |x^4 - 9| + \frac{1}{12} \log \left| \frac{x^2 - 3}{x^2 + 3} \right| + c$

(c) $\frac{1}{4} \log |x^4 - 9| - \frac{1}{12} \log \left| \frac{x-3}{x+3} \right| + c$

(d) None

16. Evaluate $I = \int \left(\sqrt{\frac{a+x}{a-x}} + \sqrt{\frac{a-x}{a+x}} \right) dx$

(a) $2 \sin^{-1} \left(\frac{x}{a} \right) + c$

(b) $2a \sin^{-1} \left(\frac{x}{a} \right) + c$

(c) $2 \cos^{-1} \left(\frac{x}{a} \right) + c$

(d) $2a \cos^{-1} \left(\frac{x}{a} \right) + c$

17. $\int \sqrt{\frac{e^x + a}{e^x - a}} dx$

(a) $\ln(e^x + \sqrt{e^{2x} - a^2}) + \sin^{-1}(ae^{-x}) + c$

(b) $\ln(e^x - \sqrt{e^{2x} - a^2}) + \cos^{-1}(ae^{-x}) + c$

(c) $\ln(e^x + \sqrt{e^{2x} - a^2}) + \cos^{-1}(ae^{-x}) + c$

(d) None

18. $\int \frac{x^2 - 1}{(x^4 + 3x^2 + 1) \tan^{-1} \left(\frac{x^2 + 1}{x} \right)} dx$ is equal to

(a) $\tan^{-1} \left(x + \frac{1}{x} \right) + c$

(b) $\log_e \left| \tan^{-1} \left(x + \frac{1}{x} \right) \right| + c$

(c) $\log_e \left| \tan \left(\frac{x^2 + 1}{x} \right) \right| + c$

(d) $\left(x + \frac{1}{x} \right) \tan^{-1} \left(x + \frac{1}{x} \right) + c$



19. $\int \sqrt{e^{2x}-1} dx$ is equal to
- (a) $\sqrt{e^{2x}-1} + \sec^{-1} e^{2x} + c$
 (b) $\sqrt{e^{2x}-1} - \sec^{-1} e^{2x} + c$
 (c) $\sqrt{e^{2x}-1} - \sec^{-1} e^x + c$
 (d) none of these
20. Evaluate : $\int \frac{(1+x)}{x(1+xe^x)^2} dx$
- (a) $\ln\left(\frac{xe^x}{1+xe^x}\right) + \frac{1}{1+xe^x} + c$
 (b) $\ln\left(\frac{1+xe^x}{xe^x}\right) + \frac{1}{xe^x+1} + c$
 (c) $\ln\left(\frac{xe^x}{xe^x+1}\right) - \frac{1}{xe^x+1} + c$
 (d) None of these
21. If $f(x)$ is a polynomial function of the n^{th} degree, then $\int e^x f(x) dx$ is equal to
- (a) $e^x \{f(x) - f'(x) - f''(x) - f'''(x) - \dots - (-1)^n f^{(n)}(x)\}$
 (b) $e^x \{f(x) - f'(x) + f''(x) - f'''(x) + \dots + (-1)^n f^{(n)}(x)\}$
 (c) $e^x \{f(x) - f'(x) + f''(x) - f'''(x) + \dots + (-1)^n f^{(n)}(x)\}$
 (d) none of these
22. If $I = \int e^x (x \cos x + \sin x) dx$ then I equals :
- (a) $\frac{1}{2} e^x (x \sin x - \cos x) + c$
 (b) $\frac{1}{2} e^x (x \sin x + \cos x) + c$
 (c) $\frac{1}{2} e^x (x \cos x - \sin x) + c$
 (d) None
23. $\int \sqrt[3]{x} (\log_e x)^2 dx =$
- (a) $\frac{3}{4} (x)^{4/3} \left[(\log_e x)^2 - \frac{3}{2} \log_e x - \frac{9}{8} \right] + c$
 (b) $\frac{3}{4} (x)^{4/3} \left[(\log_e x)^2 + \frac{3}{2} \log_e x + \frac{9}{8} \right] + c$
 (c) $\frac{3}{4} (x)^{4/3} \left[(\log_e x)^2 - \frac{3}{2} \log_e x + \frac{9}{8} \right] + c$
 (d) $\frac{3}{4} (x)^{1/3} \left[(\log_e x)^2 + \frac{3}{2} \log_e x - \frac{9}{8} \right] + c$
24. If $I = \int \frac{\sqrt{\sin^3 2x}}{\sin^5 x} dx$, and $f(x) = (\cot x)^{3/2}$, $g(x) = (\cot x)^{5/2}$, then I equals
- (a) $\frac{2\sqrt{3}}{3} f(x) - \frac{1}{5} g(x) + c$ (b) $-\frac{4\sqrt{2}}{5} g(x) + c$
 (c) $\frac{1}{2\sqrt{3}} f(x) + c$ (d) $\frac{2\sqrt{2}}{3} f(x) + \frac{1}{5} g(x) + c$
25. $\int \frac{dx}{\tan x + \cot x + \sec x + \operatorname{cosec} x}$ is equal to
- (a) $\frac{1}{2} (\sin x - \cos x + x) + c$
 (b) $\frac{1}{2} (\sin x - \cos x - \tan x + \cot x) + c$
 (c) $\frac{1}{2} (\sin x - \cos x - x) + c$
 (d) $\frac{1}{2} (\sin x + \cos x - \tan x - \cot x + x) + c$
26. $\int \frac{\sin x - \cos x}{(\sin x + \cos x) \sqrt{\sin x \cos x + \sin^2 x \cos^2 x}} dx =$
- (a) $-\sin(\sin 2x + 1) + c$
 (b) $\operatorname{cosec}(\sin 2x + 1)$
 (c) $-\sec^{-1}(\sin 2x + 1) + c$
 (d) $\tan^{-1}(\sin 2x + 1) + c$



27. If $y = \sqrt{x^2 - x + 1}$ and for $n \geq 1$, $I_n = \int x^n / y dx$ and $aI_3 + bI_2 + cI_1 = x^2y$, then (a, b, c) is equal to

(a) $\left(\frac{3}{2}, \frac{1}{2}, -1\right)$ (b) $(1, -1, 1)$

(c) $\left(3, -\frac{5}{2}, 2\right)$ (d) $\left(\frac{1}{2}, -\frac{1}{2}, 1\right)$

28. If $\lim_{x \rightarrow 0} \frac{f(x)}{x^2}$ exists finitely and

$$\lim_{x \rightarrow 0} \left(1 + x + \frac{f(x)}{x}\right)^{1/x} = e^3, \text{ where } f(x) = ax^2 + bx + c$$

then $\int f(x) \log_e x dx$ is equal to

(a) $\frac{2}{3}x^3 \left(\log_e x - \frac{1}{3}\right) + c$ (b) $\frac{x^3}{3} \left(\log_e x - \frac{1}{3}\right) + c$

(c) $\frac{2}{3}x^3 (\log_e x + 1) + c$ (d) $\frac{2}{3}x^3 (\log_e x - 1) + c$

29. For $0 < x < 1$, let

$$f(x) = \lim_{n \rightarrow \infty} (1+x)(1+x^2)(1+x^4) \dots (1+x^{2^n})$$

then $\int \frac{f(x)}{1-x} \log_e x dx$ equals

(a) $\log_e \left(\frac{x}{1-x}\right) + c$

(b) $-\log_e \left(\frac{x}{1-x}\right) + \frac{\log_e x}{1-x} + c$

(c) $\frac{\log_e x}{1-x} + \log_e (1-x) + c$

(d) $x \log_e x + \log_e (1-x) + c$

30. If $\int \left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)^{1/2} \frac{dx}{x} = 2 \cos^{-1} \sqrt{x} - f(x) + c$, then $f(x)$ equals.

(a) $\log_e \left(\frac{1+\sqrt{1-x}}{\sqrt{x}}\right)$ (b) $\frac{1}{2} \log_e \left(\frac{1+\sqrt{1-x}}{\sqrt{x}}\right)$

(c) $2 \log_e \left(\frac{1-\sqrt{1-x}}{\sqrt{x}}\right)$ (d) $2 \log_e \left(\frac{1+\sqrt{1-x}}{\sqrt{x}}\right)$

31. $\int \frac{\sqrt{x} - \sqrt{a}}{\sqrt{x+a}} dx$

(a) $\sqrt{ax+x^2} - 2\sqrt{a^2+ax} - a/\ln(\sqrt{x} + \sqrt{a+x}) + c$

(b) $\sqrt{ax+x^2} - 2\sqrt{a+x} - a/\ln(\sqrt{x} + \sqrt{a+x}) + c$

(c) $\sqrt{ax+x^2} - 2\sqrt{a^2+ax} + a/\ln(\sqrt{x} + \sqrt{a-x}) + c$

(d) None of these

32. If $I = \int \frac{x^2 + a^2}{x^4 - a^2x^2 + a^4} dx$, then $I =$

(a) $\frac{1}{a} \tan^{-1} \left(\frac{ax}{x^2 - a^2}\right) + c$ (b) $\frac{1}{a} \tan^{-1} \left(\frac{x^2 - a^2}{ax}\right) + c$

(c) $\log |x + \sqrt{x^2 - a^2}| + c$ (d) None of these

33. Evaluate: $\int \sqrt{x + \sqrt{x^2 + 2}} dx$

(a) $\frac{2}{3} (x + \sqrt{x^2 + 2})^{3/2} - \frac{1}{\sqrt{x + \sqrt{x^2 + 2}}} + c$

(b) $\frac{1}{3} (x + \sqrt{x^2 + 2})^{3/2} - \frac{2}{\sqrt{x + \sqrt{x^2 + 2}}} + c$

(c) $\frac{1}{3} (x + \sqrt{x^2 + 2})^{1/2} - \frac{2}{\sqrt{x + \sqrt{x^2 + 2}}} + c$

(d) None of these



34. $\int \frac{x-1}{(x+1)\sqrt{x(x^2+x+1)}} dx$ is

(a) $\tan^{-1}\left(\frac{x^2+x+1}{x}\right) + c$

(b) $2\tan^{-1}\left(\frac{x^2+x+1}{x}\right) + c$

(c) $\tan^{-1}\left(\frac{\sqrt{x^2+x+1}}{x}\right) + c$

(d) $2\tan^{-1}\sqrt{x+\frac{1}{x}+1} + c$

35. $\int \frac{(1+x)\sin x}{(x^2+2x)\cos^2 x - (1+x)\sin 2x} dx = \frac{1}{2} \log_e \left| \frac{t+1}{t-1} \right| + c$

where t is

(a) $(x+1)\cos x - \sin x$ (b) $(x+1)\sin x - \cos x$

(c) $(x+1)\sin x + \cos x$ (d) $(x+1)\cos x + \sin x$

36. $\int \frac{(x^2-1)dx}{2x\sqrt{x^4+4x^3-6x^2+4x+1}}$ is

(a) $\frac{1}{2} \ln \left| x + \frac{1}{x} + 2 + \sqrt{\left(x + \frac{1}{x} + 2\right)^2 - 12} \right| + c$

(b) $\frac{1}{2} \ln \left| x - \frac{1}{x} + 2 + \sqrt{\left(x - \frac{1}{x} + 2\right)^2 - 12} \right| + c$

(c) $\frac{1}{2} \ln \left| x + \frac{1}{x} - 2 + \sqrt{\left(x + \frac{1}{x} - 2\right)^2 - 12} \right| + c$

(d) None of these

37. Evaluate : $\int \sqrt{\frac{1+x^{2n}}{x^{2n}}} \cdot \frac{\ln(1+x^{2n}) - 2n/\ln x}{x^{2n+1}} dx$

(a) $\frac{2P^3}{9n}(1-3/\ln P) + C$ (b) $\frac{P^3}{3n}(3/\ln P - 1) + C$

(c) $\frac{2P^3}{3n}(3/\ln P - 1) + C$ (d) None of these

where $P = \left(1 + \frac{1}{x^{2n}}\right)^{1/2}$

38. Evaluate $I = \int \frac{dx}{\sin^3 x + \cos^3 x}$

(a) $\frac{1}{3\sqrt{2}} \log \left| \frac{\sqrt{2}+t}{\sqrt{2}-t} \right| + \tan^{-1} t + c$

(b) $\frac{1}{3\sqrt{2}} \log \left| \frac{\sqrt{2}-t}{\sqrt{2}+t} \right| + \frac{1}{3} \tan^{-1} t + c$

(c) $\frac{1}{3\sqrt{2}} \log \left| \frac{\sqrt{2}+t}{\sqrt{2}-t} \right| + \frac{2}{3} \tan^{-1} t + c$

(d) None of these

where $t = \sin x - \cos x$

39. Evaluate : $\int \frac{\sec x dx}{\sqrt{\sin(2x+A) + \sin A}}$

(a) $2 \cos A \sqrt{2 \cos A + \sin A \tan x} + c$

(b) $\sqrt{2} \sec A \sqrt{2 \cos A \tan x + 2 \sin A} + c$

(c) $\sqrt{2} \sec A \sqrt{\cos A \tan x + \sin A} + c$

(d) None of these

40. $\int \sqrt{\sec x - 1} dx$

(a) $2 \ln(\sqrt{\cos x} + \sqrt{1 + \cos x}) + c$

(b) $2 \ln(\sqrt{\sin x} + \sqrt{1 + \sin x}) + c$

(c) $-2 \ln(\sqrt{\cos x} + \sqrt{1 + \cos x}) + c$

(d) None of these



41. $\int (\sin 4x) e^{\tan^2 x} dx =$
- (a) $-2e^{\tan^2 x} \cos^4 x + c$ (b) $2e^{\tan^2 x} \sec^4 x + c$
- (c) $-2e^{\tan^2 x} \sec^2 x + c$ (d) $2e^{\tan^2 x} \cos^2 x + c$

42. If $I = \int \frac{(2x+3)dx}{(x^2+2x+3)\sqrt{x^2+2x+4}}$, then I equals

- (a) $\log \left| \frac{\sqrt{x^2+2x+4}-1}{\sqrt{x^2+2x+4}+1} \right| + c$
- (b) $\log \left| \frac{\sqrt{x^2+2x+4}-1}{\sqrt{x^2+2x+4}+1} \right| + \tan^{-1} \left(\frac{x+2}{3} \right) + c$
- (c) $\log \tan^{-1} \left(\frac{\sqrt{x+3}}{2} \right) + c$
- (d) None

43. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a function satisfying the following :

- (i) $f(-x) = -f(x)$
- (ii) $f(x+1) = f(x) + 1$
- (iii) $f\left(\frac{1}{x}\right) = \frac{f(x)}{x^2} \quad \forall x \neq 0$

then $\int e^x f(x) dx$ is equal to

- (a) $e^x (x-1) + c$ (b) $e^x \log x + c$
- (c) $\frac{e^x}{x} + c$ (d) $\frac{e^x}{x+1} + c$

Objective Questions II [One or more than one correct option]

44. If primitive of $\sin(\log x)$ is $f(x) (\sin(g(x)) - \cos(h(x))) + c$ then

- (a) $\lim_{x \rightarrow 2} f(x) = 1$ (b) $\lim_{x \rightarrow 1} \frac{g(x)}{h(x)} = 1$
- (c) $g(e^3) = 3$ (d) $h(e^5) = 5$

45. $\int \frac{dx}{5+4\cos x} = \lambda \tan^{-1} \left(m \tan \frac{x}{2} \right) + c$ then

- (a) $\lambda = \frac{2}{3}$ (b) $m = \frac{1}{3}$
- (c) $\lambda = \frac{1}{3}$ (d) $m = \frac{2}{3}$

46. $\int \frac{1}{x^2-1} \ln \left(\frac{x-1}{x+1} \right) dx$ equals

- (a) $\frac{1}{2} \ln \left(\frac{x-1}{x+1} \right) + c$ (b) $\frac{1}{4} \ln^2 \left(\frac{x-1}{x+1} \right) + c$
- (c) $\frac{1}{2} \ln^2 \left(\frac{x+1}{x-1} \right) + c$ (d) $\frac{1}{4} \ln^2 \left(\frac{x+1}{x-1} \right) + c$

47. If $f(x) = \lim_{n \rightarrow \infty} e^{x \tan(1/n) \log(1/n)}$ and $\int \frac{f(x) dx}{\sqrt[3]{\sin^{11} x \cos x}} = g(x) + c$

then

- (a) $g\left(\frac{\pi}{4}\right) = \frac{3}{2}$
- (b) $g(x)$ is continuous for all x
- (c) $g\left(\frac{\pi}{4}\right) = \frac{-15}{8}$
- (d) $g(x)$ is non-differentiable at infinitely many points

Numerical Value Type Questions

48. Let f be a function satisfying $f''(x) = x^{-3/2}$, $f'(4) = 2$ and $f(0) = 0$ then $f(784)$ is equal to

49. If the graph of the antiderivative $F(x)$ of $f(x) = \log(\log x) + (\log x)^{-2}$ passes through $(e, 1998-e)$, then the term independent of x in $F(x)$ is

50. Let $F(x)$ be the antiderivative of

$$f(x) = \frac{1}{(3+5\sin x + 3\cos x)} \text{ whose graph passes through}$$

the point $(0, 0)$ then the value of $F\left(\frac{\pi}{2}\right) - \frac{1}{5} \log \frac{8}{3} + 1982$ is equal to

51. $f(x)$ is the integral of $\frac{2\sin x - \sin 2x}{x^3}$, $x \neq 0$ find $\lim_{x \rightarrow 0} f'(x)$.

52. If $\int \frac{x^2-1}{(x^2+1)\sqrt{x^4+1}} dx$ is equal to

$$\frac{\sqrt{2}}{4} A \tan^{-1} \left(\frac{1}{\sqrt{2}} \sqrt{x^2 + \frac{1}{x^2}} \right) + c \text{ then } A \text{ is equal to}$$



Assertion & Reason

- (A) If ASSERTION is true, REASON is true, REASON is a correct explanation for ASSERTION.
 (B) If ASSERTION is true, REASON is true, REASON is not a correct explanation for ASSERTION.
 (C) If ASSERTION is true, REASON is false.
 (D) If ASSERTION is false, REASON is true.

53. Assertion : If $\int \frac{1}{f(x)} dx = 2 \log |f(x)| + c$, then $f(x) = \frac{x}{2}$

Reason : When $f(x) = \frac{x}{2}$, then

$$\int \frac{1}{f(x)} dx = \int \frac{2}{x} dx = 2 \log |x| + c$$

- (a) A (b) B
 (c) C (d) D

54. Assertion : $\int \frac{x^2-1}{x^2} e^{\frac{x^2+1}{x}} dx = e^{\frac{x^2+1}{x}} + c$

Reason : $\int f(x) e^{f(x)} dx = e^{f(x)} + c$

- (a) A (b) B
 (c) C (d) D

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

55. $\int f(x) dx$ when

- (A) $f(x) = \frac{1}{e^x + 1}$ (P) $2 \log (e^{x/2} + e^{-x/2}) + c$
 (B) $f(x) = e^{e^x + x}$ (Q) $\log \frac{e^x}{e^x + 1} + c$
 (C) $f(x) = \frac{e^x - 1}{e^x + 1}$ (R) $\frac{1}{4} e^{2x^2} + c$
 (D) $f(x) = e^{2x^2 + \log x}$ (S) $e^{e^x} + c$

The correct matching is :

- (a) (A-S, B-Q, C-P, D-R)
 (b) (A-Q, B-S, C-R, D-P)
 (c) (A-Q, B-S, C-P, D-R)
 (d) (A-R, B-S, C-P, D-Q)

56. $\int f(x) dx$, when

(A) $f(x) = \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}$ (P) $c - \frac{2}{\sqrt{15}} \log \left| \frac{x+6+\sqrt{60x-15x^2}}{2x-3} \right|$

(B) $f(x) = \sqrt{\frac{1-x}{1+x}} \cdot \frac{1}{x}$ (Q) $\log \left| \frac{\sqrt{x+1} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \right| + 2$

$$\tan^{-1} \sqrt{\frac{1-x}{1+x}} + c$$

(C) $f(x) = \frac{1}{\left(x - \frac{3}{2}\right) \sqrt{4x-x^2}}$ (R) $(\sqrt{x}-2) \sqrt{1-x} - \sin^{-1} \sqrt{x} + c$

(D) $f(x) = \sqrt{\frac{1+\sqrt{x}}{x}}$ (S) $\frac{4}{3} (1+\sqrt{x})^{3/2} + c$

The correct matching is :

- (a) (A-Q, B-R, C-P, D-S)
 (b) (A-R, B-Q, C-P, D-S)
 (c) (A-R, B-P, C-Q, D-S)
 (d) (A-R, B-Q, C-S, D-P)

Using the following passage, solve Q.57 to Q.60

Passage – 1

We can derive reduction formula for the integration of the form $\int \sin^n x dx$, $\int \cos^n x dx$, $\int \tan^n x dx$ and other integrals of these form using integration by parts. In turn these reduction formulas can be used to compute integrals of higher power of $\sin x$ and $\cos x$.

57. If $I = \int \sec^6 x dx = \frac{1}{5} \tan^5 x + A \tan^3 x + \tan x + c$ then A is equal to

- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
 (c) $-\frac{1}{3}$ (d) $-\frac{2}{3}$



58. If $\int \sin^5 x \, dx = -\frac{1}{5} \sin^4 x \cos x + A \sin^2 x \cos x - \frac{8}{15} \cos x + c$

then A is equal to

- (a) $-\frac{2}{15}$ (b) $-\frac{3}{5}$
(c) $-\frac{4}{15}$ (d) $-\frac{1}{15}$

59. If $\int \tan^6 x \, dx = \frac{1}{5} \tan^5 x + A \tan^3 x + \tan x - x + c$ then A is

equal to

- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $-\frac{2}{3}$ (d) $-\frac{1}{3}$

60. If $\int \operatorname{cosec}^n x \, dx = -\frac{\operatorname{cosec}^{n-2} x \cot x}{n-1} + A \int \operatorname{cosec}^{n-2} x \, dx$

then A is equal to

- (a) $\frac{1}{n-2}$ (b) $\frac{n}{n-2}$
(c) $\frac{n-1}{n-2}$ (d) $\frac{n-2}{n-1}$

Using the following passage, solve Q.61 and Q.62

Passage – 2

If the integrand is a rational function of x and fractional

powers of a linear fractional function of the form $\frac{ax+b}{cx+d}$.

Then rationalization of the integral is affected by the

substitution $\frac{ax+b}{cx+d} = t^m$, where m is the L.C.M. of

fractional powers of $\frac{ax+b}{cx+d}$.

61. If $I = \int \frac{dx}{\sqrt[4]{(x-1)^3 (x+2)^5}} = A \sqrt[4]{\frac{x-1}{x+2}} + c$ then A is equal to

- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $\frac{3}{4}$ (d) $\frac{4}{3}$

62. If $I = \int \frac{(2x-3)^{1/2}}{(2x-3)^{1/3} + 1} dx = 3 \left[\frac{1}{7} (2x-3)^{7/6} - \frac{1}{5} (2x-3)^{5/6} + \right.$

$\left. \frac{1}{3} (2x-3)^{1/2} - (2x-3)^{1/6} + g(x) \right] - 1$ then g(x) is equal to

- (a) $\tan^{-1} (2x-3)^{1/6}$ (b) $(2x-3)^{1/2}$
(c) $3 \tan^{-1} (2x-3)^{1/6}$ (d) $4 (2x-3)^{1/6}$

Text

63. Integrate $\int \frac{x^3 + 3x + 2}{(x^2 + 1)^2 (x + 1)} dx$

64. Evaluate $\int \sin^{-1} \left(\frac{2x+2}{\sqrt{4x^2 + 8x + 13}} \right) dx$

65. Evaluate $\int \frac{dx}{(x+1) \sqrt{x^2 + x + 1}}$



EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTIONS

Objective Questions I [Only one correct option]

1. Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and

$$g(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{f \text{ occurs } n \text{ times}}(x)$$

Then $\int x^{n-2} g(x) dx$ equals (2007)

(a) $\frac{1}{n(n-1)}(1+nx^n)^{1-\frac{1}{n}} + c$

(b) $\frac{1}{n-1}(1+nx^n)^{1-\frac{1}{n}} + c$

(c) $\frac{1}{n(n+1)}(1+nx^n)^{1+\frac{1}{n}} + c$

(d) $\frac{1}{n+1}(1+nx^n)^{1+\frac{1}{n}} + c$

2. Let $I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx$, $J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx$

Then, for an arbitrary constant c , the value of $J-I$ equals

(2008)

(a) $\frac{1}{2} \log \left(\frac{e^{4x} - e^{2x} + 1}{e^{4x} + e^x + 1} \right) + c$

(b) $\frac{1}{2} \log \left(\frac{e^{2x} + e^x + 1}{e^{2x} - e^x + 1} \right) + c$

(c) $\frac{1}{2} \log \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + c$

(d) $\frac{1}{2} \log \left(\frac{e^{4x} + e^{2x} + 1}{e^{4x} - e^{2x} + 1} \right) + c$

3. The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ equals to (for some arbitrary constant C) (2012)

(a) $\frac{-1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + C$

(b) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + C$

(c) $\frac{-1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + C$

(d) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + C$

Objective Questions II [One or more than one correct option]

4. Let b be a nonzero real number. Suppose $f: R \rightarrow R$ is a differentiable function such that $f(0) = 1$. If the derivative f' of f satisfies the equation $f'(x) = \frac{f(x)}{b^2 + x^2}$

for all $x \in R$, then which of the following statements is/are TRUE? (2020)

(a) If $b > 0$, then f is an increasing function

(b) If $b < 0$, then f is a decreasing function

(c) $f(x) \cdot f(-x) = 1$ for all $x \in R$

(d) $f(x) - f(-x) = 0$ for all $x \in R$



Assertion & Reason

- (A) If ASSERTION is true, REASON is true, REASON is a correct explanation for ASSERTION.
- (B) If ASSERTION is true, REASON is true, REASON is not a correct explanation for ASSERTION.
- (C) If ASSERTION is true, REASON is false.
- (D) If ASSERTION is false, REASON is true.

5. Let $F(x)$ be an indefinite integral of $\sin^2 x$.

Assertion : The function $F(x)$ satisfies $F(x + \pi) = F(x)$ for all real x .

Reason : $\sin^2(x + \pi) = \sin^2 x$ for all real x . (2007)

- (a) A (b) B
(c) C (d) D

Find Answer Key and Detailed Solutions at the end of this book



INDEFINITE INTEGRATION



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06

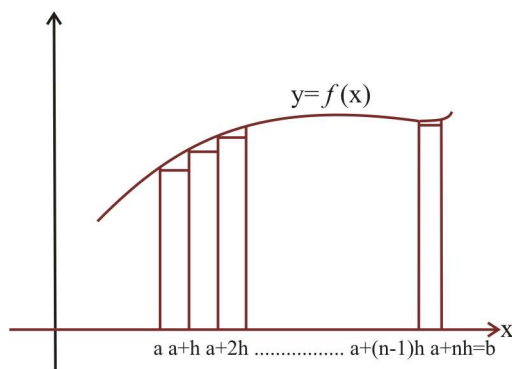
DEFINITE INTEGRATION, AREA UNDER CURVES

DEFINITE INTEGRATION, AREA UNDER CURVES



1. DEFINITION

Let $f(x)$ be a continuous real valued function defined on the closed interval $[a, b]$ which is divided into n parts as shown in figure.



The point of division on x-axis are

$a, a+h, a+2h, \dots, a+(n-1)h, a+nh$, where $\frac{b-a}{n} = h$.

Let S_n denotes the area of these n rectangles.

Then, $S_n = hf(a) + hf(a+h) + hf(a+2h) + \dots + hf(a+(n-1)h)$

Clearly, S_n is area very close to the area of the region bounded by curve $y=f(x)$, x-axis and the ordinates $x=a, x=b$.

$$\text{Hence } \int_a^b f(x) dx = \lim_{n \rightarrow \infty} S_n$$

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} hf(a+rh)$$

$$= \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \left(\frac{b-a}{n} \right) f \left(a + \frac{(b-a)r}{n} \right)$$

NOTES :

1. We can also write

$$S_n = hf(a+h) + hf(a+2h) + \dots + hf(a+nh) \text{ and}$$

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{b-a}{n} \right) f \left(a + \left(\frac{b-a}{n} \right) r \right)$$

$$2. \text{ If } a=0, b=1, \int_0^1 f(x) dx = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{1}{n} f \left(\frac{r}{n} \right)$$

2. DEFINITE INTEGRAL AS LIMIT OF SUM/SUM OF INFINITE SERIES USING DEFINITE INTEGRAL

Step 1. Replace $\frac{r}{n}$ by x , $\frac{1}{n}$ by dx and $\lim_{n \rightarrow \infty} \sum$ by $\int$

Step 2. Evaluate $\lim_{n \rightarrow \infty} \left(\frac{r}{n} \right)$ by putting least and greatest values of r as lower and upper limits respectively.

$$\text{For example } \lim_{n \rightarrow \infty} \sum_{r=1}^{pn} \frac{1}{n} f \left(\frac{r}{n} \right) = \int_0^p f(x) dx$$

$$\left[\lim_{n \rightarrow \infty} \left(\frac{r}{n} \right) \right]_{r=1} = 0, \left[\lim_{n \rightarrow \infty} \left(\frac{r}{n} \right) \right]_{r=np} = p$$

3. SECOND FUNDAMENTAL THEOREM OF CALCULUS

Let $F(x)$ be any antiderivative of $f(x)$, then for any two values of the independent variable x , say a and b , the difference $F(b) - F(a)$ is called the definite integral of $f(x)$ from a to b and is

denoted by $\int_a^b f(x) dx$. Thus $\int_a^b f(x) dx = F(b) - F(a)$,





The numbers a and b are called the limits of integration; a is the lower limit and b is the upper limit. Usually $F(b) - F(a)$ is abbreviated by writing $F(x) \Big|_a^b$.

4. GEOMETRICAL INTERPRETATION OF THE DEFINITE INTEGRAL

4.1

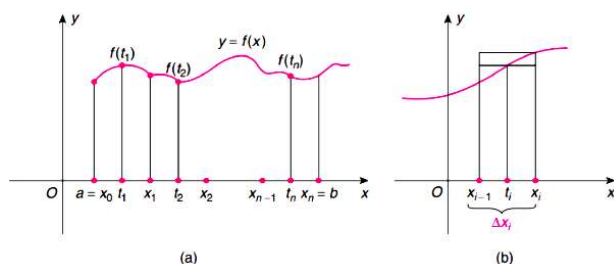


Figure 4.1

Let $f: [a, b] \rightarrow [0, \infty)$ be a function, $P = \{x_0, x_1, x_2, \dots, x_n\}$ be a partition of $[a, b]$ and $t_i \in [x_{i-1}, x_i]$ for $i = 1, 2, \dots, n$. Then (see fig. 4.1) $f(t_i) \Delta x_i = f(t_i)(x_i - x_{i-1})$ = Area of the rectangle with width Δx_i and height $f(t_i)$.

Hence

$$S(f, P) = \sum_{i=1}^n f(t_i) \Delta x_i$$

= Sum of the areas of the rectangles with width $\Delta x_i = x_i - x_{i-1}$ and height $f(t_i)$.

Thus, the area A enclosed by the x -axis, the lines $x = a$, $x = b$ and the curve $y = f(x)$ is approximately equal to $S(f, P)$. When the width of the rectangles becomes smaller, that is when $\text{Max} \{\Delta x_1, \Delta x_2, \dots, \Delta x_n\}$ is small, the sum of the areas or $S(f, P)$ is very nearly equal to A . If f is integrable, then

$S(f, P)$ converges to $\int_a^b f(x) dx$ and hence

$$A = \int_a^b f(x) dx$$

Thus, definite integral of a non-negative function f , when integrable, may be interpreted over $[a, b]$ as the area enclosed by the curve $y = f(x)$, the lines $x = a$, $x = b$ and the x -axis.

4.2 Geometrical Interpretation of the Definite Integral

If $y = f(x)$ is continuous and $\int_a^b f(x) dx = 0$,

then $f(x) = 0$ has at least one real root in (a, b) .

5. PROPERTIES OF DEFINITE INTEGRALS

- $\int_a^b f(x) dx = - \int_b^a f(x) dx$
- $\int_a^b f(x) dx = \int_a^b f(y) dy$
- $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, where c may or may not lie between a and b .
- $\int_0^a f(x) dx = \int_0^a f(a-x) dx$
- $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

NOTES :

$$1. \int_0^a \frac{f(x)}{f(x) + f(a-x)} dx = \frac{a}{2}$$

$$2. \int_a^b \frac{f(x)}{f(x) + f(a+b-x)} dx = \frac{b-a}{2}$$

$$6. \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a-x) dx$$

$$= \begin{cases} 0 & \text{if } f(2a-x) = -f(x) \\ 2 \int_0^a f(x) dx & \text{if } f(2a-x) = f(x) \end{cases}$$



$$7. \quad \int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx & \text{if } f(-x) = f(x) \text{ i.e. } f(x) \text{ is even} \\ 0 & \text{if } f(-x) = -f(x) \text{ i.e. } f(x) \text{ is odd} \end{cases}$$

8. If $f(x)$ is a periodic function of period 'a', i.e. $f(a+x) = f(x)$, then

$$(a) \quad \int_0^{na} f(x) dx = n \int_0^a f(x) dx$$

$$(b) \quad \int_a^{a+na} f(x) dx = n \int_a^{a+a} f(x) dx$$

$$(c) \quad \int_{na}^{b+na} f(x) dx = \int_0^b f(x) dx, \text{ where } b \in \mathbb{R}$$

$$(d) \quad \int_b^{b+a} f(x) dx \text{ independent of } b.$$

$$(e) \quad \int_b^{b+na} f(x) dx = n \int_0^a f(x) dx, \text{ where } n \in \mathbb{I}$$

9. If $f(x) \geq 0$ on the interval $[a, b]$, then $\int_a^b f(x) dx \geq 0$.

10. If $f(x) \leq g(x)$ on the interval $[a, b]$, then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx$$

$$11. \quad \left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

12. If $f(x)$ is continuous on $[a, b]$, m is the least and M is the greatest value of $f(x)$ on $[a, b]$, then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

13. For any two functions $f(x)$ and $g(x)$, integrable on the interval $[a, b]$, the Schwarz – Bunyakovsky inequality holds

$$\left| \int_a^b f(x) \cdot g(x) dx \right| \leq \sqrt{\int_a^b f^2(x) dx \cdot \int_a^b g^2(x) dx}$$

14. If a function $f(x)$ is continuous on the interval $[a, b]$, then there exists a point $c \in (a, b)$ such that

$$\int_a^b f(x) dx = f(c)(b-a), \text{ where } a < c < b.$$

6. DIFFERENTIATION UNDER INTEGRAL SIGN

NEWTON LEIBNITZ'S THEOREM:

If f is continuous on $[a, b]$ and $g(x)$ & $h(x)$ are differentiable functions of x whose values lie in $[a, b]$, then

$$\frac{d}{dx} \left[\int_{g(x)}^{h(x)} f(t) dt \right] = \frac{d}{dx} [h(x)] \cdot f[h(x)] - \frac{d}{dx} [g(x)] \cdot f[g(x)]$$

7. REDUCTION FORMULAE IN DEFINITE INTEGRALS

7.1 If $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$, then show that $I_n = \left(\frac{n-1}{n} \right) I_{n-2}$

Proof: $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$

$$I_n = \left[-\sin^{n-1} x \cos x \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} (n-1) \sin^{n-2} x \cdot \cos^2 x dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cdot (1 - \sin^2 x) dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x dx - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$I_n = \left(\frac{n-1}{n} \right) I_{n-2}$$





NOTES :

$$1. \int_0^{\frac{\pi}{2}} \sin^n x \, dx = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$$

$$2. I_n = \left(\frac{n-1}{n}\right) \left(\frac{n-3}{n-2}\right) \left(\frac{n-5}{n-4}\right) \dots I_0 \text{ or } I_1$$

according as n is even or odd. $I_0 = \frac{\pi}{2}$, $I_1 = 1$

$$\text{Hence } I_n = \begin{cases} \left(\frac{n-1}{n}\right) \left(\frac{n-3}{n-2}\right) \left(\frac{n-5}{n-4}\right) \dots \left(\frac{1}{2}\right) \cdot \frac{\pi}{2} & \text{if } n \text{ is even} \\ \left(\frac{n-1}{n}\right) \left(\frac{n-3}{n-2}\right) \left(\frac{n-5}{n-4}\right) \dots \left(\frac{2}{3}\right) \cdot 1 & \text{if } n \text{ is odd} \end{cases}$$

$$7.2 \quad \text{If } I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx, \text{ then show that } I_n + I_{n-2} = \frac{1}{n-1}$$

$$\text{Proof. } I_n = \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} \cdot \tan^2 x \, dx$$

$$= \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} (\sec^2 x - 1) \, dx$$

$$= \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} \sec^2 x \, dx - \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} \, dx$$

$$= \left[\frac{(\tan x)^{n-1}}{n-1} \right]_0^{\frac{\pi}{4}} - I_{n-2}$$

$$I_n = \frac{1}{n-1} - I_{n-2}$$

$$\therefore I_n + I_{n-2} = \frac{1}{n-1}$$

$$7.3 \quad \text{If } I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^m x \cdot \cos^n x \, dx, \text{ then show that}$$

$$I_{m,n} = \frac{m-1}{m+n} I_{m-2,n}$$

$$\text{Proof. } I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^{m-1} x (\sin x \cos^n x) \, dx$$

$$= \left[-\frac{\sin^{m-1} x \cdot \cos^{n+1} x}{n+1} \right]_0^{\frac{\pi}{2}} +$$

$$\int_0^{\frac{\pi}{2}} \frac{\cos^{n+1} x}{n+1} (m-1) \sin^{m-2} x \cos x \, dx$$

$$= \left(\frac{m-1}{n+1} \right) \int_0^{\frac{\pi}{2}} \sin^{m-2} x \cdot \cos^n x \cdot \cos^2 x \, dx$$

$$= \left(\frac{m-1}{n+1} \right) \int_0^{\frac{\pi}{2}} (\sin^{m-2} x \cdot \cos^n x - \sin^m x \cdot \cos^n x) \, dx$$

$$= \left(\frac{m-1}{n+1} \right) I_{m-2,n} - \left(\frac{m-1}{n+1} \right) I_{m,n}$$

$$\Rightarrow \left(1 + \frac{m-1}{n+1} \right) I_{m,n} = \left(\frac{m-1}{n+1} \right) I_{m-2,n}$$

$$I_{m,n} = \left(\frac{m-1}{m+n} \right) I_{m-2,n}$$





NOTES :

1. $I_{m,n} = \left(\frac{m-1}{m+n} \right) \left(\frac{m-3}{m+n-2} \right) \left(\frac{m-5}{m+n-4} \right) \dots \dots \dots I_{0,n}$ or $I_{1,n}$
according as m is even or odd.

$$I_{0,n} = \int_0^{\frac{\pi}{2}} \cos^n x \, dx \text{ and } I_{1,n} = \int_0^{\frac{\pi}{2}} \sin x \cdot \cos^n x \, dx = \frac{1}{n+1}$$

2. Walli's Formula

$$I_{m,n} = \begin{cases} \frac{(m-1)(m-3)(m-5) \dots (n-1)(n-3)(n-5) \dots \pi}{(m+n)(m+n-2)(m+n-4) \dots \dots \dots 2} & \text{when both } m, n \text{ are even} \\ \frac{(m-1)(m-3)(m-5) \dots (n-1)(n-3)(n-5) \dots \dots \dots}{(m+n)(m+n-2)(m+n-4) \dots \dots \dots} & \text{otherwise} \end{cases}$$

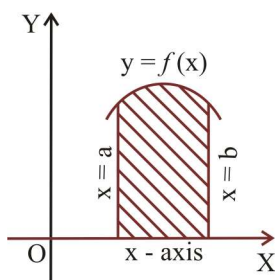
8. AREA OF PLANE REGIONS

1. The area bounded by the curve $y = f(x)$, x -axis and the ordinates $x = a$ and $x = b$ (where $b > a$) is given by

$$A = \int_a^b |y| \, dx = \int_a^b |f(x)| \, dx$$

- (i) If $f(x) > 0 \, \forall x \in [a, b]$

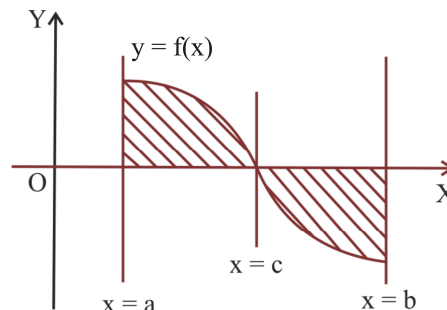
$$\text{Then } A = \int_a^b f(x) \, dx$$



- (ii) If $f(x) > 0 \, \forall x \in [a, c]$ &
 $< 0 \, \forall x \in (c, b]$ Then

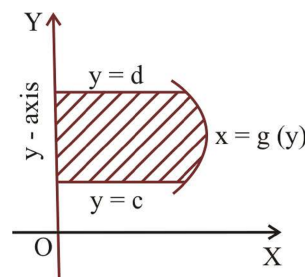
$$A = \left| \int_a^c y \, dx \right| + \left| \int_c^b y \, dx \right| = \int_a^c f(x) \, dx - \int_c^b f(x) \, dx$$

where c is a point in between a and b .



2. The area bounded by the curve $x = g(y)$, y -axis and the abscissae $y = c$ and $y = d$ (where $d > c$) is given by

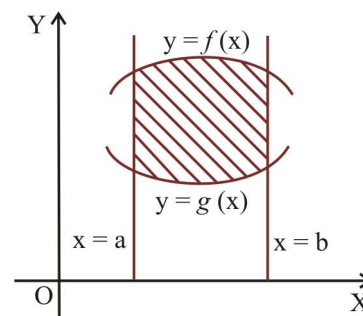
$$A = \int_c^d |x| \, dy = \int_c^d |g(y)| \, dy$$



3. If we have two curve $y = f(x)$ and $y = g(x)$, such that $y = f(x)$ lies above the curve $y = g(x)$ then the area bounded between them and the ordinates $x = a$ and $x = b$ ($b > a$), is given by

$$A = \int_a^b f(x) \, dx - \int_a^b g(x) \, dx$$

i.e. upper curve area – lower curve area.

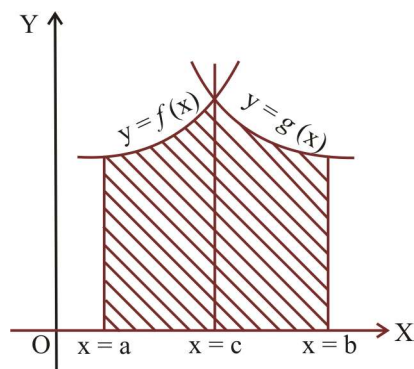




4. The area bounded by the curves $y = f(x)$ and $y = g(x)$ between the ordinates $x = a$ and $x = b$ is given by

$$A = \int_a^c f(x) dx + \int_c^b g(x) dx,$$

where $x = c$ is the point of intersection of the two curves.



9. CURVE TRACING

In order to find the area bounded by several curves, it is important to have rough sketch of the required portion. The following steps are very useful in tracing a cartesian curve $f(x, y) = 0$.

Step 1 : Symmetry

- The curve is symmetrical about x-axis if all powers of y in the equation of the given curve are even.
- The curve is symmetrical about y-axis if all powers of x in the equation of the given curve are even.
- The curve is symmetrical about the line $y = x$, if the equation of the given curve remains unchanged on interchanging x and y .
- The curve is symmetrical in opposite quadrants, if the equation of the given curve remains unchanged when x and y are replaced by $-x$ and $-y$ respectively.

Step 2 : Origin

If there is no constant term in the equation of the algebraic curve, then the curve passes through the origin.

In that case, the tangents at the origin are given by equating to zero the lowest degree terms in the equation of the given algebraic curve.

For example, the curve $y^3 = x^3 + axy$ passes through the origin and the tangents at the origin are given by $axy = 0$ i.e. $x = 0$ and $y = 0$.

Step 3 : Intersection with the Co-ordinate Axes

- To find the points of intersection of the curve with X-axis, put $y = 0$ in the equation of the given curve and get the corresponding values of x .
- To find the points of intersection of the curve with Y-axis, put $x = 0$ in the equation of the given curve and get the corresponding values of y .

Step 4 : Asymptotes

Find out the asymptotes of the curve.

- The vertical asymptotes or the asymptotes parallel to y-axis of the given algebraic curve are obtained by equating to zero the coefficient of the highest power of y in the equation of the given curve.
- The horizontal asymptotes or the asymptotes parallel to x-axis of the given algebraic curve are obtained by equating to zero the coefficient of the highest power of x in the equation of the given curve.

Step 5 : Region

Find out the regions of the plane in which no part of the curve lies. To determine such regions we solve the given equation for y in terms of x or vice-versa. Suppose that y becomes imaginary for $x > a$, the curve does not lie in the region $x > a$.

Step 6 : Critical Points

Find out the values of x at which $\frac{dy}{dx} = 0$.

At such points y generally changes its character from an increasing function of x to a decreasing function of x or vice-versa.

Step 7 : Trace the curve with the help of the above points.



SOLVED EXAMPLES

Example – 1

Let $f(x)$ be a function satisfying $f'(x) = f(x)$ with $f(0) = 1$ and $g(x)$ be a function that satisfies $f(x) + g(x) = x^2$. Then the value of the integral

$$\int_0^1 f(x)g(x)dx \text{ is}$$

$$(a) e + \frac{e^2}{2} - \frac{2}{2} \quad (b) e - \frac{e^2}{2} - \frac{3}{2}$$

$$(c) e + \frac{e^2}{2} + \frac{5}{2} \quad (d) e - \frac{e^2}{2} - \frac{5}{2}$$

Ans. (b)

Sol. As $f(x) = f'(x)$ and $f(0) = 1$

$$\Rightarrow \frac{f'(x)}{f(x)} = 1$$

$$\Rightarrow \log(f(x)) = x + c \Rightarrow f(x) = e^x + k$$

$$\Rightarrow f(x) = e^x \text{ as } f(0) = 1$$

$$\text{Now } g(x) = x^2 - e^x$$

$$\therefore \int_0^1 f(x)g(x)dx = \int_0^1 e^x (x^2 - e^x) dx$$

$$= \int_0^1 x^2 e^x dx - \int_0^1 e^{2x} dx$$

$$= \left[(x^2 - 2x + 2)e^x \right]_0^1 - \left(\frac{e^{2x}}{2} \right)_0^1$$

$$= (e - 2) - \left(\frac{e^2 - 1}{2} \right) = e - \frac{e^2}{2} - \frac{3}{2}$$

$$\text{Using } f^n(x)e^x dx = e^x [f^n(x) - f_1^n(x) + f_2^n(x) - \dots + (-1)^n f_n(x)]$$

Where $f_1, f_2, \dots, f_n$ are derivatives of first, second $\dots n^{\text{th}}$ order.

Example – 2

Evaluate the following integrals :

$$(i) \int_2^3 x^2 dx \quad (ii) \int_1^3 \frac{x}{(x+1)(x+2)} dx$$

$$\text{Sol. (i) } \int_2^3 x^2 dx = \left[\frac{x^3}{3} \right]_2^3$$

$$= \frac{27}{3} - \frac{8}{3} = \frac{19}{3}$$

$$(ii) \frac{x}{(x+1)(x+2)} = \frac{-1}{x+1} + \frac{2}{x+2}$$

$$\therefore \int_1^3 \frac{x}{(x+1)(x+2)} dx = [-\log|x+1| + 2\log|x+2|]_1^3$$

$$= [-\log|4| + 2\log|5|] - [-\log|2| + 2\log|3|]$$

$$= [-\log 4 + 2\log 5] - [-\log 2 + 2\log 3]$$

$$= -2\log 2 + 2\log 5 + \log 2 - 2\log 3$$

$$= -\log 2 + \log 25 - \log 9 = \log 25 - \log 18$$

$$= \log \frac{25}{18}$$

Example – 3

$$\text{Evaluate : } \int_0^{\pi/4} \sec x \cdot \sqrt{\frac{1-\sin x}{1+\sin x}} dx$$

$$\text{Sol. } I = \int_0^{\pi/4} \sec x \cdot \sqrt{\frac{1-\sin x}{1+\sin x}} dx$$

$$= \int_0^{\pi/4} \sec x \cdot \sqrt{\frac{1-\sin x}{1+\sin x}} \cdot \sqrt{\frac{1-\sin x}{1-\sin x}} dx$$

$$= \int_0^{\pi/4} \sec x \frac{1-\sin x}{\sqrt{1-\sin^2 x}} dx$$



$$= \int_0^{\pi/4} \sec x \frac{1 - \sin x}{\cos x} dx$$

$$= \int_0^{\pi/4} (\sec^2 x - \sec x \tan x) dx$$

$$= \int_0^{\pi/4} \sec^2 x dx - \int_0^{\pi/4} \sec x \tan x dx$$

$$= [\tan x]_0^{\pi/4} - [\sec x]_0^{\pi/4}$$

$$= \left(\tan \frac{\pi}{4} - \tan 0 \right) - \left(\sec \frac{\pi}{4} - \sec 0 \right)$$

$$= (1 - 0) - (\sqrt{2} - 1) = 2 - \sqrt{2}.$$

Example - 4

Evaluate : $\int_{-1}^1 5x^4 \sqrt{x^5 + 1} dx.$

Sol. Let $I = \int_{-1}^1 5x^4 \sqrt{x^5 + 1} dx$

Put $x^5 = t$ so that $5x^4 dx = dt$.

When $x = -1$, $t = -1$. When $x = 1$, $t = 1$.

$$\therefore I = \int_{-1}^1 \sqrt{t+1} dt$$

$$= \left[\frac{(t+1)^{3/2}}{3/2} \right]_{-1}^1 = \frac{2}{3} \left[(t+1)^{3/2} \right]_{-1}^1$$

$$= \frac{2}{3} [2^{3/2} - 0] = \frac{4\sqrt{2}}{3}.$$

Example - 5

Prove that $\int_0^{\pi/2} \sqrt{\sin \phi} \cos^5 \phi d\phi = \frac{64}{231}.$

Sol. $I = \int_0^{\pi/2} \sqrt{\sin \phi} \cos^5 \phi d\phi$

$$= \int_0^{\pi/2} \sqrt{\sin \phi} \cos^4 \phi \cos \phi d\phi$$

$$= \int_0^{\pi/2} \sqrt{\sin \phi} (1 - \sin^2 \phi)^2 \cos \phi d\phi$$

Put $\sin \phi = t$ so that $\cos \phi d\phi = dt$.

When $\phi = 0$, $\sin 0 = t \Rightarrow t = 0$.

When $\phi = \frac{\pi}{2}$, $\sin \frac{\pi}{2} = t \Rightarrow t = 1$

$$\therefore I = \int_0^1 \sqrt{t} (1 - t^2)^2 dt = \int_0^1 \sqrt{t} (1 - 2t^2 + t^4) dt$$

$$= \int_0^1 (t^{1/2} - 2t^{5/2} + t^{9/2}) dt$$

$$= \left[\frac{t^{3/2}}{3/2} - 2 \frac{t^{7/2}}{7/2} + \frac{t^{11/2}}{11/2} \right]_0^1$$

$$= \left[\frac{2}{3} t^{3/2} - \frac{4}{7} t^{7/2} + \frac{2}{11} t^{11/2} \right]_0^1$$

$$= \left[\frac{2}{3} (1) - \frac{4}{7} (1) + \frac{2}{11} (1) \right] - [0 - 0 + 0]$$

$$= \frac{2}{3} - \frac{4}{7} + \frac{2}{11}$$

$$= \frac{154 - 132 + 42}{231} = \frac{64}{231}.$$



Example – 6

Evaluate : $\int_1^2 \left(\frac{x-1}{x^2} \right) e^x dx$

Or

$$\int_1^2 e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$$

Sol. $\int \left(\frac{x-1}{x^2} \right) e^x dx = \int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$

$$= \int \frac{1}{x} \cdot e^x dx - \int \frac{1}{x^2} \cdot e^x dx$$

$$= \frac{1}{x} \cdot e^x - \int \left(-\frac{1}{x^2} \right) e^x dx - \int \frac{1}{x^2} \cdot e^x dx$$

[Integrating first integral by parts]

$$= \frac{1}{x} \cdot e^x = F(x)$$

$$\therefore \int_1^2 \left(\frac{x-1}{x^2} \right) e^x dx = \left[\frac{e^x}{x} \right]_1^2$$

$$= \frac{1}{2} \cdot e^2 - \frac{1}{1} e^1 = \frac{1}{2} e^2 - e$$

Example – 7

Evaluate : $\int_a^b x^2 dx$ using limit of a sum formula.

Sol. Let $I = \int_a^b x^2 dx = \lim_{\substack{n \rightarrow \infty \\ h \rightarrow 0}} h [(a+h)^2 + (a+2h)^2 + \dots + (a+nh)^2]$

$$= \lim_{\substack{n \rightarrow \infty \\ h \rightarrow 0}} h \left[(a^2 + a^2 + \dots) + (h^2 + 4h^2 + \dots + n^2 h^2) \right. \\ \left. + (2ah + 4ah + \dots + 2anh) \right]$$

$$\Rightarrow I = \lim_{\substack{n \rightarrow \infty \\ h \rightarrow 0}} \left[nha^2 + \frac{2ah^2 n(n+1)}{2} + \frac{h^3 n(n+1)(2n+1)}{6} \right]$$

Using $nh = b - a$, we get :

$$\Rightarrow I = \lim_{n \rightarrow \infty} \left[a^2(b-a) + a(b-a)^2 \left(1 + \frac{1}{n} \right) \right. \\ \left. + (b-a)^3 \frac{1}{6} \left(1 + \frac{1}{n} \right) \left(2 + \frac{1}{n} \right) \right]$$

$$\Rightarrow I = a^2(b-a) + a(b-a)^2 + \frac{(b-a)^3}{6} (2)$$

$$\Rightarrow I = (b-a) \left[a^2 + ab - a^2 + \frac{b^2 + a^2 - 2ab}{3} \right]$$

$$\Rightarrow I = \frac{(b-a)}{3} [a^2 + b^2 + ab] = \frac{b^3 - a^3}{3}$$

Example – 8

Evaluate : $\int_0^1 |3x-1| dx$

Sol. We know that $|3x-1| = 3x-1$,

when $x \geq \frac{1}{3}$ i.e. when $3x-1 \geq 0$

and $|3x-1| = -(3x-1)$

when $x < \frac{1}{3}$ i.e. when $3x-1 < 0$.

$$\therefore \int_0^1 |3x-1| dx = \int_0^{1/3} |3x-1| dx + \int_{1/3}^1 |3x-1| dx$$

$$= \int_0^{1/3} -(3x-1) dx + \int_{1/3}^1 (3x-1) dx$$

$$= \left[-\left(\frac{3}{2} x^2 - x \right) \right]_0^{1/3} + \left[\frac{3}{2} x^2 - x \right]_{1/3}^1$$

$$= -\left(\frac{3}{2} \cdot \frac{1}{9} - \frac{1}{3} \right) + \left[\left(\frac{3}{2} (1) - 1 \right) - \left(\frac{3}{2} \cdot \frac{1}{9} - \frac{1}{3} \right) \right]$$

$$= -\left[\frac{1}{6} - \frac{1}{3} \right] + \left[\frac{3}{2} - 1 \right] - \left[\frac{1}{6} - \frac{1}{3} \right]$$

$$= -2 \left(\frac{1}{6} - \frac{1}{3} \right) + \frac{1}{2} = -2 \left(-\frac{1}{6} \right) + \frac{1}{2} = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$$



Example – 9

Evaluate : $\int_{0.5}^{3.5} [x] dx$ where $[.]$ is GIF

Sol. Here $f(x) = [x] = \begin{cases} 0, 0.5 \leq x < 1 \\ 1, 1 \leq x < 2 \\ 2, 2 \leq x < 3 \\ 3, 3 \leq x < 3.5 \end{cases}$

$$\begin{aligned} \therefore \int_{0.5}^{3.5} [x] dx &= \int_{0.5}^1 [x] dx + \int_1^2 [x] dx + \int_2^3 [x] dx + \int_3^{3.5} [x] dx \\ &= \int_{0.5}^1 0 dx + \int_1^2 1 dx + \int_2^3 2 dx + \int_3^{3.5} 3 dx \\ &= 0 + [x]_1^2 + 2[x]_2^3 + 3[x]_3^{3.5} \\ &= (2-1) + 2(3-2) + 3(3.5-3) \\ &= 1 + 2 + 1.5 = 4.5. \end{aligned}$$

Example – 10

Evaluate : $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$

Sol. Let : $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \quad \dots (i)$

Using property – 4, we have :

$$I = \int_0^{\pi/2} \frac{\sqrt{\sin(\pi/2 - x)}}{\sqrt{\sin(\pi/2 - x)} + \sqrt{\cos(\pi/2 - x)}} dx$$

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx \quad \dots (ii)$$

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx + \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} dx = \frac{\pi}{2} \quad \Rightarrow I = \frac{\pi}{4}$$

Example – 11

Prove that $\int_0^{\pi/2} \sin 2x \log \tan x dx = 0$.

Sol. Let $I = \int_0^{\pi/2} \sin 2x \log \tan x dx \quad \dots (1)$

$$\therefore I = \int_0^{\pi/2} \sin 2\left(\frac{\pi}{2} - x\right) \log \tan\left(\frac{\pi}{2} - x\right) dx$$

$$= \int_0^{\pi/2} \sin(\pi - 2x) \log \tan\left(\frac{\pi}{2} - x\right) dx$$

$$= \int_0^{\pi/2} \sin 2x \log \cot x dx \quad \dots (2)$$

Adding (1) and (2) :

$$2I = \int_0^{\pi/2} \sin 2x [\log \tan x + \log \cot x] dx$$

$$= \int_0^{\pi/2} \sin 2x \log (\tan x \cot x) dx$$

$$= \int_0^{\pi/2} \sin 2x \log 1 dx = 0$$

$$[\because \log 1 = 0]$$

Hence $I = 0$.



Example – 12

If $f(a+b-x)=f(x)$, then $\int_a^b x f(x) dx$ is equal to

(a) $\frac{a+b}{2} \int_a^b f(x) dx$

(b) $\frac{b-a}{2} \int_a^b f(x) dx$

(c) $\frac{a+b}{2} \int_a^b f(a+b-x) dx$

(d) $\frac{a+b}{2} \int_a^b f(b-x) dx$

Ans. (a,c)

Sol. Let $I = \int_a^b x f(x) dx$

$$I = \int_a^b (a+b-x) f(a+b-x) dx$$

$$I = \int_a^b (a+b) f(a+b-x) dx - \int_a^b x f(a+b-x) dx$$

$$I = \int_a^b (a+b) f(x) dx - \int_a^b x f(x) dx$$

$$\therefore I = \frac{a+b}{2} \int_a^b f(x) dx = \frac{a+b}{2} \int_a^b f(a+b-x) dx$$

Example – 13

Evaluate : $\int_{-2}^2 f(x) dx$;

$$\text{where } f(x) = \begin{cases} 2x-1, & -2 \leq x < 1 \\ 3x-2, & 1 \leq x < 2 \end{cases}$$

Sol. $\int_{-2}^2 f(x) dx = \int_{-2}^1 f(x) dx + \int_1^2 f(x) dx$

$$= \int_{-2}^1 (2x-1) dx + \int_1^2 (3x-2) dx$$

$$= \left[2 \cdot \frac{x^2}{2} - x \right]_{-2}^1 + \left[\frac{3x^2}{2} - 2x \right]_1^2$$

$$= \left[x^2 - x \right]_{-2}^1 + \left[\frac{3}{2} x^2 - 2x \right]_1^2$$

$$= [(1-1) - (4+2)] + \left[\left(\frac{3}{2}(4) - 2(2) \right) - \left(\frac{3}{2}(1) - 2(1) \right) \right]$$

$$= (0-6) + \left(2 + \frac{1}{2} \right) = -6 + \frac{5}{2} = -\frac{7}{2}$$

Example – 14

Evaluate : $\int_0^{\pi/2} \sqrt{1-\sin 2x} dx$.

Sol. Let $I = \int_0^{\pi/2} \sqrt{1-\sin 2x} dx$

$$\Rightarrow I = \int_0^{\pi/2} \sqrt{(\sin x - \cos x)^2} dx$$

$$\Rightarrow I = \int_0^{\pi/2} |\sin x - \cos x| dx$$

$$\Rightarrow I = \int_0^{\pi/4} |\sin x - \cos x| dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx$$

$$\Rightarrow I = \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx$$

$$\Rightarrow I = [\sin x + \cos x]_0^{\pi/4} + [-\cos x - \sin x]_{\pi/4}^{\pi/2}$$

$$\Rightarrow I = \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1 \right) + (-1) - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right)$$

$$\Rightarrow I = 2\sqrt{2} - 2$$



Example – 15

Evaluate : $\int_0^{\pi} \frac{x}{1 + \cos^2 x} dx$.

Sol. Let $I = \int_0^{\pi} \frac{x}{1 + \cos^2 x} dx$... (i)

$\Rightarrow I = \int_0^{\pi} \frac{(\pi - x)}{1 + \cos^2(\pi - x)} dx$ [using property – 4]

... (ii)

Adding (i) and (ii), we get :

$\Rightarrow 2I = \int_0^{\pi} \frac{\pi}{1 + \cos^2 x} dx$

$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi} \frac{dx}{1 + \cos^2 x} = \frac{2\pi}{2} \int_0^{\pi/2} \frac{dx}{1 + \cos^2 x}$

[using property – 6]

Divide N^r and D^r by $\cos^2 x$ to get :

$\Rightarrow I = \pi \int_0^{\pi/2} \frac{\sec^2 x}{\sec^2 x + 1} dx$

Put $\tan x = t \Rightarrow \sec^2 x dx = dt$

[$\sec^2 x = 1 + \tan^2 x$]

For $x = \pi/2$, $t \rightarrow \infty$ and for $x = 0$, $t = 0$

$\Rightarrow I = \pi \int_0^{\infty} \frac{dt}{2 + t^2}$

$\Rightarrow I = \frac{\pi}{\sqrt{2}} \tan^{-1} \frac{t}{\sqrt{2}} \Big|_0^{\infty} = \frac{\pi}{\sqrt{2}} \times \frac{\pi}{2} = \frac{\pi^2}{2\sqrt{2}}$

Example – 16

Evaluate : $\int_0^{\pi} \frac{x \sin(2x) \sin\left(\frac{\pi}{2} \cos x\right) dx}{2x - \pi}$

Sol. Let $I = \int_0^{\pi} \frac{x \sin(2x) \sin\left(\frac{\pi}{2} \cos x\right) dx}{2x - \pi}$... (i)

Apply property – 4 to get

$\Rightarrow I = \int_0^{\pi} \frac{(\pi - x) \sin(2\pi - 2x) \sin\left(\frac{\pi}{2} \cos(\pi - x)\right) dx}{2(\pi - x) - \pi}$

$= \int_0^{\pi} \frac{(\pi - x) \sin 2x \sin\left(\frac{\pi}{2} \cos x\right) dx}{2x - \pi}$... (ii)

Add (i) and (ii) to get

$2I = \int_0^{\pi} 2 \sin x \cos x \sin\left[\frac{\pi}{2} \cos x\right] dx$

Let $\frac{\pi}{2} \cos x = t \Rightarrow -\frac{\pi}{2} \sin x dx = dt$

$\Rightarrow I = \frac{4}{\pi^2} \int_{-\pi/2}^{\pi/2} t \sin t dt = \frac{8}{\pi^2} \int_0^{\pi/2} t \sin t dt$

$\Rightarrow I = \frac{8}{\pi^2} \left[(-t \cos t) \Big|_0^{\pi/2} + (\sin t) \Big|_0^{\pi/2} \right]$

$= \frac{8}{\pi^2} [0 + 1] = \frac{8}{\pi^2}$



Example – 17

Evaluate : $\int_0^{\pi/2} \log \sin x \, dx$.

Sol. Let $I = \int_0^{\pi/2} \log \sin x \, dx$... (i)

$$\Rightarrow I = \int_0^{\pi/2} \log \sin \left(\frac{\pi}{2} - x \right) dx \quad [\text{using property-4}]$$

$$\Rightarrow I = \int_0^{\pi/2} \log \cos x \, dx \quad \dots (ii)$$

Adding (i) and (ii) we get :

$$2I = \int_0^{\pi/2} \log (\sin x \cos x) \, dx = \int_0^{\pi/2} \log \left(\frac{\sin 2x}{2} \right) dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \log \sin 2x \, dx - \int_0^{\pi/2} \log 2 \, dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \log \sin 2x \, dx - \frac{\pi}{2} \log 2 \quad \dots (iii)$$

$$\text{Let } I_1 = \int_0^{\pi/2} \log \sin 2x \, dx$$

$$\text{Put } t = 2x \Rightarrow dt = 2dx$$

$$\text{For } x = \frac{\pi}{2}, t = \pi \text{ and for } x = 0, t = 0$$

$$\Rightarrow I_1 = \frac{1}{2} \int_0^{\pi} \log \sin t \, dt = \frac{2}{2} \int_0^{\pi/2} \log \sin t \, dt$$

$$\Rightarrow I_1 = \int_0^{\pi/2} \log \sin x \, dx \quad [\text{using property-2}]$$

$$\Rightarrow I_1 = I$$

Substituting in (iii) we get :

$$2I = I - \pi/2 \log 2$$

$$\Rightarrow I = -\pi/2 \log 2$$

[learn this result so that you can directly apply it in other difficult problem]

Example – 18

Show that $\int_0^{n\pi+v} |\sin x| \, dx = 2n + 1 - \cos v$, where n is a +ve

integer and $0 \leq v \leq \pi$.

Sol. Let $I = \int_0^{n\pi+v} |\sin x| \, dx = \int_0^{n\pi} |\sin x| \, dx + \int_{n\pi}^{n\pi+v} |\sin x| \, dx$

[using property – 3]

$$\Rightarrow I = I_1 + I_2 \quad \dots (i)$$

Consider I_1 :

$$I_1 = \int_0^{n\pi} |\sin x| \, dx = n \int_0^{\pi} |\sin x| \, dx$$

[using property and period of $|\sin x|$ is π]

$$\Rightarrow I_1 = n \int_0^{\pi} \sin x \, dx$$

[As $\sin x \geq 0$ in $[0, \pi]$, $|\sin x| = \sin x$]

$$\Rightarrow I_1 = -n \cos x \Big|_0^{\pi} = -n [-1 - 1] = 2n$$

$$\text{Consider } I_2 : I_2 = \int_{n\pi}^{n\pi+v} |\sin x| \, dx = \int_0^v |\sin x| \, dx$$

[as period of $|\sin x| = \pi$]

$$\Rightarrow I_2 = \int_0^v \sin x \, dx$$

[as for $0 \leq x \leq \pi$, $\sin x$ is positive]

$$= -\cos x \Big|_0^v = 1 - \cos v$$

On substituting the values of I_1 and I_2 in (i), we get

$$I = 2n + (1 - \cos v) = 2n + 1 - \cos v.$$

Hence proved.



Example – 19

Evaluate $\frac{d}{dx} \left(\int_{1/x}^{\sqrt{x}} \cos t^2 dt \right)$

Sol. Let, $f(x) = \int_{1/x}^{\sqrt{x}} \cos t^2 dt$

$$\therefore \frac{d}{dx} (f(x)) = \cos(\sqrt{x})^2 \cdot \left\{ \frac{d}{dx} (\sqrt{x}) \right\} - \cos\left(\frac{1}{x}\right)^2 \left\{ \frac{d}{dx} \left(\frac{1}{x}\right) \right\}$$

$$= \frac{1}{2\sqrt{x}} \cos x + \frac{1}{x^2} \cdot \cos\left(\frac{1}{x^2}\right) \text{ (Using Leibnitz Rule)}$$

$$\Rightarrow \frac{d}{dx} \left(\int_{1/x}^{\sqrt{x}} \cos t^2 dt \right) = \frac{1}{2\sqrt{x}} \cos x + \frac{1}{x^2} \cos\left(\frac{1}{x^2}\right).$$

Example – 20

Find the points of local minimum and local maximum of the

function $\int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t} dt$.

Sol. Let $y = \int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t} dt = \int_0^{x^2} \frac{(t-1)(t-4)}{2 + e^t} dt$

For the points of Extremes,

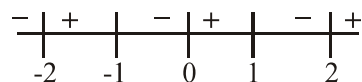
$$\frac{dy}{dx} = 0$$

$$\left[\frac{(x^2-1)(x^2-4)}{2+e^{x^2}} \right] (2x) = 0 \quad [\text{using property under point 3}]$$

$$\Rightarrow x=0 \text{ or } x^4 - 5x^2 + 4 = 0$$

$$\Rightarrow x=0 \text{ or } (x-1)(x+1)(x-2)(x+2) = 0$$

$$\Rightarrow x=0, x=\pm 1 \text{ and } x=\pm 2$$



$f'(x)$ changes sign from $(-)$ to $(+)$ at $x = -2, 0, 2$ where as

$f'(x)$ changes sign from $(+)$ to $(-)$ at $x = -1, 1$

$\Rightarrow x = -2, 0, 2$ are points of local minimum and $x = -1, 1$ are points of local maximum.

Example – 21

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function having

$f(2) = 6, f'(2) = \left(\frac{1}{48}\right)$. Then $\lim_{x \rightarrow 2} \int_6^{f(x)} \frac{4t^3}{x-2} dt$ equals

(a) 36 (b) 24

(c) 18 (d) 12

Ans. (c)

Sol. $\lim_{x \rightarrow 2} \int_6^{f(x)} \frac{4t^3}{x-2} dt$

$$\therefore \left(\frac{0}{0} \right) \text{ form}$$

$$= \lim_{x \rightarrow 2} \frac{f'(x) \times 4(f(x))^3}{1}$$

$$= 4f'(2) \times (f(2))^3 = \frac{1}{48} \times 4 \times 6 \times 6 \times 6 = 18$$

Example – 22

If for a continuous function

$\int_{-\pi}^t (f(x) + x) dx = \pi^2 - t^2$, for all $t \geq -\pi$, then $f\left(-\frac{\pi}{3}\right)$ is equal

to:

(a) π (b) $\frac{\pi}{2}$

(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

Ans. (a)

Sol. $\int_{-\pi}^t (f(x) + x) dx = \pi^2 - t^2$

Applying Newton-Leibnitz theorem,

$$f(t) + t = -2t$$

$$f\left(-\frac{\pi}{3}\right) - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\Rightarrow f\left(-\frac{\pi}{3}\right) = \pi$$



Example – 23

Find the sum of the series :

$$\lim_{n \rightarrow \infty} \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{6n}.$$

Sol. Let $S = \lim_{n \rightarrow \infty} \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+5n}$

Take $1/n$ common from the series i.e.

$$S = \lim_{n \rightarrow \infty} \frac{1}{n} \left[1 + \frac{1}{1 + \frac{1}{n}} + \frac{1}{1 + \frac{2}{n}} + \dots + \frac{1}{1 + \frac{5n}{n}} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{5n} \frac{1}{1 + r/n}$$

For the definite integral,

$$\text{Lower limit} = a = \lim_{n \rightarrow \infty} \left(\frac{r}{n} \right) = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\text{Upper limit} = b = \lim_{n \rightarrow \infty} \left(\frac{r}{n} \right) = \lim_{n \rightarrow \infty} \frac{5n}{n} = 5$$

Therefore,

$$S = \frac{1}{n} \lim_{n \rightarrow \infty} \sum_{r=0}^{5n} \frac{1}{1 + (r/n)}$$

$$= \int_0^5 \frac{dx}{1+x} = [\ln |1+x|]_0^5 = \ln 6 - \ln 1 = \ln 6$$

Example – 24

If $S_n = \frac{1}{1+\sqrt{n}} + \frac{1}{2+\sqrt{2n}} + \frac{1}{3+\sqrt{3n}} + \dots + \frac{1}{n+\sqrt{n^2}}$, then find

$$\lim_{n \rightarrow \infty} S_n.$$

- (a) $\log 2$ (b) $2 \log 2$
(c) $3 \log 2$ (d) $4 \log 2$

Ans. (b)

Sol. $S_n = \frac{1}{1+\sqrt{n}} + \frac{1}{2+\sqrt{2n}} + \frac{1}{3+\sqrt{3n}} + \dots + \frac{1}{n+\sqrt{n^2}}$

$$S_n = \sum_{r=1}^n \frac{1}{n} \left(\frac{1}{\frac{r}{n} + \sqrt{\frac{r}{n}}} \right)$$

$$\therefore \lim_{n \rightarrow \infty} S_n = \int_0^1 \frac{1}{x + \sqrt{x}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{x}(\sqrt{x}+1)} dx$$

$$= 2 \ln (\sqrt{x}+1) \Big|_0^1$$

$$= 2 \ln 2$$

Example – 25

If $I_1 = \int_0^1 2^{x^2} dx$, $I_2 = \int_0^1 2^{x^3} dx$,

$I_3 = \int_1^2 2^{x^2} dx$ and $I_4 = \int_1^2 2^{x^3} dx$

then

- (a) $I_1 > I_2$ (b) $I_2 > I_1$
(c) $I_3 > I_4$ (d) $I_3 = I_4$

Ans. (a)

Sol. For $0 < x < 1$,

$$x^2 > x^3$$

$$\therefore 2^{x^2} > 2^{x^3}$$

$$\therefore \int_0^1 2^{x^2} dx > \int_0^1 2^{x^3} dx$$

$$\therefore I_1 > I_2$$

and for $1 < x < 2$, $x^3 > x^2 \therefore 2^{x^3} > 2^{x^2}$

i.e. $\int 2^{x^2} < \int 2^{x^3} \Rightarrow I_3 < I_4$



Example – 26

$I_n = \int_0^{\pi/4} \tan^n x \, dx$, then $\lim_{n \rightarrow \infty} n [I_n + I_{n-2}]$ equals

- (a) $1/2$ (b) 1
(c) ∞ (d) 0

Ans. (b)

Sol. $I_n = \int_0^{\pi/4} \tan^n x \, dx$

$$I_{n-2} = \int_0^{\pi/4} \tan^{n-2} x \, dx$$

$$\therefore I_n + I_{n-2} = \int_0^{\pi/4} \tan^n x \, dx + \int_0^{\pi/4} \tan^{n-2} x \, dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x (\sec^2 x - 1) \, dx + \int_0^{\pi/4} \tan^{n-2} x \, dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x \sec^2 x \, dx = \int_0^1 t^{n-2} \, dt$$

$$\therefore I_n + I_{n-2} = \frac{1}{n-1}$$

$$\therefore n(I_n + I_{n-2}) = \frac{1}{1 - \frac{1}{n}}$$

$$\therefore \lim_{n \rightarrow \infty} n(I_n + I_{n-2}) = 1$$

Example – 27

If for $n \geq 1$, $P_n = \int_1^e (\log x)^n \, dx$, then $P_{10} - 90P_8$ is equal to:

- (a) -9 (b) $10e$
(c) $-9e$ (d) 10

Ans. (c)

Sol. $P_n = \int_1^e \underbrace{(\log x)^n}_I \cdot \underbrace{1}_{II} \cdot dx$

$$= x(\log x)^n \Big|_1^e - \int_1^e \underbrace{n(\log x)^{n-1}}_I \cdot \underbrace{1}_{II} \cdot dx$$

$$= e - n \left[x(\log x)^{n-1} \Big|_1^e - \int_1^e (n-1)(\log x)^{n-2} \, dx \right]$$

$$P_n = e - n[e - (n-1)P_{n-2}]$$

Put $n = 10$,

$$P_{10} = e - 10e + 90P_8$$

$$\Rightarrow P_{10} - 90P_8 = -9e$$

Example – 28

Find the area bounded by the curve

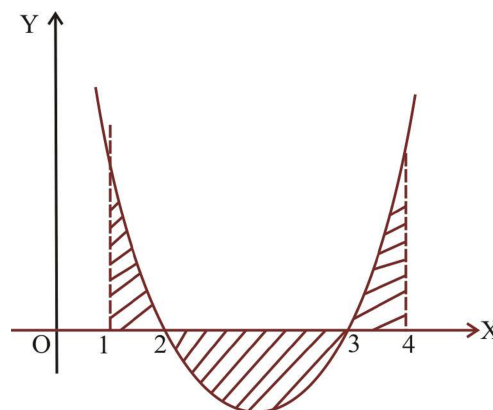
$y = x^2 - 5x + 6$, X-axis and the lines $x = 1$ and 4 .

Sol. For $y = 0$, we get $x^2 - 5x + 6 = 0$

$$\Rightarrow x = 2, 3$$

Hence the curve crosses X-axis at $x = 2, 3$ in the interval $[1, 4]$.

$$\text{Bounded Area} = \left| \int_1^2 y \, dx \right| + \left| \int_2^3 y \, dx \right| + \left| \int_3^4 y \, dx \right|$$



$$\Rightarrow A = \left| \int_1^2 (x^2 - 5x + 6) \, dx \right| + \left| \int_2^3 (x^2 - 5x + 6) \, dx \right| + \left| \int_3^4 (x^2 - 5x + 6) \, dx \right|$$

$$A_1 = \left[\frac{2^3 - 1^3}{3} \right] - 5 \left[\frac{2^2 - 1^2}{2} \right] + [6(2-1)] = \frac{5}{6}$$

$$A_2 = \frac{3^3 - 2^3}{3} - 5 \left[\frac{3^2 - 2^2}{2} \right] + 6(3-2) = -\frac{1}{6}$$

$$A_3 = \frac{4^3 - 3^3}{3} - 5 \left[\frac{4^2 - 3^2}{2} \right] + 6(4-3) = \frac{5}{6}$$

$$\Rightarrow A = \frac{5}{6} + \left| -\frac{1}{6} \right| + \frac{5}{6} = \frac{11}{6} \text{ sq. units.}$$



Example – 29

The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines $x = 4$, $y = 4$ and the coordinate axes. If S_1 , S_2 , S_3 are respectively the areas of these parts numbered from top to bottom; then $S_1 : S_2 : S_3$ is

- (a) 1 : 2 : 3 (b) 1 : 2 : 1
(c) 1 : 1 : 1 (d) 2 : 1 : 2

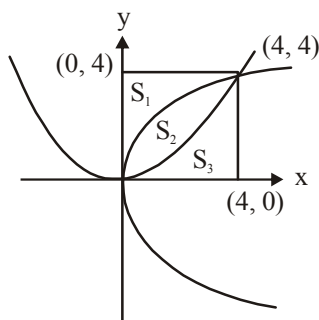
Ans. (c)

Sol. Total area = $4 \times 4 = 16$ sq. units

$$\text{Area of } S_3 = \int_0^4 \frac{x^2}{4} dx = \frac{16}{3} = S_1$$

$$\therefore S_2 = 16 - \frac{16}{3} \times 2 = \frac{16}{3}$$

$$\therefore S_1 : S_2 : S_3 \text{ is } 1 : 1 : 1$$



Example – 30

Find the area bounded by the circle $x^2 + y^2 = a^2$.

Sol. $x^2 + y^2 = a^2 \Rightarrow y = \pm \sqrt{a^2 - x^2}$

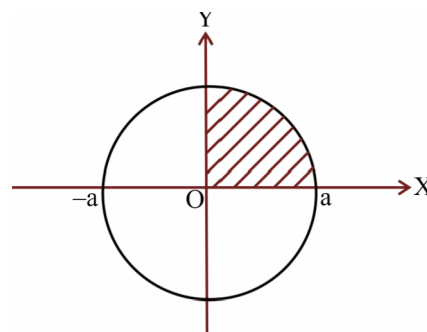
Equation of semicircle above X-axis is $y = + \sqrt{a^2 - x^2}$

Area of circle = 4 (shaded area)

$$= 4 \int_0^a \sqrt{a^2 - x^2} dx$$

$$= 4 \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_0^a$$

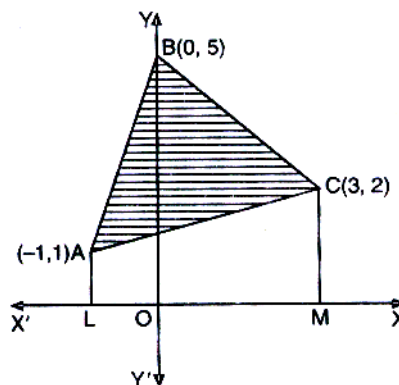
$$= 4 \frac{a^2}{2} \left(\frac{\pi}{2} \right) = \pi a^2$$



Example – 31

Using integration, find the area of the region bounded by: $(-1, 1)$, $(0, 5)$ and $(3, 2)$.

Sol. Let A $(-1, 1)$, B $(0, 5)$ and C $(3, 2)$ be the vertices of the triangle as shown in the following figure :



Equation of AB is :

$$y - 1 = \frac{5 - 1}{0 - (-1)} (x + 1)$$

$$\left[\text{Using } y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) \right]$$

$$\Rightarrow y - 1 = 4(x + 1)$$

$$\Rightarrow y = 4x + 5$$

... (1)

Equation of BC is :

$$y - 5 = \frac{2 - 5}{3 - 0} (x - 0)$$

$$\Rightarrow 3y - 15 = -3x$$

$$\Rightarrow 3y = 15 - 3x$$

$$\Rightarrow y = 5 - x$$

... (2)

Equation of AC is :



$$y-1 = \frac{2-1}{3-(-1)}(x+1)$$

$$\Rightarrow y-1 = \frac{1}{4}(x+1)$$

$$\Rightarrow 4y-4 = x+1$$

$$\Rightarrow 4y = x+5$$

$$\Rightarrow y = \frac{x}{4} + \frac{5}{4}$$

Now ar (ΔABC) = ar (ALOB) + ar (OMCB) - ar (ALMC)

$$= \int_{-1}^0 (4x+5) dx + \int_0^3 (5-x) dx - \int_{-1}^3 \left(\frac{1}{4}x + \frac{5}{4} \right) dx$$

$$= \left[\frac{4x^2}{2} + 5x \right]_{-1}^0 + \left[5x - \frac{x^2}{2} \right]_0^3 - \left[\frac{x^2}{8} + \frac{5}{4}x \right]_{-1}^3$$

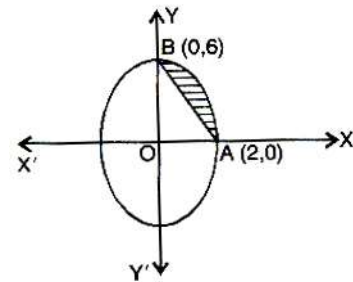
$$= \left[2x^2 + 5x \right]_{-1}^0 + \left[5x - \frac{x^2}{2} \right]_0^3 - \left[\frac{x^2}{8} + \frac{5}{4}x \right]_{-1}^3$$

$$= [(0+0) - (2-5)] + \left[\left(15 - \frac{9}{2} \right) - (0-0) \right]$$

$$- \left[\left(\frac{9}{8} + \frac{15}{4} \right) - \left(\frac{1}{8} - \frac{5}{4} \right) \right]$$

$$= 3 + \frac{21}{2} - \left(\frac{9+30-1+10}{8} \right) = 3 + \frac{21}{2} - \frac{48}{8}$$

$$= 3 + \frac{21}{2} - 6 = \frac{21}{2} - 3 = \frac{15}{2} = 7.5 \text{ sq. units.}$$



$\therefore$ A is (2, 0) and B is (0, 6).

$\therefore$ The equation of chord AB is :

$$y-0 = \frac{6-0}{0-2}(x-2)$$

$$\Rightarrow y = -3x + 6.$$

$\therefore$ Reqd. area (shown shaded)

$$= \int_0^2 \sqrt{4-x^2} dx - \int_0^2 (6-3x) dx$$

$$= 3 \left[\frac{x\sqrt{4-x^2}}{2} + \frac{4}{2} \sin^{-1} \frac{x}{2} \right]_0^2 - \left[6x - \frac{3x^2}{2} \right]_0^2$$

$$= 3 \left[\frac{2}{2}(0) + 2 \sin^{-1}(1) \right] - \left[6(2) - \frac{3(4)}{2} \right]$$

$$= 3 \left[2 \times \frac{\pi}{2} \right] - [12-6]$$

$$= (3\pi - 6) \text{ sq. units.}$$

Example - 33

Find the area bounded by the curves $y = x^2$ and $x^2 + y^2 = 2$ above X-axis.

Sol. Let us first find the points of intersection of curves.

Solving $y = x^2$ and $x^2 + y^2 = 2$ simultaneously, we get :

$$x^2 + x^4 = 2$$

$$\Rightarrow (x^2-1)(x^2+2) = 0$$

$$\Rightarrow x^2 = 1 \text{ and } x^2 = -2 \text{ [reject]}$$

$$\Rightarrow x = \pm 1$$

$$\Rightarrow A = (-1, 1) \text{ and } B = (1, 1)$$

$$\text{Shaded Area} = \int_{-1}^{+1} \left(\sqrt{2-x^2} - x^2 \right) dx$$

Example - 32

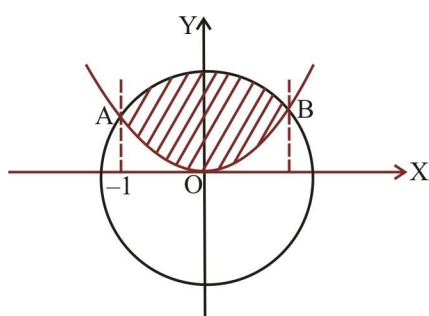
AOBA is the part of the ellipse $9x^2 + y^2 = 36$ in the first quadrant such that OA = 2 and OB = 6. Find the area between the arc AB and the chord AB.

Sol. The given equation of the ellipse can be written as

$$\frac{x^2}{4} + \frac{y^2}{36} = 1 \text{ i.e. } \frac{x^2}{2^2} + \frac{y^2}{6^2} = 1$$



$$\begin{aligned}
 &= \int_{-1}^{+1} \sqrt{2-x^2} dx - \int_{-1}^{+1} x^2 dx \\
 &= 2 \int_0^1 \sqrt{2-x^2} dx - 2 \int_0^1 x^2 dx \\
 &= 2 \left[\frac{x}{2} \sqrt{2-x^2} + \frac{2}{2} \sin^{-1} \frac{x}{\sqrt{2}} \right]_0^1 - 2 \left(\frac{1}{3} \right) \\
 &= 2 \left(\frac{1}{2} + \frac{\pi}{4} \right) - \frac{2}{3} = \frac{1}{3} + \frac{\pi}{2} \text{ sq. units.}
 \end{aligned}$$



Example – 34

Find the area of the region included between :

The parabola $y = \frac{3}{4}x^2$ and the line $3x - 2y + 12 = 0$

Sol. The given parabola is $4y = 3x^2$.

i.e. $y = \frac{3}{4}x^2$... (1)

and the given line is $3x - 2y + 12 = 0$... (2)

Putting the value of y from (1) in (2), we get :

$$3x - 2\left(\frac{3}{4}x^2\right) + 12 = 0 \Rightarrow 3x - \frac{3}{2}x^2 + 12 = 0$$

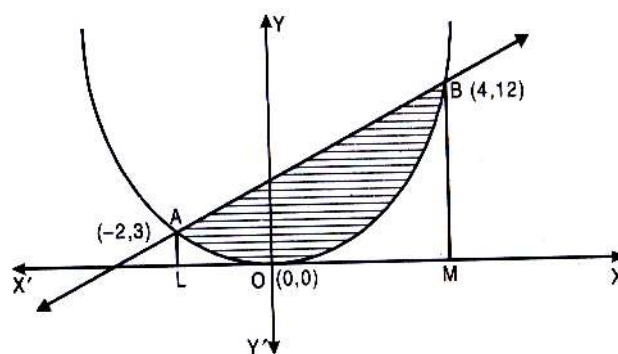
$$\Rightarrow 6x - 3x^2 + 24 = 0$$

$$\Rightarrow x^2 - 2x - 8 = 0$$

$$\Rightarrow (x-4)(x+2) = 0$$

$$\Rightarrow x = 4, -2.$$

Putting in (1), $y = \frac{3}{4}(4)^2 = 12$ and $y = \frac{3}{4}(-2)^2 = 3$.



Hence the line (2) intersects parabola (1) in the points A (-2, 3) and B (4, 12).

$\therefore$ Reqd. area = area ALMB – (area ALO + area OMB)

$$= \int_{-2}^4 \frac{3x+12}{2} dx - \left(\int_{-2}^0 \frac{3}{4}x^2 dx + \int_0^4 \frac{3}{4}x^2 dx \right)$$

$$\left[\because \text{From (2), } y = \frac{3x+12}{2} \right]$$

$$= \frac{1}{2} \left[\frac{3x^2}{2} + 12x \right]_{-2}^4 - \left[\frac{3}{4} \left\{ \frac{x^3}{3} \right\}_{-2}^0 + \frac{3}{4} \left\{ \frac{x^3}{3} \right\}_0^4 \right]$$

$$= \frac{1}{2} [(24+48) - (6-24)] - \left[\frac{3}{4} \left(0 + \frac{8}{3} \right) + \frac{3}{4} \left(\frac{64}{3} - 0 \right) \right]$$

$$= \frac{1}{2} [72+18] - [2+16] = 45 - 18$$

$$= 27 \text{ sq. units.}$$

Example – 35

Using integration, find the area of the region :

$$\{(x, y) : |x-1| \leq y \leq \sqrt{5-x^2}\}$$

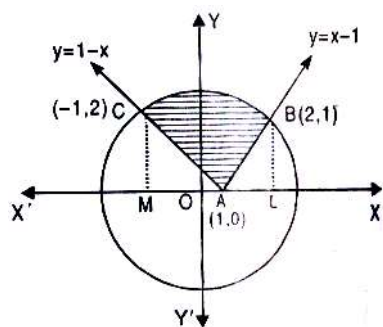
Sol. The given curves are :

$$x^2 + y^2 = 5$$

$$[\because y = \sqrt{5-x^2} \Rightarrow y^2 = 5-x^2 \Rightarrow x^2 + y^2 = 5]$$

and $y = \begin{cases} 1-x, & \text{if } x < 1 \\ x-1, & \text{if } x \geq 1 \end{cases}$

The reqd. region is shown as shaded in the following figure :



$y = x - 1$ meets $x^2 + y^2 = 5$ at B (2, 1)

$y = 1 - x$ meets $x^2 + y^2 = 5$ at C (-1, 2)

$y = x - 1$ and $y = 1 - x$ meet at A (1, 0).

∴ Reqd. area = ar. (MCBML) – ar (CMAC) – ar (ALBA)

$$= \int_{-1}^2 \sqrt{5-x^2} \, dx - \int_{-1}^1 (1-x) \, dx - \int_1^2 (x-1) \, dx$$

$$= \left[\frac{x\sqrt{5-x^2}}{2} + \frac{5}{2} \sin^{-1} \frac{x}{\sqrt{5}} \right]_{-1}^2 - \left[x - \frac{x^2}{2} \right]_{-1}^1 - \left[\frac{x^2}{2} - x \right]_1^2$$

$$= \left[\left(1 + \frac{5}{2} \sin^{-1} \frac{2}{\sqrt{5}} \right) - \left(-\frac{1}{2} \times 2 + \frac{5}{2} \sin^{-1} \left(-\frac{1}{\sqrt{5}} \right) \right) \right]$$

$$- \left[\left(1 - \frac{1}{2} \right) - \left(-1 - \frac{1}{2} \right) \right] - \left[(2-2) - \left(\frac{1}{2} - 1 \right) \right]$$

$$= 1 + \frac{5}{2} \sin^{-1} \frac{2}{\sqrt{5}} + 1 - \frac{5}{2} \sin^{-1} \left(-\frac{1}{\sqrt{5}} \right) - 2 - \frac{1}{2}$$

$$= -\frac{1}{2} + \frac{5}{2} \left[\sin^{-1} \frac{2}{\sqrt{5}} + \sin^{-1} \left(\frac{1}{\sqrt{5}} \right) \right]$$

$$= -\frac{1}{2} + \frac{5}{2} \times \frac{\pi}{2} = \frac{5\pi}{4} - \frac{1}{2} \text{ sq. units}$$



EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Definite Integration

Definite integration by methods of indefinite integration

- Let $\frac{d}{dx} F(x) = \left(\frac{e^{\sin x}}{x} \right)$, $x > 0$. If $\int_1^4 \frac{3}{x} e^{\sin x^3} dx = F(k) - F(1)$, then one of the possible values of k is
 - 16
 - 63
 - 64
 - 15
- The solution for x of the equation $\int_{\sqrt{2}}^x \frac{dt}{t\sqrt{t^2-1}} = \frac{\pi}{12}$ is
 - $\frac{\sqrt{3}}{2}$
 - $2\sqrt{2}$
 - 2
 - π
- If $\int_0^{\pi/2} \frac{d\theta}{9\sin^2\theta + 4\cos^2\theta} = k\pi$, then the value of k is :
 - $\frac{1}{16}$
 - $\frac{1}{12}$
 - $\frac{1}{8}$
 - $\frac{1}{3}$
- The value of $\int_0^{\infty} \frac{dx}{(1+x)^3}$ is:
 - $\frac{1}{3}$
 - $\frac{1}{2}$
 - 0
 - ∞
- $\int_0^{a/2} \frac{a dx}{(x-a)(x-2a)}$ equals to
 - $\ln \frac{2}{3}$
 - $\ln \frac{3}{2}$
 - $\ln 6$
 - none
- $\int_0^{\pi/2} \cos^5 \left(\frac{x}{2} \right) \cdot \sin x dx$ is equal to
 - $\frac{2}{7} \left(1 - \frac{1}{8\sqrt{2}} \right)$
 - $\frac{-4}{7} \left(1 - \frac{1}{8\sqrt{2}} \right)$
 - $\frac{4}{7} \left(1 - \frac{1}{8\sqrt{2}} \right)$
 - None of these
- For $0 < x < \frac{\pi}{2}$, $\int_{1/\sqrt{2}}^{1/2} \cot x d(\cos x)$ equals to
 - $\frac{\sqrt{3}-\sqrt{2}}{2}$
 - $\frac{\sqrt{2}-\sqrt{3}}{2}$
 - $\frac{1-\sqrt{3}}{2}$
 - none
- $\int_0^{\pi/4} \frac{x \cdot \sin x}{\cos^3 x} dx$ equal to
 - $\frac{\pi}{4} + \frac{1}{2}$
 - $\frac{\pi}{4} - \frac{1}{2}$
 - $\frac{\pi}{4}$
 - none
- The value of the integral $I = \int_0^1 x(1-x)^n dx$ is:
 - $\frac{1}{n+2}$
 - $\frac{1}{n+1} - \frac{1}{n+2}$
 - $\frac{1}{n+1} + \frac{1}{n+2}$
 - $\frac{1}{n+1}$
- $\int_{-3\pi/2}^{-\pi/2} [(x+\pi)^3 + \cos^2(x+3\pi)] dx$ is equal to
 - $\frac{\pi^4}{32}$
 - $\frac{\pi^4}{32} + \frac{\pi}{2}$
 - $\frac{\pi}{2}$
 - $\frac{\pi}{2} - 1$



Properties of Definite Integrals

11. The integral $\int_0^{\frac{1}{2}} \frac{\ln(1+2x)}{1+4x^2} dx$, equals:

- (a) $\frac{\pi}{4} \ln 2$ (b) $\frac{\pi}{8} \ln 2$
(c) $\frac{\pi}{16} \ln 2$ (d) $\frac{\pi}{32} \ln 2$

12. $\int_0^{\pi} x f(\sin x) dx$ is equal to :

- (a) $\pi \int_0^{\pi} x f(\cos x) dx$ (b) $\pi \int_0^{\pi} f(\sin x) dx$
(c) $\frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$ (d) $\pi \int_0^{\pi} f(\cos x) dx$

13. The value of the integral $\int_0^{\pi} x (\sin^4 x \cos^4 x) dx$ is

- (a) $\frac{3\pi^2}{64}$ (b) $\frac{3\pi^2}{128}$
(c) $\frac{3\pi^2}{256}$ (d) none of these

14. The value of the integral $\int_{\log 1/3}^{\log 3} \log(x + \sqrt{1+x^2}) dx$ is

- (a) $\log 3$ (b) $2 \log 3$
(c) 0 (d) None

15. If $[]$ denotes the greatest integer function, then the integral $\int_0^{\pi} [\cos x] dx$ is equal to:

- (a) $\frac{\pi}{2}$ (b) 0
(c) -1 (d) $-\frac{\pi}{2}$

16. $\int_0^{\pi} \frac{x}{a^2 \cos^2 x + b^2 \sin^2 x} dx$ is equal to

- (a) $\frac{\pi^2}{ab}$ (b) $\frac{\pi^2}{2ab}$
(c) $\frac{\pi^2}{4ab}$ (d) none of these

17. The value of $\int_{-1}^1 \frac{1+\sin x}{1+x^2} dx$ is :

- (a) $\frac{\pi}{4}$ (b) $-\frac{\pi}{4}$
(c) $\frac{\pi}{2}$ (d) $-\frac{\pi}{2}$

18. Let $f(x) = x - [x]$, for every real number x , where $[x]$ is the integral part of x . Then $\int_{-1}^1 f(x) dx$ is:

- (a) 1 (b) 2
(c) 0 (d) $\frac{1}{2}$

19. $\int_0^{100\pi} \sqrt{1 - \cos 2x} dx$ is equal to

- (a) $100\sqrt{2}$ (b) $50\sqrt{2}$
(c) $200\sqrt{2}$ (d) $400\sqrt{2}$

20. $\int_{\pi}^{10\pi} |\sin x| dx$ is equal to

- (a) 18 (b) 20
(c) 40 (d) None

21. $\int_0^{10} e^{x-[x]} dx$ ($[]$ denotes GIF) is equal to

- (a) $10(e-1)$ (b) $\frac{e-1}{10}$
(c) $\frac{e^{10}-1}{10}$ (d) $\frac{e^{10}-1}{e-1}$



22. $\int_0^{\sqrt{2}} [x^2] dx$ where $[\cdot]$ is GIF is :
- (a) $2 - \sqrt{2}$ (b) $2 + \sqrt{2}$
 (c) $\sqrt{2} - 1$ (d) $\sqrt{2} - 2$
23. The integral $\int_0^{\pi} \sqrt{1 + 4 \sin^2 \frac{x}{2} - 4 \sin \frac{x}{2}} dx$ equals :
- (a) $4\sqrt{3} - 4 - \frac{\pi}{3}$ (b) $\pi - 4$
 (c) $\frac{2\pi}{3} - 4 - 4\sqrt{3}$ (d) $4\sqrt{3} - 4$
24. If $f(x) = |x| + |x-1| + |x-2|, x \in R$ then $\int_0^3 f(x) dx$ equals to
- (a) $9/2$ (b) $15/2$
 (c) $19/2$ (d) none
25. Let $y = \{x\}^{[x]}$ where $\{x\}$ denotes the fractional part of x & $[x]$ denotes greatest integer $\leq x$ then $\int_0^3 y dx =$
- (a) $5/6$ (b) $2/3$
 (c) 1 (d) $11/6$
26. $\int_0^{\pi} |1 + 2 \cos x| dx$ equals to
- (a) $\frac{2\pi}{3}$ (b) π
 (c) 2 (d) $\frac{\pi}{3} + 2\sqrt{3}$
27. The value of $\int_1^a [x] f'(x) dx, a > 1$, where $[x]$ denotes the greatest integer not exceeding x , is
- (a) $a f(a) - \{f(1) + f(2) + \dots + f([a])\}$
 (b) $[a] f(a) - \{f(1) + f(2) + \dots + f([a])\}$
 (c) $[a] f([a]) - \{f(1) + f(2) + \dots + f(a)\}$
 (d) $a f([a]) - \{f(1) + f(2) + \dots + f(a)\}$
28. $\int_{-1}^3 \left(\tan^{-1} \frac{x}{x^2+1} + \tan^{-1} \frac{x^2+1}{x} \right) dx$ equals to
- (a) 2π (b) π
 (c) 4π (d) none
29. $\int_{-\pi}^{\pi} \frac{2x(1 + \sin x)}{1 + \cos^2 x} dx$ is
- (a) $\pi^2/4$ (b) π^2
 (c) $5/4$ (d) $-3/4$
30. Let function F be defined as $F(x) = \int_1^x \frac{e^t}{t} dt, x > 0$ then the value of the integral $\int_1^x \frac{e^t}{t+a} dt$, where $a > 0$, is:
- (a) $e^a[F(x) - F(1+a)]$ (b) $e^a[F(x+a) - F(a)]$
 (c) $e^a[F(x+a) - F(1+a)]$ (d) $e^a[F(x+a) - F(1+a)]$

Differentiation of Definite Integrals: Newton leibniz Theorem

31. If $f(x) = \int_0^x t \sin t dt$, then $f'(x)$ is
- (a) $\cos x + x \sin x$ (b) $x \cos x + \sin x$
 (c) $x \cos x$ (d) $x \sin x$
32. If $f(x) = \int_1^{x^3} \frac{dt}{1+t^4}$, then $f''(x)$ is equal to
- (a) $\frac{6x(1-5x^{12})}{(1+x^{12})^2}$ (b) $\frac{6x(1+5x^{12})}{(1+x^{12})^2}$
 (c) $-\frac{6x(1-5x^{12})}{(1+x^{12})^2}$ (d) none of these
33. If the variables x and y are connected by the relation

$$x = \int_1^y \frac{dz}{\sqrt{1+6z^3}}, \text{ then } \frac{d^2y}{dx^2} \text{ is proportional to}$$

- (a) y (b) y^2
 (c) y^3 (d) none of these



34. $\lim_{x \rightarrow 0^+} \frac{1}{x^3} \int_0^{x^2} \sin \sqrt{t} \, dt$ is equal to

- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $-\frac{1}{3}$ (d) $-\frac{2}{3}$

35. The value of $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sec^2 t \, dt}{x \sin x}$ is

- (a) 2 (b) 1
(c) 0 (d) 3

36. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Then

$\lim_{x \rightarrow \pi/4} \frac{\frac{\pi}{4} \int_2^{\sec^2 x} f(x) \, dx}{x^2 - \frac{\pi^2}{16}}$ is equal to

- (a) $4f(2)$ (b) $f(2)$
(c) $2f(\sqrt{2})$ (d) $2f(2)$

Summation of series using integration

37. $\lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \sec^2 \frac{1}{n^2} + \frac{2}{n^2} \sec^2 \frac{4}{n^2} + \dots + \frac{1}{n} \sec^2 1 \right]$ equals

- (a) $\frac{1}{2} \operatorname{cosec} 1$ (b) $\frac{1}{2} \sec 1$
(c) $\frac{1}{2} \tan 1$ (d) $\tan 1$

38. $\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \dots + \frac{1}{8n} \right]$ is equal to

- (a) $\frac{3}{8}$ (b) $\frac{1}{4}$
(c) $\frac{1}{8}$ (d) none of these

39. The value of the

$\lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2-1}} + \frac{1}{\sqrt{n^2-2^2}} + \dots + \frac{1}{\sqrt{n^2-(n-1)^2}} \right]$ is

- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{2}$ (d) none of these

40. If $S_n = \left[\frac{1}{2n} + \frac{1}{\sqrt{4n^2-1}} + \frac{1}{\sqrt{4n^2-4}} + \dots + \frac{1}{\sqrt{3n^2+2n-1}} \right]$

then $\lim_{n \rightarrow \infty} S_n$ is equal to

- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{6}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

41. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} e^{\frac{r}{n}}$ is :

- (a) e (b) $e-1$
(c) $1-e$ (d) $e+1$

Bounds of definite integrals

42. Let $I = \int_0^1 \frac{\sin x}{\sqrt{x}} \, dx$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}} \, dx$. Then which one of the following is true ?

- (a) $I > \frac{2}{3}$ and $J < 2$ (b) $I > \frac{2}{3}$ and $J > 2$
(c) $I < \frac{2}{3}$ and $J < 2$ (d) $I < \frac{2}{3}$ and $J > 2$

43. $I_1 = \int_1^2 e^x \, dx$ and $I_2 = \int_1^2 \log_e x \, dx$. Then

- (a) $I_1 > I_2$ (b) $I_1 < I_2$
(c) $I_1 = I_2$ (d) None of these



44. $I = \int_1^2 \frac{e^x}{x} dx$. Then

- (a) $I < e$ (b) $I > \frac{e^2}{2}$
(c) $e < I < \frac{e^2}{2}$ (d) None of these

45. $I_1 = \int_0^{\pi/4} \tan^3 x dx$ and $I_2 = \int_0^{\pi/4} \tan^5 x dx$
 $I_3 = \int_0^{\pi/4} \tan^{1/2} x dx$ $I_4 = \int_0^{\pi/4} \tan^{1/3} x dx$ then

- (a) $I_1 < I_2$ (b) $I_1 > I_3$
(c) $I_3 > I_4$ (d) $I_1 > I_2$

46. $\int_0^2 (1 + 2^{-x^2}) dx =$

- (a) 4 (b) $\frac{17}{2}$
(c) 2 (d) None of these

Reduction formula

47. If $I_n = \int_0^{\pi/4} \tan^n \theta d\theta$, where n is a positive integer, then

$n(I_{n-1} + I_{n+1})$ is equal to

- (a) 1 (b) $n-1$
(c) $\frac{1}{n-1}$ (d) none of these

48. If $a_n = \int_{\pi/4}^{\pi/2} \cot^n x dx$, then $a_2 + a_4, a_3 + a_5, a_4 + a_6$ are in

- (a) G.P. (b) A.P.
(c) H.P. (d) None

49. Let $I_n = \int_0^{\pi/4} \tan^n x dx$,

then $\frac{1}{I_2 + I_4}, \frac{1}{I_3 + I_5}, \frac{1}{I_4 + I_6}, \dots$ are in

- (a) A.P. (b) G.P.
(c) H.P. (d) none

50. The value of the integral $\int_0^{\pi} \frac{\sin(2kx)}{\sin x} dx$, when $k \in \mathbb{I}$, is

- (a) $\frac{\pi}{2}$ (b) π
(c) 0 (d) none of these

Numerical Value Type Questions

51. $\int_0^{\pi} \left[\cos^2 \left(\frac{3\pi}{8} - \frac{x}{4} \right) - \cos^2 \left(\frac{11\pi}{8} + \frac{x}{4} \right) \right] dx$ equals $\sqrt{k}$.

Then the value of k is.

52. $\int_0^{\pi/4} \frac{\sqrt{\tan x}}{\sin x \cos x} dx$ equals to

53. $\int_0^{(\pi/2)^{1/3}} 3x^5 \cdot \sin x^3 dx$ equals to

54. If $\int_0^1 \cot^{-1}(1-x+x^2) dx = K \cdot \int_0^1 \tan^{-1} x dx$, then K equals to

55. If the value of $\int_0^{\pi/2} \frac{dx}{1+\tan^3 x}$ is $\frac{\pi}{k}$. Then the value of k is

56. The value of the integral, $2 \int_3^6 \frac{\sqrt{x}}{\sqrt{9-x} + \sqrt{x}} dx$ is :

57. If $f(x)$ and $\phi(x)$ are continuous functions on the interval $[0, 4]$ satisfying

$$f(x) = f(4-x), \phi(x) + \phi(4-x) = 3$$

and $\int_0^4 f(x) dx = 2$, then $\int_0^4 f(x) \phi(x) dx =$

58. If $f(x) = \frac{x^7 - 3x^5 + 7x^3 - x + 1}{\cos^2 x}$ then, $\int_{-\pi/4}^{\pi/4} f(x) dx$ equals to

59. $\int_0^{100} (x - [x]) dx$ is equal to



60. $\int_0^{1.5} x [x^2] dx = 3/k$, where $[]$ denotes greatest integer function. Then the value of k is.
61. Suppose for every integer n , $\int_n^{n+1} f(x) dx = n^2$. The value of $\int_{-2}^4 f(x) dx$ is
62. If $\int_{-1}^{-4} f(x) dx = 4$ and $\int_2^{-4} (3 - f(x)) dx = 7$ then the value of $\int_{-2}^1 f(-x) dx$ is
63. $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} (\tan^{-1} t)^2 dt}{\int_0^{x^4} \sin \sqrt{t} dt}$ is equal to $1/k$. Then the value of k is.
64. Area bounded by the curve $y = x^3$, the x -axis and the ordinates $x = -2$ and $x = 1$ is :
- (a) -9 (b) $-\frac{15}{4}$
(c) $\frac{15}{4}$ (d) $\frac{17}{4}$
65. The area of the region bounded by the curve $y = x - x^2$, x -axis between $x = 0$ and $x = 1$ is :
- (a) $\frac{1}{6}$ (b) $\frac{1}{3}$
(c) $\frac{1}{2}$ (d) $\frac{5}{6}$
66. The area of the region bounded by the curve $y = \sqrt{1 - \cos x}$ between $x = 0$ and $x = \pi$ is :
- (a) $\frac{\sqrt{2}}{2}$ (b) $2\sqrt{2}$
(c) $\sqrt{2}$ (d) 2
67. The area of the region bounded by the curve $y = x \sin x$ between $x = 0$ and $x = 2\pi$ is :
- (a) π (b) 2π
(c) 3π (d) 4π
68. Area of the region bounded by the curve $y^2 = 4x$, y -axis and the line $y = 3$ is :
- (a) 2 (b) $\frac{9}{4}$
(c) $\frac{9}{3}$ (d) $\frac{9}{2}$
69. The area enclosed between the curve $y = \log_e (x + e)$ and the coordinate axes is
- (a) 2 (b) 1
(c) 4 (d) 3
70. The area bounded by the curves $y = \ln x$, $y = \ln |x|$, $y = |\ln x|$ and $y = |\ln |x||$ is
- (a) 4 sq. units (b) 6 sq. units
(c) 10 sq. units (d) none of these
71. The area of the region bounded by the curves $y = |x-1|$ and $y = 3 - |x|$ is
- (a) 3 sq. units (b) 4 sq. units
(c) 6 sq. units (d) 2 sq. units
72. The area enclosed between the curves $y^2 = x$ and $y = |x|$ is
- (a) $\frac{1}{6}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d)
73. The area bounded between the parabolas $x^2 = \frac{y}{4}$ and $x^2 = 9y$ and the straight line $y = 2$ is.
- (a) $20\sqrt{2}$ (b) $\frac{10\sqrt{2}}{3}$
(c) $\frac{20\sqrt{2}}{3}$ (d) $10\sqrt{2}$



74. The area (in square units) bounded by the curves $y = \sqrt{x}$, $2y - x + 3 = 0$, x-axis and lying in the first quadrant is
(a) 9 (b) 6
(c) 18 (d) $\frac{27}{4}$
75. The area of the region described by $A = \{(x, y) : x^2 + y^2 \leq 1 \text{ and } y^2 \leq 1 - x\}$ is
(a) $\frac{\pi}{2} + \frac{2}{3}$ (b) $\frac{\pi}{2} + \frac{4}{3}$
(c) $\frac{\pi}{2} - \frac{4}{3}$ (d) $\frac{\pi}{2} - \frac{2}{3}$
76. Let $A = \{(x, y) : y^2 \leq 4x, y - 2x \geq -4\}$. The area (in square units) of the region A is
(a) 8 (b) 9
(c) 10 (d) 11
77. The area of the circle $x^2 + y^2 = 16$ exterior to the parabola $y^2 = 6x$ is :
(a) $\frac{4}{3}(4\pi - \sqrt{3})$ (b) $\frac{4}{3}(4\pi + \sqrt{3})$
(c) $\frac{4}{3}(8\pi - \sqrt{3})$ (d) $\frac{4}{3}(8\pi + \sqrt{3})$
78. The area between the curves $y = \cos x$ and the line $y = x + 1$ in the second quadrant is –
(a) 1 (b) 2
(c) $3/2$ (d) $1/2$
79. The area bounded by the curves $y = \sin x$, $y = \cos x$ and y – axis in first quadrant is –
(a) $\sqrt{2} - 1$ (b) $\sqrt{2}$
(c) $\sqrt{2} + 1$ (d) None of these
80. The area bounded by $y = x^2 - 4$ and $x + y = 2$ is
(a) $\frac{75}{6}$ (b) $\frac{100}{6}$
(c) $\frac{125}{6}$ (d) $\frac{150}{6}$
81. Smaller area enclosed by the circle $x^2 + y^2 = 4$ and the line $x + y = 2$ is :
(a) $2(\pi - 2)$ (b) $\pi - 2$
(c) $2\pi - 1$ (d) $2(\pi + 2)$
- Numerical Value Type Questions**
82. The area between the curves $y = \tan x$, $y = \cot x$ and x – axis in the interval $[0, \pi/2]$ is $\log k$. Then the value of k is.
83. If $0 \leq x \leq \pi$; then the area bounded by the curve $y = x$ and $y = x + \sin x$ is –
84. The area bounded by the curves : $y^2 = 4a(x + a)$ and $y^2 = 4b(b - x)$ is $\frac{k}{3}(a + b)\sqrt{4ab}$. Then the value of k is.
85. If the area of the region $\{(x, y) : x^2 \leq y \leq |x|\}$ is $1/k$ sq. units, then the value of k is
86. If the line $y = mx$ bisects the area enclosed by the lines $x = 0$, $y = 0$, $x = \frac{3}{2}$ and the curve $y = 1 + 4x - x^2$, and the value of m is equal to $13/k$. then the value of k is
87. If the area enclosed between the parabolas $y^2 = 4ax$ and $x^2 = 4by$ is $\frac{16}{3}$, then value of ab is
88. AOB is the positive quadrant of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$; where $OA = a$, $OB = b$. Area between arc AB and chord AB of ellipse is $\frac{(\pi - p)}{q} ab$, then $p + q$ equals
89. If the area bounded by the parabola $x^2 = 4y$, the x-axis and the line $x = 4$ is divided into two equal areas by the line $x = \alpha$, and the value of α is $(32)^{1/k}$, then the value of k is
90. The value of m for which the area included between the curves $y^2 = 4ax$ and $y = mx$ equals $\frac{a^2}{3}$ is



EXERCISE - 2 : PREVIOUS YEAR JEE MAIN QUESTIONS

Definite Integration

1. The integral

$$\int_2^4 \frac{\log x^2}{\log x^2 + \log(36 - 12x + x^2)} dx \text{ is equal to: } \quad (2015)$$

- (a) 1 (b) 6
(c) 2 (d) 4

2. The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x} dx$, $a > 0$ is

(2015/Online Set-2)

- (a) π (b) $a\pi$
(c) $\pi/2$ (d) 2π

3. If $2\int_0^1 \tan^{-1} x dx = \int_0^1 \cot^{-1}(1-x+x^2) dx$,

then $\int_0^1 \tan^{-1}(1-x+x^2) dx$ is equal to :

(2016/Online Set-1)

- (a) $\log 4$ (b) $\frac{\pi}{2} + \log 2$
(c) $\log 2$ (d) $\frac{\pi}{2} - \log 4$

4. The value of the integral

$$\int_4^{10} \frac{[x^2] dx}{[x^2 - 28x + 196] + [x^2]}, \text{ where } [x] \text{ denotes the greatest}$$

integer less than or equal to x , is :

(2016/Online Set-2)

- (a) 6 (b) 3
(c) 7 (d) $\frac{1}{3}$

5. For $x \in \mathbb{R}$, $x \neq 0$, if $y(x)$ is a differentiable function such

that $x \int_1^x y(t) dt = (x+1) \int_1^x t y(t) dt$, then $y(x)$ equals :

(where C is a constant.)

(2016/Online Set-2)

- (a) $\frac{C}{x} e^{-\frac{1}{x}}$ (b) $\frac{C}{x^2} e^{-\frac{1}{x}}$
(c) $\frac{C}{x^3} e^{-\frac{1}{x}}$ (d) $Cx^3 e^{\frac{1}{x}}$

6. The integral $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{dx}{1+\cos x}$ is equal to: (2017)

- (a) -2 (b) 2
(c) 4 (d) -1

7. The integral $\int_{\frac{\pi}{12}}^{\frac{\pi}{4}} \frac{8\cos 2x}{(\tan x + \cot x)^3} dx$ equals :

(2017/Online Set-1)

- (a) $\frac{15}{128}$ (b) $\frac{15}{64}$
(c) $\frac{13}{32}$ (d) $\frac{13}{256}$

8. If $\int_1^2 \frac{dx}{(x^2 - 2x + 4)^{\frac{3}{2}}} = \frac{k}{k+5}$, then k is equal to:

(2017/Online Set-2)

- (a) 1 (b) 2
(c) 3 (d) 4

9. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1+2^x} dx$ is: (2018)

- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{8}$
(c) $\frac{\pi}{2}$ (d) 4π



10. The value of the integral

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^4 x \left(1 + \log \left(\frac{2 + \sin x}{2 - \sin x} \right) \right) dx \text{ is}$$

(2018/Online Set-1)

- (a) 0 (b) $\frac{3}{4}$
(c) $\frac{3}{8}\pi$ (d) $\frac{3}{16}\pi$

11. The value of integral $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{x}{1 + \sin x} dx$ is :

(2018/Online Set-2)

- (a) $\pi\sqrt{2}$ (b) $\pi(\sqrt{2}-1)$
(c) $\frac{\pi}{2}(\sqrt{2}+1)$ (d) $2\pi(\sqrt{2}-1)$

12. If $I_1 = \int_0^1 e^{-x} \cos^2 x dx$, $I_2 = \int_0^1 e^{-x^2} \cos^2 x dx$ and

$$I_3 = \int_0^1 e^{-x^3} dx; \text{ then:}$$

(2018/Online Set-2)

- (a) $I_2 > I_3 > I_1$ (b) $I_2 > I_1 > I_3$
(c) $I_3 > I_2 > I_1$ (d) $I_3 > I_1 > I_2$

13. If $f(x) = \int_0^x t(\sin x - \sin t) dt$ then:

(2018/Online Set-3)

- (a) $f'''(x) + f''(x) = \sin x$
(b) $f'''(x) + f''(x) - f'(x) = \cos x$
(c) $f'''(x) + f'(x) = \cos x - 2x \sin x$
(d) $f'''(x) - f''(x) = \cos x - 2x \sin x$

14. If $f(x) = \frac{2 - x \cos x}{2 + x \cos x}$ and $g(x) = \log_e x, (x > 0)$ then the

value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} g(f(x)) dx$ is:

(8-04-2019/Shift-1)

- (a) $\log_e 3$ (b) $\log_e e$
(c) $\log_e 2$ (d) $\log_e 1$

15. Let $f(x) = \int_0^x g(t) dt$, where g is a non-zero even function.

If $f(x+5) = g(x)$, then $\int_0^x f(t) dt$ equals :

(8-04-2019/Shift-2)

- (a) $\int_{x+5}^5 g(t) dt$ (b) $\int_5^{x+5} g(t) dt$
(c) $2 \int_5^{x+5} g(t) dt$ (d) $5 \int_5^{x+5} g(t) dt$

16. The value of the integral $\int_0^1 x \cot^{-1}(1 - x^2 + x^4) dx$ is :

(9-04-2019/Shift-2)

- (a) $\frac{\pi}{2} - \frac{1}{2} \log_e 2$ (b) $\frac{\pi}{4} - \log_e 2$
(c) $\frac{\pi}{2} - \log_e 2$ (d) $\frac{\pi}{4} - \frac{1}{2} \log_e 2$

17. If $f: R \rightarrow R$ is a differentiable function and

$f(2) = 6$, then $\lim_{x \rightarrow 2} \int_6^{f(x)} \frac{2t dt}{(x-2)}$ is: (9-04-2019/Shift-2)

- (a) $24f'(2)$ (b) $2f'(2)$
(c) 0 (d) $12f'(2)$

18. The value of $\int_0^{2\pi} [\sin 2x(1 + \cos 3x)] dx$, where $[]$ denotes the greatest integer function, is: (10-04-2019/Shift-1)

- (a) π (b) $-\pi$
(c) -2π (d) 2π



19. $\lim_{n \rightarrow \infty} \left(\frac{(n+1)^{1/3}}{n^{4/3}} + \frac{(n+2)^{1/3}}{n^{4/3}} + \dots + \frac{(2n)^{1/3}}{n^{4/3}} \right)$ is equal to:

(10-04-2019/Shift-1)

(a) $\frac{3}{4}(2)^{4/3} - \frac{3}{4}$ (b) $\frac{4}{3}(2)^{4/3}$

(c) $\frac{3}{2}(2)^{4/3} - \frac{4}{3}$ (d) $\frac{4}{3}(2)^{3/4}$

20. The integral $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec^{\frac{2}{3}} x \operatorname{cosec}^{\frac{4}{3}} x dx$ is equal to:

(10-4-2019/Shift-2)

(a) $3^{\frac{5}{6}} - 3^{\frac{2}{3}}$ (b) $3^{\frac{4}{3}} - 3^{\frac{1}{3}}$

(c) $3^{\frac{7}{6}} - 3^{\frac{5}{6}}$ (d) $3^{\frac{5}{3}} - 3^{\frac{1}{3}}$

21. If $\int_0^{\frac{\pi}{2}} \frac{\cot x}{\cot x + \operatorname{cosec} x} dx = m(\pi + n)$, then m. n is equal to

(12-04-2019/Shift-1)

(a) $-\frac{1}{2}$ (b) 1

(c) $\frac{1}{2}$ (d) -1

22. Let $f: R \rightarrow R$ be a continuously differentiable function such that $f(2) = 6$ and $f'(2) = \frac{1}{48}$.

If $\int_6^{f(x)} 4t^3 dt = (x-2)g(x)$, then $\lim_{x \rightarrow 2} g(x)$ is equal to

(12-04-2019/Shift-1)

23. A value of α such that

$\int_{\alpha}^{\alpha+1} \frac{dx}{(x+\alpha)(x+\alpha+1)} = \log_e \left(\frac{9}{8} \right)$ is _____.

(12-04-2019/Shift-2)

(a) -2 (b) $\frac{1}{2}$

(c) $-\frac{1}{2}$ (d) 2

24. The value of $\int_0^{\pi} |\cos x|^3 dx$ is: (9-01-2019/Shift-1)

(a) 0 (b) $\frac{4}{3}$

(c) $\frac{2}{3}$ (d) $\frac{-4}{3}$

25. If $\int_0^{\pi/3} \frac{\tan \theta}{\sqrt{2k \sec \theta}} d\theta = 1 - \frac{1}{\sqrt{2}}$, ($k > 0$) then the value of k is: (9-01-2019/Shift-2)

(a) 4 (b) $\frac{1}{2}$

(c) 1 (d) 2

26. Let $I = \int_a^b (x^4 - 2x^2) dx$. If I is minimum then the ordered pair (a, b) is: (10-1-2019/Shift-1)

(a) $(0, \sqrt{2})$ (b) $(-\sqrt{2}, 0)$

(c) $(\sqrt{2}, -\sqrt{2})$ (d) $(-\sqrt{2}, \sqrt{2})$

27. If $\int_0^x f(t) dt = x^2 + \int_x^1 t^2 f(t) dt$, then $f'(1/2)$ is:

(10-01-2019/Shift-2)

(a) $\frac{24}{25}$ (b) $\frac{18}{25}$

(c) $\frac{4}{5}$ (d) $\frac{6}{25}$

28. The value of $\int_{-\pi/2}^{\pi/2} \frac{dx}{[x] + [\sin x] + 4}$, where [t] denotes the greatest integer less than or equal to t, is:

(10-01-2019/Shift-2)

(a) $\frac{1}{12}(7\pi + 5)$ (b) $\frac{1}{12}(7\pi - 5)$

(c) $\frac{3}{20}(4\pi - 3)$ (d) $\frac{3}{10}(4\pi - 3)$



29. The value of the integral $\int_{-2}^2 \frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} dx$ (where $[x]$ denotes the greatest integer less than or equal to x) is :
(11-01-2019/Shift-1)
- (a) 0 (b) $\sin 4$
(c) 4 (d) $4 - \sin 4$
30. The integral $\int_{\pi/6}^{\pi/4} \frac{dx}{\sin 2x (\tan^5 x + \cot^5 x)}$ equals :
(11-01-2019/Shift-2)
- (a) $\frac{1}{20} \tan^{-1} \left(\frac{1}{9\sqrt{3}} \right)$ (b) $\frac{1}{10} \left(\frac{\pi}{4} - \tan^{-1} \left(\frac{1}{9\sqrt{3}} \right) \right)$
(c) $\frac{\pi}{40}$ (d) $\frac{1}{5} \left(\frac{\pi}{4} - \tan^{-1} \left(\frac{1}{3\sqrt{3}} \right) \right)$
31. Let f and g be continuous functions on $[0, a]$ such that $f(x) = f(a-x)$ and $g(x) + g(a-x) = 4$ then $\int_0^a f(x)g(x)dx$ is equal to (12-01-2019/Shift-1)
- (a) $4 \int_0^a f(x)dx$ (b) $\int_0^a f(x)dx$
(c) $2 \int_0^a f(x)dx$ (d) $-3 \int_0^a f(x)dx$
32. The integral $\int_1^e \left\{ \left(\frac{x}{e} \right)^{2x} - \left(\frac{e}{x} \right)^x \right\} \log_e x dx$ is equal to (12-01-2019/Shift-2)
- (a) $\frac{1}{2} - e - \frac{1}{e^2}$ (b) $-\frac{1}{2} + \frac{1}{e} - \frac{1}{2e^2}$
(c) $\frac{3}{2} - \frac{1}{e} - \frac{1}{2e^2}$ (d) $\frac{3}{2} - e - \frac{1}{2e^2}$
33. $\lim_{n \rightarrow \infty} \left(\frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \frac{n}{n^2+3^2} + \dots + \frac{1}{5n} \right)$ is equal to (12-01-2019/Shift-2)
- (a) $\frac{\pi}{4}$ (b) $\tan^{-1}(3)$
(c) $\frac{\pi}{2}$ (d) $\tan^{-1}(2)$
34. The integral $\int_0^2 ||x-1| - x| dx$ is equal to : (2-9-2020/Shift-1)
35. Let $[t]$ denote the greatest integer less than or equal to t . Then the value of $\int_1^2 |2x - [3x]| dx$ is (2-9-2020/Shift-2)
36. $\int_{-\pi}^{\pi} |\pi - |x|| dx$ is equal to : (3-09-2020/Shift-1)
- (a) π^2 (b) $\frac{\pi^2}{2}$
(c) $\sqrt{2}\pi^2$ (d) $2\pi^2$
37. If the value of the integral $\int_0^{1/2} \frac{x^2}{(1-x^2)^{3/2}} dx$ is $\frac{k}{6}$, then k is equal to : (03-09-2020/Shift-2)
- (a) $2\sqrt{3} + \pi$ (b) $3\sqrt{2} + \pi$
(c) $3\sqrt{2} - \pi$ (d) $2\sqrt{3} - \pi$
38. Let $f(x) = \int \frac{\sqrt{x}}{(1+x)^2} dx (x \geq 0)$. Then $f(3) - f(1)$ is equal to : (04-09-2020/Shift-1)
- (a) $-\frac{\pi}{6} + \frac{1}{2} + \frac{\sqrt{3}}{4}$ (b) $\frac{\pi}{6} + \frac{1}{2} - \frac{\sqrt{3}}{4}$
(c) $-\frac{\pi}{12} + \frac{1}{2} + \frac{\sqrt{3}}{4}$ (d) $\frac{\pi}{12} + \frac{1}{2} - \frac{\sqrt{3}}{4}$
39. Let $f(x) = |x-2|$ and $g(x) = f(f(x))$, $x \in [0, 4]$. Then $\int_0^3 (g(x) - f(x)) dx$ is equal to : (4-09-2020/Shift-1)
- (a) $\frac{1}{2}$ (b) 0
(c) 1 (d) $\frac{3}{2}$



40. Let f be a twice differentiable function on $(1, 6)$. If $f(2) = 8, f'(2) = 5, f'(x) \geq 1$ and $f''(x) \geq 4$ for all $x \in (1, 6)$ then : (04-09-2020/Shift-1)
- (a) $f(5) + f'(5) \geq 28$ (b) $f'(5) + f''(5) \leq 20$
- (c) $f(5) \leq 10$ (d) $f(5) + f'(5) \leq 26$
41. The integral $\int_{\pi/6}^{\pi/3} \tan^3 x \cdot \sin^2 3x (2 \sec^2 x \cdot \sin^2 3x + 3 \tan x \cdot \sin 6x) dx$ is equal to: (4-9-2020/Shift-2)
- (a) $\frac{9}{2}$ (b) $-\frac{1}{18}$
- (c) $-\frac{1}{9}$ (d) $\frac{7}{18}$
42. Let $\{x\}$ and $[x]$ denote the fractional part of x and the greatest integer $\leq x$ respectively of real number x . If $\int_0^n \{x\} dx, \int_0^n [x] dx$ and $10(n^2 - n)$, ($n \in N, n > 1$) are three consecutive terms of a G.P., then n is equal to (4-9-2020/Shift-2)
43. The value of $\int_{-\pi/2}^{\pi/2} \frac{1}{1 + e^{\sin x}} dx$ is: (5-09-2020/Shift-1)
- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
- (c) π (d) $\frac{3\pi}{2}$
44. If $I_1 = \int_0^1 (1 - x^{50})^{100} dx$ and $I_2 = \int_0^1 (1 - x^{50})^{101} dx$ such that $I_2 = \alpha I_1$ then α equal to : (6-09-2020/Shift-1)
- (a) $\frac{5050}{5049}$ (b) $\frac{5050}{5051}$
- (c) $\frac{5051}{5050}$ (d) $\frac{5049}{5050}$
45. The integral $\int_1^2 e^x \cdot x^x (2 + \log_e x) dx$ equal : (6-09-2020/Shift-2)
- (a) $e(4e - 1)$ (b) $e(4e + 1)$
- (c) $4e^2 - 1$ (d) $e(2e - 1)$
46. If $f(a+b+1-x) = f(x) \forall x$ where a and b are fixed positive real numbers, then $\frac{1}{(a+b)} \int_a^b x(f(x) + f(x+1)) dx$ is equal to (7-01-2020/Shift-1)
- (a) $\int_{a-1}^{b-a} f(x) dx$ (b) $\int_{a+1}^{b+1} f(x+1) dx$
- (c) $\int_{a-1}^{b-1} f(x+1) dx$ (d) $\int_{a+1}^{b+1} f(x) dx$
47. The value of α for which $4\alpha \int_{-1}^2 e^{-\alpha|x|} dx = 5$, is: (07-01-2020/Shift-2)
- (a) $\log_e 2$ (b) $\log_e \sqrt{2}$
- (c) $\log_e \left(\frac{4}{3}\right)$ (d) $\log_e \left(\frac{3}{2}\right)$
48. If θ_1 and θ_2 be respectively the smallest and the largest values of θ in $[0, 2\pi) - \{\pi\}$ which satisfy the equation, $2 \cot^2 \theta - \frac{5}{\sin \theta} + 4 = 0$ then $\int_{\theta_1}^{\theta_2} \cos^2 3\theta d\theta$ is equal to (07-01-2020/Shift-2)
- (a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{3}$
- (c) $\frac{\pi}{3} + \frac{1}{6}$ (d) $\frac{\pi}{9}$
49. If $I = \int_1^2 \frac{dx}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$, then: (8-01-2020/Shift-2)
- (a) $\frac{1}{6} < I^2 < \frac{1}{2}$ (b) $\frac{1}{8} < I^2 < \frac{1}{4}$
- (c) $\frac{1}{9} < I^2 < \frac{1}{8}$ (d) $\frac{1}{16} < I^2 < \frac{1}{9}$



50. The value of $\int_0^{2\pi} \frac{x \sin^8 x}{\sin^8 x + \cos^8 x} dx$ is equal to:

(9-01-2020/Shift-1)

- (a) 2π (b) 4π
(c) $2\pi^2$ (d) π^2

51. If for all real triplets (a, b, c) , $f(x) = a + bx + cx^2$

then $\int_0^1 f(x) dx$ is equal to:

(9-01-2020/Shift-1)

(a) $2 \left(3f(1) + 2f\left(\frac{1}{2}\right) \right)$

(b) $\frac{1}{3} \left(f(0) + f\left(\frac{1}{2}\right) \right)$

(c) $\frac{1}{2} \left(f(1) + 3f\left(\frac{1}{2}\right) \right)$

(d) $\frac{1}{6} \left(f(0) + f(1) + 4f\left(\frac{1}{2}\right) \right)$

52. The value of the integral $\int_{-1}^1 \log_e (\sqrt{1-x} + \sqrt{1+x}) dx$ is equal to:

(20-07-21/Shift-1)

(a) $2 \log_e 2 + \frac{\pi}{4} - 1$ (b) $\frac{1}{2} \log_e 2 + \frac{\pi}{4} - \frac{3}{2}$

(c) $2 \log_e 2 + \frac{\pi}{2} - \frac{1}{2}$ (d) $\log_e 2 + \frac{\pi}{2} - 1$

53. Let a be a positive real number such that $\int_0^a e^{x-[x]} dx = 10e - 9$, where $[x]$ is the greatest integer less than or equal to x . Then a is equal to: (20-07-21/Shift-1)

- (a) $10 + \log_e 3$ (b) $10 - \log_e (1+e)$
(c) $10 + \log_e 2$ (d) $10 + \log_e (1+e)$

54. If $[x]$ denotes the greater integer less than or equal to

x , then the value of the integral $I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [[x] - \sin x] dx$ is

equal to ?

(20-07-21/Shift-2)

- (a) 0 (b) π
(c) 1 (d) $-\pi$

55. Let $g(t) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos\left(\frac{\pi}{4}t + f(x)\right) dx$, where

$f(x) = \log_e (x + \sqrt{x^2 + 1})$, $x \in \mathbb{R}$. Then which one of the following is correct ? (20-07-21/Shift-2)

- (a) $g(1) + g(0) = 0$ (b) $g(1) = \sqrt{2}g(0)$
(c) $g(1) = g(0)$ (d) $\sqrt{2}g(1) = g(0)$

56. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = x + 1$, then the value of

$\lim_{n \rightarrow \infty} \frac{1}{n} \left[f(0) + f\left(\frac{5}{n}\right) + f\left(\frac{10}{n}\right) + \dots + f\left(\frac{5(n-1)}{n}\right) \right]$, is:

(20-07-21/Shift-2)

- (a) $\frac{3}{2}$ (b) $\frac{7}{2}$
(c) $\frac{5}{2}$ (d) $\frac{1}{2}$

57. If the real part of the complex number

$(1 - \cos \theta + 2i \sin \theta)^{-1}$ is $\frac{1}{5}$ for $\theta \in (0, \pi)$, then the value

of the integral $\int_0^\theta \sin x dx$ is equal to ?

(20-07-21/Shift-2)

- (a) 2 (b) -1
(c) 0 (d) 1



58. Let $f : [0, \infty) \rightarrow [0, \infty)$ be defined as $f(x) = \int_0^x [y] dy$

Where $[x]$ is the greatest integer less than or equal to x .
Which of the following is true ? (25-07-21/Shift-1)

- (a) f is differentiable at every point in $[0, \infty)$
- (b) f is continuous everywhere except at the integer points in $[0, \infty)$
- (c) f is continuous at every point in $[0, \infty)$ and differentiable except at the integer points.
- (d) f is both continuous and differentiable except at the integer points in $[0, \infty)$.

59. The value of the definite integral $\int_{\pi/24}^{5\pi/24} \frac{dx}{1 + \sqrt[3]{\tan 2x}}$ is (25-07-21/Shift-1)

- (a) $\frac{\pi}{18}$
- (b) $\frac{\pi}{3}$
- (c) $\frac{\pi}{6}$
- (d) $\frac{\pi}{12}$

60. The value of the definite integral

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{dx}{(1 + e^{x \cos x})(\sin^4 x + \cos^4 x)}$$
 is equal to :

(27-07-21/Shift-1)

- (a) $\frac{\pi}{\sqrt{2}}$
- (b) $-\frac{\pi}{4}$
- (c) $\frac{\pi}{2\sqrt{2}}$
- (d) $-\frac{\pi}{2}$

61. The value of $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \frac{(2j-1) + 8n}{(2j-1) + 4n}$ is equal to:

(27-07-21/Shift-1)

- (a) $2 - \log_e \left(\frac{2}{3} \right)$
- (b) $3 + 2 \log_e \left(\frac{2}{3} \right)$
- (c) $1 + 2 \log_e \left(\frac{3}{2} \right)$
- (d) $5 + \log_e \left(\frac{3}{2} \right)$

62. Let the domain of the function (27-07-21/Shift-1)

$$f(x) = \log_4 \left(\log_5 \left(\log_3 (18x - x^2 - 77) \right) \right) \text{ be } (a, b).$$

Then the value of the integral

$$\int_a^b \frac{\sin^3 x}{(\sin^3 x + \sin^3 (a+b-x))} dx \text{ is equal to.}$$

63. If $\int_0^\pi (\sin^3 x) e^{-\sin^2 x} dx = \alpha - \frac{\beta}{e} \int_0^1 \sqrt{t} e^t dt$, then $\alpha + \beta$ is equal to _____. (27-07-21/Shift-2)

64. If $\int_0^{100\pi} \frac{\sin^2 x}{e^{\left(\frac{x}{\pi} \left[\frac{x}{\pi}\right]\right)}} dx = \frac{a\pi^2}{1 + 4\pi^2}$, $a \in \mathbb{R}$, where $[x]$ is the

greatest integer less than or equal to x , then the value of α is: (22-07-21/Shift-2)

- (a) $100(1-e)$
- (b) $200(1-e^{-1})$
- (c) $150(e^{-1}-1)$
- (d) $50(e-1)$

65. The value of the integral $\int_{-1}^1 \log(x + \sqrt{x^2 + 1}) dx$ is:

(25-07-21/Shift-2)

- (a) 1
- (b) 0
- (c) -1
- (d) 2

66. If $f(x) = \begin{cases} \int_0^x (5 + |1-t|) dt, & x > 2 \\ 5x + 1, & x \leq 2 \end{cases}$ (25-07-21/Shift-2)

- (a) $f(x)$ is not differentiable at $x = 1$
- (b) $f(x)$ is continuous but not differentiable at $x = 2$
- (c) $f(x)$ is not continuous at $x = 2$
- (d) $f(x)$ is everywhere differentiable

67. The function $f(x)$, that satisfies the condition

$$f(x) = x + \int_0^{\pi/2} \sin x \cdot \cos y f(y) dy, \text{ is } (01-09-21/Shift-2)$$

- (a) $x + (\pi - 2) \sin x$
- (b) $x + \frac{\pi}{2} \sin x$
- (c) $x + \frac{2}{3} (\pi - 2) \sin x$
- (d) $x + (\pi + 2) \sin x$



68. If the value of the integral $\int_0^5 \frac{x + [x]}{e^{x-[x]}} dx = \alpha e^{-1} + \beta$, where $\alpha, \beta \in \mathbb{R}$, $5\alpha + 6\beta = 0$, and $[x]$ denotes the greatest integer less than to x , then the value of $(\alpha + \beta)^2$ is equal to: **(26-08-21/Shift-2)**

- (a) 36 (b) 100
(c) 16 (d) 25

69. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1 + \sin^2 x}{1 + \pi^{\sin x}} \right) dx$ is: **(26-08-21/Shift-2)**

- (a) $\frac{3\pi}{2}$ (b) $\frac{\pi}{2}$
(c) $\frac{3\pi}{4}$ (d) $\frac{5\pi}{4}$

70. $\int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$ is equal to:

(27-08-21/Shift-1)

- (a) 10 (b) 8
(c) 6 (d) 5

71. If $U_n = \left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \dots \left(1 + \frac{n^2}{n^2}\right)$, then

$\lim_{n \rightarrow \infty} (U_n)^{\frac{-4}{n^2}}$ is equal to: **(27-08-21/Shift-1)**

- (a) $\frac{4}{e^2}$ (b) $\frac{4}{e}$
(c) $\frac{16}{e^2}$ (d) $\frac{e^2}{16}$

72. The value of $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{2n-1} \frac{n^2}{n^2 + 4r^2}$ is: **(26-08-21/Shift-1)**

- (a) $\frac{1}{4} \tan^{-1}(4)$ (b) $\tan^{-1}(4)$
(c) $\frac{1}{2} \tan^{-1}(2)$ (d) $\frac{1}{2} \tan^{-1}(4)$

73. The value of $\int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left(\left(\frac{x+1}{x-1} \right)^2 + \left(\frac{x-1}{x+1} \right)^2 - 2 \right)^{\frac{1}{2}} dx$ is:

(26-08-21/Shift-1)

- (a) $\log_e 4$ (b) $\log_e 16$
(c) $4 \log_e (3 + 2\sqrt{2})$ (d) $2 \log_e 16$

74. The value of the integral $\int_0^1 \frac{\sqrt{x} dx}{(1+x)(1+3x)(3+x)}$ is:

(27-08-21/Shift-2)

- (a) $\frac{\pi}{8} \left(1 - \frac{\sqrt{3}}{2} \right)$ (b) $\frac{\pi}{4} \left(1 - \frac{\sqrt{3}}{6} \right)$
(c) $\frac{\pi}{4} \left(1 - \frac{\sqrt{3}}{2} \right)$ (d) $\frac{\pi}{8} \left(1 - \frac{\sqrt{3}}{6} \right)$

75. Let $[t]$ denote the greatest integer $\leq t$. Then the value of $\int_{-\frac{1}{2}}^1 ([2x] + |x|) dx$ is _____ ?

(31-08-21/Shift-1)

76. If $x\phi(x) = \int_5^x (3t^2 - 2\phi'(t)) dt$, $x > -2$, and $\phi(0) = 4$, then, $\phi(2)$ is _____ ? **(31-08-21/Shift-1)**

77. If $[x]$ is the greatest integer $\leq x$, then

$\pi^2 \int_0^2 \left(\sin \frac{\pi x}{2} \right) (x - [x])^{[x]} dx$ is equal to:

(31-08-21/Shift-2)

- (a) $4(\pi - 1)$ (b) $2(\pi - 1)$
(c) $2(\pi + 1)$ (d) $4(\pi + 1)$



78. Let f be a non-negative function in $[0,1]$ and twice differentiable in $(0,1)$. If

$$\int_0^x \sqrt{1-(f'(t))^2} dt = \int_0^x f(t) dt, 0 \leq x \leq 1 \text{ and}$$

$$f(0) = 0, \text{ then } \lim_{x \rightarrow 0} \frac{1}{x^2} \int_0^x f(t) dt : \quad (31-08-21/\text{Shift-1})$$

- (a) Equals 1 (b) Does not exist

- (c) Equals $\frac{1}{2}$ (d) Equals 0

79. $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} (\sin \sqrt{t}) dt}{x^3}$ is equal to (24-02-21/Shift-1)

- (a) $\frac{1}{15}$ (b) 0

- (c) $\frac{2}{3}$ (d) $\frac{3}{2}$

80. If $\int_{-a}^a (|x| + |x-2|) dx = 22, (a > 2)$ 40. and $[x]$ denotes the greatest integer $\leq x$, then $\int_a^{-a} (x + [x]) dx$ is equal to _____ (24-02-21/Shift-1)

81. Let $f(x)$ be a differentiable function defined on $[0,2]$ such that $f'(x) = f'(2-x)$ for all $x \in (0,2), f(0) = 1$ and $f(2) = e^2$. Then the value of $\int_0^2 f(x) dx$ is

(24-02-21/Shift-2)

- (a) $2(1-e^2)$ (b) $1-e^2$
(c) $2(1+e^2)$ (d) $1+e^2$

82. The value of the integral $\int_1^3 [x^2 - 2x - 2] dx$, where $[x]$ denotes the greatest integer less than or equal to x , is (24-02-21/Shift-2)

- (a) -5 (b) $-\sqrt{2} - \sqrt{3} - 1$
(c) -4 (d) $-\sqrt{2} - \sqrt{3} + 1$

83. The value of $\int_{-1}^1 x^2 e^{[x^3]} dx$, where $[t]$ denotes the greatest integer $\leq t$, is: (25-02-21/Shift-1)

- (a) $\frac{1}{3e}$ (b) $\frac{e+1}{3}$

- (c) $\frac{e-1}{3e}$ (d) $\frac{e+1}{3e}$

84. If $I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x dx$, then (25-02-21/Shift-2)

- (a) $I_2 + I_4, (I_3 + I_5)^2, I_4 + I_6$ are in G.P.

- (b) $\frac{1}{I_2 + I_4}, \frac{1}{I_3 + I_5}, \frac{1}{I_4 + I_6}$ are in A.P.

- (c) $\frac{1}{I_2 + I_4}, \frac{1}{I_3 + I_5}, \frac{1}{I_4 + I_6}$ are in G.P.

- (d) $I_2 + I_4, I_3 + I_5, I_4 + I_6$ are in A.P.

85. $\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n}{(n+1)^2} + \frac{n}{(n+2)^2} + \dots + \frac{n}{(2n-1)^2} \right]$ is equal to: (25-02-21/Shift-2)

- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$

- (c) $\frac{1}{3}$ (d) 1

86. The value of $\int_{-2}^2 |3x^2 - 3x - 6| dx$ is: (25-02-21/Shift-2)

87. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos^2 x}{1+3^x} dx$ is (26-02-21/Shift-1)

- (a) 2π (b) $\frac{\pi}{2}$

- (c) 4π (d) $\frac{\pi}{4}$



88. The Value of $\sum_{n=1}^{100} \int_{n-1}^n e^{x-[x]} dx$ where $[x]$ is the greatest integer $\leq x$, is

(26-02-21/Shift-1)

- (a) $100(e-1)$ (b) $100(1+e)$
(c) $100(1-e)$ (d) $100e$

89. The value of the integral $\int_0^{\pi} |\sin 2x| dx$ is _____.

(26-02-21/Shift-1)

90. For $x > 0$, if $f(x) = \int_1^x \frac{\log_e t}{(1+t)} dt$, then $f(e) + f\left(\frac{1}{e}\right)$ is equal to:

(26-02-21/Shift-2)

- (a) 1 (b) $\frac{1}{2}$
(c) 0 (d) -1

91. If $I_{m,n} = \int_0^1 x^{m-1} (1-x)^{n-1} dx$, for $m, n \geq 1$ and

$$\int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx = \alpha I_{m,n}, \alpha \in \mathbb{R}, \text{ then } \alpha \text{ equals } \underline{\hspace{2cm}}.$$

(26-02-21/Shift-2)

92. If the normal to the curve $y(x) = \int_0^x (2t^2 - 15t + 10) dt$ at a point (a, b) is parallel to the line $x + 3y = -5, a > 1$, then the value of $|a + 6b|$ is equal to _____.

(16-03-21/Shift-1)

93. Let $f : (0, 2) \rightarrow \mathbb{R}$ be defined as

$$f(x) = \log_2 \left(1 + \tan \left(\frac{\pi x}{4} \right) \right).$$

$$\text{Then } \lim_{n \rightarrow \infty} \frac{2}{n} \left(f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots + f(1) \right) \text{ is equal to } \underline{\hspace{2cm}}.$$

(16-03-21/Shift-1)

94. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $f(x) + f(x+1) = 2$, for all $x \in \mathbb{R}$. If $I_1 = \int_0^8 f(x) dx$ and

$$I_2 = \int_{-1}^3 f(x) dx, \text{ then the value of } I_1 + 2I_2 \text{ is equal to}$$

(16-03-21/Shift-1)

95. Let $P(x) = x^2 + bx + c$ be a quadratic polynomial with real coefficients such that $\int_0^1 P(x) dx = 1$ and $P(x)$ leaves remainder 5 when it is divided by $(x-2)$. Then the value of $9(b+c)$ is equal to

(16-03-21/Shift-2)

- (a) 11 (b) 9
(c) 7 (d) 15

96. Consider the integral $I = \int_0^{10} \frac{[x]e^{[x]}}{e^{x-1}} dx$ where $[x]$ denotes the greatest integer less than or equal to x . Then the value of I is equal to :

(16-03-21/Shift-2)

- (a) $9(e-1)$ (b) $9(e+1)$
(c) $45(e+1)$ (d) $45(e-1)$

97. Which of the following statements is incorrect for the function $g(\alpha)$ for $\alpha \in \mathbb{R}$ such that

$$g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^\alpha x}{\cos^\alpha x + \sin^\alpha x} dx$$

(17-03-21/Shift-1)

- (a) $g(\alpha)$ has an inflection point at $\alpha = -\frac{1}{2}$
(b) $g(\alpha)$ is a strictly decreasing function
(c) $g(\alpha)$ is a strictly increasing function
(d) $g(\alpha)$ is an even function

98. If $[x]$ represents the greatest integer function, then the value

$$\text{of } \left| \int_0^{\sqrt{\frac{\pi}{2}}} \left[[x^2] - \cos x \right] dx \right| \text{ is } \dots\dots\dots$$

(17-03-21/Shift-1)



99. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = e^{-x} \sin x$. If $F: [0, 1] \rightarrow \mathbb{R}$ is a differentiable function such that $F(x) = \int_0^x f(t) dt$, then the value of $\int_0^1 (F'(x) + f(x)) e^x dx$ lies in the interval

(17-03-21/Shift-2)

- (a) $\left[\frac{330}{360}, \frac{331}{360}\right]$ (b) $\left[\frac{327}{360}, \frac{329}{360}\right]$
(c) $\left[\frac{331}{360}, \frac{334}{360}\right]$ (d) $\left[\frac{335}{360}, \frac{336}{360}\right]$

100. If the integral $\int_0^{10} \frac{[\sin 2\pi x]}{e^{x-[x]}} dx = \alpha e^{-1} + \beta e^{-\frac{1}{2}} + \gamma$, where α, β, γ are integers and $[x]$ denotes the greatest integer less than or equal to x , then the value of $\alpha + \beta + \gamma$ is equal to :
- (17-03-21/Shift-2)
- (a) 0 (b) 20
(c) 10 (d) 25

101. Let $I_n = \int_1^e x^{19} (\log |x|)^n dx$, where $n \in \mathbb{N}$. If $(20)I_{10} = \alpha I_9 + \beta I_8$, for natural numbers α and β , then $\alpha - \beta$ equals to
- (17-03-21/Shift-2)

102. Let $f(x)$ and $g(x)$ be two functions satisfying $f(x^2) + g(4-x) = 4x^3$ and $g(4-x) + g(x) = 0$, then the value of $\int_{-4}^4 f(x^2) dx$ is
- (18-03-21/Shift-1)

103. Let $g(x) = \int_0^x f(t) dt$, where f is continuous function in $[0, 3]$ such that $\frac{1}{3} \leq f(t) \leq 1$ for all $t \in [0, 1]$ and $0 \leq f(t) \leq \frac{1}{2}$ for all $t \in (1, 3]$. The largest possible interval in which $g(3)$ lies is:
- (18-03-21/Shift-2)

- (a) $[1, 3]$ (b) $\left[-\frac{3}{2}, -1\right]$
(c) $\left[\frac{1}{3}, 2\right]$ (d) $\left[-1, -\frac{1}{2}\right]$

104. Let $P(x)$ be a real polynomial of degree 3 which vanishes at $x = -3$. Let $P(x)$ have Local minima at $x = 1$ local maxima at $x = -1$ and $\int_{-1}^1 P(x) dx = 18$, then the sum of all the coefficients of the polynomial $P(x)$ is equal to
- (18-03-21/Shift-2)

Area Under Curves

105. The area (in sq. units) of the region described by $\{(x, y) : y^2 \leq 2x \text{ and } y \geq 4x - 1\}$ is:
- (2015)

- (a) $\frac{15}{64}$ (b) $\frac{9}{32}$
(c) $\frac{7}{32}$ (d) $\frac{5}{64}$

106. The area (in square units) of the region bounded by the curves $y + 2x^2 = 0$ and $y + 3x^2 = 1$, is equal to :
- (2015/Online Set-1)

- (a) $\frac{1}{3}$ (b) $\frac{3}{4}$
(c) $\frac{3}{5}$ (d) $\frac{4}{3}$

107. The area of the plane region bounded by the curves $x + 2y^2 = 0$ and $x + 3y^2 = 1$ is equal to
- (2015/Online Set-2)

- (a) $\frac{4}{3}$ (b) $\frac{5}{3}$
(c) $\frac{1}{3}$ (d) $\frac{2}{3}$

108. The area (in sq. units) of the region $\{(x, y) : y^2 \geq 2x \text{ and } x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$ is :
- (2016)

- (a) $\pi - \frac{8}{3}$ (b) $\pi - \frac{4\sqrt{2}}{3}$
(c) $\frac{\pi}{2} - \frac{2\sqrt{2}}{3}$ (d) $\pi - \frac{4}{3}$



109. The area (in sq. units) of the region described by

$$A = \{(x, y) | y \geq x^2 - 5x + 4, x + y \geq 1, y \leq 0\} \text{ is:}$$

(2016/Online Set-1)

- (a) $\frac{7}{2}$ (b) $\frac{19}{6}$
(c) $\frac{13}{6}$ (d) $\frac{17}{6}$

110. The area (in sq. units) of the region

$$\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y \text{ and } y \leq 1 + \sqrt{x}\} \text{ is:}$$

(2017)

- (a) $\frac{59}{12}$ (b) $\frac{2}{3}$
(c) $\frac{7}{3}$ (d) $\frac{5}{2}$

111. The area (in sq. units) of the smaller portion enclosed between the curves, $x^2 + y^2 = 4$ and $y^2 = 3x$, is :

(2017/Online Set-1)

- (a) $\frac{1}{2\sqrt{3}} + \frac{\pi}{3}$ (b) $\frac{1}{\sqrt{3}} + \frac{2\pi}{3}$
(c) $\frac{1}{2\sqrt{3}} + \frac{2\pi}{3}$ (d) $\frac{1}{\sqrt{3}} + \frac{4\pi}{3}$

112. Let $g(x) = \cos x^2$, $f(x) = \sqrt{x}$, and $\alpha, \beta (\alpha < \beta)$ be the roots of the quadratic equation $18x^2 - 9\pi x + \pi^2 = 0$. Then the area (in sq. units) bounded by the curve $y = (g \circ f)(x)$ and the lines $x = \alpha, x = \beta$ and $y = 0$, is :

(2018)

- (a) $\frac{1}{2}(\sqrt{2} - 1)$ (b) $\frac{1}{2}(\sqrt{3} - 1)$
(c) $\frac{1}{2}(\sqrt{3} + 1)$ (d) $\frac{1}{2}(\sqrt{3} - \sqrt{2})$

113. The area (in sq. units) of the region

$$\{x \in \mathbb{R} : x \geq 0, y \geq 0, y \geq x - 2 \text{ and } y \leq \sqrt{x}\} \text{ is :}$$

(2018/Online Set-1)

- (a) $\frac{13}{3}$ (b) $\frac{8}{3}$
(c) $\frac{10}{3}$ (d) $\frac{5}{3}$

114. If the area of the region bounded by the curves, $y = x^2$, $y = \frac{1}{x}$ and the lines $y = 0$ and $x = t$ ($t > 1$) is 1 sq. unit, then t is equal to :

(2018/Online Set-3)

- (a) e^2 (b) $\frac{4}{3}$
(c) $\frac{3}{2}$ (d) $e^{\frac{2}{3}}$

115. The area (in sq. units) of the region

$$A = \{(x, y) \in \mathbb{R} \times \mathbb{R} | 0 \leq x \leq 3, 0 \leq y \leq 4, y \leq x^2 + 3x\} \text{ is}$$

(8-04-2019/Shift-1)

- (a) $\frac{53}{6}$ (b) 8
(c) $\frac{59}{6}$ (d) $\frac{26}{3}$

116. Let $S(\alpha) = \{(x, y) : y^2 \leq x, 0 \leq x \leq \alpha\}$ and $A(\alpha)$ is area of the region $S(\alpha)$. If for a λ , $0 < \lambda < 4$, $A(\lambda) : A(4) = 2:5$, then λ equals :

(8-04-2019/Shift-2)

- (a) $2\left(\frac{4}{5}\right)^{\frac{1}{3}}$ (b) $2\left(\frac{2}{5}\right)^{\frac{1}{3}}$
(c) $4\left(\frac{2}{5}\right)^{\frac{1}{3}}$ (d) $4\left(\frac{4}{25}\right)^{\frac{1}{3}}$

117. The area (in sq. units) of the region

$$A = \{(x, y) : x^2 \leq y \leq x + 2\} \text{ is:}$$

(09-04-2019/Shift-1)

- (a) $\frac{10}{3}$ (b) $\frac{9}{2}$
(c) $\frac{31}{6}$ (d) $\frac{13}{6}$



118. The area (in sq. units) of the region

$$A = \left\{ (x, y) : \frac{y^2}{2} \leq x \leq y + 4 \right\} \text{ is: } \quad (9-04-2019/\text{Shift-2})$$

- (a) $\frac{53}{3}$ (b) 30
(c) 16 (d) 18

119. The area (in sq. units) of the region bounded by the curves $y = 2^x$ and $y = |x + 1|$, in the first quadrant is:

(10-4-2019/Shift-2)

- (a) $\ln 2 + \frac{3}{2}$ (b) $\frac{3}{2}$
(c) $\frac{1}{2}$ (d) $\frac{3}{2} - \frac{1}{\ln 2}$

120. If the area (in sq. units) of the region $\{(x, y) : y^2 \leq 4x, x + y \leq 1, x \geq 0, y \geq 0\}$ is $a\sqrt{2} + b$, then $a - b$ is equal to :

(12-04-2019/Shift-1)

- (a) $\frac{10}{3}$ (b) 6
(c) $\frac{8}{3}$ (d) $-\frac{2}{3}$

121. If the area (in sq. units) bounded by the parabola $y^2 = 4\lambda x$ and the line $y = \lambda x, \lambda > 0$ is $\frac{1}{9}$ then λ is equal to _____.

(12-04-2019/Shift-2)

- (a) $2\sqrt{6}$ (b) 48
(c) 24 (d) $4\sqrt{3}$

122. The area of the region

$$A = \{(x, y) : 0 \leq y \leq x |x| + 1 \text{ and } -1 \leq x \leq 1\} \text{ in sq. units is:}$$

(09-01-2019/Shift-2)

- (a) $\frac{2}{3}$ (b) 2
(c) $\frac{4}{3}$ (d) $\frac{1}{3}$

123. If the area enclosed between the curves $y = kx^2$ and $x = ky^2, (k > 0)$, is 1 square unit. Then k is:

(10-1-2019/Shift-1)

- (a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{\sqrt{3}}$
(c) $\sqrt{3}$ (d) $\frac{2}{\sqrt{3}}$

124. The area (in sq. units) of the region bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$ is

(11-01-2019/Shift-1)

- (a) $\frac{5}{4}$ (b) $\frac{9}{8}$
(c) $\frac{7}{8}$ (d) $\frac{3}{4}$

125. The area (in sq. units) in the first quadrant bounded by the parabola, $y = x^2 + 1$, the tangent to it at the point (2, 5) and the coordinate axes is :

(11-01-2019/Shift-2)

- (a) $\frac{8}{3}$ (b) $\frac{37}{24}$
(c) $\frac{187}{24}$ (d) $\frac{14}{3}$

126. The maximum area (in sq. units) of a rectangle having its base on the X-axis and its other two vertices on the parabola, $y = 12 - x^2$ such that the rectangle lies inside the parabola, is _____.

(12-01-2019/Shift-1)

- (a) 36 (b) $20\sqrt{2}$
(c) 32 (d) $18\sqrt{3}$

127. The area (in sq. units) of the region bounded by the parabola, $y = x^2 + 2$ and the lines, $y = x + 1, x = 0$ and $x = 3$, is :

(12-01-2019/Shift-1)

- (a) $\frac{15}{4}$ (b) $\frac{21}{2}$
(c) $\frac{17}{4}$ (d) $\frac{15}{2}$



128. Area (in sq. units) of the region outside $\frac{|x|}{2} + \frac{|y|}{3} = 1$

and inside the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ is :

(2-9-2020/Shift-1)

- (a) $3(\pi - 2)$ (b) $6(\pi - 2)$
(c) $6(4 - \pi)$ (d) $3(4 - \pi)$

129. Consider a region $R = \{(x, y) \in \mathbb{R}^2 : x^2 \leq y \leq 2x\}$. If a line $y = \alpha$ divides the area of region R into two equal parts, then which of the following is true ?

(2-09-2020/Shift-2)

- (a) $\alpha^3 - 6\alpha^2 + 16 = 0$ (b) $3\alpha^2 - 8\alpha^{3/2} + 8 = 0$
(c) $\alpha^3 - 6\alpha^{3/2} - 16$ (d) $3\alpha^2 - 8\alpha + 8 = 0$

130. The area (in sq. units) of the region

$$\left\{ (x, y) : 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, \frac{1}{2} \leq x \leq 2 \right\} \text{ is}$$

(3-09-2020/Shift-1)

- (a) $\frac{23}{16}$ (b) $\frac{79}{16}$
(c) $\frac{23}{6}$ (d) $\frac{79}{24}$

131. The area (in sq. units) of the region

$$A = \{(x, y) : (x-1)[x] \leq y \leq 2\sqrt{x}, 0 \leq x \leq 2\}$$

where $[t]$ denotes the greatest integer function, is:

(5-09-2020/Shift-2)

- (a) $\frac{4}{3}\sqrt{2} - \frac{1}{2}$ (b) $\frac{8}{3}\sqrt{2} - \frac{1}{2}$
(c) $\frac{8}{3}\sqrt{2} - 1$ (d) $\frac{4}{3}\sqrt{2} + 1$

132. The area (in sq. units) of the region

$$A = \{(x, y) : |x| + |y| \leq 1, 2y^2 \geq |x|\} \text{ (6-09-2020/Shift-1)}$$

- (a) $\frac{1}{6}$ (b) $\frac{5}{6}$
(c) $\frac{1}{3}$ (d) $\frac{7}{6}$

133. The area (in sq. units) of the region enclosed by the curves $y = x^2 - 1$ and $y = 1 - x^2$ is equal to :

(6-09-2020/Shift-2)

- (a) $\frac{4}{3}$ (b) $\frac{7}{2}$
(c) $\frac{16}{3}$ (d) $\frac{8}{3}$

134. The area of the region, enclosed by the circle $x^2 + y^2 = 2$ which is not common to the region bounded by the parabola $y^2 = x$ and the straight line $y = x$

(7-01-2020/Shift-1)

- (a) $\frac{1}{3}(12\pi - 1)$ (b) $\frac{1}{3}(6\pi - 1)$
(c) $\frac{1}{6}(12\pi - 1)$ (d) $\frac{1}{6}(24\pi - 1)$

135. The area (in sq. units) of the region $\{(x, y) \in \mathbb{R}^2 : 4x^2 \leq y \leq 8x + 12\}$ is: (07-01-2020/Shift-2)

- (a) $\frac{125}{3}$ (b) $\frac{128}{3}$
(c) $\frac{124}{3}$ (d) $\frac{127}{3}$

136. For $a > 0$, let the curves $C_1 : y^2 = ax$ and

$C_2 : x^2 = ay$ intersect at origin O and a point P. let the line $x = b$ ($0 < b < a$) intersect the chord OP and the x-axis at points Q and R, respectively. If the line $x = b$ bisects the area bounded by the curves, C_1 and C_2 , and the area of

$\Delta OQR = \frac{1}{2}$, then 'a' satisfies the equation:

(8-01-2020/Shift-1)

- (a) $x^6 - 12x^3 + 4 = 0$ (b) $x^6 - 12x^3 - 4 = 0$
(c) $x^6 + 6x^3 - 4 = 0$ (d) $x^6 - 6x^3 + 4 = 0$



137. The area (in sq. units) of the region $\{(x, y) \in \mathbb{R}^2 : x^2 \leq y \leq 3 - 2x\}$, is:

(8-01-2020/Shift-2)

- (a) $\frac{31}{3}$ (b) $\frac{32}{3}$
(c) $\frac{29}{3}$ (d) $\frac{34}{3}$

138. Given:

$$f(x) = \begin{cases} x, & 0 \leq x < \frac{1}{2} \\ \frac{1}{2}, & x = \frac{1}{2} \\ 1-x, & \frac{1}{2} < x \leq 1 \end{cases}$$

and $g(x) = \left(x - \frac{1}{2}\right)^2, x \in \mathbb{R}$. Then the area (in sq. units)

of the region bounded by the curves $y = f(x)$

and $y = g(x)$ between the lines $2x = 1$ to $2x = \sqrt{3}$ is

(9-1-2020/Shift-2)

- (a) $\frac{\sqrt{3}}{4} - \frac{1}{3}$ (b) $\frac{1}{3} + \frac{\sqrt{3}}{4}$
(c) $\frac{1}{2} + \frac{\sqrt{3}}{4}$ (d) $\frac{1}{2} - \frac{\sqrt{3}}{4}$

139. Let T be the tangent to the ellipse $E : x^2 + 4y^2 = 5$ at the point $P(1, 1)$. If the area of the region bounded by the tangent T, ellipse E, lines $x = 1$ and $x = \sqrt{5}$ is

$\alpha\sqrt{5} + \beta + \gamma \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$, then $\left|\frac{4}{5}(\alpha + \beta + \gamma)\right|$ is equal to

_____.

(20-07-21/Shift-1)

140. The area (in sq. units) of the region, given by the set $\{(x, y) \in \mathbb{R} \times \mathbb{R} : x \geq 0, 2x^2 \leq y \leq 4 - 2x\}$

(25-07-21/Shift-1)

- (a) $\frac{7}{3}$ (b) $\frac{13}{3}$
(c) $\frac{17}{3}$ (d) $\frac{8}{3}$

141. If the area of the bounded region

$$R = \left\{ (x, y) : \max\{0, \log_e x\} \leq y \leq 2^x, \frac{1}{2} \leq x \leq 2 \right\}$$

$\alpha(\log_e 2)^{-1} + \beta(\log_e 2) + \gamma$, then the value of

$(\alpha + \beta - 2\gamma)^2$ is equal to :

(27-07-21/Shift-1)

- (a) 4 (b) 1
(c) 8 (d) 2

142. The area of the region bounded by $y - x = 2$ and $x^2 = y$ is equal to

(27-07-21/Shift-2)

- (a) $\frac{4}{3}$ (b) $\frac{9}{2}$
(c) $\frac{16}{3}$ (d) $\frac{2}{3}$

143. The area (in sq. units) of the region bounded by the curves $x^2 + 2y - 1 = 0, y^2 + 4x - 4 = 0$ and $y^2 - 4x - 4 = 0$, in the upper half plane is ____.

(22-07-21/Shift-2)

144. The area, enclosed by curves $y = \sin x + \cos x$ and

$y = |\cos x - \sin x|$ and the lines $x = 0, x = \frac{\pi}{2}$, is:

(01-09-21/Shift-2)

- (a) $2\sqrt{2}(\sqrt{2} + 1)$ (b) $4(\sqrt{2} - 1)$
(c) $2(\sqrt{2} + 1)$ (d) $2\sqrt{2}(\sqrt{2} - 1)$



145. Let a and b respectively be the points of local maximum and local minimum of the function $f(x) = 2x^3 - 3x^2 - 12x$.
If A is the total area of the region bounded by $y = f(x)$, the x -axis and the lines $x = a$ and $x = b$, then $4A$ is equal to _____. (26-08-21/Shift-2)
146. The area of the region $S = \{(x, y) : 3x^2 \leq 4y \leq 6x + 24\}$ is _____. (26-08-21/Shift-1)
147. The area of the region bounded by the parabola $(y - 2)^2 = (x - 1)$, the tangent to it at the point whose ordinate is 3 and the x -axis is : (27-08-21/Shift-2)
(a) 6 (b) 4
(c) 10 (d) 9
148. If the line $y = mx$ bisects the area enclosed by the lines $x = 0, y = 0, x = \frac{3}{2}$ and the curve $y = 1 + 4x - x^2$ then $12m$ is equal to _____. (31-08-21/Shift-2)
149. The area (in sq. units) of the part of the circle $x^2 + y^2 = 36$, which is outside the parabola $y^2 = 9x$, is (24-02-21/Shift-1)
(a) $24\pi + 3\sqrt{3}$ (b) $12\pi - 3\sqrt{3}$
(c) $12\pi + 3\sqrt{3}$ (d) $24\pi - 3\sqrt{3}$
150. The area of the region $R = \{(x, y) : 5x^2 \leq y \leq 2x^2 + 9\}$ is (24-02-21-Shift/2)
(a) $11\sqrt{3}$ square units (b) $12\sqrt{3}$ square units
(c) $6\sqrt{3}$ square units (d) $9\sqrt{3}$ square units
151. The graphs of sine and cosine functions, intersect each other at a number of points and between two consecutive points of intersection, the two graphs enclose the same area A . Then A^4 is equal to _____. (25-02-21/Shift-1)
152. The area bounded by the lines $y = |x - 1| - 2$ is _____. (26-02-21/Shift-1)
153. Let A_1 be the area of the region bounded by the curves $y = \sin x, y = \cos x$ and y -axis in the first quadrant. Also, let A_2 be the area of the region bounded by the curves $y = \sin x, y = \cos x, x$ -axis and $x = \frac{\pi}{2}$ in the first quadrant. Then (26-02-21/Shift-2)
(a) $A_1 = A_2$ and $A_1 + A_2 = \sqrt{2}$
(b) $A_1 : A_2 = 1 : \sqrt{2}$ and $A_1 + A_2 = 1$
(c) $2A_1 = A_2$ and $A_1 + A_2 = 1 + \sqrt{2}$
(d) $A_1 : A_2 = 1 : 2$ and $A_1 + A_2 = 1$
154. Let $f : [-3, 1] \rightarrow \mathbb{R}$ be given as
$$f(x) = \begin{cases} \min \{(x + 6), x^2\}, & -3 \leq x \leq 0 \\ \max \{\sqrt{x}, x^2\}, & 0 \leq x \leq 1. \end{cases}$$

If the area bounded by $y = f(x)$ and x -axis is A , then the value of $6A$ is equal to (17-03-21/Shift-2)
155. The area bounded by the curve $4y^2 = x^2(4 - x)(x - 2)$ is equal to: (18-03-21/Shift-2)
(a) $\frac{\pi}{8}$ (b) $\frac{3\pi}{2}$
(c) $\frac{3\pi}{8}$ (d) $\frac{\pi}{16}$



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

Definite Integration

Objective Questions I [Only one correct option]

1. If $f(x) = A \sin\left(\frac{\pi x}{2}\right) + B$, $f'\left(\frac{1}{2}\right) = \sqrt{2}$ and

$\int_0^1 f(x) dx = \frac{2A}{\pi}$, then constants A and B are :

(a) $\frac{\pi}{2}$ and $\frac{\pi}{2}$ (b) $\frac{2}{\pi}$ and $\frac{3}{\pi}$

(c) 0 and $-\frac{4}{\pi}$ (d) $\frac{4}{\pi}$ and 0

2. The value of $\int_{\pi}^{2\pi} [2 \sin x] dx$ where $[.]$ represents the greatest integral functions, is :

(a) $-\frac{5\pi}{3}$ (b) $-\pi$

(c) $\frac{5\pi}{3}$ (d) -2π

3. $\int_3^{29} \frac{\sqrt[3]{(x-2)^2}}{3 + \sqrt[3]{(x-2)^2}} dx =$

(a) $4 + \frac{3\sqrt{3}}{2}\pi$ (b) $2 + \frac{3\sqrt{3}}{2}\pi$

(c) $4 + \frac{\sqrt{3}}{2}\pi$ (d) $8 + \frac{3\sqrt{3}}{2}\pi$

4. $\int_0^{1/\sqrt{3}} \frac{dx}{(2x^2+1)\sqrt{x^2+1}} =$

(a) $\frac{\pi}{2}$ (b) $\tan^{-1} 2$

(c) $\tan^{-1} 1/2$ (d) π

5. If $a \neq b$ and $af\left(\frac{1}{x}\right) + bf\left(\frac{1}{x}\right) = \frac{1}{x} - 5$

for all $x \neq 0$, then

$\int_1^2 f(x) dx = \frac{1}{a^2 - b^2} \left[a(\log 2 - \alpha) + \beta \left(\frac{b}{2} \right) \right]$

where $\beta - \alpha$ is equal to

(a) 12 (b) 5

(c) 7 (d) 2

6. $\int_{-\pi}^{\pi} (\cos px - \sin qx)^2 dx$ is equal to (where $p, q \in \mathbb{Z}$)

(a) 0 (b) $\frac{\pi}{2}$

(c) π (d) 2π

7. $\int_1^e \frac{1 + \log_{10} x}{x} dx =$

(a) $\frac{1}{2} \log_{10} e$ (b) $\frac{1 + \log_{10} e}{2}$

(c) $\frac{1}{2} \log_{10} e + 1$ (d) $2 \log_{10} e$

8. $\int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_{1/e}^{\cot x} \frac{dt}{t(1+t^2)} =$

(a) $2(\tan e - 1)$ (b) $2 \tan e$

(c) 1 (d) $\tan e + \cot e$

9. $\int_0^{\log_e 5} \frac{e^x \sqrt{e^x - 1}}{e^x + 3} dx =$

(a) $3 + \pi$ (b) $3 - \pi$

(c) $4 + \pi$ (d) $4 - \pi$



10. $\int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx = \pi \frac{\cos \alpha}{1 + \sin^2 \alpha}$

(a) for no value of α

(b) for exactly two values of α in $(0, \pi)$

(c) for at least one α in $\left(\frac{\pi}{2}, \pi\right)$

(d) for exactly one α in $\left(0, \frac{\pi}{2}\right)$

11. $\int_{-2}^2 |x(x-1)| dx =$

(a) $\frac{17}{3}$

(b) $\frac{11}{3}$

(c) $\frac{13}{3}$

(d) $\frac{16}{3}$

12. If $[t]$ stands for the integral part of t , then

$\int_0^{5\pi/12} [\tan x] dx =$

(a) $\frac{\pi}{2}$

(b) π

(c) $\frac{\pi}{4}$

(d) 2π

13. If $[t]$ denotes the integral part of t , then

$\int_0^1 \cos(\pi x) \cos([2x]\pi) dx =$

(a) 1

(b) -1

(c) $-\frac{2}{\pi}$

(d) $\frac{2}{\pi}$

14. $\int_0^1 \frac{dx}{(5+2x-2x^2)(1-e^{2-4x})} =$

(a) $\frac{1}{\sqrt{11}} \log_e \left(\frac{\sqrt{11}+2}{\sqrt{11}} \right)$

(b) $\frac{1}{\sqrt{11}} \log_e \left(\frac{\sqrt{11}+1}{\sqrt{10}} \right)$

(c) $\frac{1}{\sqrt{10}} \log_e \left(\frac{\sqrt{10}+2}{\sqrt{11}} \right)$

(d) $\frac{1}{\sqrt{10}} \log_e \left(\frac{\sqrt{10}+1}{\sqrt{11}} \right)$

15. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be continuous functions. Then the value of the integral

$\int_{-\pi/2}^{\pi/2} [f(x) + f(-x)][g(x) - g(-x)] dx$ is :

(a) π

(b) 1

(c) -1

(d) 0

16. $\int_0^{2\pi} \frac{e^{|\sin x|} \cos x}{1 + e^{\tan x}} dx =$

(a) e^{π}

(b) 1

(c) $e^{\pi} - 1$

(d) 0

17. If $m = \int_{-2}^0 \frac{|\sin x|}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} dx$ and $n = \int_0^2 \frac{|\sin x|}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} dx$, where $[.]$

represents greatest integer function, then

(a) $m = n$

(b) $m = -n$

(c) $m = 2n$

(d) $m = -2n$

18. The maximum value of $\int_{a-1}^{a+1} e^{-(x-1)^2} dx$ is attained (a is real)

at

(a) $a = 2$

(b) $a = 1$

(c) $a = -1$

(d) $a = 0$

19. If $f(x)$ is differentiable & defined on $\mathbb{R}^+$ such that

$\int_0^{t^2} xf(x) dx = \frac{2}{3} t^5$ then $f(4/25) =$

(a) $\frac{2}{3}$

(b) $-\frac{3}{2}$

(c) 1

(d) $\frac{3}{2}$



20. The function $F(x) = \int_{\pi/6}^x (4 \sin t + 3 \cos t) dt$ attains least value on $[\pi/4, 3\pi/4]$ at x equals.
- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{3}$
(c) $\frac{3\pi}{4}$ (d) $\frac{\pi}{4}$
21. If $f(x) = \int_{\pi^2/16}^{x^2} \frac{\cos x \cos \sqrt{t}}{1 + \sin^2 \sqrt{t}} dt$ then $f'(\pi)$ is equal to
- (a) 0 (b) π
(c) 2π (d) $\frac{\pi}{2}$
22. If $f(t) = \begin{cases} at - 1 & t < 1 \\ t^2 + b & t \geq 1 \end{cases}$ then possible set of values of (a, b) so that $\int_0^x f(x) dx$ is differentiable for all $x \geq 0$ is
- (a) $(5, 1)$ (b) $(1, 3)$
(c) $(4, 2)$ (d) none of these
23. If $f(x) = e^{g(x)}$ and $g(x) = \int_2^x \frac{t}{1+t^4} dt$ then $f'(2)$ has the value equal to
- (a) $2/17$ (b) 0
(c) 1 (d) cannot be determined
24. The value of the function $f(x) = 1 + x + \int_1^x (\ln^2 t + 2 \ln t) dt$ where $f'(x)$ vanishes is
- (a) e^{-1} (b) 0
(c) $2e^{-1}$ (d) $1 + 2e^{-1}$
25. $\lim_{h \rightarrow 0} \frac{\int_a^{x+h} \ln^2 t dt - \int_a^x \ln^2 t dt}{h}$ equals to
- (a) 0 (b) $\ln^2 x$
(c) $\frac{2 \ln x}{x}$ (d) does not exist
26. If $\int_0^x f(t) dt = x + \int_x^1 t f(t) dt$ then $f(1)$ is
- (a) $\frac{1}{2}$ (b) 0
(c) 1 (d) $\frac{-1}{2}$
27. $\int_0^{1/2} e^x \left(\sin^{-1} x - \frac{x}{(1-x^2)^{3/2}} \right) dx =$
- (a) $\sqrt{e} \left(\frac{\pi}{6} - \frac{1}{\sqrt{3}} \right) + 1$ (b) $\sqrt{e} \left(\frac{\pi}{6} + \frac{2}{\sqrt{3}} \right) + 1$
(c) $\sqrt{e} \left(\frac{\pi}{6} - \frac{2}{\sqrt{3}} \right) + 1$ (d) $\sqrt{e} \left(\frac{\pi}{6} + \frac{1}{\sqrt{3}} \right) + 1$
28. Let $f(x) = \begin{cases} \int_0^x \{5 + |1-y|\} dy & \text{if } x > 2 \\ 5x + 1 & \text{if } x \leq 2 \end{cases}$
- Then
- (a) $f(x)$ is continuous but not differentiable at $x = 2$
(b) $f(x)$ is not continuous at $x = 2$
(c) $f(x)$ is differentiable everywhere
(d) The right derivative of $f(x)$ at $x = 2$ does not exist



29. Consider the function $f(x) = \int_0^x [t] dt$ where $x > 0$ and $[t]$ is the integral part of t . Then
- (a) $f(x)$ is not defined for $x = 1, 2, 3, \dots$
 (b) $f(x)$ is defined for all $x > 0$ but is not continuous at $x = 1, 2, 3, \dots$
 (c) $f(x)$ is continuous for all $x > 0$
 (d) $f(x)$ is differentiable for all $x > 0$
30. $\int_0^\pi \frac{x^2 \sin 2x \sin[(\pi/2)\cos x]}{2x - \pi} dx =$
- (a) $\frac{4}{\pi^2}$ (b) $\frac{\pi^2}{4}$
 (c) $\frac{\pi^2}{8}$ (d) $\frac{8}{\pi}$
31. Let $f(x) = \frac{e^x}{1+e^x}$
- $$I_1 = \int_{f(-a)}^{f(a)} x g(x(1-x)) dx$$
- $$I_2 = \int_{f(-a)}^{f(a)} g(x(1-x)) dx$$
- then I_2 / I_1 is
- (a) 1 (b) -3
 (c) -1 (d) 2
32. $\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{\cos^{-1}\left(\frac{2x}{1+x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right)}{e^x + 1} dx =$
- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
 (c) $\frac{\pi}{4\sqrt{3}}$ (d) $\frac{\pi}{2\sqrt{3}}$
33. For any integer n , the integral $\int_0^\pi e^{\cos^2 x} \cos^3(2n+1)x dx$ has the value :
- (a) π (b) 1
 (c) 0 (d) none of these
34. If $I_1 = \int_{-4}^{-5} e^{(x+5)^2} dx$ and $I_2 = 3 \int_{1/3}^{2/3} e^{9[x-(2/3)]^2} dx$ then the value of $I_1 + I_2$ is
- (a) 0 (b) 1
 (c) e^{-1} (d) e
35. If $I = \int_0^1 \frac{e^t}{t+1} dt$, then $\int_{a-1}^a \frac{e^{-t}}{t-a-1} dt =$
- (a) Ie^a (b) $(-I)e^a$
 (c) $(-I)e^{-a}$ (d) Ie^{-a}
36. Let $\lambda = \int_0^1 \frac{dx}{1+x^3}$, $p = \lim_{n \rightarrow \infty} \left[\frac{\prod_{r=1}^n (n^3 + r^3)}{n^{3n}} \right]^{1/n}$, then $\ln p$ is equal to
- (a) $\ln 2 - 1 + \lambda$ (b) $\ln 2 - 3 + 3\lambda$
 (c) $2 \ln 2 - \lambda$ (d) $\ln 4 - 3 + 3\lambda$
37. Consider the integrals
- $$I_1 = \int_0^1 e^{-x} \cos^2 x dx, I_2 = \int_0^1 e^{-x^2} \cos^2 x dx,$$
- $$I_3 = \int_0^1 e^{-\frac{x^2}{2}} \cos^2 x dx, I_4 = \int_0^1 e^{-\frac{x^2}{2}} dx$$
- Then
- (a) $I_2 > I_4 > I_1 > I_3$ (b) $I_2 < I_4 < I_1 < I_3$
 (c) $I_1 < I_2 < I_3 < I_4$ (d) $I_1 > I_2 > I_3 > I_4$
38. If $I_n = \int_1^e (\log_e x)^n dx$ (n is a positive integer), then
- $$I_{2012} + (2012)I_{2011} =$$
- (a) $I_{2011} + (2010)I_{2010}$ (b) $I_{2013} + (2013)I_{2012}$
 (c) $I_{2011} + (2010)I_{2009}$ (d) $I_{2012} - (2012)I_{2011}$



39. A function $f(x)$ which satisfies the relation

$$f(x) = e^x + \int_0^1 e^x f(t) dt, \text{ then } f(x) \text{ is}$$

- (a) $\frac{e^x}{2-e}$ (b) $(e-2)e^x$
(c) $2e^x$ (d) $\frac{e^x}{2}$

40. If $\int_0^\pi \frac{x^2}{(1+\sin x)^2} dx = A$ then $\int_0^\pi \frac{2x^2 \cos^2(x/2)}{(1+\sin x)^2} dx =$

- (a) $A + \pi - \pi^2$ (b) $A - \pi + \pi^2$
(c) $A - \pi - \pi^2$ (d) $A + 2\pi - \pi^2$

41. $\int_0^{\sin^2 x} (\sin^{-1} \sqrt{t}) dt + \int_0^{\cos^2 x} (\cos^{-1} \sqrt{t}) dt =$

- (a) π (b) $\frac{\pi}{2}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2\sqrt{2}}$

42. For each positive integer n , define

$$f_n(x) = \min\left(\frac{x^n}{n!}, \frac{(1-x)^n}{n!}\right) \text{ for } 0 \leq x \leq 1. \text{ Let}$$

$$I_n = \int_0^1 f_n(x) dx \text{ for } n \geq 1. \text{ Then the value of } \sum_{n=1}^{\infty} I_n \text{ is}$$

- (a) $2\sqrt{e} - 3$ (b) $2\sqrt{e} - 2$
(c) $2\sqrt{e} - 1$ (d) $2\sqrt{e}$

43. $\int_0^{\pi/2} f(\sin 2x) \sin x dx = k \int_0^{\pi/4} f(\cos 2x) \cos x dx$ where k equals

- (a) 2 (b) 4
(c) $\sqrt{2}$ (d) $2\sqrt{2}$

44. The value of the definite integral $\int_0^1 (1 + e^{-x^2}) dx$ is :

- (a) -1 (b) 2
(c) $1 + e^{-1}$ (d) none of these

45. If $\int_0^x e^{zx} \cdot e^{-z^2} dz = f(x) \int_0^x e^{-z^2/4} dz$

$$\text{then } \int e^x \left(\log_e(f(x)) + \frac{x}{2} \right) dx =$$

- (a) $\frac{xe^x}{2} + c$ (b) $\frac{x^2 e^x}{4} + c$
(c) $\frac{x^2 e^x}{2} + c$ (d) $\frac{xe^x}{4} + c$

Objective Questions II [One or more than one correct option]

46. $\int_0^\infty \frac{x}{(1+x)(1+x^2)} dx$

- (a) $\pi/4$
(b) $\pi/2$

$$(c) \text{ is same as } \int_0^\infty \frac{dx}{(1+x)(1+x^2)}$$

- (d) cannot be evaluated

47. If $f(2-x) = f(2+x)$ and $f(4-x) = f(4+x)$ and

$$f(x) \text{ is a function for which } \int_0^2 f(x) dx = 5, \text{ then}$$

$$\int_0^{50} f(x) dx \text{ is equal to}$$

- (a) 125 (b) $\int_{-4}^{46} f(x) dx$
(c) $\int_1^{51} f(x) dx$ (d) $\int_2^{52} f(x) dx$



48. If $F(x) = \frac{1}{x^2} \int_4^x (4t^2 - 2F'(t)) dt$, then $F'(4)$ equals –

(a) $\frac{32}{9}$ (b) $\frac{64}{9}$

(c) $\frac{(2F(4))^5}{9}$ (d) $\frac{11F(8)}{28}$

49. If $f(x)$ is integrable over $[1, 2]$, then $\int_1^2 f(x) dx$ is equal to

(a) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right)$ (b) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=n+1}^{2n} f\left(\frac{r}{n}\right)$

(c) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r+n}{n}\right)$ (d) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} f\left(\frac{r}{n}\right)$

50. If $I = \int_0^{\pi/2} e^{-\alpha \sin x} dx$, where $\alpha \in (0, \infty)$, then

(a) $I < \frac{\pi}{2}$ (b) $I > \frac{\pi}{2} (e^{-\alpha} + 1)$

(c) $I > \frac{\pi}{2} e^{-\alpha}$ (d) $I > 0$

51. If $I_n = \int_0^1 \frac{dx}{(1+x^2)^n}$, $n \in N$, then which of the following

statements hold good ?

(a) $2nI_{n+1} = 2^{-n} + (2n-1)I_n$

(b) $I_2 = \frac{\pi}{8} + \frac{1}{4}$

(c) $I_2 = \frac{\pi}{8} - \frac{1}{4}$

(d) $I_3 = \frac{\pi}{16} - \frac{5}{48}$

Numerical Value Type Questions

52. Determine a positive integer $n \leq 5$, such that

$$\int_0^1 e^x (x-1)^n dx = 16 - 6e$$

Assertion & Reason

(A) If ASSERTION is true, REASON is true, REASON is a correct explanation for ASSERTION.

(B) If ASSERTION is true, REASON is true, REASON is not a correct explanation for ASSERTION.

(C) If ASSERTION is true, REASON is false.

(D) If ASSERTION is false, REASON is true.

53. Assertion : $\int_0^{\pi} x \sin x \cos^2 x dx = \frac{\pi}{2} \int_0^{\pi} \sin x \cos^2 x dx$

Reason : $\int_a^b x f(x) dx = \frac{a+b}{2} \int_a^b f(x) dx$

- (a) A (b) B
(c) C (d) D

54. Assertion :

$$\int e^{\sin^{-1} x} \left(1 - \frac{x}{\sqrt{1-x^2}} \right) dx = e^{\sin^{-1} x} \cdot \sqrt{1-x^2} + c$$

Reason : $\int e^{g(x)} (g'(x)f(x) + f'(x)) dx = e^{g(x)} f(x) + c$

- (a) A (b) B
(c) C (d) D

55. Assertion : $1 \leq \int_0^{\pi/2} \frac{\sin x}{x} dx \leq \frac{\pi}{2}$

Reason : If $f(x)$ is continuous in $[a, b]$ and m and l are greatest and least value of $f(x)$ in $[a, b]$, then

$$l(b-a) \leq \int_a^b f(x) dx \leq m(b-a)$$

- (a) A (b) B
(c) C (d) D



Match the Following

Using the following passage, solve Q.57 to Q.59

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

Passage

$$\text{Using integral } \int_0^{\pi/2} \ln(\sin x) dx$$

$$= - \int_0^{\pi/2} \ln(\sec x) dx = -\frac{\pi}{2} \ln 2,$$

$$\int_0^{\pi/2} \ln(\tan x) dx = 0 \text{ and } \int_0^{\pi/4} \ln(1 + \tan x) dx = \frac{\pi}{8} \ln 2.$$

56. Column - I Column - II

(A) If $\int_0^1 \frac{\cos x}{1+x} dx = k$ and

(P) 4

$$\int_{6\pi-3}^{6\pi} \frac{\cos(x/3)}{6\pi+3-x} dx = mk, \text{ then m is}$$

(B) $\int_{-1}^1 \left(\sin^{-1} \left[x + \frac{3}{4} \right] \right) dx = \frac{\pi}{k}$, then

(Q) 1

k is {where $[.]$ denotes greatest integer function}

(C) If $f(x) = \max(x - |x|, x + [x])$ and

(R) 3

$$\int_{-3}^3 f(x) dx = -\frac{3}{k}, \text{ then k is}$$

{where $[.]$ denotes greatest integer function}

(D) If $\int_0^{20} \sqrt{1 - \cos \pi x} dx = \frac{10k\sqrt{2}}{\pi}$,

(S) 2

then k is

The correct matching is :

- (a) A-Q; B-P; C-S; D-P
- (b) A-Q; B-P; C-S; D-R
- (c) A-Q; B-R; C-Q; D-P
- (d) A-S; B-P; C-S; D-P

57. Evaluate $\int_0^{\pi/4} \ln(\sin 2x) dx =$

(a) $\frac{-\pi \ln 2}{2}$ (b) $\pi \ln 2$

(c) $\frac{\pi \ln 2}{4}$ (d) none of these

58. Evaluate $\int_{-\pi/4}^{\pi/4} \ln \left(\frac{\sin x + \cos x}{\cos x - \sin x} \right) dx =$

(a) $\pi \ln 2$ (b) $\frac{\pi \ln 2}{2}$

(c) 0 (d) $-\pi \ln 2$

59. Evaluate $\int_{-\pi/4}^{\pi/4} \ln(\sin x + \cos x) dx =$

(a) $\frac{\pi \ln 2}{2}$ (b) $\frac{-\pi \ln 2}{4}$

(c) $\pi \ln 2$ (d) 0

Text

60. Determine the value of $\int_{-\pi}^{\pi} \frac{2x(1 + \sin x)}{1 + \cos^2 x} dx$

61. Evaluate $\int_2^3 \frac{2x^5 + x^4 - 2x^3 + 2x^2 + 1}{(x^2 + 1)(x^4 - 1)} dx$.



Area Under Curves

Objective Questions I [Only one correct option]

62. The area bounded by the curve $y = 3 + 2x - x^2$, $y = 0$ & the ordinate at $x = 1$ & $x = 4$ is
 (a) $25/3$ (b) $23/3$
 (c) $19/3$ (d) none
63. The area bounded by the curve $y = x(1 - \ln x)$ and positive x -axis between $x = e^{-1}$ and $x = e$ is
 (a) $\left(\frac{e^2 - 4e^{-2}}{5}\right)$ (b) $\left(\frac{e^2 - 5e^{-2}}{4}\right)$
 (c) $\left(\frac{4e^2 - e^{-2}}{5}\right)$ (d) $\left(\frac{5e^2 - e^{-2}}{4}\right)$
64. The area enclosed by $y = x^3$, its normal at $(1, 1)$ and x -axis is equal to
 (a) $\frac{7}{4}$ (b) $\frac{9}{4}$
 (c) $\frac{5}{4}$ (d) $\frac{8}{4}$
65. Area of the region bounded by $x = 0$, $y = 0$, $x = 2$, $y = 2$, $y \leq e^x$ and $y \geq \ln x$, is
 (a) $6 - 4 \ln 2$ (b) $4 \ln 2 - 2$
 (c) $2 \ln 2 - 4$ (d) $6 - 2 \ln 2$
66. The area bounded by the curve $y = e^x$ and the lines $y = |x - 1|$, $x = 2$ is given by
 (a) $e^2 + 1$ (b) $e^2 - 1$
 (c) $e^2 - 2$ (d) none
67. The area of the closed figure bounded by the curves $y = \sqrt{x}$, $y = \sqrt{4 - 3x}$ & $y = 0$ is
 (a) $4/9$ (b) $8/9$
 (c) $16/9$ (d) none
68. The area of the closed figure bounded by the curves $y = \cos x$, $y = 1 + \frac{2}{\pi}x$ & $x = \frac{\pi}{2}$ is
 (a) $\frac{\pi + 4}{4}$ (b) $\frac{3\pi}{4}$
 (c) $\frac{3\pi + 4}{4}$ (d) $\frac{3\pi - 4}{4}$
69. The ratio in which the curve $y = x^2$ divides the region bounded by the curve; $y = \sin\left(\frac{\pi x}{2}\right)$ & the x -axis as x varies from 0 to 1, is
 (a) $2 : \pi$ (b) $1 : 3$
 (c) $3 : \pi$ (d) $(6 - \pi) : \pi$
70. The area bounded by $y = 2 - |2 - x|$ and $y = \frac{3}{|x|}$ is
 (a) $\frac{4 + 3 \ln 3}{2}$ (b) $\frac{4 - 3 \ln 3}{2}$
 (c) $\frac{3}{2} + \ln 3$ (d) $\frac{1}{2} + \ln 3$
71. The area of the region enclosed between the curves $7x^2 + 9y + 9 = 0$ and $5x^2 + 9y + 27 = 0$ is
 (a) 2 (b) 4
 (c) 8 (d) 16
72. Value of the parameter a such that the area bounded by $y = a^2x^2 + ax + 1$, co-ordinate axes and the line $x = 1$, attains its least value, is equal to
 (a) $-\frac{1}{4}$ (b) $-\frac{1}{2}$
 (c) $-\frac{3}{4}$ (d) -1
73. The area enclosed by the curves $y = \sqrt{4 - x^2}$, $y \geq \sqrt{2} \sin\left(\frac{x\pi}{2\sqrt{2}}\right)$ and x -axis is divided by y -axis in the ratio
 (a) $\frac{\pi^2 - 8}{\pi^2 + 8}$ (b) $\frac{\pi^2 - 4}{\pi^2 + 4}$
 (c) $\frac{\pi - 4}{\pi + 4}$ (d) $\frac{2\pi^2}{2\pi + \pi^2 - 8}$



74. The area of the plane figure bounded in first quadrant by $y = x^{1/3}$; $y = -x^2 + 2x + 3$; $y = 2x - 1$ and the axis of ordinates is

- (a) 12 / 55 (b) 55 / 12
(c) 32 / 55 (d) none

75. If $f(x) = \sin x \forall x \in \left[0, \frac{\pi}{2}\right]$, $f(x) + f(\pi - x) =$
 $2 \forall x \in \left(\frac{\pi}{2}, \pi\right]$ and $f(x) = f(2\pi - x) \forall x \in (\pi, 2\pi]$,

then the area enclosed by $y = f(x)$ and x -axis is

- (a) π (b) 2π
(c) 2 (d) 4

76. The area bounded by curve $y = ex \log x$ and $y = \frac{\log x}{ex}$ is -

- (a) $\frac{e^2 - 5}{4e}$ (b) $\frac{e^2 + 5}{4e}$
(c) $\frac{e^2}{4} - \frac{5}{e}$ (d) None of these

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

77. Column - I Column - II

- (A) The area bounded by the curve (P) 2

$y = x + \sin x$ and its inverse function
between the ordinates $x = 0$ to $x = 2\pi$
is $4s$. Then the value of s is

- (B) The area bounded by $y = xe^{|x|}$ (Q) 1

and lines $|x| = 1, y = 0$ is

- (C) The area bounded by the curves (R) $\frac{32}{5}$

$y^2 = x^3$ and $|y| = 2x$ is

- (D) The smaller area included

(S) $\frac{1}{3}$

between the curves $\sqrt{x} + \sqrt{y} = 1$

and $|x| + |y| = 1$ is

The correct matching is :

- (a) A-P; B-Q; C-R; D-S
(b) A-Q; B-P; C-R; D-S
(c) A-P; B-R; C-S; D-P
(d) A-P; B-P; C-R; D-S

Using the following passage, solve Q.78 to Q.80

Passage

Consider the curve defined implicitly by the equation

$$y^2 - 2ye^{\sin^{-1}x} + x^2 - 1 + [x] + e^{2\sin^{-1}x} = 0, \text{ Where } [x]$$

denotes the greatest integer function

78. The area of the region bounded by the curve between the lines $x = -1$ and $x = 0$ is

(a) $\frac{\pi}{2} + 1$ (b) $\pi - 1$

(c) $\pi + 1$ (d) $\frac{\pi}{2} - 1$

79. The area of the region bounded by the curve between the lines $x = 0$ & $x = 1$ is

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{2} - 1$

(c) $\frac{\pi}{2} + 1$ (d) $\pi + 1$

80. The Area of the region bounded by the curve between the lines $x = 0$ & $x = \frac{1}{2}$ is

(a) $\frac{\sqrt{3}}{4} + \frac{\pi}{6}$ (b) $\frac{\sqrt{3}}{2} + \frac{\pi}{6}$

(c) $\frac{\sqrt{3}}{4} - \frac{\pi}{6}$ (d) $\frac{\sqrt{3}}{2} - \frac{\pi}{6}$



EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTIONS

Definite Integration

Objective Questions I [Only one correct option]

- Let $g(x) = \int_0^x f(t) dt$, where f is such that $\frac{1}{2} \leq f(t) \leq 1$ for $t \in [0, 1]$ and $0 \leq f(t) \leq \frac{1}{2}$ for $t \in [1, 2]$. Then $g(2)$ satisfies the inequality. (2000)
 - $-\frac{3}{2} \leq g(2) < \frac{1}{2}$
 - $0 \leq g(2) < 2$
 - $\frac{3}{2} < g(2) \leq 5/2$
 - $2 < g(2) < 4$
- The value of the integral $\int_{e^{-1}}^{e^2} \left| \frac{\log_e x}{x} \right| dx$ is : (2000)
 - $3/2$
 - $5/2$
 - 3
 - 5
- If $f(x) = \begin{cases} e^{\cos x} \sin x, & \text{for } |x| \leq 2, \\ 2, & \text{otherwise} \end{cases}$ then $\int_{-2}^3 f(x) dx$, is equal to (2000)
 - 0
 - 1
 - 2
 - 3
- The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x} dx$, $a > 0$ is : (2001)
 - π
 - $a\pi$
 - $\pi/2$
 - 2π
- Let $f: (0, \infty) \rightarrow \mathbb{R}$ and $F(x) = \int_0^x f(t) dt$. If $F(x^2) = x^2(1+x)$, then $f(4)$ equals : (2001)
 - $5/4$
 - 7
 - 4
 - 2
- Let $f(x) = \int_1^x \sqrt{2-t^2} dt$. Then the real roots of the equation $x^2 - f'(x) = 0$ are : (2002)
 - ± 1
 - $\pm \frac{1}{\sqrt{2}}$
 - $\pm \frac{1}{2}$
 - 0 and 1
- Let $T > 0$ be a fixed real number, Suppose f is a continuous function such that for all $x \in \mathbb{R}$, $f(x+T) = f(x)$. If $I = \int_0^T f(x) dx$, then the value of $\int_3^{3+3T} f(2x) dx$ is : (2002)
 - $3/2 I$
 - I
 - $3 I$
 - $6 I$
- The integral $\int_{-1/2}^{1/2} \left([x] + \ln \left(\frac{1+x}{1-x} \right) \right) dx$ equals (2002)
 - $-\frac{1}{2}$
 - 0
 - 1
 - $2 \ln \left(\frac{1}{2} \right)$
- If $I(m, n) = \int_0^1 t^m (1+t)^n dt$, then the expression for $I(m, n)$ in terms of $I(m+1, n-1)$ is : (2003)
 - $\frac{2^n}{m+1} - \frac{n}{m+1} I(m+1, n-1)$
 - $\frac{n}{m+1} I(m+1, n-1)$
 - $\frac{2^n}{m+1} + \frac{n}{m+1} I(m+1, n-1)$
 - $\frac{m}{m+1} I(m+1, n-1)$



10. If $f(x) = \int_{x^2}^{x^2+1} e^{-t^2} dt$, then $f(x)$ increases in : (2003)

- (a) (2, 2) (b) no value of x
(c) $(0, \infty)$ (d) $(-\infty, 0)$

11. The value of the integral $\int_0^1 \sqrt{\frac{1-x}{1+x}} dx$ is (2004)

- (a) $\frac{\pi}{2} + 1$ (b) $\frac{\pi}{2} - 1$
(c) -1 (d) 1

12. If $f(x)$ is differentiable and $\int_0^{t^2} x f(x) dx = \frac{2}{5} t^5$, then $f(4/25)$ equals : (2004)

- (a) $2/5$ (b) $-5/2$
(c) 1 (d) $5/2$

13. The value of $\int_{-2}^0 [x^3 + 3x^2 + 3x + 3 + (x+1) \cos(x+1)] dx$ is: (2005)

- (a) 0 (b) 3
(c) 4 (d) 1

14. If $\int_{\sin x}^1 t^2 f(t) dt = 1 - \sin x \forall x \in (0, \pi/2)$ then $f\left(\frac{1}{\sqrt{3}}\right)$ is : (2005)

- (a) 3 (b) $\sqrt{3}$
(c) $1/3$ (d) none of these

15. Let f be a non-negative function defined on the interval $[0, 1]$. If $\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt$, $0 \leq x \leq 1$ and $f(0) = 0$, then (2009)

- (a) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$
(b) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$
(c) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$
(d) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$

16. The value of the integral

$$\int_{-\pi/2}^{\pi/2} \left(x^2 + \log \frac{\pi-x}{\pi+x} \right) \cos x dx \text{ is (2012)}$$

- (a) 0 (b) $\frac{\pi^2}{2} - 4$
(c) $\frac{\pi^2}{2} + 4$ (d) $\frac{\pi^2}{2}$

17. Let $f: \left[\frac{1}{2}, 1 \right] \rightarrow \mathbb{R}$ (the set of all real numbers) be a positive, non-constant and differentiable function such that $f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right) = 1$. Then, the value of $\int_{1/2}^1 f(x) dx$ lies in the interval (2013)

- (a) $(2e - 1, 2e)$ (b) $(e - 1, 2e - 1)$
(c) $\left(\frac{e-1}{2}, e-1 \right)$ (d) $\left(0, \frac{e-1}{2} \right)$

18. The following integral $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2 \operatorname{cosec} x)^{17} dx$ is equal to (2014)

- (a) $\int_0^{\log(1+\sqrt{2})} 2(e^u + e^{-u})^{16} du$
(b) $\int_0^{\log(1+\sqrt{2})} (e^u + e^{-u})^{17} du$
(c) $\int_0^{\log(1+\sqrt{2})} (e^u - e^{-u})^{17} du$
(d) $\int_0^{\log(1+\sqrt{2})} 2(e^u - e^{-u})^{16} du$

19. Let $f'(x) = \frac{192x^3}{2 + \sin^4 \pi x}$ for all $x \in \mathbb{R}$ with $f\left(\frac{1}{2}\right) = 0$. If $m \leq \int_{1/2}^1 f(x) dx \leq M$, then the possible values of m and M are (2015)

- (a) $m = 13, M = 24$ (b) $m = \frac{1}{4}, M = \frac{1}{2}$
(c) $m = -11, M = 0$ (d) $m = 1, M = 12$



20. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \cos x}{1+e^x} dx$ is equal to (2016)

- (a) $\frac{\pi^2}{4} - 2$ (b) $\frac{\pi^2}{4} + 2$
(c) $\pi^2 - e^{\frac{\pi}{2}}$ (d) $\pi^2 + e^{\frac{\pi}{2}}$

Objective Questions II [One or more than one correct option]

21. Let $S_n = \sum_{k=1}^n \frac{n}{n^2 + kn + k^2}$ and $T_n = \sum_{k=0}^{n-1} \frac{n}{n^2 + kn + k^2}$, for $n = 1, 2, 3, \dots$, then (2008)

- (a) $S_n < \frac{\pi}{3\sqrt{3}}$ (b) $S_n > \frac{\pi}{3\sqrt{3}}$
(c) $T_n < \frac{\pi}{3\sqrt{3}}$ (d) $T_n > \frac{\pi}{3\sqrt{3}}$

22. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1+\pi^x) \sin x} dx$, $n = 0, 1, 2, \dots$, then (2009)

- (a) $I_n = I_{n+2}$ (b) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$
(c) $\sum_{m=1}^{10} I_{2m} = 0$ (d) $I_n = I_{n+1}$

23. For $a \in \mathbb{R}$ (the set of all real numbers), $a \neq -1$,

$$\lim_{n \rightarrow \infty} \frac{(1^a + 2^a + \dots + n^a)}{(n+1)^{a-1} [(na+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60}$$

Then, a is equal to

- (a) 5 (b) 7
(c) $\frac{-15}{2}$ (d) $\frac{-17}{2}$

24. Let $f: (0, \infty) \rightarrow \mathbb{R}$ be given by

$$f(x) = \int_{\frac{1}{x}}^x e^{-\left(t+\frac{1}{t}\right)} \frac{dt}{t}. \text{ Then (2014)}$$

- (a) $f(x)$ is monotonically increasing on $[1, \infty)$
(b) $f(x)$ is monotonically decreasing on $(0, 1)$
(c) $f(x) + f\left(\frac{1}{x}\right) = 0$, for all $x \in (0, \infty)$
(d) $f(2^x)$ is an odd function of x on $\mathbb{R}$

25. The option(s) with the value of a and L that satisfy the following equation is(are)

$$\frac{\int_0^{4\pi} e^t (\sin^6 at + \cos^4 at) dt}{\int_0^{\pi} e^t (\sin^6 at + \cos^4 at) dt} = L ? \quad (2015)$$

- (a) $a = 2, L = \frac{e^{4\pi}-1}{e^{\pi}-1}$ (b) $a = 2, L = \frac{e^{4\pi}+1}{e^{\pi}+1}$
(c) $a = 4, L = \frac{e^{4\pi}-1}{e^{\pi}-1}$ (d) $a = 4, L = \frac{e^{4\pi}+1}{e^{\pi}+1}$

26. Let $f(x) = \lim_{n \rightarrow \infty} \left(\frac{n^n (x+n) \left(x+\frac{n}{2}\right) \dots \left(x+\frac{n}{n}\right)}{n! (x^2+n^2) \left(x^2+\frac{n^2}{4}\right) \dots \left(x^2+\frac{n^2}{n^2}\right)} \right)^{\frac{x}{n}}$,

for all $x > 0$. Then

(2016)

- (a) $f\left(\frac{1}{2}\right) \geq f(1)$ (b) $f\left(\frac{1}{3}\right) \leq f\left(\frac{2}{3}\right)$
(c) $f'(2) \leq 0$ (d) $\frac{f'(3)}{f(3)} \geq \frac{f'(2)}{f(2)}$

27. If $I = \sum_{k=1}^{98} \int_k^{k+1} \frac{k+1}{x(x+1)} dx$, then (2017)

- (a) $I > \frac{49}{50}$ (b) $I < \frac{49}{50}$
(c) $I < \log_e 99$ (d) $I > \log_e 99$



Numerical Value Type Questions

28. For $\lim_{n \rightarrow \infty} \frac{1 + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{n^{7/3} \left(\frac{1}{(na+1)^2} + \frac{1}{(na+2)^2} + \dots + \frac{1}{(na+n)^2} \right)} = 54$

$a \in \mathbb{R}$, $|a| > 1$, Then possible value(s) of a is/are –

(2019)

- (a) 8 (b) -9
(c) 7 (d) -6

29. Which of the following inequalities is/are TRUE ?

(2020)

(a) $\int_0^1 x \cos x \, dx \geq \frac{3}{8}$ (b) $\int_0^1 x \sin x \, dx \geq \frac{3}{10}$

(c) $\int_0^1 x^2 \cos x \, dx \geq \frac{1}{2}$ (d) $\int_0^1 x^2 \sin x \, dx \geq \frac{2}{9}$

30. Let $f: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$ be a continuous function such that

$f(0) = 1$ and $\int_0^{\frac{\pi}{3}} f(t) \, dt = 0$ Then which of the following

statements is (are) TRUE ?

(2021)

(a) The equation $f(x) - 3 \cos 3x = 0$ has a least one solution

in $\left(0, \frac{\pi}{3}\right)$

(b) The equation $f(x) - 3 \sin 3x = -\frac{6}{\pi}$ has at least one

solution in $\left(0, \frac{\pi}{3}\right)$

(c) $\lim_{x \rightarrow 0} \frac{x \int_0^x f(t) \, dt}{1 - e^{x^2}} = -1$

(d) $\lim_{x \rightarrow 0} \frac{\sin x \int_0^x f(t) \, dt}{x^2} = -1$

31. The value of $\frac{(5050) \int_0^1 (1-x^{50})^{100} \, dx}{\int_0^1 (1-x^{50})^{101} \, dx}$ is (2006)

32. The value of $\int_0^1 4x^3 \left\{ \frac{d^2}{dx^2} (1-x^2)^5 \right\} \, dx$ is (2014)

33. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \begin{cases} [x], & x \leq 2 \\ 0, & x > 2 \end{cases}$ where $[x]$ is the greatest integer less than or equal to x , if $I = \int_{-1}^2 \frac{xf(x^2)}{2+f(x+1)} \, dx$, then the value of $(4I-1)$ is

(2015)

34. If $\alpha = \int_0^1 \left(e^{9x+3 \tan^{-1} x} \right) \left(\frac{12+9x^2}{1+x^2} \right) \, dx$ where $\tan^{-1} x$ takes only principal values, then the value of

$\left(\log_e |1+\alpha| - \frac{3\pi}{4} \right)$ is (2015)

35. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous odd function, which vanishes exactly at one point and $f(1) = \frac{1}{2}$. Suppose that

$F(x) = \int_{-1}^x f(t) \, dt$ for all $x \in [-1, 2]$ and

$G(x) = \int_{-1}^x t |f(f(t))| \, dt$ for all $x \in [-1, 2]$. If $\lim_{x \rightarrow 1} \frac{F(x)}{G(x)} = \frac{1}{14}$,

then the value of $f\left(\frac{1}{2}\right)$ is (2015)

36. The total number of distinct $x \in [0, 1]$ for which

$\int_0^x \frac{t^2}{1+t^4} \, dt = 2x-1$ is (2016)

37. The value of the integral $\int_0^{\frac{1}{2}} \frac{1+\sqrt{3}}{\left((x+1)^2(1-x)^6\right)^{\frac{1}{4}}} \, dx$ is ____.

(2018)



38. $I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1 + e^{\sin x})(2 - \cos 2x)}$ then find $27I^2$ equals

(2019)

39. The value of the integral

$$\int_0^{\pi/2} \frac{3\sqrt{\cos \theta}}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5} d\theta \text{ equals}$$

(2019)

40. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that its derivative f' is continuous and $f(\pi) = -6$.

If $F: [0, \pi] \rightarrow \mathbb{R}$ is defined by $F(x) = \int_0^x f(t) dt$, and if

$$\int_0^{\pi} (f'(x) + F(x)) \cos x dx = 2 \text{ then the value of } f(0) \text{ is}$$

(2020)

41. For any real number x , let $[x]$ denote the largest integer

less than or equal to x . If $I = \int_0^{10} \left[\sqrt{\frac{10x}{x+1}} \right] dx$, then the

value of $9I$ is ____

(2021)

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

42. Match the conditions/expressions in Column I with statement in Column II. (2007)

Column-I

Column-II

(A) $\int_{-1}^1 \frac{dx}{1+x^2}$

(P) $\frac{1}{2} \log\left(\frac{2}{3}\right)$

(B) $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

(Q) $2 \log\left(\frac{2}{3}\right)$

(C) $\int_2^3 \frac{dx}{1-x^2}$

(R) $\frac{\pi}{3}$

(D) $\int_1^2 \frac{dx}{x\sqrt{x^2-1}}$

(S) $\frac{\pi}{2}$

The correct matching is:

(a) (A-S; B-S; C-P; D-R)

(b) (A-Q; B-S; C-P; D-R)

(c) (A-S; B-Q; C-P; D-R)

(d) (A-S; B-R; C-P; D-S)

43. List I

List II

P. The number of polynomials $f(x)$

1. 8

with non-negative integer

coefficients of degree ≤ 2 ,

satisfying $f(0) = 0$ and

$$\int_0^1 f(x) dx = 1, \text{ is}$$

Q. The number of points in the

2. 2

interval $[-\sqrt{13}, \sqrt{13}]$ at which

$f(x) = \sin(x^2) + \cos(x^2)$ attains

its maximum value is

R. $\int_{-2}^2 \frac{3x^2}{(1+e^x)} dx$ equals

3. 4

S. $\frac{\left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}{\left(\int_0^2 \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}$ equals 4. 0

(2014)

	P	Q	R	S
(a)	3	2	4	1
(b)	2	3	4	1
(c)	3	2	1	4
(d)	2	3	1	4



Using the following passage, solve Q.44 and 45

Passage – 1

Read the following passage and answer the questions.

Suppose we define definite integral using formula

$$\int_a^b f(x) dx = \left(\frac{b-a}{2} \right) \{f(a) + f(b)\},$$

for more accurate results for $c \in (a, b)$,

$$F(c) = \frac{c-a}{2} [f(a) - f(c)] + \frac{b-c}{2} [f(b) - f(c)]$$

$$\text{When } c = \frac{a+b}{2}$$

$$\int_a^b f(x) dx = \frac{b-a}{4} \{f(a) + f(b) + 2f(c)\} \quad (2006)$$

44. Good approximation of $\int_0^{\pi/2} \sin x dx$, is

- (a) $\pi/4$ (b) $\pi(\sqrt{2}+1)/4$
(c) $\pi(\sqrt{2}+1)/8$ (d) $\pi/8$

45. If $\lim_{t \rightarrow a} \frac{\int_a^t f(x) dx - \frac{(t-a)}{2} \{f(t) + f(a)\}}{(t-a)^3} = 0$,

then degree of polynomial function $f(x)$ at-most is

- (a) 0 (b) 1
(c) 3 (d) 2

Using the following passage, solve Q.46 and 47

Passage – 2

Given that for each $a \in (0, 1)$, $\lim_{h \rightarrow 0^+} \int_h^{1-h} t^{-a} (1-t)^{a-1} dt$

exists. Let this limit be $g(a)$. In addition, it is given that the function $g(a)$ is differentiable on $(0, 1)$. (2014)

46. The value of $g\left(\frac{1}{2}\right)$ is

- (a) π (b) 2π
(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$

47. The value of $g\left(\frac{1}{2}\right)$ is

- (a) $\frac{\pi}{2}$ (b) π
(c) $-\frac{\pi}{2}$ (d) 0

Using the following passage, solve Q.48 and 49

Passage – 3

Let $F: R \rightarrow R$ be a thrice differentiable function.

Suppose that $F(1) = 0$, $F(3) = -4$ and $F'(x) < 0$ for all

$x \in \left(\frac{1}{2}, 3\right)$. Let $f(x) = xF(x)$ for all $x \in R$. (2015)

48. The correct statement(s) is (are).

- (a) $f'(1) < 0$
(b) $f(2) < 0$
(c) $f'(x) \neq 0$ for any $x \in (1, 3)$
(d) $f'(x) = 0$ for some $x \in (1, 3)$

49. If $\int_1^3 x^2 F'(x) dx = -12$ and $\int_1^3 x^3 F''(x) dx = 40$, then

the correct expression(s) is (are)

- (a) $9f'(3) + f'(1) - 32 = 0$
(b) $\int_1^3 f(x) dx = 12$
(c) $9f'(3) - f'(1) + 32 = 0$
(d) $\int_1^3 f(x) dx = -12$



Using the following passage, solve Q.50 and 51

Passage – 4

Let $g_i : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \rightarrow \mathbb{R}$, $i = 1, 2$, and $f : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \rightarrow \mathbb{R}$ be functions such that $g_1(x) = 1$, $g_2(x) = |4x - \pi|$ and $f(x) = \sin^2 x$, for all $x \in \left[\frac{\pi}{8}, \frac{3\pi}{8}\right]$

Define $S_i = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} f(x) \cdot g_i(x) dx$, $i = 1, 2$ (2021)

50. The value of $\frac{16S_1}{\pi}$ is -----.

51. The value of $\frac{48S_2}{\pi^2}$ is -----.

Using the following passage, solve Q.52 and 53

Passage – 5

Let $\psi_1 : [0, \infty) \rightarrow \mathbb{R}$, $\psi_2 : [0, \infty) \rightarrow \mathbb{R}$, $f : [0, \infty) \rightarrow \mathbb{R}$, and $g : [0, \infty) \rightarrow \mathbb{R}$ be functions such that $f(0) = g(0) = 0$, $\psi_1(x) = e^{-x} + x$, $x \geq 0$, $\psi_2(x) = x^2 - 2x - 2e^{-x} + 2$,

$x \geq 0$, $f(x) = \int_{-x}^x (|t| - t^2) e^{-t^2} dt$, $x > 0$ and

$g(x) = \int_0^{x^2} \sqrt{t} e^{-t} dt$, $x > 0$ (2021)

52. Which of the following statements are TRUE?

(a) $f(\sqrt{\ln 3}) + g(\sqrt{\ln 3}) = \frac{1}{3}$

(b) For every $x > 1$, there exists an $\alpha \in (1, x)$ such that $\psi_1(x) = 1 + \alpha x$

(c) For every $x > 0$, there exists a $\beta \in (0, x)$ such that $\psi_2(x) = 2x(\psi_1(\beta) - 1)$

(d) f is an increasing function on the interval $\left[0, \frac{3}{2}\right]$

53. Which of the following statements is TRUE?

(a) $\psi_1(x) \leq 1$, for all $x > 0$

(b) $\psi_2(x) \leq 0$, for all $x > 0$

(c) $f(x) \geq 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5$, for all $x \in \left(0, \frac{1}{2}\right)$

(d) $g(x) \leq \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{1}{7}x^7$, for all $x \in \left(0, \frac{1}{2}\right)$

Text

54. For $x > 0$, let $f(x) = \int_1^x \frac{\ln t}{1+t} dt$. Find the function $f(x) + f(1/x)$ and show that $f(e) + f(1/e) = 1/2$. Here, $\ln t = \log_e t$ (2000)

55. Evaluate $\int_{-\pi/3}^{\pi/3} \frac{\pi + 4x^3}{2 - \cos\left(x + \frac{\pi}{3}\right)} dx$. (2004)

56. Evaluate

$$\int_0^{\pi} e^{|\cos x|} \left(2 \sin\left(\frac{1}{2} \cos x\right) + 3 \cos\left(\frac{1}{2} \cos x\right) \right) \sin x \, dx$$
 (2005)

Area Under Curves

Objective Questions I [Only one correct option]

57. The area enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ over the interval $\left[0, \frac{\pi}{2}\right]$ is (2013)

(a) $4(\sqrt{2} - 1)$ (b) $2\sqrt{2}(\sqrt{2} - 1)$

(c) $4(\sqrt{2} + 1)$ (d) $2\sqrt{2}(\sqrt{2} + 1)$

58. Area of the region

$$\{(x, y) \in \mathbb{R}^2 : y \geq \sqrt{x+3}, 5y \leq x+9 \leq 15\}$$
 is equal to (2016)

(a) $\frac{1}{6}$ (b) $\frac{4}{3}$

(c) $\frac{3}{2}$ (d) $\frac{5}{3}$



59. The area of the region $\{(x, y) : xy \leq 8, 1 \leq y \leq x^2\}$ is

(2019)

- (a) $16 \log_2 2 - 6$ (b) $8 \log_e 2 - \frac{7}{3}$
(c) $16 \log_e 2 - \frac{14}{3}$ (d) $8 \log_e 2 - \frac{14}{3}$

60. Let the functions $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = e^{x-1} - e^{-|x-1|} \text{ and } g(x) = \frac{1}{2}(e^{x-1} + e^{1-x}).$$

Then the area of the region in the first quadrant bounded by the curves $y = f(x)$, $y = g(x)$ and $x = 0$ is (2020)

- (a) $(2 - \sqrt{3}) + \frac{1}{2}(e - e^{-1})$ (b) $(2 + \sqrt{3}) + \frac{1}{2}(e - e^{-1})$
(c) $(2 - \sqrt{3}) + \frac{1}{2}(e + e^{-1})$ (d) $(2 + \sqrt{3}) + \frac{1}{2}(e + e^{-1})$

61. The area of the region

$$\{(x, y) : 0 \leq x \leq \frac{9}{4}, 0 \leq y \leq 1, x \geq 3y, x + y \geq 2\}$$

(2021)

- (a) $\frac{11}{32}$ (b) $\frac{35}{96}$
(c) $\frac{37}{96}$ (d) $\frac{13}{32}$

Objective Questions II [One or more than one correct option]

62. If S be the area of the region enclosed by $y = e^{-x^2}$, $y = 0$, $x = 0$ and $x = 1$. Then, (2012)

- (a) $S \geq \frac{1}{e}$ (b) $S \geq 1 - \frac{1}{e}$

- (c) $S \leq \frac{1}{4} \left(1 + \frac{1}{\sqrt{e}}\right)$ (d) $S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}} \left(1 - \frac{1}{\sqrt{2}}\right)$

63. If the line $x = \alpha$ divided the area of region

$$R = \{(x, y) \in \mathbb{R}^2 : x^3 \leq y \leq x, 0 \leq x \leq 1\}$$

into two equal parts, then

(2017)

- (a) $0 < \alpha \leq \frac{1}{2}$ (b) $2\alpha^4 - 4\alpha^2 + 1 = 0$
(c) $\alpha^4 + 4\alpha^2 - 1 = 0$ (d) $\frac{1}{2} < \alpha < 1$

Numerical Value Type Questions

64. Let $F(x) = \int_x^{x^2 + \frac{\pi}{6}} 2 \cos^2 t \, dt$ for all $x \in \mathbb{R}$ and

$f: \left[0, \frac{1}{2}\right] \rightarrow [0, \infty)$ be a continuous function. For

$a \in \left[0, \frac{1}{2}\right]$, if $F'(a) + 2$ is the area of the region bounded by $x = 0$, $y = 0$, $y = f(x)$ and $x = a$, then $f(0)$ is

(2015)

65. A farmer F_1 has a land in the shape of a triangle with vertices at $P(0, 0)$, $Q(1, 1)$ and $R(2, 0)$. From this land, a neighbouring farmer F_2 takes away the region which lies between the side PQ and a curve of the form $y = x^n$ ($n > 1$). If the area of the region taken away by the farmer F_2 is exactly 30% of the area of ΔPQR , then the value of n is (2018)

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DIFFERENTIAL EQUATIONS

DIFFERENTIAL EQUATIONS

1. INTRODUCTION

Differential equation constitute a very important part of mathematics as it has many applications in real life. Various laws of physics are often in the form of equations involving rate of change of one quantity with respect to another. As the mathematical equivalent of a rate is a derivative, differential equation arise very naturally in real life and methods for solving them acquire paramount importance.

1.1 Definition

An equation involving the dependent variable and independent variable and also the derivatives of the dependable variable is known as differential equation.

For example:

$$\begin{aligned} \text{(i)} \quad \frac{dy}{dx} &= \frac{x}{y^{1/3}(1+x^{1/3})} & \text{(ii)} \quad \frac{d^2y}{dx^2} &= -p^2y \\ \text{(iii)} \quad \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} &= 3 \frac{d^2y}{dx^2} & \text{(iv)} \quad x^2 \left(\frac{dy}{dx}\right)^2 &= y^2 + 1 \end{aligned}$$

Differential equations which involve only one independent variable are called ordinary differential equation.

2. ORDER AND DEGREE OF DIFFERENTIAL EQUATIONS

2.1 Order

The order of a differential equation is the order of the highest derivative involved in the differential equation

For example:

$$\text{(i)} \quad \left(\frac{dy}{dx}\right)^3 + \left(\frac{dy}{dx}\right)^2 + 4x = 0 \text{ is the differential equation of the first order because maximum derivative of } y \text{ with respect to } x \text{ is } \frac{dy}{dx}$$

$$\text{(ii)} \quad \frac{d^2y}{dx^2} = -p^2y \text{ is the differential equation of the second order because maximum derivative of } y \text{ w.r.t } x \text{ is } \frac{d^2y}{dx^2}$$

$$\text{(iii)} \quad \left(\frac{d^3y}{dx^3}\right)^2 - 3\left(\frac{dy}{dx}\right)^3 + 2 = 0 \text{ is the differential equation of the third order because maximum derivative of } y \text{ w.r.t } x \text{ is } \frac{d^3y}{dx^3}$$

2.2 Degree

The degree of a differential equation is the degree of the highest differential coefficient when the equation has been made rational and integral as far as the differential coefficients are concerned.

For example:

$$\text{(i)} \quad \frac{dy}{dx} = \frac{x}{y^{1/3}(1+x^{1/3})} \text{ is the differential equation of first degree, because power of the highest order derivative } \frac{dy}{dx} \text{ is } 1.$$

$$\text{(ii)} \quad \left(\frac{d^3y}{dx^3}\right)^2 - 3\left(\frac{dy}{dx}\right)^3 + 2 = 0 \text{ is the differential equation of second degree, because power of highest order derivative } \frac{d^3y}{dx^3} \text{ is } 2.$$

$$\text{(iii)} \quad \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{2/3} = 3 \frac{d^2y}{dx^2} \text{ is the differential equation of third degree, because power of highest}$$



order derivative $\frac{d^2y}{dx^2}$ is 3 (after cubing)

Illustration 1:

Find the order and degree of the following differential equations.

$$(i) \sqrt{\frac{d^2y}{dx^2}} = \sqrt[3]{\frac{dy}{dx} + 3}$$

$$(ii) \frac{d^2y}{dx^2} = \left\{ 1 + \left(\frac{dy}{dx} \right)^4 \right\}^{5/3}$$

$$(iii) y = px + \sqrt{a^2p^2 + b^2} \text{ where } p = \frac{dy}{dx}$$

Sol. (i) The given differential equation can be written as

$$\left(\frac{d^2y}{dx^2} \right)^3 = \left(\frac{dy}{dx} + 3 \right)^2$$

Hence order = 2, degree = 3

(ii) The given differential equation can be written as

$$\left(\frac{d^2y}{dx^2} \right)^3 = \left[1 + \left(\frac{dy}{dx} \right)^4 \right]^{5/3}$$

Hence order = 2, degree = 3

(iii) The given differential equation can be written as

$$\left(y - x \frac{dy}{dx} \right)^2 = a^2 \left(\frac{dy}{dx} \right)^2 + b^2$$

Hence order = 1, degree = 2

3. FORMATION OF ORDINARY DIFFERENTIAL EQUATION

An ordinary differential equation is formed in an attempt to eliminate certain arbitrary constants from a relation in the variables and constants. Consider an equation containing n arbitrary constants. Differentiating this equation n times we get n additional equations containing n arbitrary constants and derivatives. Eliminating n arbitrary constants from the above $(n + 1)$ equations, we obtain differential equation involving n^{th} derivative.

Thus if an equation contains n arbitrary constants, the resulting differential equation obtained by eliminating these constants will be a differential equation of n^{th} order. i.e., an equation of the form

$$\phi \left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n} \right) = 0$$

Illustration 2 :

Find the differential equation of the family of all circles which pass through the origin and whose centre lie on y -axis

Sol. Let the equation of the circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

If it passes through $(0, 0)$, then $c = 0$

$\therefore$ The equation of circle is $x^2 + y^2 + 2gx + 2fy = 0$

Since the centre of the circle lies on y -axis then $g = 0$

$\therefore$ The equation of the circle is

$$x^2 + y^2 + 2fy = 0 \quad \dots(i)$$

This represents family of circles.

Differentiating, we get

$$2x + 2y \frac{dy}{dx} + 2f \frac{dy}{dx} = 0 \quad \dots(ii)$$

From (i) and (ii), we get

or, $(x^2 - y^2) \frac{dy}{dx} - 2xy = 0$ Which is the required differential equation.

4. SOLUTION OF A DIFFERENTIAL EQUATION

The solution of the differential equation is a relation between the independent and dependent variable free from derivatives satisfying the given differential equation.

Thus the solution of $dy/dx = m$ could be obtained by simply integrating both sides i.e., $y = mx + c$, where c is arbitrary constant.

(a) **General solution (or complete primitive)**

The general solution of a differential equation is the relation between the variables (not involving the derivatives) which contain the same number of the arbitrary constants as the order of the differential equation.



Thus the general solution of the differential equation

$\frac{d^2 y}{dx^2} = 4y$ is $y = A \sin 2x + B \cos 2x$, where A and B are the constants.

(b) Particular solution or Integral

A solution which is obtained by giving particular values to the arbitrary constants in the general solution is called a particular solution.

Illustration 3:

Show that $v = \frac{A}{r} + B$ is the general solution of the second

order differential equation $\frac{d^2 v}{dr^2} + \frac{2}{r} \frac{dv}{dr} = 0$, where A and

B are arbitrary constant.

Sol. Given $v = \frac{A}{r} + B$

Differentiating twice $\frac{d^2 v}{dr^2} = \frac{2A}{r^3}$... (i)

From (i) $\frac{d^2 v}{dr^2} + \frac{2}{r} \frac{dv}{dr} = \frac{2A}{r^3} + \frac{2}{r} \left(-\frac{A}{r^2} \right) = \frac{2A}{r^3} - \frac{2A}{r^3} = 0$

Putting $A = 4$, $B = 5$ in $v = \frac{A}{r} + B$ we get a particular solution of the differential equation

$\frac{d^2 v}{dr^2} + \frac{2}{r} \frac{dv}{dr} = 0$ is $v = \frac{4}{r} + 5$.

Illustration 4:

Show that $y = ae^x + be^{2x} + ce^{-3x}$ is a solution of the

equation $\frac{d^3 y}{dx^3} - 7 \frac{dy}{dx} + 6y = 0$

Sol. We have

$y = ae^x + be^{2x} + ce^{-3x}$... (i)

Differentiating, we get

$y_1 = ae^x + 2be^{2x} - 3ce^{-3x}$... (ii)

$\therefore y_1 - y = be^{2x} - 4ce^{-3x}$... (iii)

Differentiating (ii), we get

$y_2 = ae^x + 4be^{2x} + 9ce^{-3x}$
 $\therefore y_2 - y_1 = 2be^{2x} + 12ce^{-3x}$... (iv)

Now, (iv) - 2 (iii) $\Rightarrow y_2 - y_1 - 2(y_1 - y) = 20ce^{-3x}$

or, $y_2 - 3y_1 + 2y = 20ce^{-3x}$... (v)

Differentiating, $y_2 = ae^x + 4be^{2x} + 9ce^{-3x}$, we get

$y_3 = ae^x + 8be^{2x} - 27ce^{-3x}$
 Now $y_3 - 3y_2 + 2y_1 = -60ce^{-3x}$... (vi)

And (vi) + 3(v) $\Rightarrow y_3 - 3y_2 + 2y_1 + 3(y_2 - 3y_1 + 2y) = 0$

or, $y_3 - 7y_1 + 6y = 0$

i.e., $\frac{d^3 y}{dx^3} - 7 \frac{dy}{dx} + 6y = 0$, which is the required differential equation

5. METHOD OF SOLVING AN EQUATION OF THE FIRST ORDER AND FIRST DEGREE

A differential equation of the first order and first degree can be written in the form

$$\frac{dy}{dx} = f(x, y)$$

or, $M dx + N dy = 0$, where M and N are functions of x and y

5.1 Variable Separation:

The general form of such an equation is

$$f(x)dx + f(y)dy = 0 \quad \dots (i)$$

Integrating, we get

$\int f(x)dx + \int f(y)dy = c$ which is the solution of (i)

5.2 Solution of differential equation of the type

$$\frac{dy}{dx} = f(ax + by + c) :$$

Consider the differential equation $\frac{dy}{dx} = f(ax + by + c)$

... (i)

Where $f(ax + by + c)$ is some function of $ax + by + c$.

Let $z = ax + by + c$



$$\therefore \frac{dz}{dx} = a + b \frac{dy}{dx}$$

$$\text{or, } \frac{dy}{dx} = \frac{\frac{dz}{dx} - a}{b}$$

$$\text{From (i) } \frac{\frac{dz}{dx} - a}{b} = f(z) \quad \text{or, } \frac{dz}{dx} = b f(z) + a$$

$$\text{or, } \frac{dz}{b f(z) + a} = dx \quad \dots(\text{ii})$$

In the differential equation (ii), the variables x and z are separated.

Integrating, we get

$$\int \frac{dx}{b f(z) + a} = \int dx + c$$

$$\text{or, } \int \frac{dx}{b f(z) + a} = x + c, \text{ where } z = ax + by + c$$

This represents the general solution of the differential equation (i)

Illustration 5:

$$\text{Solve } (x - y)^2 \frac{dy}{dx} = a^2$$

Sol. Putting $x - y = v$

$$\Rightarrow \frac{dy}{dx} = 1 - \frac{dv}{dx} \Rightarrow dx = \frac{v^2}{v^2 - a^2} dv, \text{ variable have been separated}$$

$$\text{Integrating, we get } \int dx = \int \frac{v^2}{v^2 - a^2} dv$$

$$\text{or, } 2y + k = a \log \frac{x - y - a}{x - y + a}$$

Illustration 6:

$$\text{Solve, } \frac{dy}{dx} = \sin(x + y) + \cos(x + y)$$

Sol. Let $z = x + y$

$$\therefore \frac{dz}{dx} = 1 + \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{dz}{dx} - 1$$

$$\frac{dz}{dx} - 1 = \sin z + \cos z$$

$$\text{or, } dx = \frac{dz}{\sin z + \cos z + 1}$$

Integrating, we get

$$\int dx = \int \frac{dz}{\sin z + \cos z + 1} = \int \frac{dt}{t + 1}, \text{ putting } t = \tan \frac{z}{2}$$

i.e., $x + c = \log |t + 1|$ This is the required general solution.

5.3 Solution of differential equation of the type

$$\frac{dy}{dx} = \frac{a_1 x + b_1 y + c_1}{a_2 x + b_2 y + c_2}, \text{ where } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\text{Here } \frac{dy}{dx} = \frac{a_1 x + b_1 y + c_1}{a_2 x + b_2 y + c_2} \text{ where } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \quad \dots(\text{i})$$

$$\text{Let } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \lambda \text{ (say)}$$

$$\therefore a_1 = \lambda a_2, b_1 = \lambda b_2$$

$$\begin{aligned} \text{From (i), } \frac{dy}{dx} &= \frac{\lambda a_2 x + \lambda b_2 y + c_1}{a_2 x + b_2 y + c_2} \\ &= \frac{\lambda (a_2 x + b_2 y) + c_1}{a_2 x + b_2 y + c_2} \quad \dots(\text{ii}) \end{aligned}$$

$$\text{Let } z = a_2 x + b_2 y$$

$$\therefore \frac{dz}{dx} = a_2 + b_2 \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{\frac{dz}{dx} - a_2}{b_2} \quad \dots(\text{iii})$$





From (ii) and (iii), we get $\frac{dz}{dx} - a_2 = \frac{\lambda z + c_1}{z + c_2}$

or, $\frac{dz}{dx} = \frac{b_2(\lambda z + c_1)}{z + c_2} + a_2 = \frac{\lambda b_2 z + b_2 c_1 + a_2 z + a_2 c_2}{z + c_2}$

or, $dx = \frac{z + c_2}{(\lambda b_2 + a_2)z + b_2 c_1 + a_2 c_2} dz$, where x and z are

separated

Integrating, we get

$$x + c = \int \frac{z + c_2}{(\lambda b_2 + a_2)z + b_2 c_1 + a_2 c_2} dz \text{ where } z = a_2 x + b_2 y$$

6. HOMOGENEOUS DIFFERENTIAL EQUATION

A function $f(x, y)$ is called homogeneous function of degree n if

$$f(\lambda x, \lambda y) = \lambda^n f(x, y)$$

For example:

- (a) $f(x, y) = x^2 y^2 - xy^3$ is a homogeneous function of degree four, since

$$\begin{aligned} f(\lambda x, \lambda y) &= (\lambda^2 x^2)(\lambda^2 y^2) - (\lambda x)(\lambda^3 y^3) \\ &= \lambda^4 (x^2 y^2 - xy^3) \\ &= \lambda^4 f(x, y) \end{aligned}$$

- (b) $f(x, y) = x^2 e^{x/y} + \frac{x^3}{y} + y^2 \log\left(\frac{y}{x}\right)$ is a homogeneous function of degree two, since

$$\begin{aligned} f(\lambda x, \lambda y) &= (\lambda^2 x^2) e^{\lambda x / \lambda y} + \frac{\lambda^3 x^3}{\lambda y} + (\lambda^2 y^2) \log\left(\frac{\lambda y}{\lambda x}\right) \\ &= \lambda^2 \left[x^2 e^{x/y} + \frac{x^3}{y} + y^2 \log\left(\frac{y}{x}\right) \right] \\ &= \lambda^2 f(x, y) \end{aligned}$$

A differential equation of the form $\frac{dy}{dx} = f(x, y)$, where

$f(x, y)$ is a homogeneous polynomial of degree zero is

called a homogeneous differential equation. Such equations are solved by substituting $v = \frac{y}{x}$ or $\frac{x}{y}$ and then separating the variables.

Illustration 7 :

Solve $\frac{dy}{dx} = \frac{y(2y - x)}{x(2y + x)}$... (i)

Sol. Since each of the functions $y(2y - x)$ and $x(2y + x)$ is a homogeneous function of degree 2, so the given equation is a homogeneous differential equation.

$\therefore$ Putting $y = vx$

Differentiating w.r.t x , we get $\frac{dy}{dx} = v + x \frac{dv}{dx}$

From (i),

$$v + x \frac{dv}{dx} = \frac{vx(2vx - x)}{x(2vx + x)} = \frac{v(2v - 1)}{2v + 1}$$

$$\Rightarrow 2dv + \frac{dv}{v} + 2 \frac{dx}{x} = 0$$

Integrating, we get

$$2v + \log v + \log x^2 = \log k \quad \text{or,} \quad xy = ke^{-2y/x}$$

6.1 Differential equation reducible to homogeneous forms:

Equation of the form $\frac{dy}{dx} = \frac{ax + by + c}{a'x + b'y + c'}$ where $\frac{a}{a'} \neq \frac{b}{b'}$

can be reduced to homogeneous form by changing the variables x, y to x', y' by equations $x = x' + h$ and $y = y' + k$ where h and k are constants to be chosen so as to make the given equation homogeneous, we have $dx = dx'$ and $dy = dy'$

$\therefore$ The given equation becomes

$$\frac{dy'}{dx'} = \frac{a(x' + h) + b(y' + k) + c}{a'(x' + h) + b'(y' + k) + c'}$$



$$= \frac{ax' + by' + (ah + bk + c)}{a'x' + b'y' + (a'h + b'k + c')}$$

Now, we choose h and k so that

$$ah + bk + c = 0$$

$$\text{and } a'h + b'k + c' = 0$$

From these equation we get the values of h and k in terms of the coefficients.

Then the given equation reduces to

$$\frac{dy'}{dx'} = \frac{ax' + by'}{a'x' + b'y'}$$

Which is the homogeneous form.

7. LINEAR DIFFERENTIAL EQUATION

A differential equation is said to be linear if the dependent variable y and its derivative occur in the first degree.

$$\text{An equation of the form } \frac{dy}{dx} + Py = Q \quad \dots(i)$$

where P and Q are functions of x only or constant is called a linear equation of the first order

Similarly $\frac{dx}{dy} + Px = Q$ is a linear differential equation

where P and Q are functions of y only. To get the general solution of the above equations we shall determine a function R of x called Integrating function (I.F). We shall multiply both sides of the given equation by R

$$\therefore \text{ where, } R = e^{\int P dx} = \text{I.F} \quad \dots(ii)$$

From (i) and (ii), we get

$$e^{\int P dx} \cdot \frac{dy}{dx} + P y e^{\int P dx} = Q e^{\int P dx} \quad \text{or,}$$

$$\frac{d}{dx} \left(y e^{\int P dx} \right) = Q e^{\int P dx}$$

Integrating, we get

$$y e^{\int P dx} = \int Q e^{\int P dx} dx + c \text{ is the required solution.}$$

NOTES :

We remember the solution of the above equation as

$$y(\text{I.F}) = \int Q(\text{I.F}) dx + c$$

Illustration 8 :

$$\text{Solve } 2x \frac{dy}{dx} = y + 6x^{5/2} - 2\sqrt{x}$$

Sol. The given equation can be written as

$$\frac{dy}{dx} + \left(\frac{-1}{2x} \right) y = 3x^{3/2} - \frac{1}{\sqrt{x}} \quad \dots(i)$$

This is the form of $\frac{dy}{dx} + Py = Q$

$$\text{Hence I.F} = e^{\int \frac{-1}{2x} dx} = e^{\frac{-1}{2} \log x} = \frac{1}{\sqrt{x}} \quad \dots(ii)$$

$$\text{From (i) and (ii), we get } y = \frac{3}{2} x^{5/2} - \sqrt{x} \log x + c\sqrt{x}$$

7.1 Differential equation reducible to the linear form:

Sometimes equations which are not linear can be reduced to the linear form by suitable transformation.

$$\text{Here, } f'(y) \frac{dy}{dx} + f(y) P(x) = Q(x) \quad \dots(i)$$

$$\text{Let, } f(y) = u \Rightarrow f'(y) dy = du$$

Then (i) reduces to

$$\frac{du}{dx} + u P(x) = Q(x) \text{ Which is of the linear differential equation form.}$$

Illustration 9:

$$\text{Solve } \sec^2 \theta d\theta + \tan \theta (1 - r \tan \theta) dr = 0$$

Sol. The given equation can be written as

$$\frac{d\theta}{dr} + \frac{\tan \theta}{\sec^2 \theta} = \frac{r \tan^2 \theta}{\sec^2 \theta}$$





$$\text{or, } \left(\frac{\sec^2 \theta}{\tan^2 \theta} \right) \frac{d\theta}{dr} + \frac{1}{\tan \theta} = r$$

$$\text{or, } \cos \sec^2 \theta \frac{d\theta}{dr} + \cot \theta = r \quad \dots(i)$$

$$\text{Let } \cot \theta = u$$

$$\Rightarrow -\cos \sec^2 \theta d\theta = du$$

Then (i) reduces to

$$-\frac{du}{dr} + u = r \quad \text{or, } \frac{du}{dr} - u = -r \quad \dots(ii)$$

Which is a linear differential equation.

$$\text{So, } I.F = e^{\int -1 dr} = e^{-r} \quad \dots(iii)$$

Form (ii) and (iii), we get

$$u e^{-r} = -\int r e^{-r} dr = r e^{-r} + \int e^{-r} dr, \text{ by parts}$$

$$= r e^{-r} - e^{-r} + c \text{ or } u = r - 1 + c e^r$$

$$\cot \theta = r - 1 + c e^r$$

7.2 Extended form of linear equations :

Bernoulli's equation:

An equation of the form $\frac{dy}{dx} + P y = Q y^n$, where P and

Q are function of x alone or constants and n is constant, other than 0 and 1, is called a Bernoulli's equation.

$$\text{Here } \frac{dy}{dx} + P y = Q y^n$$

Dividing by y^n , we get

$$\frac{1}{y^n} \frac{dy}{dx} + P \cdot \frac{1}{y^{n-1}} = Q$$

$$\text{Putting } \frac{1}{y^{n-1}} = v \text{ and differentiating w.r.t } x,$$

$$\text{we get } -\frac{(n-1)}{y^n} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{or, } \frac{1}{y^n} \frac{dy}{dx} = \frac{-1}{n-1} \frac{dv}{dx}$$

the equation becomes

$$\frac{dv}{dx} + (1-n) P v = Q(1-n)$$

Which is a linear equation with v as independent variable.

8. EXACT DIFFERENTIAL EQUATION

A differential equation is said to be exact if it can be derived from its solution (primitive) directly by differentiation, without any elimination, multiplication etc.

For example, the differential equation $x dy + y dx = 0$ is an exact differential equation as it is derived by direct differentiation for its solution, the function $xy = c$

Illustration 10 :

$$\text{Solve } (1 + xy) y dx + (1 - xy) x dy = 0$$

Sol. The given equation can be written as

$$y dx + xy^2 dx + x dy - x^2 y dy = 0$$

$$\text{or, } (y dx + x dy) + xy (y dx - x dy) = 0$$

$$\text{or, } d(xy) + xy (y dx - x dy) = 0$$

Dividing by $x^2 y^2$, we get

$$\frac{d(xy)}{x^2 y^2} + \frac{y dx - x dy}{xy} = 0 \quad \text{or,}$$

$$\frac{d(xy)}{x^2 y^2} + \frac{dx}{x} - \frac{dy}{y} = 0$$

Integrating, we get

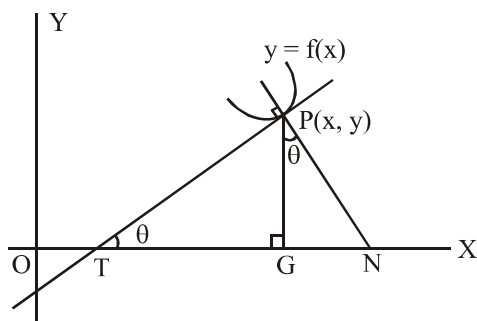
$$-\frac{1}{xy} + \log x - \log y = c$$

Which is the required solution.



9. APPLICATION OF DIFFERENTIAL EQUATIONS

In solving some geometrical problems, the following results are very helpful.



Let PT and PN be the tangent and the normal at $P(x, y)$.
Let the tangent at P makes an angle θ with the x-axis.

Then the slope of the tangent at $P = \tan \theta = \left(\frac{dy}{dx}\right)_P$ and

the slope of the normal at $P = -\frac{1}{\left(\frac{dy}{dx}\right)_P}$.

Equation of the tangent at $P(x, y)$ is

$$Y - y = \left(\frac{dy}{dx}\right)_P (X - x)$$

Equation of the normal at $P(x, y)$ is

$$Y - y = -\frac{1}{\left(\frac{dy}{dx}\right)_P} (X - x)$$

From ΔPGT $\sin \theta = \frac{PG}{PT} = \frac{y}{PT}$

$\therefore$ $PT = y \operatorname{cosec} \theta$ (length of the tangent)

$$= y \frac{\sqrt{1 + \tan^2 \theta}}{\tan \theta} = y \frac{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}{\frac{dy}{dx}}$$

And, $\tan \theta = \frac{PG}{TG} = \frac{y}{TG}$

$$\Rightarrow TG = y \cot \theta \text{ (length of the sub tangent)} = \frac{y}{\frac{dy}{dx}}$$

From ΔPGN $\cos \theta = \frac{PG}{PN} = \frac{y}{PN}$

$$\Rightarrow PN = y \sec \theta \text{ (length of the normal)}$$

$$= y \sqrt{1 + \tan^2 \theta} = y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

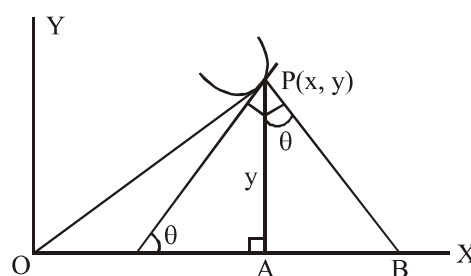
$$\tan \theta = \frac{GN}{y}$$

$$\Rightarrow GN = y \tan \theta = y \frac{dy}{dx} \text{ (length of the sub normal)}$$

Illustration 11 :

If the length of the sub-normal at any point P on the curve is directly proportional to OP^2 , where O is the origin, then form the differential equation of the family of curves and hence find the family of curves.

Sol. Here $AB = y \tan \theta = y \frac{dy}{dx}$



Also $OP^2 = x^2 + y^2$

Given, length of the subnormal = k. OP^2

or, $y \frac{dy}{dx} = k(x^2 + y^2)$

or, $2y \frac{dy}{dx} - 2ky^2 = 2kx^2 \dots (i)$

Let $y^2 = t \Rightarrow 2y \frac{dy}{dx} = \frac{dt}{dx} \dots (ii)$

From (i) and (ii), we get $\frac{dt}{dx} - 2kt = 2kx^2$

Which is a linear differential equation.





$$\therefore I.F = e^{\int -2k \, dx} = e^{-2kx}$$

$\therefore$ The solution is

$$t \cdot e^{-2kx} = \int 2kx^2 e^{-2kx} \, dx + c$$

$$= 2k \left[x^2 \frac{e^{-2kx}}{-2k} + \frac{2}{2k} \int x e^{-2kx} \, dx \right]$$

$$= 2k \left[x^2 \frac{e^{-2kx}}{-2k} + \frac{1}{k} \left\{ x \frac{e^{-2kx}}{-2k} + \frac{1}{2k} \int e^{-2kx} \, dx \right\} \right]$$

$$= -x^2 e^{-2kx} - \frac{x e^{-2kx}}{k} - \frac{1}{k} \frac{e^{-2kx}}{2k} + c \text{ or,}$$

$$y^2 = -x^2 - \frac{x}{k} - \frac{1}{2k^2} + c e^{2kx}$$

10. MISCELLANEOUS

10.1 A special type of second order differential equation

$$\frac{d^2 y}{dx^2} = f(x) \quad \dots (1)$$

Equation (1) may be re-written as

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = f(x) \Rightarrow d \left(\frac{dy}{dx} \right) = f(x) \, dx$$

$$\text{Integrating, } \frac{dy}{dx} = \int f(x) \, dx + c_1 \text{ i.e. } \frac{dy}{dx} = F(x) + c_1 \dots (ii)$$

$$\text{Where } F(x) = \int f(x) \, dx$$

$$\text{From (ii), } dy = \int F(x) \, dx + c_1 \, dx$$

$$\text{Integrating, } y = \int F(x) \, dx + c_1 x + c_2$$

$$\therefore y = H(x) + c_1 x + c_2$$

where $H(x) = \int F(x) \, dx$ c_1 and c_2 are arbitrary constants.

10.2 Particular solution type problems

To solve such a problem, we proceed according to the type of the problem (i.e. variable - separable, linear, exact, homogeneous etc.) and then we apply the given conditions to find the particular values of the arbitrary constants.



SOLVED EXAMPLES

Example – 1

Find the order and degree of the differential equation

$$\left[4 + \left(\frac{dy}{dx}\right)^2\right]^{2/3} = \frac{d^2y}{dx^2}$$

Sol. Here power on the differential coefficient is fractional, therefore change it into positive integer, so

$$\begin{aligned} \left[4 + \left(\frac{dy}{dx}\right)^2\right]^{2/3} &= \frac{d^2y}{dx^2} \\ \Rightarrow \left[4 + \left(\frac{dy}{dx}\right)^2\right]^2 &= \left[\frac{d^2y}{dx^2}\right]^3 \end{aligned}$$

Hence order is 2 and degree is 3.

Example – 2

Find the degree of the differential equation

$$\frac{d^2y}{dx^2} - \sqrt{\frac{dy}{dx}} - 3 = x$$

Sol. $\frac{d^2y}{dx^2} - \sqrt{\frac{dy}{dx}} - 3 = x$

$$\Rightarrow \frac{d^2y}{dx^2} - x = \sqrt{\frac{dy}{dx}} - 3$$

Squaring both sides, we get

$$\left(\frac{d^2y}{dx^2} - x\right)^2 = \left(\frac{dy}{dx} - 3\right)$$

$$\Rightarrow \left(\frac{d^2y}{dx^2}\right)^2 + x^2 - 2x \frac{d^2y}{dx^2} = \frac{dy}{dx} - 3.$$

Clearly, degree = 2.

Example – 3

The differential equation for the family of curves $x^2 + y^2 - 2ay = 0$, where a is an arbitrary constant, is

- (a) $2(x^2 - y^2)y' = xy$ (b) $2(x^2 + y^2)y' = xy$
(c) $(x^2 - y^2)y' = 2xy$ (d) $(x^2 + y^2)y' = 2xy$

Ans: (c)

Sol: Given family of curve is $x^2 + y^2 - 2ay = 0$... (i)

$$\Rightarrow 2a = \frac{x^2 + y^2}{y}$$

Also from (i), $2x + 2yy' - 2ay' = 0$

$$\Rightarrow 2x + 2yy' - \left(\frac{x^2 + y^2}{y}\right)y' = 0$$

$$\Rightarrow 2x + y' \left(\frac{2y^2 - x^2 - y^2}{y}\right) = 0 \Rightarrow y'(x^2 - y^2) = 2xy$$

Example – 4

If the differential equation representing the family of all circles touching x -axis at the origin is

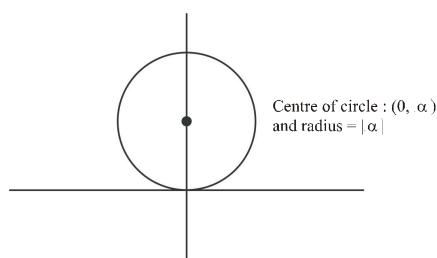
$(x^2 - y^2) \frac{dy}{dx} = g(x)y$, then $g(x)$ equals:

- (a) $\frac{1}{2}x$ (b) $2x^2$
(c) $2x$ (d) $\frac{1}{2}x^2$

Ans: (c)



Sol:



Center of circle = $(0, \alpha)$

and radius = $|\alpha|$

Equation of circle :

$$x^2 + (y - \alpha)^2 = \alpha^2$$

$$\Rightarrow x^2 + y^2 - 2\alpha y = 0 \quad \dots(1)$$

Differentiate w.r.t.x

$$2x + 2y \frac{dy}{dx} - 2\alpha \frac{dy}{dx} = 0$$

$$\Rightarrow \alpha = \frac{x + y \frac{dy}{dx}}{\frac{dy}{dx}}$$

put in (1)

$$x^2 + y^2 - 2y \left(\frac{x + y \frac{dy}{dx}}{\frac{dy}{dx}} \right) = 0$$

$$\Rightarrow (x^2 + y^2) \frac{dy}{dx} - 2xy - 2y^2 \frac{dy}{dx} = 0$$

$$\Rightarrow (x^2 - y^2) \frac{dy}{dx} = 2xy$$

$$\Rightarrow g(x) = 2x$$

Example – 5

Find the differential equation for all the straight lines which are at a unit distance from the origin

Sol. Since the equation of lines whose distance from origin is unit, is given by $x \cos \alpha + y \sin \alpha = 1 \dots (i)$

Differentiate w.r.t. x, we get $\cos \alpha + \frac{dy}{dx} \sin \alpha = 0 \dots (ii)$

On eliminating the ' α ' with the help of (i) and (ii) i.e.,

(i) $-(x \times (ii))$

$$\Rightarrow \sin \alpha \left(y - x \frac{dy}{dx} \right) = 1 \Rightarrow \left(y - x \frac{dy}{dx} \right) = \operatorname{cosec} \alpha \dots (iii)$$

$$\text{Also (ii)} \Rightarrow \frac{dy}{dx} = -\cot \alpha \Rightarrow \left(\frac{dy}{dx} \right)^2 = \cot^2 \alpha \dots (iv)$$

$$\text{Therefore by (iii) and (iv), } 1 + \left(\frac{dy}{dx} \right)^2 = \left(y - x \frac{dy}{dx} \right)^2.$$

Example – 6

Form the differential equation represented by family of curves $y = e^x (A \cos x + B \sin x)$.

Sol. Given $y = e^x A \cos x + e^x B \sin x$

$$\frac{dy}{dx} = Ae^x \cos x - Ae^x \sin x + Be^x \sin x + Be^x \cos x$$

$$\frac{dy}{dx} = (A + B) e^x \cos x + (B - A) e^x \sin x$$

$$\frac{d^2y}{dx^2} = (A + B) e^x \cos x - e^x \sin x (A + B) +$$

$$(B - A) e^x \sin x + (B - A) e^x \cos x$$

$$\frac{d^2y}{dx^2} = 2Be^x \cos x - 2Ae^x \sin x.$$

$$\text{Hence } \frac{d^2y}{dx^2} = 2 \frac{dy}{dx} - 2y.$$



Example – 7

Find the solution of the differential equation
 $(\sin x + \cos x) dy + (\cos x - \sin x) dx = 0$

Sol. $\frac{dy}{dx} = -\frac{\cos x - \sin x}{\sin x + \cos x} \Rightarrow dy = -\left(\frac{\cos x - \sin x}{\sin x + \cos x}\right) dx$

On integrating both sides, we get

$$\Rightarrow y = -\log(\sin x + \cos x) + \log c$$

$$\Rightarrow y = \log\left(\frac{c}{\sin x + \cos x}\right) \Rightarrow e^y(\sin x + \cos x) = c.$$

Example – 8

Find the solution of differential equation $x \frac{dy}{dx} + y = y^2$

Sol. $x \frac{dy}{dx} + y = y^2 \Rightarrow x \frac{dy}{dx} = y^2 - y$

$$\Rightarrow \frac{dy}{y^2 - y} = \frac{dx}{x} \Rightarrow \left[\frac{1}{y-1} - \frac{1}{y} \right] dy = \frac{dx}{x}$$

On integrating, we get

$$\log(y-1) - \log y = \log x + \log c$$

$$\Rightarrow \frac{y-1}{y} = xc \Rightarrow y = 1 + cxy.$$

Example – 9

Find the solution of the equation $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$

Sol. $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0 \Rightarrow \int \frac{dy}{\sqrt{1-y^2}} = -\int \frac{dx}{\sqrt{1-x^2}}$

$$\Rightarrow \sin^{-1} y = -\sin^{-1} x + \sin^{-1} c$$

$$\Rightarrow \sin^{-1} \left[x\sqrt{1-y^2} + y\sqrt{1-x^2} \right] = \sin^{-1} c$$

$$\Rightarrow x\sqrt{1-y^2} + y\sqrt{1-x^2} = c$$

Example – 10

Find the solution of the differential equation
 $(1+x^2)(1+y) dy + (1+x)(1+y^2) dx = 0$

Sol. Given equation $(1+x^2)(1+y) dy + (1+x)(1+y^2) dx = 0$

$$\Rightarrow \frac{(1+y)}{(1+y^2)} dy = -\frac{(1+x)}{(1+x^2)} dx$$

$$\Rightarrow \int \left[\frac{1}{1+y^2} + \frac{y}{1+y^2} \right] dy + \int \left[\frac{1}{1+x^2} + \frac{x}{1+x^2} \right] dx + c = 0$$

$$\Rightarrow \tan^{-1} y + \frac{1}{2} \log(1+y^2) + \tan^{-1} x + \frac{1}{2} \log(1+x^2) = c.$$

Example – 11

Find the solution of the differential equation

$$\sqrt{a+x} \frac{dy}{dx} + xy = 0 \text{ is}$$

Sol. Given $\frac{dy}{dx} + \frac{xy}{\sqrt{a+x}} = 0 \Rightarrow \frac{dy}{y} = \frac{-x dx}{\sqrt{a+x}}$

Integrating both sides, $\int \frac{dy}{y} = \int \frac{-x}{\sqrt{x+a}} dx$

$$\log y = -\int \frac{x+a-a}{\sqrt{x+a}} dx = -\int \sqrt{x+a} dx + \int \frac{a}{\sqrt{x+a}} dx$$

$$\Rightarrow \log y = -\frac{2}{3}(x+a)^{\frac{3}{2}} + 2a\sqrt{x+a} + \log A$$

$$= 2\sqrt{x+a} \left(\frac{-(x+a)}{3} + a \right) + \log A$$

$$= 2\sqrt{x+a} \left(\frac{2a-x}{3} \right) + \log A$$

$$\Rightarrow y = Ae^{\left[\frac{2}{3}\sqrt{x+a}(2a-x) \right]}.$$



Example – 12

Find the solution of $(x + y - 1)dx + (2x + 2y - 3)dy = 0$

Sol. Given equation is $\frac{dy}{dx} = -\left(\frac{x + y - 1}{2x + 2y - 3}\right)$

Put $x + y = t \Rightarrow \frac{dy}{dx} = \frac{dt}{dx} - 1$

$$\therefore \frac{dt}{dx} - 1 = \frac{1-t}{2t-3} \Rightarrow \frac{dt}{dx} = \frac{t-2}{2t-3}$$

$$\Rightarrow \frac{2t-3}{t-2} dt = dx.$$

Integrating both sides, we get

$$\int \frac{2t-4}{t-2} dt - \int \frac{3-4}{t-2} dt = \int 1 dx$$

$$\Rightarrow 2t + \log(t-2) = x + c$$

Substituting $t = x + y$, we get,

$$\Rightarrow 2(x + y) + \log(x + y - 2) = x + c$$

$$\Rightarrow 2y + x + \log(x + y - 2) = c.$$

Example – 13

Find the solution of the differential equation

$$(x + y)^2 \frac{dy}{dx} = a^2$$

Sol. Put $x + y = v \Rightarrow 1 + \frac{dy}{dx} = \frac{dv}{dx} \Rightarrow \frac{dy}{dx} = \frac{dv}{dx} - 1$

$$\therefore v^2 \left(\frac{dv}{dx} - 1 \right) = a^2$$

$$\Rightarrow \frac{dv}{dx} = \frac{a^2}{v^2} + 1 = \frac{a^2 + v^2}{v^2} \Rightarrow \frac{v^2}{a^2 + v^2} dv = dx$$

$$\Rightarrow \left(1 - \frac{a^2}{a^2 + v^2} \right) dv = dx \Rightarrow v - a \tan^{-1} \frac{v}{a} = x + c$$

Substituting $v = x + y$, we get,

$$\Rightarrow y = a \tan^{-1} \left(\frac{x + y}{a} \right) + c$$

Example – 14

Find the solution of the equation $\frac{dy}{dx} = \frac{x + y}{x - y}$

Sol. Given equation, $\frac{dy}{dx} = \frac{x + y}{x - y}$

It is a homogeneous equation as $f(\lambda x, \lambda y) = f(x, y)$ so putting $y = vx$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}, \text{ we get}$$

$$v + x \frac{dv}{dx} = \frac{x + vx}{x - vx} = \frac{1 + v}{1 - v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^2}{1 - v}$$

$$\Rightarrow \frac{1}{x} dx = \left(\frac{1}{1 + v^2} - \frac{v}{1 + v^2} \right) dv$$

$$\Rightarrow \log_e x = \tan^{-1} v - \frac{1}{2} \log(1 + v^2) - \log_e c$$

Substituting $v = \frac{y}{x}$, we get

$$\log_e x = \tan^{-1} \frac{y}{x} - \frac{1}{2} \log \left[1 + \left(\frac{y}{x} \right)^2 \right] - \log_e c$$

$$\Rightarrow c(x^2 + y^2)^{1/2} = e^{\tan^{-1}(y/x)}.$$

Example – 15

Find the solution of the differential equation

$$x + y \frac{dy}{dx} = 2y \text{ is}$$

Sol. Given $x + y \frac{dy}{dx} = 2y \Rightarrow \frac{x}{y} + \frac{dy}{dx} = 2$

$$\text{Put } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\therefore \frac{1}{v} + v + x \frac{dv}{dx} = 2 \Rightarrow v + x \frac{dv}{dx} = \frac{2v - 1}{v}$$



$$\Rightarrow \frac{v}{(v-1)^2} dv = -\frac{dx}{x} \Rightarrow \frac{v-1+1}{(v-1)^2} dv = -\frac{dx}{x}$$

$$\left[\frac{1}{(v-1)} + \frac{1}{(v-1)^2} \right] dv = -\frac{dx}{x}$$

Integrating both sides,

$$\Rightarrow \log(v-1) - \frac{1}{v-1} = -\log x + c$$

$$\text{Substituting } v = \frac{y}{x} \Rightarrow \log(y-x) = \frac{x}{y-x} + c$$

Example - 16

Find the general solution of the differential equation

$$(2x-y+1)dx + (2y-x+1)dy = 0$$

Sol. $(2x-y+1)dx + (2y-x+1)dy = 0$

$$\frac{dy}{dx} = \frac{2x-y+1}{x-2y-1}, \text{ put } x = X+h, y = Y+k$$

$$\frac{dY}{dX} = \frac{2X-Y+2h-k+1}{X-2Y+h-2k-1}$$

$$\text{Put } 2h-k+1=0 \text{ and } h-2k-1=0$$

$$\text{On solving } h=-1, k=-1;$$

$$\text{So, } X=x+1, Y=y+1$$

$$\therefore \frac{dY}{dX} = \frac{2X-Y}{X-2Y}$$

$$\text{Put } Y = vX; \Rightarrow \frac{dY}{dX} = v + X \frac{dv}{dX}$$

$$v + X \frac{dv}{dX} = \frac{2X-vX}{X-2vX} = \frac{2-v}{1-2v}$$

$$X \frac{dv}{dX} = \frac{2-2v+2v^2}{1-2v} = \frac{2(v^2-v+1)}{1-2v}$$

$$\therefore \frac{dX}{X} = \frac{(1-2v)}{2(v^2-v+1)} dv$$

$$\text{Put } v^2-v+1=t \Rightarrow (2v-1)dv=dt$$

$$\therefore \frac{dX}{X} = -\frac{dt}{2t}$$

Integrating both sides,

$$\log X = \log t^{-1/2} + \log c$$

$$\therefore X = t^{-1/2} c \Rightarrow X = (v^2-v+1)^{-1/2} \cdot c$$

$$X^2(v^2-v+1) = \text{constant}$$

$$\text{Plugging } X=x+1, v = \frac{Y}{X} = \frac{y+1}{x+1}, \text{ we get}$$

$$(x+1)^2 \left(\frac{(y+1)^2}{(x+1)^2} - \frac{(y+1)}{x+1} + 1 \right) = \text{constant}$$

$$(y+1)^2 - (y+1)(x+1) + (x+1)^2 = \text{constant}$$

$$y^2 + x^2 - xy + x + y = c.$$

Example - 17

$$\text{Solve the differential equation } (1+x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}.$$

Sol. Given differential equation is $(1+x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}$

Dividing throughout by $1+x^2$, we get

$$\frac{dy}{dx} + \frac{1}{1+x^2} y = \frac{1}{1+x^2} e^{\tan^{-1} x},$$

$$\text{which is of the form } \frac{dy}{dx} + Py = Q.$$

$$\text{Here, } P = \frac{1}{1+x^2} \text{ and } Q = \frac{1}{1+x^2} e^{\tan^{-1} x}.$$

So, the given differential equation is linear in y.

$$I.F. = e^{\int P dx} = e^{\int \frac{1}{1+x^2} dx} = e^{\tan^{-1} x}$$

$$\therefore \text{Solution is given by } ye^{\tan^{-1} x} = \int e^{\tan^{-1} x} \cdot \frac{e^{\tan^{-1} x}}{1+x^2} dx + C$$

$$\text{On R.H.S., substitute } \tan^{-1} x = t \Rightarrow \frac{1}{1+x^2} dx = dt$$

$$\text{So, we obtain } ye^{\tan^{-1} x} = \int e^t e^t dt + C$$

$$\Rightarrow ye^{\tan^{-1} x} = \int e^{2t} dt + C$$

$$\Rightarrow ye^{\tan^{-1} x} = \frac{e^{2t}}{2} + C$$



$$\Rightarrow ye^{\tan^{-1} x} = \frac{1}{2} e^{2 \tan^{-1} x} + C$$

$$\text{or } y = \frac{1}{2} e^{\tan^{-1} x} + C e^{-\tan^{-1} x}, \text{ } C \text{ being an arbitrary constant.}$$

Example – 18

$$\text{Solve } \frac{dy}{dx} - 2y \cos x = -2 \sin 2x.$$

$$\text{Sol. Given } \frac{dy}{dx} - 2 \cos x \cdot y = -2 \sin 2x. \quad \dots (1)$$

It is a linear in y with 'P' = $-2 \cos x$ and 'Q' = $-2 \sin 2x$.

$$\text{I.F.} = e^{\int -2 \cos x \, dx} = e^{-2 \sin x}, \quad \left(\because \text{I.F.} = e^{\int P \, dx} \right)$$

$\therefore$ Solution of (1) is

$$y \cdot e^{-2 \sin x} = \int -2 \sin 2x \cdot e^{-2 \sin x} \, dx + C$$

$$= -4 \int e^{-2 \sin x} \sin x \cos x \, dx + C,$$

$$\text{On R.H.S. put } -2 \sin x = t \Rightarrow \cos x \, dx = -\frac{1}{2} dt$$

$$\therefore ye^{-2 \sin x} = -4 \int e^t \left(-\frac{t}{2} \right) \left(-\frac{1}{2} dt \right) + C$$

$$= -\int t e^t \, dt + C = -\left(te^t - \int 1 \cdot e^t \, dt \right) + C$$

$$= -(te^t - e^t) + C = -(-2 \sin x - 1) e^{-2 \sin x} + C$$

$$\Rightarrow y = 2 \sin x + 1 + C \cdot e^{2 \sin x}.$$

Example – 19

$$\text{Solve the equation } \frac{dy}{dx} + \frac{1}{x} y = x^2 y^6, x > 0$$

$$\text{Sol. Given equation is } \frac{dy}{dx} + \frac{1}{x} y = x^2 y^6,$$

dividing throughout by y^6 ,

$$\text{we get } y^{-6} \frac{dy}{dx} + \frac{1}{x} y^{-5} = x^2 \quad \dots (1)$$

Substitute $y^{-5} = z$

$$\Rightarrow -5 y^{-6} \frac{dy}{dx} = \frac{dz}{dx} \Rightarrow y^{-6} \frac{dy}{dx} = -\frac{1}{5} \frac{dz}{dx}$$

The given equation becomes

$$-\frac{1}{5} \left(\frac{dz}{dx} \right) + \frac{1}{x} z = x^2 \text{ or } \frac{dz}{dx} - \frac{5}{x} z = -5x^2,$$

which is linear in z with $P = -\frac{5}{x}$ and $Q = -5x^2$.

$$\text{I.F.} = e^{\int P \, dx} = e^{\int -\frac{5}{x} \, dx} = e^{-5 \log x} = (e^{\log x})^{-5} = x^{-5}$$

Hence, the solution of the transformed equation is given by

$$zx^{-5} = \int x^{-5} (-5x^2) \, dx + C \quad \text{or}$$

$$y^{-5} x^{-5} = -5 \int x^{-3} \, dx + C \quad \left(\because z = y^{-5} \right)$$

$$\text{or } \frac{1}{(xy)^5} = -5 \left(\frac{x^{-2}}{-2} \right) + C$$

$$\text{or } \frac{1}{y^5} = \frac{5}{2} x^3 + Cx^5,$$

which is the required solution of the given differential equation.

Example – 20

Solve the differential equation

$$\sec^2 y \frac{dy}{dx} + 2x \tan y = x^3$$

$$\text{Sol. Given } \sec^2 y \cdot \frac{dy}{dx} + 2x \tan y = x^3 \quad \dots (1)$$

$$\text{Put } \tan y = z \Rightarrow \sec^2 y \frac{dy}{dx} = \frac{dz}{dx},$$

$$\therefore (1) \text{ becomes } \frac{dz}{dx} + 2x \cdot z = x^3 \quad \dots (2)$$

which is linear in z with 'P' = $2x$.

$$\text{I.F.} = e^{\int 2x \, dx} = e^{x^2},$$

$\therefore$ solution of (2) is given by

$$ze^{x^2} = \int x^3 \cdot e^{x^2} \, dx + C,$$

$$\left(\text{put } x^2 = t \Rightarrow x \, dx = \frac{1}{2} dt \right)$$



$$= \int t.e^t \cdot \frac{1}{2} dt + C = \frac{1}{2} \left(te^t - \int 1.e^t dt \right) + C$$

$$= \frac{1}{2} (te^t - e^t) + C = \frac{1}{2} (x^2 - 1) e^{x^2} + C$$

$$\therefore z = \frac{1}{2} (x^2 - 1) + C.e^{-x^2}$$

$$\therefore \text{Solution of (1) is } \tan y = \frac{1}{2} (x^2 - 1) + C.e^{-x^2}.$$

Example – 21

Find the solution of $(1 + xy)y dx + (1 - xy)x dy = 0$ is

Sol. $ydx + xdy + xy^2 dx - x^2 y dy = 0$

$$\frac{ydx + xdy}{x^2 y^2} + \frac{dx}{x} - \frac{dy}{y} = 0. \text{ On integrating, we get}$$

$$\Rightarrow \frac{d(xy)}{x^2 y^2} + \frac{dx}{x} - \frac{dy}{y} = 0$$

$$\Rightarrow -\frac{1}{xy} + \log x - \log y = k \Rightarrow \log \frac{x}{y} = \frac{1}{xy} + k.$$

Example – 22

Find the solution of $ye^{-x/y} dx - (xe^{-x/y} + y^3) dy = 0$

Sol. $ye^{-x/y} dx - (xe^{-x/y} + y^3) dy = 0$

$$e^{-x/y} (ydx - xdy) = y^3 dy \Rightarrow e^{-x/y} \frac{(ydx - xdy)}{y^2} = y dy$$

$$\Rightarrow e^{-x/y} d\left(\frac{x}{y}\right) = y dy.$$

Integrating both sides, we get

$$k - e^{-x/y} = \frac{y^2}{2}$$

$$\Rightarrow \frac{y^2}{2} + e^{-x/y} = k$$

Example – 23

Find the equation of a curve passing through the point $(0, -2)$ given that at any point (x, y) on the curve the product of the slope of its tangent and y coordinate of the point is equal to the x coordinate of the point.

Sol. We are given that $y \frac{dy}{dx} = x$... (1)

$\Rightarrow y dy = x dx$, integrating, we obtain

$$\frac{y^2}{2} = \frac{x^2}{2} + C \quad \dots (2)$$

Since $(0, -2)$ lies on this curve, therefore,

$$\frac{(-2)^2}{2} = 0 + C \Rightarrow C = \frac{4}{2} = 2$$

Substituting this value of C in (2), we obtain the required equation of the curve as

$$\frac{y^2}{2} = \frac{x^2}{2} + 2 \quad \text{or} \quad x^2 - y^2 + 4 = 0.$$

Example – 24

If the slope of the tangent at (x, y) to a curve passing through $\left(1, \frac{\pi}{4}\right)$ is given by $\frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$, then find the equation of the curve is

Sol. We have $\frac{dy}{dx} = \frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$

Putting $y = vx$ so that $\frac{dy}{dx} = v + x \frac{dv}{dx}$, we get

$$v + x \frac{dv}{dx} = v - \cos^2 v$$

$$\Rightarrow \frac{dv}{\cos^2 v} = -\frac{dx}{x}$$

On integrating, we get

$$\tan v = -\log x + \log c$$

$$\Rightarrow \tan\left(\frac{y}{x}\right) = -\log x + \log c$$



This passes through $\left(1, \frac{\pi}{4}\right)$, therefore $1 = \log c$

or $\log c = \log e$

$$\Rightarrow \tan\left(\frac{y}{x}\right) = -\log x + \log e \Rightarrow y = x \tan^{-1}\left[\log\left(\frac{e}{x}\right)\right].$$

Example – 25

The population $p(t)$ at time t of a certain mouse species

satisfies the differential equation $\frac{dp(t)}{dt} = 0.5 p(t) - 450$. If

$p(0) = 850$, then the time at which the population becomes zero is

(a) $2 \log 18$ (b) $\log 9$

(c) $\frac{1}{2} \log 18$ (d) $\log 18$

Ans: (a)

Sol: $\frac{dp(t)}{dt} = \frac{1}{2} p(t) - 450$

$$\frac{dp(t)}{dt} = \frac{P(t) - 900}{2}$$

$$\therefore 2 \int \frac{d(p(t))}{p(t) - 900} = \int dt$$

$$\therefore 2 \ln |p(t) - 900| = t + c$$

putting $t = 0$

$$\Rightarrow 2 \ln 50 = c$$

$$\therefore 2 \ln |p(t) - 900| = t + 2 \ln 50$$

$$p(t) = 0$$

$$\Rightarrow 2 \ln 900 = t + 2 \ln 50$$

$$t = 2(\ln 900 - \ln 50)$$

$$\Rightarrow t = 2 \ln \left(\frac{900}{50}\right)$$

$$\Rightarrow t \Rightarrow 2 \ln 18$$



EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Order and degree of Differential Equation

1. The order and degree of the following differential equation

$$\left(\frac{d^2s}{dt^2}\right)^2 + 3\left(\frac{ds}{dt}\right)^3 + 4 = 0 \text{ are}$$

- (a) 2, 2 (b) 2, 3
(c) 3, 2 (d) none of these

2. The degree of the differential equation

$$\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 = x \sin\left(\frac{d^2y}{dx^2}\right) \text{ is}$$

- (a) 1 (b) 2
(c) 3 (d) none of these

3. Which of the following differential equations has the same order and degree-

(a) $\frac{d^4y}{dx^4} + 8\left(\frac{dy}{dx}\right)^6 + 5y = e^x$

(b) $5\left(\frac{d^3y}{dx^3}\right)^4 + 8\left(1 + \frac{dy}{dx}\right)^2 + 5y = x^8$

(c) $\left[1 + \left(\frac{dy}{dx}\right)^3\right]^{2/3} = 4\frac{d^3y}{dx^3}$

(d) $y = x^2 \frac{dy}{dx} + \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

4. The degree of the differential equation corresponding to the family of curves $y = a(x + a)^2$, where a is an arbitrary constant, is

- (a) 1 (b) 2
(c) 3 (d) none of these

The formation of differential equations

5. The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$, where c_1 and c_2 are arbitrary constants is

- (a) $y' = y^2$ (b) $y'' = y' y$
(c) $yy'' = y'$ (d) $yy'' = (y')^2$

6. The differential equation of all circles passing through the origin and having their centres on the x-axis is

(a) $x^2 = y^2 + xy \frac{dy}{dx}$ (b) $x^2 = y^2 + 3xy \frac{dy}{dx}$

(c) $y^2 = x^2 + 2xy \frac{dy}{dx}$ (d) $y^2 = x^2 - 2xy \frac{dy}{dx}$

7. The differential equation of all non-vertical lines in a plane is

(a) $\frac{d^2y}{dx^2} = 0$ (b) $\frac{d^2x}{dy^2} = 0$

(c) $\frac{dy}{dx} = 0$ (d) $\frac{dx}{dy} = 0$

8. Form the differential equation representing the family of curves $y = A \cos 2x + B \sin 2x$, where A and B are constants.

(a) $\frac{d^2y}{dx^2} + 4y = 0$ (b) $\frac{d^2y}{dx^2} - 4y = 0$

(c) $y \cdot \frac{dy}{dx} + 4x = 0$ (d) $\frac{dy}{dx} + 4x = 0$

9. The differential equation from $y = ke^{\sin^{-1} x} + 3$ is

(a) $\sqrt{1-x^2} \frac{dy}{dx} = y + 3$ (b) $\sqrt{1-x^2} \frac{dy}{dx} = y - 3$

(c) $\sqrt{1+x^2} \frac{dy}{dx} = y + 3$ (d) none of these



Variable separable method

10. The general solution of the differential equation

$$(1 + y^2) dx + (1 + x^2) dy = 0 \text{ is}$$

- (a) $x - y = C(1 - xy)$ (b) $x - y = C(1 + xy)$
(c) $x + y = C(1 - xy)$ (d) $x + y = C(1 + xy)$

11. The solution of the differential equation $\frac{dy}{dx} + \frac{1+x^2}{x} = 0$ is

- (a) $y = -\frac{1}{2} \tan^{-1} x + c$ (b) $y + \log x + \frac{x^2}{2} + c = 0$
(c) $y = \frac{1}{2} \tan^{-1} x + c$ (d) $y - \log x - \frac{x^2}{2} = c$

12. The general solution of the differential equation

$$(e^x + e^{-x}) \frac{dy}{dx} = (e^x - e^{-x}) \text{ is}$$

- (a) $y = \log |e^x + e^{-x}| + c$ (b) $y = \log |e^x - e^{-x}| - c$
(c) $y = -\log |e^x - e^{-x}| + c$ (d) none of these

13. The solution of $\frac{dy}{dx} = 1 + x + y + xy$ is

- (a) $\log(1 - y) = x + \frac{x^3}{2} + C$
(b) $\log(1 + y) = x + \frac{x^2}{2} + C$
(c) $\log(1 + y) = x - \frac{x^2}{2} - C$
(d) none of these

14. The solution of the equation $\frac{dy}{dx} = \frac{(1+x)y}{(y-1)x}$ is :

- (a) $\log xy + x + y = c$ (d) $\log \left(\frac{x}{y} \right) + x - y = c$
(c) $\log xy + x - y = c$ (d) none of these

15. The solution of the differential equation

$$\cos y \log(\sec x + \tan x) dx = \cos x \log(\sec y + \tan y) dy \text{ is}$$

- (a) $\sec^2 x + \sec^2 y = c$ (b) $\sec x + \sec y = c$
(c) $\sec x - \sec y = c$ (d) None of these

16. The solution of $\frac{dy}{dx} = \frac{e^x (\sin^2 x + \sin 2x)}{y(2 \log y + 1)}$ is-

- (a) $y^2 (\log y) - e^x \sin^2 x + c = 0$
(b) $y^2 (\log y) - e^x \cos^2 x + c = 0$
(c) $y^2 (\log y) + e^x \cos^2 x + c = 0$
(d) None of these

17. The solution of $(\operatorname{cosec} x \log y) dy + (x^2 y) dx = 0$ is -

- (a) $\frac{\log y}{2} + (2 - x^2) \cos x + 2 \sin x = c$
(b) $\left(\frac{\log y}{2} \right)^2 + (2 - x^2) \cos x + 2x \sin x = c$
(c) $\frac{(\log y)^2}{2} + (2 - x^2) \cos x + 2x \sin x = c$
(d) None of these

18. The solution of the differential equation

$$xy \frac{dy}{dx} = \frac{(1+y^2)(1+x+x^2)}{(1+x^2)} \text{ is -}$$

- (a) $\frac{1}{2} \log(1 + y^2) = \log x - \tan^{-1} x + c$
(b) $\frac{1}{2} \log(1 + y^2) = \log x + \tan^{-1} x + c$
(c) $\log(1 + y^2) = \log x - \tan^{-1} x + c$
(d) $\log(1 + y^2) = \log x + \tan^{-1} x + c$

19. The solution of

$$(x \sqrt{1+y^2}) dx + (y \sqrt{1+x^2}) dy = 0 \text{ is -}$$

- (a) $\sqrt{1+x^2} + \sqrt{1+y^2} = c$
(b) $\sqrt{1+x^2} - \sqrt{1+y^2} = c$
(c) $(1+x^2)^{3/2} + (1+y^2)^{3/2} = c$
(d) None of these

20. The Solution of the equation

$$(1 - x^2) dy + xy dx = xy^2 dx \text{ is-}$$

- (a) $(y - 1)^2 (1 - x^2) = 0$
(b) $(y - 1)^2 (1 - x^2) = c^2 y^2$
(c) $(y - 1)^2 (1 + x^2) = c^2 y^2$
(d) None of these



21. The solution of differential equation $\frac{dy}{dx} = \sec(x+y)$ is

- (a) $y - \tan \frac{x+y}{2} = c$ (b) $y + \tan \frac{x+y}{2} = c$
(c) $y + 2 \tan \frac{x+y}{2} = c$ (d) none of these

Homogeneous differential equation

22. The solution of the differential equation

$$\frac{dy}{dx} = \frac{x+y}{x}$$

satisfying the condition $y(1) = 1$ is

- (a) $y = x \log x + x$ (b) $y = \log x + x$
(c) $y = x \log x + x^2$ (d) $y = xe^{(x-1)}$

23. If $x \frac{dy}{dx} = y (\log y - \log x + 1)$, then the solution of the equation is

- (a) $\log \left(\frac{x}{y} \right) = cy$ (b) $\log \left(\frac{y}{x} \right) = cx$
(c) $x \log \left(\frac{y}{x} \right) = cy$ (d) $y \log \left(\frac{x}{y} \right) = cx$

24. The solution of differential equation

$$x^2 y dx - (x^3 + y^3) dy = 0$$

- (a) $\frac{1}{3} \frac{x^3}{y^3} + \log y = C$ (b) $\frac{1}{3} \frac{x^3}{y^3} - \log y = C$
(c) $\frac{x^3}{y^3} + \log y = -C$ (d) none of these

25. The solution of Differential equation $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$ is

- (a) $\frac{(x-1) \left(1 + \frac{y-1}{x-1} \right)^{1/2}}{\left(1 - \frac{y-1}{x-1} \right)^{3/2}} = C$ (b) $\frac{\left(1 + \frac{y-1}{x-1} \right)^{1/2}}{\left(1 - \frac{y-1}{x-1} \right)^{3/2}} = C$
(c) $\frac{\left(1 + \frac{y-1}{x-1} \right)^{1/2}}{(x-1) \left(1 - \frac{y-1}{x-1} \right)^{3/2}} = C$ (d) none of these

Linear differential equation

26. The solution of differential equation

$$(x^2-1) \frac{dy}{dx} + 2xy = \frac{1}{x^2-1}$$

- (a) $y(x^2-1) = \frac{1}{2} \log \left| \frac{x-1}{x+1} \right| + C$
(b) $y(x^2+1) = \frac{1}{2} \log \left| \frac{x-1}{x+1} \right| - C$
(c) $y(x^2-1) = \frac{5}{2} \log \left| \frac{x-1}{x+1} \right| + C$
(d) none of these

27. The general solution of the differential equation,

$$\sin 2x \left(\frac{dy}{dx} - \sqrt{\tan x} \right) - y = 0, \text{ is:}$$

- (a) $y\sqrt{\tan x} = x + c$ (b) $y\sqrt{\cot x} = \tan x + c$
(c) $y\sqrt{\tan x} = \cot x + c$ (d) $y\sqrt{\cot x} = x + c$

28. If $\phi(x)$ is a differentiable function, then the solution of the differential equation

$$dy + \{y\phi'(x) - \phi(x)\phi'(x)\} dx = 0$$

- (a) $y = \{\phi(x) - 1\} + ce^{-\phi(x)}$ (b) $y\phi(x) = \{\phi(x)\}^2 + c$
(c) $ye^{\phi(x)} = \phi(x)e^{\phi(x)} + c$ (d) none of these

29. Consider the differential equation $y^2 dx + \left(x - \frac{1}{y} \right) dy = 0$.

If $y(1) = 1$, then x is given by

- (a) $1 - \frac{1}{y} + \frac{e^{1/y}}{e}$ (b) $4 - \frac{2}{y} - \frac{e^{1/y}}{e}$
(c) $3 - \frac{1}{y} + \frac{e^{1/y}}{e}$ (d) $1 + \frac{1}{y} - \frac{e^{1/y}}{e}$

30. The solution of the differential equation

$$(1+y^2) + (x - e^{\tan^{-1} y}) \frac{dy}{dx} = 0, \text{ is}$$

- (a) $(x-2) = ce^{-2 \tan^{-1} y}$ (b) $2xe^{\tan^{-1} y} = e^{2 \tan^{-1} y} + c$
(c) $xe^{\tan^{-1} y} = \tan^{-1} y + c$ (d) $xe^{2 \tan^{-1} y} = e^{\tan^{-1} y} + c$



31. The solution of differential equation

$$(2x - 10y^3) \frac{dy}{dx} + y = 0 \text{ is :}$$

- (a) $xy^2 = y^5 + c$ (b) $xy^2 + 2y^5 = c$
(c) $xy^2 = 2y^5 + c$ (d) none of these

32. The integrating factor of the differentiable equation

$$(xy - 1) \frac{dy}{dx} + y^2 = 0 \text{ is}$$

- (a) y (b) $\frac{1}{y}$
(c) $\frac{1}{xy}$ (d) xy

33. If $\sin x$ is an integrating factor of the differential equation

$$\frac{dy}{dx} + Py = Q, \text{ then } P \text{ can be}$$

- (a) $\log \sin x$ (b) $\cot x$
(c) $\sin x$ (d) $\log \cos x$

34. The solution of differential equation

$$\frac{dy}{dx} - y \tan x = -y^2 \sec x \text{ is :}$$

- (a) $y^{-1} \sec x = \cot x + c$ (b) $y^{-1} \cos x = \tan x + c$
(c) $y^{-1} \sec x = \tan x + c$ (d) none of these

35. The solution of the equation $x^3 \frac{dy}{dx} + 4x^2 \tan y = e^x \sec y$ satisfying $y(1) = 0$, is

- (a) $\tan y = (x - 2) e^x \log 3$ (b) $\sin y = e^x (x - 1) x^{-4}$
(c) $\tan y = (x - 1) e^x x^{-3}$ (d) $\sin y = e^x (x - 1) x^{-3}$

36. The general solution of the differential equation

$$[2\sqrt{xy} - x] dy + y dx = 0 \text{ is}$$

- (a) $\log x + \sqrt{\frac{y}{x}} = c$ (b) $\log y - \sqrt{\frac{x}{y}} = c$
(c) $\log y + \sqrt{\frac{x}{y}} = c$ (d) none of these

37. The general solution of the differential equation

$$y(x^2y + e^x) dx - e^x dy = 0 \text{ is}$$

- (a) $x^3y - 3e^x = cy$ (b) $x^3y + 3e^x = 3cy$
(c) $y^3x - 3e^y = cx$ (d) $y^3x + 3e^y = cx$

Exact form

38. Solution of the differential equation

$$\cos x dy = y (\sin x - y) dx, 0 < x < \frac{\pi}{2}, \text{ is}$$

- (a) $\sec x = (\tan x + c) y$ (b) $y \sec x = \tan x + c$
(c) $y \tan x = \sec x + c$ (d) $\tan x = (\sec x + c) y$

39. The solution of the differential equation $y dx + (x + x^2y) dy = 0$ is

- (a) $-\frac{1}{xy} = c$ (b) $-\frac{1}{xy} + \log y = c$
(c) $\frac{1}{xy} + \log y = c$ (d) $\log y = cx$

40. The solution of the differential equation

$$\frac{x}{x^2 + y^2} dy = \left(\frac{y}{x^2 + y^2} - 1 \right) dx \text{ is}$$

- (a) $y = x \cot(c - x)$ (b) $\cos^{-1} \frac{y}{x} = (-x + c)$
(c) $y = x \tan(c - x)$ (d) $\frac{y^2}{x^2} = x \tan(c - x)$

Application of Differential Equation

41. Equation of curve through point (1, 0) which satisfies the differential equation

$$(1 + y^2) dx - xy dy = 0, \text{ is}$$

- (a) $x^2 + y^2 = 1$ (b) $x^2 - y^2 = 1$
(c) $2x^2 + y^2 = 2$ (d) none of these

42. The equation of a curve passing through $\left(2, \frac{7}{2}\right)$ and

having gradient $1 - \frac{1}{x^2}$ at (x, y) is

- (a) $y = x^2 + x + 1$ (b) $xy = x^2 + x + 1$
(c) $xy = x + 1$ (d) none of these



43. The curve passing through the point (0, 1) and satisfying the equation $\sin\left(\frac{dy}{dx}\right) = a$ is :
- (a) $\cos\left(\frac{y+1}{x}\right) = a$ (b) $\cos\left(\frac{x}{y+1}\right) = a$
 (c) $\sin\left(\frac{y-1}{x}\right) = a$ (d) $\sin\left(\frac{x}{y-1}\right) = a$
44. A curve passes through the point (0, 1) and the gradient at (x, y) on it is $y(xy - 1)$. The equation of the curve is
 (a) $y(x - 1) = 1$ (b) $y(x + 1) = 1$
 (c) $x(y + 1) = 1$ (d) $x(y - 1) = 1$
45. A tangent to the curve $y = f(x)$ cuts the line $y = x$ at a point which is at a distance of 1 unit from y-axis. The equation of the curve is
 (a) $\frac{x-1}{y-1} = c$ (b) $\frac{x}{y} = c$
 (c) $xy = c$ (d) none of these
46. The equation of the curve for which the tangent at P (x, y) cuts the y - axis at (0, y^3) is
 (a) $xy^2 = x + y$ (b) $x^2(y^2 - 1) = cy^2$
 (c) $y(x^2 - 1) = cx^2$ (d) $yx + x^2 = c$
47. The slope of a curve at any point is the reciprocal of twice the ordinate at the point and it passes through the point (4, 3). The equation of the curve is-
 (a) $x^2 = y + 5$ (b) $y^2 = x - 5$
 (c) $y^2 = x + 5$ (d) $x^2 = y - 5$
48. The curve in which the portion of the tangent included between the coordinate axes is bisected at the point of contact, is
 (a) a parabola (b) an ellipse
 (c) a circle (d) a hyperbola
49. The equation of the curve in which subnormal varies as the square of the ordinate is (λ is constant of proportionality)
 (a) $y = ce^{2\lambda x}$ (b) $y = ce^{\lambda x}$
 (c) $\frac{y^2}{2} + \lambda x = c$ (d) $y^2 + \lambda x^2 = c$
50. The slope of the tangent at (x, y) to a curve passing through (2, 1) is $\frac{x^2 + y^2}{2xy}$, then the equation of the curve is
 (a) $2(x^2 - y^2) = 3x$ (b) $2(x^2 - y^2) = 6y$
 (c) $x(x^2 - y^2) = 6$ (d) $x(x^2 + y^2) = 10$
51. Let the population of rabbits surviving at a time t be governed by the differential equation $\frac{dp(t)}{dt} = \frac{1}{2}p(t) - 200$.
 If $p(0) = 100$, then $p(t)$ equals :
 (a) $400 - 300e^{-t/2}$ (b) $400 - 300e^{t/2}$
 (c) $300 - 200e^{-t/2}$ (d) $600 - 500e^{t/2}$
52. Water is dropped at the rate of 2 m³/sec into a cone of semi vertical angle of 45°. The rate at which periphery of water surface changes when height of water in the cone is 2 meters is
 (a) 2 m/sec (b) 1 m/sec
 (c) 3 m/sec (d) 4 m/sec

Numerical Value Type Questions

53. The degree of the differential equation satisfying the relation $\sqrt{1+x^2} + \sqrt{1+y^2} = \lambda(x\sqrt{1+y^2} - y\sqrt{1+x^2})$ is
54. The sum of order and degree of the differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where $c > 0$, is a parameter, is
55. The sum of the degree and order of the differential equation of the family of all parabolas whose axis is x-axis, is.
56. If the solution of the differential equation $\frac{dy}{dx} = \frac{ax+3}{2y+f}$ represents a circle, then the value of '-a' is
57. If $\frac{dy}{dx} + y \tan x = \sin 2x$ and $y(0) = 1$ and $y(\pi)$ is equal to k. Then the value of -k is
58. At present, a firm is manufacturing 2000 items. It is estimated that the rate of change of production P with respect to additional number of workers x is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is 100 k. Then the value of k is
59. In a bank, principal increases at the rate of 5% per year. If Rs. 1000 doubles itself in $k \log_e 2$ years. Then the value of k is
60. A function $y = f(x)$ has a second order derivative $f''(x) = 6(x - 1)$. If its graph passes through the point (2, 1) and at that point the tangent to the graph is $y = 3x - 5$, then the function is $(x + a)^b$. The value of a + b is



EXERCISE - 2 : PREVIOUS YEAR JEE MAIN QUESTIONS

- Let $y(x)$ be the solution of the differential equation $(x \log x) \frac{dy}{dx} + y = 2x \log x$, ($x \geq 1$). Then $y(e)$ is equal to:
(2015)
(a) 2 (b) $2e$
(c) e (d) 0
- If $y(x)$ is the solution of the differential equation $(x+2) \frac{dy}{dx} = x^2 + 4x - 9$, $x \neq -2$ and $y(0) = 0$, then $y(-4)$ is equal to :
(2015/Online Set-1)
(a) 2 (b) 0
(c) -1 (d) 1
- The solution of the differential equation $ydx - (x + 2y^2)dy = 0$ is $x = f(y)$. If $f(-1) = 1$, then $f(1)$ is equal to :
(2015/Online Set-2)
(a) 4 (b) 3
(c) 1 (d) 2
- If a curve $y = f(x)$ passes through the point $(1, -1)$ and satisfies the differential equation, $y(1 + xy) dx = x dy$, then $f\left(-\frac{1}{2}\right)$ is equal to :
(2016)
(a) $-\frac{4}{5}$ (b) $\frac{2}{5}$
(c) $\frac{4}{5}$ (d) $-\frac{2}{5}$
- The solution of the differential equation $\frac{dy}{dx} + \frac{y}{2} \sec x = \frac{\tan x}{2y}$, where $0 \leq x < \frac{\pi}{2}$, and $y(0) = 1$, is given by :
(2016/Online Set-2)
(a) $y = 1 - \frac{x}{\sec x + \tan x}$ (b) $y^2 = 1 + \frac{x}{\sec x + \tan x}$
(c) $y^2 = 1 - \frac{x}{\sec x + \tan x}$ (d) $y = 1 + \frac{x}{\sec x + \tan x}$
- If $(2 + \sin x) \frac{dy}{dx} + (y + 1) \cos x = 0$ and $y(0) = 1$, then $y\left(\frac{\pi}{2}\right)$ is equal to :
(2017)
(a) $\frac{1}{3}$ (b) $-\frac{2}{3}$
(c) $-\frac{1}{3}$ (d) $\frac{4}{3}$
- The curve satisfying the differential equation, $ydx - (x + 3y^2) dy = 0$ and passing through the point $(1, 1)$, also passes through the point. (2017/Online Set-1)
(a) $\left(\frac{1}{4}, -\frac{1}{2}\right)$ (b) $\left(-\frac{1}{3}, \frac{1}{3}\right)$
(c) $\left(\frac{1}{3}, -\frac{1}{3}\right)$ (d) $\left(\frac{1}{4}, \frac{1}{2}\right)$
- Let $y = y(x)$ be the solution of the differential equation $\sin x \frac{dy}{dx} + y \cos x = 4x$, $x \in (0, \pi)$.
if $y\left(\frac{\pi}{2}\right) = 0$ then $y\left(\frac{\pi}{6}\right)$ is equal to :
(2018)
(a) $-\frac{4}{9}\pi^2$ (b) $\frac{4}{9\sqrt{3}}\pi^2$
(c) $\frac{-8}{9\sqrt{3}}\pi^2$ (d) $-\frac{8}{9}\pi^2$
- Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + 2y = f(x)$, where $f(x) = \begin{cases} 1, & x \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$
If $y(0) = 0$, then $y\left(\frac{3}{2}\right)$ is
(2018/Online Set-1)
(a) $\frac{e^2 + 1}{2e^4}$ (b) $\frac{1}{2e}$
(c) $\frac{e^2 - 1}{e^3}$ (d) $\frac{e^2 - 1}{2e^3}$



10. The differential equation representing the family of ellipses having foci either on the x-axis or on the y-axis, centre at the origin and passing through the point (0, 3) is:

(2018/Online Set-3)

- (a) $xy y'' + x(y')^2 - y y' = 0$
 (b) $x + y y'' = 0$
 (c) $xy y' + y^2 - 9 = 0$
 (d) $xy y' - y^2 + 9 = 0$

11. Let $y = y(x)$ be the solution of the differential equation,

$$(x^2 + 1)^2 \frac{dy}{dx} + 2x(x^2 + 1)y = 1 \text{ such that } y(0) = 0. \text{ If}$$

$$\sqrt{a}y(1) = \frac{\pi}{32}, \text{ then the value of 'a' is:}$$

(8-04-2019/Shift-1)

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
 (c) 1 (d) $\frac{1}{16}$

12. Given that the slope of the tangent to a curve $y = y(x)$ at any point (x, y) is $\frac{2y}{x^2}$. If the curve passes through the

centre of the circle $x^2 + y^2 - 2x - 2y = 0$, then its equation is:

(8-04-2019/Shift-2)

- (a) $x \log_e |y| = 2(x-1)$ (b) $x \log_e |y| = -4(x-1)$
 (c) $x \log_e |y| = -2(x-1)$ (d) $x \log_e |y| = x-1$

13. The solution of the differential equation

$$x \frac{dy}{dx} + 2y = x^2 (x \neq 0) \text{ with } y(1) = 1, \text{ is:}$$

(9-04-2019/Shift-1)

- (a) $y = \frac{4}{5}x^3 + \frac{1}{5x^2}$ (b) $y = \frac{x^3}{5} + \frac{1}{5x^2}$
 (c) $y = \frac{x^2}{4} + \frac{3}{4x^2}$ (d) $y = \frac{3}{4}x^2 + \frac{1}{4x^2}$

14. If $\cos x \frac{dy}{dx} - y \sin x = 6x, \left(0 < x < \frac{\pi}{2}\right)$

and $y\left(\frac{\pi}{3}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to:

(9-04-2019/Shift-2)

- (a) $\frac{\pi^2}{2\sqrt{3}}$ (b) $-\frac{\pi^2}{2}$
 (c) $-\frac{\pi^2}{2\sqrt{3}}$ (d) $-\frac{\pi^2}{4\sqrt{3}}$

15. If $y = y(x)$ is the solution of the differential equation

$$\frac{dy}{dx} = (\tan x - y) \sec^2 x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \text{ such that}$$

$y(0) = 0$, then $y\left(-\frac{\pi}{4}\right)$ is equal to:

(10-04-2019/Shift-1)

- (a) $e - 2$ (b) $\frac{1}{2} - e$
 (c) $2 + \frac{1}{e}$ (d) $\frac{1}{e} - 2$

16. Let $y = y(x)$ be the solution of the differential equation,

$$\frac{dy}{dx} + y \tan x = 2x + x^2 \tan x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \text{ such that}$$

$y(0) = 1$, then:

(10-04-2019/Shift-2)

- (a) $y\left(\frac{\pi}{4}\right) + y\left(-\frac{\pi}{4}\right) = \frac{\pi^2}{2} + 2$
 (b) $y\left(\frac{\pi}{4}\right) + y\left(-\frac{\pi}{4}\right) = -\sqrt{2}$
 (c) $y\left(\frac{\pi}{4}\right) - y\left(-\frac{\pi}{4}\right) = \sqrt{2}$
 (d) $y'\left(\frac{\pi}{4}\right) - y'\left(-\frac{\pi}{4}\right) = \pi - \sqrt{2}$



17. Consider the differential equation, $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$.
If value of y is 1 when $x = 1$, then the value of x for which $y = 2$, is **(12-04-2019/Shift-1)**
- (a) $\frac{5}{2} + \frac{1}{\sqrt{e}}$ (b) $\frac{3}{2} - \frac{1}{\sqrt{e}}$
(c) $\frac{1}{2} + \frac{1}{\sqrt{e}}$ (d) $\frac{3}{2} - \sqrt{e}$
18. The general solution of the differential equations $(y^2 - x^3) dx - xy dy = 0$ ($x \neq 0$) is _____. (Where c is a constant of integration) **(12-04-2019/Shift-2)**
- (a) $y^2 - 2x^2 + cx^3 = 0$ (b) $y^2 + 2x^3 + cx^2 = 0$
(c) $y^2 + 2x^2 + cx^3 = 0$ (d) $y^2 - 2x^3 + cx^2 = 0$
19. If $y = y(x)$ is the solution of the differential equation, $x \frac{dy}{dx} + 2y = x^2$ satisfying $y(1) = 1$, then $y\left(\frac{1}{2}\right)$ is equal to: **(9-01-2019/Shift-1)**
- (a) $\frac{7}{64}$ (b) $\frac{1}{4}$
(c) $\frac{49}{16}$ (d) $\frac{13}{16}$
20. Let $f: [0, 1] \rightarrow \mathbb{R}$ be such that $f(xy) = f(x).f(y)$, for all $x, y \in [0, 1]$ and $f(0) \neq 0$. If $y = y(x)$ satisfies the differential equation, $\frac{dy}{dx} = f(x)$ with $y(0) = 1$,
They $y\left(\frac{1}{4}\right) + y\left(\frac{3}{4}\right)$ is equal to: **(9-01-2019/Shift-2)**
- (a) 3 (b) 4
(c) 2 (d) 5
21. If $\frac{dy}{dx} + \frac{3}{\cos^2 x} y = \frac{1}{\cos^2 x}$, $x \in \left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$, and $y\left(\frac{\pi}{4}\right) = \frac{4}{3}$,
then $y\left(-\frac{\pi}{4}\right)$ equals: **(10-1-2019/Shift-1)**
- (a) $\frac{1}{3} + e^6$ (b) $\frac{1}{3}$
(c) $-\frac{4}{3}$ (d) $\frac{1}{3} + e^3$
22. The curve amongst the family of curves represented by the differential equation, $(x^2 - y^2) dx + 2xy dy = 0$ which passes through $(1, 1)$, is: **(10-01-2019/Shift-2)**
- (a) a circle with centre on the x -axis.
(b) an ellipse with major axis along the y -axis.
(c) a circle with centre on the y -axis.
(d) a hyperbola with transverse axis along the x -axis.
23. Let f be a differentiable such that $f'(x) = 7 - \frac{3}{4} \frac{f(x)}{x}$,
($x > 0$) and $f(1) \neq 4$. Then $\lim_{x \rightarrow 0^+} x f\left(\frac{1}{x}\right)$ **(10-01-2019/Shift-2)**
- (a) exists and equals $\frac{4}{7}$ (b) exists and equals 4
(c) does not exist (d) exists and equals 0
24. If $y(x)$ is the solution of the differential equation $\frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x}$, $x > 0$, where $y(1) = \frac{1}{2}e^{-2}$, then : **(11-01-2019/Shift-1)**
- (a) $y(\log_e 2) = \log_e 4$
(b) $y(\log_e 2) = \frac{\log_e 2}{4}$
(c) $y(x)$ is decreasing in $\left(\frac{1}{2}, 1\right)$
(d) $y(x)$ is decreasing in $(0, 1)$



25. The solution of the differential equation,

$$\frac{dy}{dx} = (x-y)^2 \text{ when } y(1) = 1, \text{ is } \quad (11-01-2019/\text{Shift-2})$$

(a) $\log_e \left| \frac{2-y}{2-x} \right| = x-y$

(b) $-\log_e \left| \frac{1-x+y}{1+x-y} \right| = 2(x-1)$

(c) $-\log_e \left| \frac{1+x-y}{1-x+y} \right| = x+y-2$

(d) $\log_e \left| \frac{2-y}{2-x} \right| = 2(y-1)$

26. Let $y = y(x)$ be the solution of the differential equation

$$x \frac{dy}{dx} + y = x \log_e x, (x > 1). \text{ If } 2y(2) = \log_e 4 - 1, \text{ then } y(e)$$

is equal to _____. (12-01-2019/Shift-1)

(a) $-\frac{e}{2}$

(b) $-\frac{e^2}{2}$

(c) $\frac{e}{4}$

(d) $\frac{e^2}{4}$

27. Let f be a differentiable function such that $f(1) = 2$ and

$$f'(x) = f(x) \text{ for all } x \in R$$

If $h(x) = f(f(x))$ then $h'(1)$ is equal

(12-01-2019/Shift-2)

(a) $2e^2$

(b) $4e$

(c) $2e$

(d) $4e^2$

28. If a curve passes through the point $(1, -2)$ and has slope

of the tangent at any point (x, y) on it as $\frac{x^2 - 2y}{x}$ then

the curve also passes through the point :

(12-01-2019/Shift-2)

(a) $(3, 0)$

(b) $(\sqrt{3}, 0)$

(c) $(-1, 2)$

(d) $(-\sqrt{2}, 1)$

29. Let $y = y(x)$ be the solution of the differential equation,

$$\frac{2 + \sin x}{y+1} \cdot \frac{dy}{dx} = -\cos x, y > 0, y(0) = 1.$$

If $y(\pi) = a$, and $\frac{dy}{dx}$ at $x = \pi$ is b , then the ordered pair

(a, b) is equal to : (2-9-2020/Shift-1)

(a) $\left(2, \frac{3}{2}\right)$

(b) $(1, 1)$

(c) $(2, 1)$

(d) $(1, -1)$

30. If a curve $y = f(x)$, passing through the point $(1, 2)$ is the solution of the differential equation,

$$2x^2 dy = (2xy + y^2) dx, \text{ then } f\left(\frac{1}{2}\right) \text{ is equal to :}$$

(2-09-2020/Shift-2)

(a) $\frac{-1}{1 + \log_e 2}$

(b) $1 + \log_e 2$

(c) $\frac{1}{1 + \log_e 2}$

(d) $\frac{1}{1 - \log_e 2}$

31. The solution curve of the differential equation,

$$(1 + e^{-x})(1 + y^2) \frac{dy}{dx} = y^2, \text{ which passes through the point}$$

$(0, 1)$, is :

(3-09-2020/Shift-1)

(a) $y^2 = 1 + y \log_e \left(\frac{1 + e^{-x}}{2} \right)$

(b) $y^2 + 1 = y \left(\log_e \left(\frac{1 + e^{-x}}{2} \right) + 2 \right)$

(c) $y^2 + 1 = y \left(\log_e \left(\frac{1 + e^x}{2} \right) + 2 \right)$

(d) $y^2 = 1 + y \left(\log_e \left(\frac{1 + e^x}{2} \right) \right)$



32. If $x^3 dy + xy dx = x^2 dy + 2y dx$; $y(2) = e$ and $x > 1$, then $y(4)$ is equal to : **(3-09-2020/Shift-2)**
- (a) $\frac{\sqrt{e}}{2}$ (b) $\frac{3}{2}\sqrt{e}$
- (c) $\frac{1}{2} + \sqrt{e}$ (d) $\frac{3}{2} + \sqrt{e}$
33. Let $y = y(x)$ be the solution of the differential equation, $xy' - y = x^2(x \cos x + \sin x)$, $x > 0$. If $y(\pi) = \pi$, then $y''\left(\frac{\pi}{2}\right) + y'\left(\frac{\pi}{2}\right)$ is equal to **(4-09-2020/Shift-1)**
- (a) $2 + \frac{\pi}{2} + \frac{\pi^2}{4}$ (b) $2 + \frac{\pi}{2}$
- (c) $1 + \frac{\pi}{2}$ (d) $1 + \frac{\pi}{2} + \frac{\pi^2}{4}$
34. The solution of the differential equation $\frac{dy}{dx} - \frac{y+3x}{\log_e(y+3x)} + 3 = 0$ is : (where c is a constant of integration) **(4-9-2020/Shift-2)**
- (a) $x - \log_e(y+3x) = C$
- (b) $x - \frac{1}{2}(\log_e(y+3x))^2 = C$
- (c) $x - 2\log_e(y+3x) = C$
- (d) $y + 3x - \frac{1}{2}(\log_e x)^2 = C$
35. If $y = y(x)$ is the solution of the differential equation $\frac{5+e^x}{2+y} \cdot \frac{dy}{dx} + e^x = 0$ satisfying $y(0) = 1$, then a value of $y(\log_e 13)$ is: **(5-09-2020/Shift-1)**
- (a) 1 (b) 0
- (c) 2 (d) -1
36. Let $y = y(x)$ be the solution of the differential equation $\cos x \frac{dy}{dx} + 2y \sin x = \sin 2x$, $x \in \left(0, \frac{\pi}{2}\right)$. If $y(\pi/3) = 0$ then $y(\pi/4)$ is equal to: **(5-09-2020/Shift-2)**
- (a) $2 + \sqrt{2}$ (b) $\sqrt{2} - 2$
- (c) $\frac{1}{\sqrt{2}} - 1$ (d) $2 - \sqrt{2}$
37. The general solution of the differential equation $\sqrt{1+x^2+y^2+x^2y^2} + xy \frac{dy}{dx} = 0$ (where C is a constant of integration) is: **(6-09-2020/Shift-1)**
- (a) $\sqrt{1+y^2} + \sqrt{1+x^2} = \frac{1}{2} \log_e \left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1} \right) + C$
- (b) $\sqrt{1+y^2} - \sqrt{1+x^2} = \frac{1}{2} \log_e \left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1} \right) + C$
- (c) $\sqrt{1+y^2} + \sqrt{1+x^2} = \frac{1}{2} \log_e \left(\frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1} \right) + C$
- (d) $\sqrt{1+y^2} - \sqrt{1+x^2} = \frac{1}{2} \log_e \left(\frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1} \right) + C$
38. If $y = \left(\frac{2}{\pi}x - 1\right) \operatorname{cosec} x$ is the solution of the differential equation, $\frac{dy}{dx} + p(x)y = \frac{2}{\pi} \operatorname{cosec} x$, $0 < x < \frac{\pi}{2}$, then the function $p(x)$ is equal to : **(6-09-2020/Shift-2)**
- (a) $\operatorname{cosec} x$ (b) $\cot x$
- (c) $\tan x$ (d) $\sec x$



39. If $y = y(x)$ is the solution of the differential equation, $e^y \left(\frac{dy}{dx} - 1 \right) = e^x$ such that $y(0) = 0$, then $y(1)$ is equal to (7-01-2020/Shift-1)
- (a) $\log_e 2$ (b) 2_e
(c) $2 + \log_e 2$ (d) $1 + \log_e 2$
40. Let $y = y(x)$ be the solution curve of the differential equation, $(y^2 - x) \frac{dy}{dx} = 1$, satisfying $y(0) = 1$. This curve intersects the x -axis at a point whose abscissa is : (7-01-2020/Shift-2)
- (a) $2 + e$ (b) 2
(c) $2 - e$ (d) $-e$
41. Let $y = y(x)$ be a solution of the differential equation, $\sqrt{1-x^2} \frac{dy}{dx} + \sqrt{1-y^2} = 0, |x| < 1$.
If $y\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$, then $y\left(\frac{-1}{\sqrt{2}}\right)$ is equal to: (8-01-2020/Shift-1)
- (a) $-\frac{1}{\sqrt{2}}$ (b) $-\frac{\sqrt{3}}{2}$
(c) $\frac{1}{\sqrt{2}}$ (d) $\frac{\sqrt{3}}{2}$
42. The differential equation of the family of curves, $x^2 = 4b(y+b), b \in R$ is (8-01-2020/Shift-2)
- (a) $xy'' = y'$ (b) $x(y')^2 = x + 2yy'$
(c) $x(y')^2 = x - 2yy'$ (d) $x(y')^2 = 2yy' - x$
43. If for $x \geq 0, y = y(x)$ is the solution of the differential equation $(1+x)dy = [(1+x)^2 + y - 3]dx, y(2) = 0$ then $y(3)$ is equal to _____. (9-01-2020/Shift-1)
44. If $\frac{dy}{dx} = \frac{xy}{x^2 + y^2} : y(1) = 1$; then a value of x satisfying $y(x) = e$ is (9-1-2020/Shift-2)
- (a) $\sqrt{3}e$ (b) $\frac{1}{2}\sqrt{3}e$
(c) $\sqrt{2}e$ (d) $\frac{e}{\sqrt{2}}$
45. If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} + (\tan x)y = \sin x, 0 \leq x \leq \frac{\pi}{3}$, with $y(0) = 0$, then $y\left(\frac{\pi}{4}\right)$ equal to: (16-03-2021/Shift-2)
- (a) $\frac{1}{2} \log_e 2$ (b) $\left(\frac{1}{2\sqrt{2}}\right) \log_e 2$
(c) $\frac{1}{4} \log_e 2$ (d) $\log_e 2$
46. Let C_1 be the curve obtained by the solution of differential equation $2xy \frac{dy}{dx} = y^2 - x^2, x > 0$. Let the curve C_2 be the solution of $\frac{2xy}{x^2 - y^2} = \frac{dy}{dx}$. If both the curves pass through $(1,1)$, then the area enclosed by the curves C_1 and C_2 is equal to (16-03-2021/Shift-2)
- (a) $\frac{\pi}{2} - 1$ (b) $\pi + 1$
(c) $\frac{\pi}{4} + 1$ (d) $\pi - 1$
47. If $y = y(x)$ is the solution of the differential equation, $\frac{dy}{dx} + 2y \tan x = \sin x, y\left(\frac{\pi}{3}\right) = 0$, then the maximum value of the function $y(x)$ over R is equal to: (16-03-2021/Shift-1)
- (a) 8 (b) $\frac{1}{8}$
(c) $\frac{1}{2}$ (d) $-\frac{15}{4}$



48. Let the curve $y = y(x)$ be the solution of the differential equation, $\frac{dy}{dx} = 2(x+1)$. If the numerical value of area bounded by the curve $y = y(x)$ and x-axis is $\frac{4\sqrt{8}}{3}$, then the value of $y(1)$ is equal to _____.

(16-03-2021/Shift-1)

49. Let $y = y(x)$ be the solution of the differential equation $\cos x (3 \sin x + \cos x + 3) dy = (1 + y \sin x (3 \sin x + \cos x + 3)) dx$, $0 \leq x \leq \frac{\pi}{2}$, $y(0) = 0$.

Then, $y\left(\frac{\pi}{3}\right)$ is equal to : (17-03-2021/Shift 2)

- (a) $2 \log_e \left(\frac{3\sqrt{3}-8}{4} \right)$ (b) $2 \log_e \left(\frac{2\sqrt{3}+10}{11} \right)$
(c) $2 \log_e \left(\frac{\sqrt{3}+7}{2} \right)$ (d) $2 \log_e \left(\frac{2\sqrt{3}+9}{6} \right)$

50. If the curve $y = y(x)$ is the solution of the differential equation $2(x^2 + x^{5/4}) dy - y(x + x^{1/4}) dx = 2x^{9/4} dx$, $x > 0$ which passes through the point $\left(1, 1 - \frac{4}{3} \log_e 2\right)$, then the value of $y(16)$ is equal to : (17-03-2021/Shift-2)

- (a) $4 \left(\frac{31}{3} - \frac{8}{3} \log_e 3 \right)$ (b) $\left(\frac{31}{3} + \frac{8}{3} \log_e 3 \right)$
(c) $\left(\frac{31}{3} - \frac{8}{3} \log_e 3 \right)$ (d) $4 \left(\frac{31}{3} + \frac{8}{3} \log_e 3 \right)$

51. Which of the following is true for $y(x)$ that satisfies the differential equation $\frac{dy}{dx} = xy - 1 + x - y$; $y(0) = 0$:

(17-03-2021/Shift-1)

- (a) $y(1) = e^{\frac{1}{2}} - e^{-\frac{1}{2}}$ (b) $y(1) = 1$
(c) $y(1) = e^{\frac{1}{2}} - 1$ (d) $y(1) = e^{-\frac{1}{2}} - 1$

52. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = (y+1)((y+1)e^{x^2/2} - x)$, $0 < x < 2.1$, with $y(2) = 0$. The value of $\frac{dy}{dx}$ at $x = 1$ is equal to:

(18-03-2021/Shift-2)

- (a) $\frac{e^{5/2}}{(1+e^2)^2}$ (b) $\frac{5e^{1/2}}{(e^2+1)^2}$
(c) $\frac{-e^{3/2}}{(e^2+1)^2}$ (d) $-\frac{2e^2}{(1+e^2)^2}$

53. The differential equation satisfied by the system of parabolas $y^2 = 4a(x+a)$ is : (18-03-2021/Shift-1)

- (a) $y \left(\frac{dy}{dx} \right)^2 + 2x \left(\frac{dy}{dx} \right) - y = 0$
(b) $y \left(\frac{dy}{dx} \right)^2 - 2x \left(\frac{dy}{dx} \right) + y = 0$
(c) $y \left(\frac{dy}{dx} \right) + 2x \left(\frac{dy}{dx} \right) - y = 0$
(d) $y \left(\frac{dy}{dx} \right)^2 - 2x \left(\frac{dy}{dx} \right) - y = 0$

54. Let f be a twice differentiable function defined on $\mathbb{R}$ such that $f(0) = 1$, $f'(0) = 2$ and $f'(x) \neq 0$ for all

$x \in \mathbb{R}$. If $\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix} = 0$, for all $x \in \mathbb{R}$, then the value

of $f(1)$ lies in the interval (24-02-2021/Shift-2)

- (a) (0,3) (b) (9,12)
(c) (3,6) (d) (6,9)



55. If a curve $y = f(x)$ passes through the point (1,2) and satisfies $x \frac{dy}{dx} + y = bx^4$, then for what value $\int_1^2 f(x)dx = \frac{62}{5}$? (24-02-2021/Shift-2)
- (a) 10 (b) 5
(c) $\frac{31}{5}$ (d) $\frac{62}{5}$
56. If a curve passes through the origin and the slope of the tangent to it any point (x, y) is $\frac{x^2 - 4x + y + 8}{x - 2}$, then this curve also passes through the point: (25-02-2021/Shift-1)
- (a) (4, 4) (b) (5, 5)
(c) (4, 5) (d) (5, 4)
57. Let $f(x) = \int_0^x e^t f(t) dt + e^x$ be a differentiable function for all $x \in \mathbb{R}$. Then $f(x)$ equals: (26-02-2021/Shift-2)
- (a) $2e^{(e^x - 1)}$ (b) $e^{e^x} - 1$
(c) $e^{(e^x - 1)}$ (d) $2e^{e^x} - 1$
58. Let slope of the tangent line to a curve at any point $P(x, y)$ be given by $\frac{xy^2 + y}{x}$. If the curve intersects the line $x + 2y = 4$ at $x = -2$, then the value of y , for which the point $(3, y)$ lies on the curve, is: (26-02-2021/Shift-2)
- (a) $-\frac{4}{3}$ (b) $\frac{18}{35}$
(c) $-\frac{18}{11}$ (d) $-\frac{18}{19}$
59. Let the normals at all the points on a given curve pass through a fixed point (a, b) . If the curve passes through $(3, -3)$ and $(4, -2\sqrt{2})$, and given that $a - 2\sqrt{2}b = 3$, then $(a^2 + b^2 + ab)$ is equal to _____. (26-02-2021/Shift-2)
60. The rate of growth of bacteria in a culture is proportional to the number of bacteria present and the bacteria count is 1000 at initial time $t = 0$. The number of bacteria is increased by 20% in 2 hours. If the population of bacteria is 2000 after $\frac{k}{\log_e \left(\frac{6}{5}\right)}$ hours, then $\left(\frac{k}{\log_e 2}\right)^2$ is equal to (26-02-2021/Shift-1)
- (a) 2 (b) 8
(c) 16 (d) 4
61. If $y = y(x)$ is the solution of the equation $e^{\sin y} \cos y \frac{dy}{dx} + e^{\sin y} \cos x = \cos x$, $y(0) = 0$; then $1 + y\left(\frac{\pi}{6}\right) + \frac{\sqrt{3}}{2}y\left(\frac{\pi}{3}\right) + \frac{1}{\sqrt{2}}y\left(\frac{\pi}{4}\right)$ is equal to (26-02-2021/Shift-1)
62. The difference between degree and order of a differential equation that represents the family of curves given by $y^2 = a\left(x + \frac{\sqrt{a}}{2}\right)$, $a > 0$ is _____. (26-02-2021/Shift-1)



63. Let $y = y(x)$ be the solution of the differential equation $x \tan\left(\frac{y}{x}\right) dy = \left(y \tan\left(\frac{y}{x}\right) - x\right) dx$,

$$-1 \leq x \leq 1, y\left(\frac{1}{2}\right) = \frac{\pi}{6}.$$

Then the area of the region bounded by the curves

$$x = 0, x = \frac{1}{\sqrt{2}} \text{ and } y = y(x) \text{ in the upper half plane is :}$$

(20-07-2021/Shift-1)

(a) $\frac{1}{12}(\pi - 3)$ (b) $\frac{1}{6}(\pi - 1)$

(c) $\frac{1}{8}(\pi - 1)$ (d) $\frac{1}{4}(\pi - 2)$

64. Let $y = y(x)$ be the solution of the differential equation $e^x \sqrt{1 - y^2} dx + \left(\frac{y}{x}\right) dy = 0, y(1) = -1$. Then the value of

$$(y(3))^2 \text{ is equal to:}$$

(20-07-2021/Shift-1)

(a) $1 + 4e^3$ (b) $1 + 4e^6$

(c) $1 - 4e^6$ (d) $1 - 4e^3$

65. Let $y = y(x)$ satisfies the equation $\frac{dy}{dx} - |A| = 0$, for all

$$x > 0, \text{ where } A = \begin{bmatrix} y & \sin x & 1 \\ 0 & -1 & 1 \\ 2 & 0 & \frac{1}{x} \end{bmatrix}. \text{ If } y(\pi) = \pi + 2, \text{ then}$$

$$\text{the value of } y\left(\frac{\pi}{2}\right) \text{ is ?}$$

(20-07-2021/Shift-2)

(a) $\frac{\pi}{2} - \frac{4}{\pi}$ (b) $\frac{3\pi}{2} - \frac{1}{\pi}$

(c) $\frac{\pi}{2} - \frac{1}{\pi}$ (d) $\frac{\pi}{2} + \frac{4}{\pi}$

66. Let a curve $y = y(x)$ be given by the solution of the differential equation $\cos\left(\frac{1}{2}\cos^{-1}(e^{-x})\right) dx = \sqrt{e^{2x} - 1} dy$.

If it intersects y-axis at $y = -1$, and the intersection point of the curve with x-axis is $(\alpha, 0)$ the e^α is equal to _____ ?

(20-07-2021/Shift-2)

67. Let $y = y(x)$ be solution of the following differential equation $e^y \frac{dy}{dx} - 2e^y \sin x + \sin x \cos^2 x = 0, y\left(\frac{\pi}{2}\right) = 0$.

If $y(0) = \log_e(\alpha + \beta e^{-2})$, then $4(\alpha + \beta)$ is equal to

(25-07-2021/Shift-1)

_____ ?

68. Let $y = y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} = 1 + xe^{y-x}, -\sqrt{2} < x < \sqrt{2}, y(0) = 0 \text{ then, the}$$

minimum value of $y(x), x \in (-\sqrt{2}, \sqrt{2})$ is equal to ?

(25-07-2021/Shift-1)

(a) $(1 - \sqrt{3}) - \log_e(\sqrt{3} - 1)$

(b) $(2 + \sqrt{3}) + \log_e 2$

(c) $(2 - \sqrt{3}) - \log_e 2$

(d) $(1 + \sqrt{3}) - \log_e(\sqrt{3} - 1)$

69. Let $y = y(x)$ be solution of the differential equation

$$\log_e\left(\frac{dy}{dx}\right) = 3x + 4y, \text{ with } y(0) = 0. \text{ If}$$

$$y\left(-\frac{2}{3}\log_e 2\right) = \alpha \log_e 2, \text{ then the value of } \alpha \text{ is equal to :}$$

(27-07-2021/Shift-1)

(a) $-\frac{1}{2}$ (b) $-\frac{1}{4}$

(c) 2 (d) $\frac{1}{4}$



70. Let $F: [3, 5] \rightarrow \mathbb{R}$ be a twice differentiable from function on $(3, 5)$ such that $F(x) = e^{-x} \int_3^x (3t^2 + 2t + 4F'(t)) dt$.
If $F'(4) = \frac{\alpha e^\beta - 224}{(e^\beta - 4)^2}$, then $\alpha + \beta$ is equal to _____.
(27-07-2021/Shift-1)
71. If $y = y(x)$, $y \in \left[0, \frac{\pi}{2}\right]$ is the solution of the differential equation $\frac{dy}{dx} - \sin(x+y) - \sin(x-y) = 0$, with $y(0) = 0$, then $5y'\left(\frac{\pi}{2}\right)$ is equal to _____.
(27-07-2021/Shift-1)
72. Let $y = y(x)$ be the solution of the differential equation $dy = e^{\alpha x + y} dx$; $\alpha \in \mathbb{N}$. If $y(\log_e 2) = \log_e 2$ and $y(0) = \log_e \left(\frac{1}{2}\right)$, then the value of α is equal to _____.
(27-07-2021/Shift-2)
73. Let $y = y(x)$ be the solution of the differential equation $(x - x^3) dy = (y + yx^2 - 3x^4) dx$, $x > 2$. If $y(3) = 3$, then $y(4)$ is equal to: (27-07-2021/Shift-2)
(a) 8 (b) 12
(c) 16 (d) 4
74. Let $y = y(x)$ be the solution of the differential equation $\operatorname{cosec}^2 x \, dy + 2dx = (1 + y \cos 2x) \operatorname{cosec}^2 x \, dx$, with $y\left(\frac{\pi}{4}\right) = 0$. Then, the value of $(y(0) + 1)^2$ is equal to :
(22-07-2021/Shift-2)
(a) $\frac{1}{e^2}$ (b) $e^{-\frac{1}{2}}$
(c) e^{-1} (d) e
75. Let $y = y(x)$ be the solution of the differential equation $\left((x+2)e^{\left(\frac{y+1}{x+2}\right)} + (y+1)\right) dx = (x+2) dy$, $y(1) = 1$.
If the domain of $y = y(x)$ is an open interval (α, β) , then $|\alpha + \beta|$ is equal to _____. (22-07-2021/Shift-2)
76. Let $y = y(x)$ be the solution of the differential equation $x dy = (y + x^3 \cos x) dx$ with $y(\pi) = 0$, then $y\left(\frac{\pi}{2}\right)$ is equal to : (25-07-2021/Shift-2)
(a) $\frac{\pi^2}{2} - \frac{\pi}{4}$ (b) $\frac{\pi^2}{4} + \frac{\pi}{2}$
(c) $\frac{\pi^2}{4} - \frac{\pi}{4}$ (d) $\frac{\pi^2}{2} + \frac{\pi}{4}$
77. Let a curve $y = f(x)$ pass through the point $(2, (\log_e 2)^2)$ and have slope $\frac{2y}{x \log_e x}$ for all positive real value of x . Then the value of $f(e)$ is equal to _____.
(25-07-2021/Shift-2)
78. If $y = y(x)$ is the solution curve of the differential equation $x^2 dy + \left(y - \frac{1}{x}\right) dx = 0$; $x > 0$ and $y(1) = 1$, then $y\left(\frac{1}{2}\right)$ is equal to (01-09-2021/Shift-2)
(a) $3 + \frac{1}{\sqrt{e}}$ (b) $\frac{3}{2} - \frac{1}{\sqrt{e}}$
(c) $3 + e$ (d) $3 - e$
79. Let $y(x)$ be the solution of the differential equation $2x^2 dy + (e^y - 2x) dx = 0$, $x > 0$. If $y(e) = 1$, then $y(1)$ is equal to: (26-08-2021/Shift-2)
(a) 0 (b) 2
(c) $\log_e(2e)$ (d) $\log_e 2$



80. Let $y = y(x)$ $\frac{dy}{dx} = 2(y + 2 \sin x - 5)x - 2 \cos x$ such that

$y(0) = 7$. Then $y(\pi)$ is equal to: (27-08-2021/Shift-1)

- (a) $3e^{\pi^2} + 5$ (b) $7e^{\pi^2} + 5$
(c) $2e^{\pi^2} + 5$ (d) $e^{\pi^2} + 5$

81. Let us consider a curve, $y = f(x)$ passing through the point $(-2, 2)$ and the slope of the tangent to the curve at any point $(x, f(x))$ is given by $f(x) + xf'(x) = x^2$. Then:

(27-08-2021/Shift-1)

- (a) $x^2 + 2xf(x) + 4 = 0$
(b) $x^3 - 3xf(x) - 4 = 0$
(c) $x^3 + xf(x) + 12 = 0$
(d) $x^2 + 2xf(x) - 12 = 0$

82. Let $y = y(x)$ be a solution curve of the differential equation

$$(y+1)\tan^2 x \, dx + \tan x \, dy + y \, dx = 0, x \in \left(0, \frac{\pi}{2}\right).$$

If $\lim_{x \rightarrow 0^+} x \, y(x) = 1$, then the value of $y\left(\frac{\pi}{4}\right)$ is:

(26-08-2021/Shift-1)

- (a) $\frac{\pi}{4} - 1$ (b) $\frac{\pi}{4} + 1$
(c) $\frac{\pi}{4}$ (d) $-\frac{\pi}{4}$

83. If the solution curve of the differential equation $(2x - 10y^3)dy + ydx = 0$, passes through the points $(0, 1)$ and $(2, \beta)$, then β is a root of the equation :

(27-08-2021/Shift-2)

- (a) $2y^5 - 2y - 1 = 0$ (b) $y^5 - y^2 - 1 = 0$
(c) $y^5 - 2y^2 - 2 = 0$ (d) $2y^5 - y^2 - 2 = 0$

84. A differential equation representing the family of parabolas with axis parallel to y-axis and whose length of latus rectum is the distance of the point $(2, -3)$ from the line $3x + 4y = 5$ is given by : (27-08-2021/Shift-2)

- (a) $10 \frac{d^2 y}{dx^2} = 11$ (b) $11 \frac{d^2 y}{dx^2} = 10$
(c) $10 \frac{d^2 x}{dy^2} = 11$ (d) $11 \frac{d^2 x}{dy^2} = 10$

85. If $\frac{dy}{dx} = \frac{2^{x+y} - 2^x}{2^y}$, $y(0) = 1$, then $y(1)$ is equal to ?

(31-08-2021/Shift-1)

- (a) $\log_2(2+e)$ (b) $\log_2(1+e)$
(c) $\log_2(2e)$ (d) $\log_2(1+e^2)$

86. If $\frac{dy}{dx} = \frac{2^x y + 2^y \cdot 2^x}{2^x + 2^{x+y} \log_e 2}$, $y(0) = 0$, then for $y = 1$, the value of x lies in the interval: (31-08-2021/Shift-2)

- (a) $\left(\frac{1}{2}, 1\right]$ (b) $\left(0, \frac{1}{2}\right]$
(c) $(1, 2)$ (d) $(2, 3)$

87. If $y \frac{dy}{dx} = x \left[\frac{y^2}{x^2} + \frac{\phi\left(\frac{y^2}{x^2}\right)}{\phi'\left(\frac{y^2}{x^2}\right)} \right]$, $x > 0, \phi > 0$ and $y(1) = -1$,

then $\phi\left(\frac{y^2}{4}\right)$ is equal to: (31-08-2021/Shift-2)

- (a) $\phi(1)$ (b) $4\phi(2)$
(c) $2\phi(1)$ (d) $4\phi(1)$



88. The population $P = P(t)$ at time 't' of a certain species follows the differential equation $\frac{dP}{dt} = 0.5P - 450$. If $P(0) = 850$, then the time at which population becomes zero is **(24-02-2021/Shift-1)**
- (a) $\log_e 9$ (b) $\frac{1}{2} \log_e 18$
 (c) $\log_e 18$ (d) $2 \log_e 18$
89. If the curve, $y = y(x)$ represented by the solution of the differential equation $(2xy^2 - y) dx + x dy = 0$, passes through the intersection of the lines, $2x - 3y = 1$ and $3x + 2y = 8$, then $|y(1)|$ is equal to: **(25-02-2021/Shift-2)**
90. Let $y = y(x)$ be the solution of the differential equation $xdy - ydx = \sqrt{(x^2 - y^2)} dx$, $x \geq 1$, with $y(1) = 0$. If the area bounded by the line $x = 1, x = e^\pi, y = 0$ and $y = y(x)$ is $\alpha e^{2\pi} + \beta$, then the value of $10(\alpha + \beta)$ is equal to _____. **(18-03-2021/Shift-2)**



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

Objective Questions I [Only one correct option]

1. If the function $y = e^{4x} + 2e^{-x}$ is a solution of the differential

equation $\frac{d^3y}{dx^3} - 13\frac{dy}{dx} = K$ then the value of K is

- (a) 4 (b) 6
(c) 9 (d) 12

2. Solution of the differential equation

$$\frac{dy}{dx} + \sin\left(\frac{x+y}{2}\right) = \sin\frac{x-y}{2} \text{ is}$$

(a) $\ln\left|\tan\frac{y}{4}\right| = C - 2\sin\frac{x}{2}$

(b) $\ln\left|\tan\frac{x}{4}\right| = C - 2\sin\frac{y}{2}$

(c) $\ln\left|\tan\frac{y}{2}\right| = C - 2\sin\frac{x}{4}$

(d) $\ln|\tan y| = C + 2\sin\frac{y}{2}$

3. The solution of the differential equation

$$x dx + y dy + \frac{xdy - ydx}{x^2 + y^2} = 0 \text{ is}$$

(a) $y = x \tan\left(\frac{x^2 + y^2 + c}{2}\right)$

(b) $x = y \tan\left(\frac{x^2 + y^2 + c}{2}\right)$

(c) $y = x \tan\left(\frac{c - x^2 - y^2}{2}\right)$

- (d) none of these

4. A curve passing through (2, 3) and satisfying the differential

equation $\int_0^x t y(t) dt = x^2 y(x)$, ($x > 0$) is

(a) $x^2 + y^2 = 13$ (b) $y^2 = \frac{9}{2}x$

(c) $\frac{x^2}{8} + \frac{y^2}{18} = 1$ (d) $xy = 6$

5. If $\int_a^x t y(t) dt = x^2 + y(x)$ then y as a function of x is

(a) $y = 2 - (2 + a^2)e^{\frac{x^2 - a^2}{2}}$ (b) $y = 1 - (2 + a^2)e^{\frac{x^2 - a^2}{2}}$

(c) $y = 2 - (1 + a^2)e^{\frac{x^2 - a^2}{2}}$ (d) none

6. The latus rectum of the conic passing through the origin and having the property that normal at each point (x, y) intersects the x-axis at ((x+1), 0) is :

- (a) 1 (b) 2
(c) 4 (d) none

7. The equation of curve for which y intercept of tangent varies as cube of the abscissa of point of contact, is

- (a) $2y + kx^3 = cx$ (b) $2x + ky^3 = cy$
(c) $2x^3 + ky = c$ (d) None

8. The solution of the differential equation

$$\frac{d^2y}{dx^2} = \sin 3x + e^x + x^2 \text{ when } y'(0) = 1 \text{ and } y(0) = 0 \text{ is}$$

(a) $-\frac{\sin 3x}{9} + e^x + \frac{x^4}{12} - 1$

(b) $-\frac{\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{x}{3}$

(c) $-\frac{\cos 3x}{3} + e^x + \frac{x^4}{12} + \frac{x}{3} + 1$

- (d) None of these



9. The differential equation of all the ellipses centred at the origin and having their axes as co-ordinate axes is :

(a) $y^2 + xy_1^2 - yy_1 = 0$ (b) $xyy_2 + xy_1^2 - yy_1 = 0$

(c) $yy_2 + xy_1^2 - xy_1 = 0$ (d) None of these

10. The differential equation whose general solution is given

by, $y = (c_1 \cos(x + c_2)) - (c_3 e^{(-x+c_4)}) + (c_5 \sin x)$,

where c_1, c_2, c_3, c_4, c_5 are arbitrary constants, is

(a) $\frac{d^4 y}{dx^4} - \frac{d^2 y}{dx^2} + y = 0$

(b) $\frac{d^3 y}{dx^3} + \frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = 0$

(c) $\frac{d^5 y}{dx^5} + y = 0$

(d) $\frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} + \frac{dy}{dx} - y = 0$

11. A curve $y = f(x)$ passing through the point $\left(1, \frac{1}{\sqrt{e}}\right)$ satisfies

the differential equation $\frac{dy}{dx} + x e^{-\frac{x^2}{2}} = 0$. Then which of the following does not hold good?

- (a) $f(x)$ is differentiable at $x = 0$.
 (b) $f(x)$ is symmetric w.r.t. the origin.
 (c) $f(x)$ is increasing for $x < 0$ and decreasing for $x > 0$.
 (d) $f(x)$ has two inflection points.

12. Solution of the differential equation

$\left(\frac{x+y-1}{x+y-2}\right) \frac{dy}{dx} = \left(\frac{x+y+1}{x+y+2}\right)$, given that $y = 1$ when $x = 1$, is

(a) $\log \left| \frac{(x-y)^2 - 2}{2} \right| = 2(x+y)$

(b) $\log \left| \frac{(x-y)^2 + 2}{2} \right| = 2(x-y)$

(c) $\log \left| \frac{(x+y)^2 + 2}{2} \right| = 2(x-y)$

(d) none of these

13. The solution of the differential equation $\frac{dy}{dx} = \frac{y}{x} + \frac{\phi(y/x)}{\phi'(y/x)}$ is

(a) $x\phi\left(\frac{y}{x}\right) = k$

(b) $\phi\left(\frac{y}{x}\right) = kx$

(c) $y\phi\left(\frac{y}{x}\right) = k$

(d) $\phi\left(\frac{y}{x}\right) = ky$

14. The substitution $y = z^\alpha$ transforms the differential equation $(x^2 y^2 - 1)dy + 2xy^3 dx = 0$ into a homogeneous differential equation for

(a) $\alpha = -1$

(b) 0

(c) $\alpha = 1$

(d) no value of α .

15. A function $y = f(x)$ satisfies the differential equation

$\frac{dy}{dx} - y = \cos x - \sin x$ with initial condition that y is bounded

when $x \rightarrow \infty$. The area enclosed by $y = f(x)$, $y = \cos x$ and the y -axis in first quadrant is

(a) $\sqrt{2} - 1$

(b) $\sqrt{2}$

(c) 1

(d) $\frac{1}{\sqrt{2}}$

16. A function $y = f(x)$ satisfies

$(x+1) \cdot f'(x) - 2(x^2+x)f(x) = \frac{e^{x^2}}{(x+1)}, \forall x > -1$

If $f(0) = 5$, then $f(x)$ is

(a) $\left(\frac{3x+5}{x+1}\right) \cdot e^{x^2}$

(b) $\left(\frac{6x+5}{x+1}\right) \cdot e^{x^2}$

(c) $\left(\frac{6x+5}{(x+1)^2}\right) \cdot e^{x^2}$

(d) $\left(\frac{5-6x}{x+1}\right) \cdot e^{x^2}$

17. Solution of differential equation

$\frac{dy}{dx} = y + \int_0^1 y dx, y(0) = 1$ is :

(a) $y = \frac{1}{3-e} (2e^x + e + 1)$

(b) $y = \frac{1}{3+e} (2e^x - e + 1)$

(c) $y = \frac{1}{3-e} (2e^x - e + 1)$

(d) none



18. Solution of the differential equation

$$\left(e^{x^2} + e^{y^2}\right)y \frac{dy}{dx} + e^{x^2}(xy^2 - x) = 0, \text{ is}$$

- (a) $e^{x^2}(y^2 - 1) + e^{y^2} = c$ (b) $e^{y^2}(x^2 - 1) + e^{x^2} = c$
 (c) $e^{y^2}(y^2 - 1) + e^{x^2} = c$ (d) $e^{x^2}(y - 1) + e^{y^2} = c$

19. Solution of the differential equation

$$y(xy + 2x^2y^2) dx + x(xy - x^2y^2) dy = 0 \text{ is given by}$$

- (a) $2 \log |x| - \log |y| - \frac{1}{xy} = c$
 (b) $2 \log |y| - \log |x| - \frac{1}{xy} = c$
 (c) $2 \log |x| + \log |y| + \frac{1}{xy} = c$
 (d) $2 \log |y| + \log |x| + \frac{1}{xy} = c$

20. Solution of differential equation

$$(x^2 + y^2 + a^2)y \frac{dy}{dx} + x(x^2 + y^2 - a^2) = 0 \text{ is}$$

- (a) $(x^2 + y^2)^2 + 2a^2(x^2 - y^2) = c$
 (b) $(x^2 + y^2)^2 + 2a^2(y^2 - x^2) = c$
 (c) $(x^2 - y^2)^2 + 2a^2(x^2 + y^2)^2 + c = 0$
 (d) None

21. Solution of the differential equation

$$3x^2y^2 + \cos(xy) - xy \sin(xy) + \frac{dy}{dx}(2x^3y - x^2 \sin(xy)) = 0 \text{ is}$$

- (a) $(x^2y^2 + \cos xy) = c$ (b) $x(x^2y^2 + \cos xy) = c$
 (c) $y(x^2y^2 + \cos xy) = c$ (d) None

22. The function satisfying the equation

$$f(x) = -\int_0^x f(t) \tan t \, dt + \int_0^x \tan(t - x) \, dt,$$

where $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ is :

- (a) $1 + \cos x$ (b) $1 - \cos x$
 (c) $\cos x - 1$ (d) none

23. The x-intercept of the tangent to a curve is equal to the ordinate of the point of contact. The equation of the curve through the point (1, 1) is

- (a) $ye^{\frac{x}{y}} = e$ (b) $xe^{\frac{x}{y}} = e$
 (c) $xe^{\frac{y}{x}} = e$ (d) $ye^{\frac{y}{x}} = e$

24. The curve for which the sum of the lengths of the tangent and subtangent at any of its point is proportional to the product of the co-ordinates of the point of tangency. The proportionality factor is equal to k.

- (a) $y = \ln |c(k^2x^2 - 1)|$ (b) $y = \frac{1}{k} \ln |c(k^2x^2 - 1)|$
 (c) $y = \ln |k(c^2x^2 + 1)|$ (d) $y = \ln |k(c^2x^2 - 1)|$

25. The equation of the curve for which any tangent intersects the y-axis at the point equidistant from the point of tangency and the origin is

- (a) $x^2 - y^2 = cx$ (b) $x^2 + y^2 = cy$
 (c) $x^2 + y^2 = cx$ (d) $x^2 - y^2 = cy$

26. Water is drained from a vertical cylindrical tank by opening a valve at the base of the tank. It is known that the rate at which the water level drops is proportional to the square root of water depth y, where the constant of proportionality $k > 0$ depends on the acceleration due to gravity and the geometry of the hole. If t is measured in minutes and

$k = \frac{1}{15}$ then the time to drain the tank if the water is 4 meter deep to start with is

- (a) 30 min (b) 45 min
 (c) 60 min (d) 80 min

27. The orthogonal trajectories of the family of curves $a^{n-1}y = x^n$ are given by (a is the arbitrary constant)

- (a) $x^n + n^2y = \text{constant}$ (b) $ny^2 + x^2 = \text{constant}$
 (c) $n^2x + y^n = \text{constant}$ (d) $n^2x - y^n = \text{constant}$

28. The orthogonal trajectories of the family of circle given by $x^2 + y^2 - 2ay = 0$ (a is parameter), is :

- (a) $x^2 + y^2 - 2kx = 0$ (b) $x^2 + y^2 - 2ky = 0$
 (c) $x^2 + y^2 - 2k_1x - 2k_2y = 0$ (d) none of these



29. The equation of the curve satisfying the differential equation

$$\frac{d^2y}{dx^2}(x^2 + 1) = 2x \frac{dy}{dx} \text{ passing through the point } (0, 1) \text{ and}$$

having slope of tangent at $x = 0$ as 3 is

- (a) $y = x^2 + 3x + 2$ (b) $y^2 = x^2 + 3x + 1$
(c) $y = x^3 + 3x + 1$ (d) None of these

30. A solution of the differential equation

$$\left(\frac{dy}{dx}\right)^2 - x \frac{dy}{dx} + y = 0 \text{ is:}$$

- (a) $y = 2$ (b) $y = 2x$
(c) $y = 2x - 4$ (d) $y = 2x^2 - 4$

31. The solution of the differential equation

$$(x^3 - 3xy^2) dx = (y^3 - 3x^2y) dy \text{ is}$$

- (a) $x^2 - y^2 = c(x^2 + y^2)$ (b) $y^2 - \frac{x^2}{2} = c(x^2 + y^2)^2$
(c) $(x^2 - y^2)^2 = c(x^2 + y^2)$ (d) none of these

32. Solution of differential equation

$$x^3 \frac{dy}{dx} = y^3 + y^2 \sqrt{y^2 - x^2} \text{ is}$$

- (a) $xy = c(y + \sqrt{y^2 - x^2})$
(b) $xy = c(y + \sqrt{x^2 - y^2})$
(c) $xy = c(x + \sqrt{x^2 + y^2})$
(d) None

33. Solution of the differential equation

$$y^2(ydx + 2xdy) - x^2(2ydx + xdy) = 0 \text{ is}$$

- (a) $x^2y^2(x^2 + y^2) = c$ (b) $x^2y^2 = (x^2 - y^2) + c$
(c) $x^2y^2 = c$ (d) None

34. The solution of the differential equation

$$\frac{dy}{dx} + \frac{\cos x (3 \cos y - 7 \sin x - 3)}{\sin y (3 \sin x - 7 \cos y + 7)} = 0 \text{ is}$$

- (a) $(\cos y - \sin x - 1)^2 (\cos y + \sin x - 1)^5 = c$
(b) $(\cos x - \sin y + 1)^2 (\cos y + \sin x - 1)^5 = c$
(c) $(\cos y - \sin x - 1)^5 (\cos y + \sin x - 1)^2 = c$
(d) none of these

35. Solution of the differential equation $\frac{\sqrt{x} dx + \sqrt{y} dy}{\sqrt{x} dx - \sqrt{y} dy} = \sqrt{\frac{y^3}{x^3}}$

is given by

$$(a) \frac{3}{2} \log\left(\frac{y}{x}\right) + \log\left|\frac{x^{3/2} + y^{3/2}}{x^{3/2}}\right| + \tan^{-1}\left(\frac{y}{x}\right)^{3/2} + c = 0$$

$$(b) \frac{2}{3} \log\left(\frac{y}{x}\right) + \log\left|\frac{x^{3/2} + y^{3/2}}{x^{3/2}}\right| + \tan^{-1} \frac{y}{x} + c = 0$$

$$(c) \frac{2}{3} \log\left(\frac{y}{x}\right) + \log\left(\frac{x+y}{x}\right) + \tan^{-1}\left(\frac{y^{3/2}}{x^{3/2}}\right) + c = 0$$

(d) none of the above

36. The equation of the curve satisfying the equation

$$x(1 - x \ln y) \frac{dy}{dx} + y = 0 \text{ and passing through } \left(1, \frac{1}{e}\right) \text{ is}$$

- (a) $x(ey + \ln y + 1) = 1$ (b) $(ey + \ln x + 1) = 1$
(c) $y(ex + \ln y + 1) = 0$ (d) None of these

37. The equation of the curve passing through origin if the middle point of the segment of its normal from any point of the curve to the x -axis lies on the parabola $2y^2 = x$, is

- (a) $y^2 = 2x + 1 - e^{2x}$ (b) $y^2 = 2x - 1 + e^x$
(c) $y^2 = 2x + 1 + e^x$ (d) None

38. An inverted conical tank of 2 m radius and 4 m height is initially full of water, has an outlet at bottom. The outlet is opened at some instant. The rate of flow through the outlet at any time t is $6h^{3/2}$, where h is height of water level above the outlet at time t . Then the time it takes to empty the tank is

$$(a) \frac{2\pi}{11} \text{ units} \quad (b) \frac{\pi}{9} \text{ units}$$

$$(c) \frac{2\pi}{8} \text{ units} \quad (d) \text{ none of these}$$



Objective Questions II [One or more than one correct option]

39. Solutions of the differential equation

$$\left(\frac{dy}{dx}\right)^2 - \frac{dy}{dx}(e^x + e^{-x}) + 1 = 0 \text{ are given by}$$

(a) $y + e^{-x} = C$ (b) $y - e^{-x} = C$

(c) $y + e^x = C$ (d) $y - e^x = C$

40. The solution of $p^2 + (2y \cot x)p = y^2$ where $p = \frac{dy}{dx}$ is

(a) $y(1 + \cos x) = c$ (b) $y(1 - \cos x) = c$

(c) $x = 2 \sin^{-1} \sqrt{\frac{c}{2y}}$ (d) $x = 2 \sin^{-1} (\sqrt{2y}) + C$

41. The solution of $\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2xy}$ satisfying $y(1) = 1$ is

given by

(a) a hyperbola (b) a circle

(c) $y^2 = x(1+x) - 1$ (d) $(x-2)^2 + (y-3)^2 = 5$

42. Solution of the differential equation :

$$(3 \tan x + 4 \cot y - 7) \sin^2 y \, dx - (4 \tan x + 7 \cot y - 5) \cos^2 x \, dy = 0 \text{ is}$$

(a) $\frac{3}{2} \cot^2 x - 7 \cot x + \frac{7}{2} \tan^2 y - 5 \tan y + 4 \cot x \cdot \tan y = c$

(b) $\frac{3}{2} \tan^2 x - 7 \tan x + \frac{7}{2} \cot^2 y - 5 \cot y + 4 \tan x \cdot \cot y = c$

(c) $3 \tan^2 y - 14 \cot x \cdot \tan^2 y + 7 \cot^2 x - 10 \tan y \cot^2 x + 8 \cot x \cdot \tan y + 2c \cot^2 x \tan^2 y = 0$

(d) $3 \cot^2 y - 14 \cot x \cdot \cot^2 y + 7 \cot^2 x + 10 \cot y \tan^2 x + 8 \tan x \cdot \cot y = 0$

Numerical Value Type Questions

43. If the solution of differential equation

$$x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} = 12y \text{ is } y = Ax^m + Bx^{-n} \text{ then find the value}$$

of $m+n$, if $m \& n \in \mathbb{N}$.

44. A rumour spreads through a population of 5000 people at a rate proportional to the product of the number of people who have heard it and the number who have not. Suppose that 100 people initiate the rumour and that a total of 500 people know the rumour after 2 days. How many days will it take for half the people to hear the rumour (round off to nearest integer)

$$(\log 9 / \log 49 = 129/229).$$

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

Column I Diff. equation	Column II Solution curve
(A) $(1+xy) \, xdy + (1-xy) \, ydx = 0$	(P) $\frac{x^3}{3} + \frac{y^3}{3} - 2x^2y - 2xy^2 = c$
(B) $(x^2 - 4xy - 2y^2) \, dx + (y^2 - 4xy - 2x^2) \, dy = 0$	(Q) $\frac{-1}{xy} + \log \left(\frac{y}{x} \right) = c$
(C) $e^y \cdot dx + (xe^y - 2y) \, dy = 0$	(R) $\sqrt{x^2 + y^2} - x/y = c$
(D) $\frac{xdx + ydy}{\sqrt{x^2 + y^2}} = \frac{ydx - xdy}{y^2}$	(S) $xe^y - y^2 = c$

The correct matching is :

(a) (A-Q; B-P; C-S; D-R)

(b) (A-P; B-S; C-S; D-R)

(c) (A-P; B-S; C-Q; D-R)

(d) (A-Q; B-P; C-R; D-S)



Using the following passage, solve Q.46 and Q.47

Passage

Consider the differential equation $e^x(ydx - dy) = e^{-x}(ydx + dy)$. Let $y = f(x)$ be a particular solution to this differential equation which passes through the point $(0, 2)$.

Let $C \equiv y = \log_{1/4} \left(x - \frac{1}{4} \right) + \frac{1}{2} \log_4(16x^2 - 8x + 1)$, be another curve

46. The range of the function $g(x) = \log_2(f(x))$ is

- (a) $[1, \infty)$ (b) $[2, \infty)$
(c) $[0, \infty)$ (d) None of these

47. If the area bounded by the curve $y = f(x)$, curve C, ordinate

$x = 1/4$ & the ordinate $x = a$ is $4 - \ln 4 + \frac{1}{e^{1/4}} - e^{1/4}$, then

value of a is

- (a) $\ln 6$ (b) $\ln 4$
(c) 4 (d) $\ln 12$

Text

48. A normal is drawn at a point $P(x, y)$ of a curve meets the x -axis at Q . If PQ is of constant length k , then show that the differential equation of the curve is

$$y \frac{dy}{dx} = \pm \sqrt{k^2 - y^2}$$

49. If $(a + bx)e^{y/x} = x$ then prove that

$$x^3 \frac{d^2y}{dx^2} = \left(x \frac{dy}{dx} - y \right)^2$$

50. A curve passing through the point $(1, 1)$ has the property that the perpendicular distance of the origin from the normal at any point P of the curve is equal to the distance of P from the x -axis. Determine the equation of the curve.



EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTIONS

Objective Questions I [Only one correct option]

1. If $x^2 + y^2 = 1$, then (2000)

- (a) $yy'' - 2(y')^2 + 1 = 0$
 (b) $yy'' + (y')^2 + 1 = 0$
 (c) $yy'' + (y')^2 - 1 = 0$
 (d) $yy'' + 2(y')^2 + 1 = 0$

2. If $y(t)$ is a solution of $(1+t)\frac{dy}{dt} - ty = 1$ and $y(0) = -1$, then $y(1)$ is equal to (2003)

- (a) $-\frac{1}{2}$ (b) $\frac{1}{2}$
 (c) 1 (d) -1

3. If $y = y(x)$ and $\frac{2 + \sin x}{y+1} \left(\frac{dy}{dx} \right) = -\cos x$, $y(0) = 1$, then

$y\left(\frac{\pi}{2}\right)$ equals : (2004)

- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
 (c) $-\frac{1}{3}$ (d) 1

4. The solution of primitive equation

$$(x^2 + y^2)dy = xy dx, \text{ is } y = y(x).$$

If $y(1) = 1$ and $y(x_0) = e$, then x_0 is

(2005)

- (a) $\sqrt{\frac{e^2 + 1}{2}}$ (b) $\sqrt{2(e^2 + 1)}$
 (c) $\sqrt{2(e^2 - 1)}$ (d) $\sqrt{3} e$

5. For the primitive integral equation

$$y dx + y^2 dy = x dy; x \in \mathbb{R}, y > 0,$$

$y = y(x)$, $y(1) = 1$, then $y(-3)$ is

(2005)

- (a) 3 (b) 5
 (c) 1 (d) 2

6. Let $f(x)$ be differentiable on the interval $(0, \infty)$ such that

$$f(1) = 1, \text{ and } \lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1 \text{ for each } x > 0. \text{ Then}$$

$f(x)$ is :

(2007)

- (a) $\frac{1}{3x} + \frac{2x^2}{3}$ (b) $-\frac{1}{3x} + \frac{4x^2}{3}$
 (c) $-\frac{1}{x} + \frac{2}{x^2}$ (d) $\frac{1}{x}$

7. The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ determines a

family of circles with :

(2007)

- (a) variable radii and a fixed centre at $(0, 1)$
 (b) variable radii and a fixed centre at $(0, -1)$
 (c) fixed radius 1 and variable centres along the x -axis
 (d) fixed radius 1 and variable centres along the y -axis

8. The function $y = f(x)$ is the solution of the differential

$$\text{equation } \frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1-x^2}}$$

in $(-1, 1)$ satisfying $f(0) = 0$. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x) dx$

is

(2014)

- (a) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$ (b) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$
 (c) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (d) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

9. If $y = y(x)$ satisfies the differential equation

$$8\sqrt{x} \left(\sqrt{9 + \sqrt{x}} \right) dy = \left(\sqrt{4 + \sqrt{9 + \sqrt{x}}} \right)^{-1} dx, \quad x > 0$$

and $y(0) = \sqrt{7}$, then $y(256) =$

(2017)

- (a) 80 (b) 9
 (c) 16 (d) 3



Objective Questions II [One or more than one correct option]

10. A curve $y = f(x)$ passes through $(1, 1)$ and tangent at $P(x, y)$ cuts the x -axis and y -axis at A and B respectively such that $BP : AP = 3 : 1$, then (2006)
- (a) equation of curve is $xy' - 3y = 0$
- (b) normal at $(1, 1)$ is $x + 3y = 4$
- (c) curve passes through $(2, 1/8)$
- (d) equation of curve is $x^3y - 1 = 0$
11. Let $y(x)$ be a solution of the differential equation $(1 + e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statement is (are) true? (2015)
- (a) $y(-4) = 0$
- (b) $y(-2) = 0$
- (c) $y(x)$ has a critical point in the interval $(-1, 0)$
- (d) $y(x)$ has no critical point in the interval $(-1, 0)$
12. Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circle is represented by the differential equation $Py'' + Qy' + 1 = 0$, where P, Q are functions of x, y and $y' \left(\text{here } y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2} \right)$, then which of the following statements is (are) true? (2015)
- (a) $P = y + x$
- (b) $P = y - x$
- (c) $P + Q = 1 - x + y + y' + (y')^2$
- (d) $P - Q = x + y - y' - (y')^2$
13. A solution curve of the differential equation $(x^2 + xy + 4x + 2y + 4) \frac{dy}{dx} - y^2 = 0, x > 0$, passes through the point $(1, 3)$. Then the solution curve (2016)
- (a) intersects $y = x + 2$ exactly at one point
- (b) intersects $y = x + 2$ exactly at two points
- (c) intersects $y = (x + 2)^2$
- (d) does **NOT** intersect $y = (x + 3)^2$

14. Let Γ denote a curve $y = f(x)$ which is in the first quadrant and let the point $(1, 0)$ lie on it. Let the tangent to Γ at a point P intersect the y -axis at Y_P . If PY_P has length 1 for each point P on Γ . Then which of the following options is/are correct? (2019)

(a) $y = -\ln \left(\frac{1 + \sqrt{1 - x^2}}{x} \right) + \sqrt{1 - x^2}$

(b) $xy' + \sqrt{1 - x^2} = 0$

(c) $xy' - \sqrt{1 - x^2} = 0$

(d) $y = \ln \left(\frac{1 + \sqrt{1 - x^2}}{x} \right) - \sqrt{1 - x^2}$

15. For any real numbers α and β , let $y_{\alpha, \beta}(x), x \in \mathbb{R}$, be the solution of the differential equation $\frac{dy}{dx} + \alpha y = xe^{\beta x}$, $y(1) = 1$. Let $S = \{y_{\alpha, \beta}(x) : \alpha, \beta \in \mathbb{R}\}$. Then which of the following functions belong(s) to the set S ? (2021)

(a) $f(x) = \frac{x^2}{2}e^{-x} + \left(e - \frac{1}{2}\right)e^{-x}$

(b) $f(x) = -\frac{x^2}{2}e^{-x} + \left(e + \frac{1}{2}\right)e^{-x}$

(c) $f(x) = \frac{e^x}{2} \left(x - \frac{1}{2}\right) + \left(e - \frac{e^2}{4}\right)e^{-x}$

(d) $f(x) = \frac{e^x}{2} \left(\frac{1}{2} - x\right) + \left(e + \frac{e^2}{4}\right)e^{-x}$



Numerical Value Type Questions

16. Let f be a real-valued differentiable function on $\mathbb{R}$ (the set of all real numbers) such that $f(1) = 1$. If the y -intercept of the tangent at any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of the abscissa of P , then the value of $f(-3)$ is equal to.... (2010)

17. Let $f: [1, \infty) \rightarrow [2, \infty)$ be a differential function such that $f(1) = 2$. If $6 \int_1^x f(t) dt = 3x f(x) - x^3 - 5$,
Then $f(2) =$ (2011)

Assertion & Reason

- (A) If ASSERTION is true, REASON is true, REASON is a correct explanation for ASSERTION.
(B) If ASSERTION is true, REASON is true, REASON is not a correct explanation for ASSERTION.
(C) If ASSERTION is true, REASON is false.
(D) If ASSERTION is false, REASON is true.

18. Let a solution $y = y(x)$ of the differential equation

$$x\sqrt{x^2-1} dy - y\sqrt{y^2-1} dx = 0 \text{ Satisfy } y(2) = \frac{2}{\sqrt{3}}.$$

(2008)

Assertion : $y(x) = \sec \left(\sec^{-1} x - \frac{\pi}{6} \right)$

Reason : $y(x)$ is given by $\frac{1}{y} = \frac{2\sqrt{3}}{x} - \sqrt{1 - \frac{1}{x^2}}$

- (a) A (b) B
(c) C (d) D

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

- | 19. Column I | Column II |
|--|----------------|
| (A) $\int_0^{\pi/2} (\sin x)^{\cos x} (\cos x \cot x - \log(\sin x)^{\sin x}) dx$ | (P) 1 |
| (B) Area bounded by $4y^2 = -x$ and $x-1 = -5y^2$ | (Q) 0 |
| (C) Cosine of the angle of intersection of curves $y = 3^{x-1} \log x$ and $y = x^x - 1$ is | (R) $Ke^{y/2}$ |
| (D) Solution of differential equation $\frac{dy}{dx} = \frac{2}{(x+y)}$ is $x + y + 2 = \dots$ | (S) $4/3$ |
- (2006)

The correct matching is :

- (a) (A-P; B-P; C-S; D-R)
(b) (A-P; B-S; C-R; D-P)
(c) (A-P; B-S; C-R; D-S)
(d) (A-P; B-S; C-P; D-R)

Text

20. A curve passes through (2, 0) and slope at point $P(x, y)$ is $\frac{(x+1)^2 + (y-3)}{(x+1)}$. Find equation of curve and area between curve and x -axis in 4th quadrant. (2004)

Find Answer Key and Detailed Solutions at the end of this book



DIFFERENTIAL EQUATIONS



Please share your valuable feedback by scanning the QR code.



Answer Key



CHAPTER - 5 | INDEFINITE INTEGRATION

EXERCISE - 1: BASIC OBJECTIVE QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|---------|---------|----------|---------|----------|
| 1. (b) | 2. (b) | 3. (a) | 4. (d) | 5. (a) |
| 6. (c) | 7. (c) | 8. (c) | 9. (c) | 10. (d) |
| 11. (a) | 12. (b) | 13. (d) | 14. (d) | 15. (b) |
| 16. (b) | 17. (c) | 18. (b) | 19. (a) | 20. (a) |
| 21. (a) | 22. (b) | 23. (a) | 24. (b) | 25. (b) |
| 26. (a) | 27. (a) | 28. (b) | 29. (b) | 30. (a) |
| 31. (b) | 32. (a) | 33. (b) | 34. (c) | 35. (b) |
| 36. (c) | 37. (a) | 38. (b) | 39. (c) | 40. (a) |
| 41. (c) | 42. (b) | 43. (a) | 44. (b) | 45. (a) |
| 46. (d) | 47. (a) | 48. (b) | 49. (b) | 50. (a) |
| 51. (b) | 52. (b) | 53. (c) | 54. (c) | 55. (c) |
| 56. (c) | 57. (a) | 58. (a) | 59. (d) | 60. (b) |
| 61. (b) | 62. (b) | 63. (b) | 64. (c) | 65. (d) |
| 66. (c) | 67. (3) | 68. (8) | 69. (1) | 70. (9) |
| 71. (4) | 72. (2) | 73. (3) | 74. (6) | 75. (8) |
| 76. (8) | 77. (2) | 78. (36) | 79. (1) | 80. (10) |

EXERCISE - 2: PREVIOUS YEAR JEE MAIN QUESTIONS

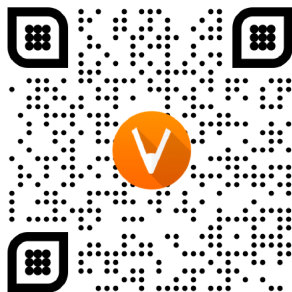


DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|---------|----------|---------|---------|---------|
| 1. (b) | 2. (b) | 3. (d) | 4. (a) | 5. (b) |
| 6. (b) | 7. (b) | 8. (b) | 9. (b) | 10. (c) |
| 11. (b) | 12. (b) | 13. (c) | 14. (c) | 15. (d) |
| 16. (a) | 17. (a) | 18. (c) | 19. (c) | 20. (b) |
| 21. (c) | 22. (d) | 23. (a) | 24. (b) | 25. (a) |
| 26. (d) | 27. (c) | 28. (b) | 29. (a) | 30. (a) |
| 31. (a) | 32. (a) | 33. (d) | 34. (a) | 35. (c) |
| 36. (b) | 37. (15) | 38. (7) | 39. (c) | 40. (3) |
| 41. (c) | 42. (b) | 43. (a) | 44. (6) | 45. (d) |

CHAPTER - 5 | INDEFINITE INTEGRATION

EXERCISE - 3 :
ADVANCED OBJECTIVE QUESTIONS**DIRECTION TO USE -**

Scan the QR code and check detailed solutions.

1. (a) 2. (d) 3. (c) 4. (c) 5. (c)
 6. (b) 7. (b) 8. (b) 9. (a) 10. (c)
 11. (b) 12. (b) 13. (b) 14. (b) 15. (b)
 16. (b) 17. (c) 18. (b) 19. (c) 20. (a)
 21. (b) 22. (d) 23. (c) 24. (b) 25. (c)
 26. (c) 27. (c) 28. (a) 29. (b) 30. (d)
 31. (a) 32. (b) 33. (b) 34. (d) 35. (a)
 36. (a) 37. (a) 38. (c) 39. (c) 40. (c)
 41. (a) 42. (d) 43. (a) 44. (a,b,c,d)
 45. (a,b) 46. (b,d) 47. (c,d) 48. (2240) 49. (1998)
 50. (1982) 51. (1) 52. (2) 53. (d) 54. (c)
 55. (c) 56. (b) 57. (b) 58. (c) 59. (d)
 60. (d) 61. (d) 62. (a)

$$63. -\frac{1}{2} \log|x+1| + \frac{1}{4} \log(x^2+1) + \frac{3}{2} \tan^{-1} x + \frac{x}{x^2+1} + c$$

$$64. (x+1) \tan^{-1} \left(\frac{2x+2}{3} \right) - \frac{3}{4} \log(4x^2+8x+13) + c$$

$$65. -\ln \left(\frac{1}{x+1} - \frac{1}{2} + \frac{x+1}{\sqrt{x^2+x+1}} \right) + C$$

EXERCISE - 4 :
PREVIOUS YEAR JEE ADVANCED QUESTIONS**DIRECTION TO USE -**

Scan the QR code and check detailed solutions.

1. (a) 2. (c) 3. (c) 4. (a,c) 5. (d)

Answer Key



CHAPTER - 6 | DEFINITE INTEGRATION, AREA UNDER CURVES

EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|----------|----------|---------|----------|---------|
| 1. (c) | 2. (c) | 3. (b) | 4. (b) | 5. (b) |
| 6. (c) | 7. (b) | 8. (b) | 9. (b) | 10. (c) |
| 11. (c) | 12. (c) | 13. (c) | 14. (c) | 15. (d) |
| 16. (b) | 17. (c) | 18. (a) | 19. (c) | 20. (a) |
| 21. (a) | 22. (c) | 23. (a) | 24. (c) | 25. (d) |
| 26. (d) | 27. (b) | 28. (b) | 29. (b) | 30. (d) |
| 31. (d) | 32. (a) | 33. (b) | 34. (b) | 35. (b) |
| 36. (d) | 37. (c) | 38. (a) | 39. (c) | 40. (b) |
| 41. (b) | 42. (c) | 43. (a) | 44. (c) | 45. (d) |
| 46. (d) | 47. (a) | 48. (c) | 49. (a) | 50. (c) |
| 51. (2) | 52. (2) | 53. (1) | 54. (2) | 55. (4) |
| 56. (3) | 57. (3) | 58. (2) | 59. (50) | 60. (4) |
| 61. (19) | 62. (29) | 63. (2) | 64. (d) | 65. (a) |
| 66. (b) | 67. (d) | 68. (b) | 69. (b) | 70. (a) |
| 71. (b) | 72. (a) | 73. (c) | 74. (a) | 75. (b) |
| 76. (b) | 77. (c) | 78. (d) | 79. (a) | 80. (c) |
| 81. (b) | 82. (2) | 83. (2) | 84. (4) | 85. (3) |
| 86. (6) | 87. (1) | 88. (6) | 89. (3) | 90. (2) |

EXERCISE - 2 : PREVIOUS YEAR JEE MAIN QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|-----------|------------|-----------|------------|------------|
| 1. (a) | 2. (c) | 3. (c) | 4. (b) | 5. (c) |
| 6. (b) | 7. (a) | 8. (a) | 9. (a) | 10. (c) |
| 11. (b) | 12. (c) | 13. (c) | 14. (d) | 15. (a) |
| 16. (d) | 17. (d) | 18. (b) | 19. (a) | 20. (c) |
| 21. (d) | 22. (18) | 23. (a) | 24. (b) | 25. (d) |
| 26. (d) | 27. (a) | 28. (c) | 29. (a) | 30. (b) |
| 31. (c) | 32. (d) | 33. (d) | 34. (1.50) | 35. (1) |
| 36. (a) | 37. (d) | 38. (d) | 39. (c) | 40. (a) |
| 41. (b) | 42. (21) | 43. (a) | 44. (b) | 45. (a) |
| 46. (c) | 47. (a) | 48. (b) | 49. (c) | 50. (d) |
| 51. (d) | 52. (d) | 53. (c) | 54. (d) | 55. (d) |
| 56. (b) | 57. (d) | 58. (c) | 59. (d) | 60. (c) |
| 61. (c) | 62. (1) | 63. (5) | 64. (b) | 65. (b) |
| 66. (b) | 67. (a) | 68. (d) | 69. (c) | 70. (d) |
| 71. (d) | 72. (d) | 73. (b) | 74. (a) | 75. (5) |
| 76. (4) | 77. (a) | 78. (c) | 79. (c) | 80. (3) |
| 81. (d) | 82. (b) | 83. (d) | 84. (b) | 85. (a) |
| 86. (19) | 87. (d) | 88. (a) | 89. (2) | 90. (b) |
| 91. (1) | 92. (406) | 93. (1) | 94. (16) | 95. (c) |
| 96. (d) | 97. (d) | 98. (1) | 99. (a) | 100. (a) |
| 101. (1) | 102. (512) | 103. (c) | 104. (8) | 105. (b) |
| 106. (d) | 107. (a) | 108. (a) | 109. (b) | 110. (d) |
| 111. (d) | 112. (b) | 113. (c) | 114. (d) | 115. (c) |
| 116. (d) | 117. (b) | 118. (d) | 119. (d) | 120. (b) |
| 121. (c) | 122. (b) | 123. (b) | 124. (b) | 125. (b) |
| 126. (c) | 127. (d) | 128. (b) | 129. (b) | 130. (d) |
| 131. (b) | 132. (b) | 133. (d) | 134. (c) | 135. (b) |
| 136. (a) | 137. (b) | 138. (a) | 139. (1) | 140. (a) |
| 141. (d) | 142. (b) | 143. (2) | 144. (d) | 145. (114) |
| 146. (27) | 147. (d) | 148. (26) | 149. (d) | 150. (b) |
| 151. (64) | 152. (4) | 153. (b) | 154. (41) | 155. (b) |

CHAPTER - 6 | DEFINITE INTEGRATION, AREA UNDER CURVES

EXERCISE - 3 :
ADVANCED OBJECTIVE QUESTIONS**DIRECTION TO USE -**

Scan the QR code and check detailed solutions.

- | | | | | |
|-------------|-------------|---|------------|---------|
| 1. (d) | 2. (a) | 3. (d) | 4. (c) | 5. (d) |
| 6. (d) | 7. (c) | 8. (c) | 9. (d) | 10. (d) |
| 11. (a) | 12. (c) | 13. (d) | 14. (b) | 15. (d) |
| 16. (d) | 17. (b) | 18. (b) | 19. (a) | 20. (d) |
| 21. (c) | 22. (c) | 23. (a) | 24. (d) | 25. (b) |
| 26. (a) | 27. (c) | 28. (a) | 29. (c) | 30. (d) |
| 31. (d) | 32. (d) | 33. (c) | 34. (a) | 35. (c) |
| 36. (b) | 37. (c) | 38. (b) | 39. (a) | 40. (d) |
| 41. (c) | 42. (a) | 43. (c) | 44. (d) | 45. (b) |
| 46. (a,c) | 47. (a,b,d) | 48. (a,d) | 49. (b, c) | |
| 50. (a,c,d) | 51. (a,b) | 52. (3) | 53. (c) | |
| 54. (a) | 55. (a) | 56. (a) | 57. (d) | 58. (c) |
| 59. (b) | 60. π^2 | 61. $\frac{1}{2} \log 6 - \frac{1}{10}$ | 62. (b) | |
| 63. (b) | 64. (a) | 65. (a) | 66. (c) | 67. (b) |
| 68. (d) | 69. (d) | 70. (b) | 71. (c) | 72. (c) |
| 73. (d) | 74. (b) | 75. (b) | 76. (a) | 77. (d) |
| 78. (a) | 79. (a) | 80. (a) | | |

EXERCISE - 4 :
PREVIOUS YEAR JEE ADVANCED QUESTIONS**DIRECTION TO USE -**

Scan the QR code and check detailed solutions.

- | | | | | |
|--|---|------------|-------------|-------------|
| 1. (b) | 2. (b) | 3. (c) | 4. (c) | 5. (c) |
| 6. (a) | 7. (c) | 8. (a) | 9. (a) | 10. (d) |
| 11. (b) | 12. (a) | 13. (c) | 14. (a) | 15. (c) |
| 16. (b) | 17. (d) | 18. (a) | 19. (d) | 20. (a) |
| 21. (a,d) | 22. (a,b,c) | 23. (b,d) | 24. (a,c,d) | |
| 25. (a,c) | 26. (b,c) | 27. (a,c) | 28. (a,b) | 29. (a,b,d) |
| 30. (a,b,c) | 31. (5051) | 32. (2) | 33. (0) | |
| 34. (9) | 35. (7) | 36. (1) | 37. (2) | 38. (4) |
| 39. (0.50) | 40. (4) | 41. (182) | 42. (a) | 43. (d) |
| 44. (c) | 45. (b) | 46. (a) | 47. (d) | 48. (a,b,c) |
| 49. (c,d) | 50. (2) | 51. (1.50) | 52. (c) | 53. (d) |
| 54. $\frac{1}{2}(\ln x)^2$ | 55. $\frac{4\pi}{\sqrt{3}} \tan^{-1}\left(\frac{1}{2}\right)$ | | | |
| 56. $\frac{24}{5} \left(\operatorname{ecsc}\left(\frac{1}{2}\right) + \frac{e}{2} \sin\left(\frac{1}{2}\right) - 1 \right)$ | 57. (b) | 58. (c) | | |
| 59. (c) | 60. (a) | 61. (a) | 62. (a,b,d) | 63. (b,d) |
| 64. (3) | 65. (4) | | | |

Answer Key



CHAPTER -7 | DIFFERENTIAL EQUATIONS

EXERCISE - 1: BASIC OBJECTIVE QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|---------|---------|----------|----------|---------|
| 1. (a) | 2. (d) | 3. (c) | 4. (c) | 5. (d) |
| 6. (c) | 7. (a) | 8. (a) | 9. (b) | 10. (c) |
| 11. (b) | 12. (a) | 13. (b) | 14. (c) | 15. (d) |
| 16. (a) | 17. (c) | 18. (b) | 19. (a) | 20. (b) |
| 21. (a) | 22. (a) | 23. (b) | 24. (b) | 25. (d) |
| 26. (a) | 27. (d) | 28. (a) | 29. (d) | 30. (b) |
| 31. (c) | 32. (a) | 33. (b) | 34. (c) | 35. (b) |
| 36. (c) | 37. (b) | 38. (a) | 39. (b) | 40. (c) |
| 41. (b) | 42. (b) | 43. (c) | 44. (b) | 45. (a) |
| 46. (b) | 47. (c) | 48. (d) | 49. (b) | 50. (a) |
| 51. (b) | 52. (b) | 53. (1) | 54. (4) | 55. (3) |
| 56. (2) | 57. (5) | 58. (35) | 59. (20) | 60. (2) |

EXERCISE - 2: PREVIOUS YEAR JEE MAIN QUESTIONS

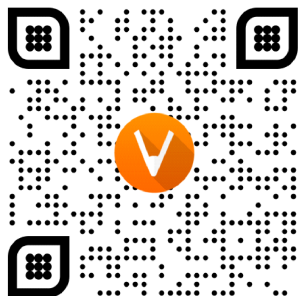


DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|---------|---------|---------|---------|----------|
| 1. (a) | 2. (b) | 3. (b) | 4. (c) | 5. (c) |
| 6. (a) | 7. (b) | 8. (d) | 9. (d) | 10. (d) |
| 11. (d) | 12. (a) | 13. (c) | 14. (c) | 15. (a) |
| 16. (d) | 17. (b) | 18. (b) | 19. (c) | 20. (a) |
| 21. (a) | 22. (a) | 23. (b) | 24. (c) | 25. (b) |
| 26. (c) | 27. (b) | 28. (b) | 29. (b) | 30. (c) |
| 31. (d) | 32. (b) | 33. (b) | 34. (b) | 35. (d) |
| 36. (b) | 37. (c) | 38. (b) | 39. (d) | 40. (c) |
| 41. (c) | 42. (b) | 43. (3) | 44. (a) | 45. (b) |
| 46. (a) | 47. (b) | 48. (2) | 49. (b) | 50. (a) |
| 51. (d) | 52. (c) | 53. (a) | 54. (d) | 55. (a) |
| 56. (b) | 57. (a) | 58. (d) | 59. (9) | 60. (d) |
| 61. (1) | 62. (2) | 63. (c) | 64. (c) | 65. (d) |
| 66. (2) | 67. (4) | 68. (a) | 69. (b) | 70. (16) |
| 71. (2) | 72. (2) | 73. (b) | 74. (c) | 75. (4) |
| 76. (b) | 77. (1) | 78. (d) | 79. (d) | 80. (c) |
| 81. (b) | 82. (c) | 83. (b) | 84. (b) | 85. (b) |
| 86. (c) | 87. (d) | 88. (d) | 89. (1) | 90. (4) |

CHAPTER - 7 | DIFFERENTIAL EQUATIONS

EXERCISE - 3 :
ADVANCED OBJECTIVE QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

1. (d) 2. (a) 3. (c) 4. (d) 5. (a)
6. (b) 7. (a) 8. (d) 9. (b) 10. (b)
11. (b) 12. (d) 13. (b) 14. (a) 15. (a)
16. (b) 17. (c) 18. (a) 19. (a) 20. (b)
21. (b) 22. (c) 23. (a) 24. (b) 25. (c)
26. (c) 27. (b) 28. (a) 29. (c) 30. (c)
31. (a) 32. (a) 33. (d) 34. (a) 35. (d)
36. (a) 37. (a) 38. (d) 39. (a,d) 40. (a,b,c)
41. (a,c) 42. (b,c) 43. (7) 44. (5) 45. (a)
46. (a) 47. (b)
50. $x^2 + y^2 = 2x$ or $x = 1$

EXERCISE - 4 :
PREVIOUS YEAR JEE ADVANCED QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

1. (b) 2. (a) 3. (a) 4. (d) 5. (a)
6. (a) 7. (c) 8. (b) 9. (d) 10. (c,d)
11. (a,c) 12. (b,c) 13. (a,d) 14. (b,d) 15. (a,c)
16. (9) 17. (6) 18. (c) 19. (d)

20. $y = x^2 - 2x, \frac{4}{3} \text{ sq. unit}$

MASTER INDEX

VOLUME 1:

Determinants & Matrices

Relations , Functions & Inverse Trigonometric Functions

Continuity, Differentiability & Differentiation

Applications of Derivatives

VOLUME 2:

Indefinite Integration

Definite Integration, Area under curves

Differential Equations

VOLUME 3:

Vectors & 3-Dimensional Geometry

Probability

Creating Impact at Scale



65 Million+

Monthly
Youtube Views



49 Million+

Monthly
Web + App Visitors



40+ Countries

Where Students
Take Live Classes



25 Million+

Monthly Active
Users



26 Million+

Hours of
Live Learning



25
MILLION+

Parents
Trust
Vedantu

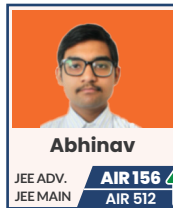
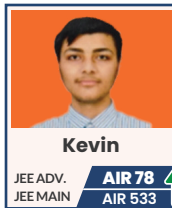
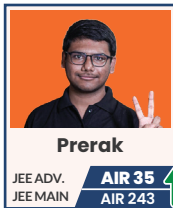


43
MILLION+

Happy
Students

Our Achievers of 2021

JEE Advanced 2021



1500+

Vedantu students ace
JEE Advanced 2021



6%

of IITs' upcoming batch
will be from Vedantu

JEE Main 2021

903

students in
Top 10,000

78

students scored
99.9+ PERCENTILE

869

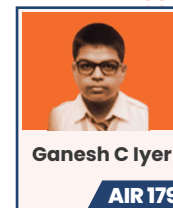
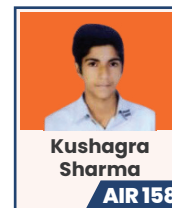
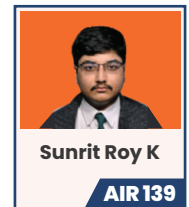
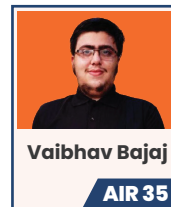
students scored
99.9+ PERCENTILE

68

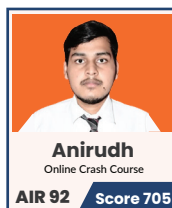
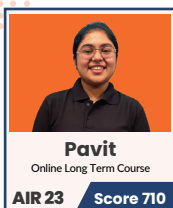
students in
Top AIR 1000

444

students in
Top AIR 5000



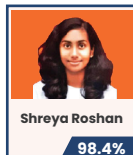
NEET (UG) 2021



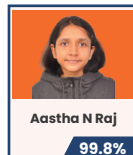
1172

NEET
Qualifiers

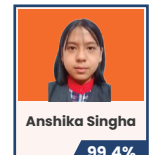
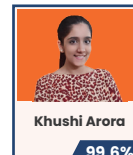
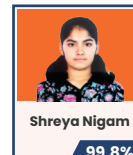
CBSE Class 12



ISC Class 12



CBSE Class 10



ICSE Class 10

